

Game Design Document: Clean Room Remake (TIM2)

Este Game Design Document (GDD) foi elaborado com base no processo de engenharia reversa "Clean Room" do jogo *The Incredible Machine 2* (1994). Toda a documentação aqui contida deriva exclusivamente das especificações do motor físico e regras de negócio extraídas dos dados originais (`.LEV`, `.PAR`, `.ANM`, `SOLVE.RES`), sem o uso de código-fonte original.

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1. Visão Geral (Pitch & Vision)

High Concept

Um jogo de puzzle e simulação física no estilo *Rube Goldberg*, onde o jogador constrói engenhocas complexas em cadeia para resolver objetivos simples (como "acender uma vela" ou "colocar a bola na cesta"). Modernizado com uma nova engine robusta (Godot 4) e ferramentas de compartilhamento online, o jogo resgata a essência de resolução criativa de problemas da era 16-bits.

Pilares de Design

1. Física Determinística ("A Regra de Ouro")

O coração do jogo não é um motor de física realista (como Box2D ou Havok), mas sim um simulador **100% determinístico** rodando em taxa fixa de 60Hz. * **O que isso significa:** Se o jogador não alterar nenhuma peça e clicar no botão "Play" 100 vezes, o resultado (colisões, reações, momento exato de cada evento) deve ser **matematicamente idêntico** em todas as 100 vezes. * **Por que importa:** Sem isso, engenhocas complexas que dependem de precisão milimétrica deixam de funcionar de forma previsível, frustrando o jogador e quebrando o design dos puzzles.

2. Nostalgia vs. Modernidade

O jogo deve manter a estética mecânica e o charme cartunesco dos anos 90, porém envelopado em uma UI/UX moderna. Resoluções amplas (4K e Ultrawide), atalhos de teclado ágeis, controles adaptados para gamepads modernos e zoom in/out livre são melhorias fundamentais que atualizam a experiência sem corromper a essência do gameplay original.

3. Criatividade Sem Limites

O jogo não julga a solução. Se o objetivo era acender uma fogueira e o jogador usou um rato em uma esteira para acionar um motor, que puxa um balde de tesouras para cortar uma corda, soltando uma bola de boliche em um interruptor de laser que incinera dinamite... a solução é válida. O ambiente sandbox (Freeplay) incentiva a criação das engenhocas mais caóticas possíveis.

Estratégia Legal (Clean Room)

Para garantir a comercialização legal (ex: via Steam), o desenvolvimento segue a arquitetura "**Clean Room**". * Toda a lógica, física e regras derivam da especificação de formato binário revertida pela equipe, garantindo não haver uso de código-fonte de terceiros. * Para os *assets* (arte e áudio), o projeto adota inicialmente o **Modelo ScummVM**, onde o jogador deve fornecer os arquivos originais legalmente adquiridos, ou operará com arte 100% nova, evitando qualquer violação de *copyright*.

Público-Alvo e Plataformas

- **Público:** Fãs de jogos retrô, amantes de puzzle (ex: *Portal*, *Baba Is You*, *Zachtronics games*), crianças com interesse em engenharia/física e entusiastas de sandbox.
- **Plataformas Iniciais:** PC (Steam - Windows, Linux, macOS) e Steam Deck.
- **Plataformas Futuras:** Consoles e Mobile (adaptando o esquema de controles *Drag and Drop* para toque).

2. Mecânicas Principais (Core Mechanics)

O funcionamento do jogo divide-se entre dois modos: o **Modo de Edição** (Build Mode) e o **Modo de Simulação** (Play Mode). Durante a simulação, o jogo confia em uma engine física altamente customizada, operando sob regras estritas extraídas do formato original.

O Core Loop

1. **Analisar Objetivo:** O jogador lê a condição de vitória e analisa o cenário pré-montado (peças fixas que não podem ser movidas).
2. **Construir Engenhoca:** Usando um inventário limitado, o jogador arrasta, posiciona, rotaciona e conecta peças no cenário (ex: ligando cordas de objetos a roldanas).
3. **Testar (Play):** O tempo começa a correr. A gravidade age sobre os itens móveis.
4. **Avaliar Resultado:**
5. Se o objetivo for cumprido, a fase é vencida.
6. Se algo der errado, o jogador aperta "Stop", a fase reseta ao estado de edição, e ele reajusta as peças.

A Física Determinística

O motor físico opera isolado dos *frames* visuais do jogador. Ele roda a exatos **60 ticks por segundo (60Hz)** usando matemática de ponto fixo (*fixed-point integer math*, convertida para pixels por $\gg 9$). O uso de inteiros em vez de *floats* previne as discrepâncias de arredondamento causadas por diferentes CPUs.

Variáveis Globais de Nível

Cada fase possui propriedades globais que alteram as leis da física: - **Gravidade (*gravity_y*)**: Pode ser normal, baixa (espaço) ou até invertida. O limite típico é de $\pm 0 \times 2600$ ou $\pm 0 \times 3000$. - **Pressão do Ar (*air_pressure*)**: Afeta balões e a física de propagação de chamas/explosões.

O Loop de Simulação (7 Passes)

O coração da mecânica. Em cada tick de 1/60s, a simulação DEVE executar exatamente os seguintes passes em ordem estrita:

1. **Contadores de Estado:** Atualiza os atrasos de estados das peças (ex: timers de detonação ou transição de animações).
2. **Integração de Gravidade:** Aplica a força gravitacional (tabela de aceleração) em objetos que não estão presos.
Exemplo: a *velocity_y* é aumentada e somada à posição *pos_y*.

3. **Comportamentos Especiais (Behaviors):** Motores giram, canhões atiram, gatos correm atrás de ratos. Regras de movimentação exclusivas de cada peça.
4. **Clamping de Viewport:** Impede que peças saiam das bordas ativas da tela, aplicando salto (`bounciness/COR`) quando batem no chão ou no teto.
5. **Colisões (AABB):** Detecção de sobreposição (*overlap*). Objetos são reposicionados e suas velocidades recebem um impulso vetorial. Exceção: peças interconectadas ignoram colisões entre si.
6. **Proximidade & Elétrica:** Condução de eletricidade através de conectores ligados. Lasers, lanternas e interruptores ativam alvos próximos ou na linha de visão.
7. **Constraint Solver (Corda/Correia):** Passo de correção. Roldanas tensionam cordas e esteiras sincronizam suas velocidades angulares de forma idêntica.

Sistemas de Restrição (Constraints)

A manipulação de transferência mecânica é uma mecânica principal.

- **Cordas e Polias (Ropes & Pulleys):** As cordas possuem comprimento definido (armazenado no campo `behavior` da peça). Um *solver* de passe único corrige a posição dos itens amarrados se eles esticarem a corda além do limite, criando o efeito de "pêndulo" ou balanço em polias.
- **Correias e Engrenagens (Belts & Gears):** Transmitem velocidade angular. O solver soma o torque das peças motrizes (como um motor ligado na tomada) e distribui a mesma velocidade angular padronizada de `±60px/s` para todas as peças escravas conectadas pelo mesmo cinto.

3. Catálogo de Peças (Part Catalog)

O coração criativo e lógico do jogo reside nas suas peças. O motor original abriga **149 peças jogáveis** e **116 peças de cenário** (decorativas).

Devido à magnitude desse catálogo, este documento fornece um guia das categorias e da lógica do sistema. A especificação cirúrgica de cada peça (massa, elasticidade, atrito, estados da animação e raio de colisão extraídos do formato `.PAR`) está catalogada nos 148 documentos individuais localizados na pasta `docs/parts/` e na base de dados `build/phase-9/part-name-catalog.json`.

As 15 Categorias de Peças

O jogo agrupa as peças em categorias que compartilham regras lógicas de simulação, colisões ou tabelas de gravidade:

1. **Força Motriz (Motores e Manivelas):** Peças que geram força angular. Ex: Moinhos de vento, motores elétricos, macaco na bicicleta, rato na roda.
2. **Mecânicos e Conectores:** Engrenagens, correias, alavancas e tesouras que transmitem movimento.
3. **Geradores de Energia / Elétricos:** Painéis solares, interruptores, tomadas e geradores.
4. **Armas e Explosivos:** Canhões, dinamites, mísseis, fogo de artifício. Respondem primariamente a chamadas ou ativação mecânica.
5. **Animais e IA Simples:** Curiosidades físicas como Mel Arnold, o Gato (perseguidor), o Rato (fujão), o Jacaré (comedor).
6. 
7. **Esferas e Projéteis:** Bolas de boliche, bolas de tênis, bolas de bilhar (variam em peso e elasticidade, "bounciness").
8. 
9. **Paredes e Plataformas (Scenery):** Tipos estáticos que servem de chão ou teto. Peças como trampolins e esteiras entram aqui se ancoradas.
10. **Fontes de Fogo:** Fósforos, velas, tochas e lupas.
11. **Fontes de Ar/Vento:** Ventiladores, secadores de cabelo, aspiradores.

12. **Recipientes e Tubos:** Baldes, cestas e o sistema de encanamentos (que transporta peças via teleporte em grafo BFS).
13. **Óptica e Lasers:** Espelhos, prismas, tomadas de laser, misturadores RGB e balões.
14. **Anti-Gravidade e Espaço:** Almofadas flutuantes e amarras espaciais.
15. **Balões e Itens de Pressão:** Balões de gás, dirigíveis (respondem diferentemente dependendo da pressão do ar da fase).
16. **Gatilhos (Triggers):** Interruptores de chão, alavancas puxáveis.
17. **Estruturas Base:** Blocos construtores.

Sistema de Máquina de Estados e Interação (SOLVE.RES)

Diferente de engines modernas onde cada peça tem scripts de colisão customizados, no TIM2 as interações ocorrem através de uma arquitetura estrita de Máquina de Estados. De acordo com a engenharia reversa do formato ANM e do motor de física, existem quatro conceitos interligados que regem isso:

1. **state_counter (Física Interna):** Um contador numérico incrementado automaticamente a cada *tick* (60 Hz) pelo motor do jogo. Ele rege timers, atrasos (como o pavo de uma dinamite) e ciclos de animação.
2. **current_state (Validação):** Uma cópia ou espelho do `state_counter`, utilizada para checar condições de vitória da fase ("Solution") e buscar os estados no ANM.
3. **state_id (Renderização ANM):** Encontrado na "Seção C" do formato `.ANM` (timgres). Ele correlaciona o estado lógico de uma peça a um `first_frame_id`, que aponta qual quadro exato de sprite desenhar na tela e se a peça é ou não animada nesse estado.
4. **Pares de Gatilho (SOLVE.RES):** A tabela de colisão global (`build/phase-16/solve-decoded.json`). Quando duas peças se encostam, o jogo cruza os dados: `{self: state_counter_A, other: state_counter_B} => anm_state_override`

A Resolução de Colisão

Ao colidir, a *engine* lê os `state_counters` de ambas as peças e busca um gatilho válido no SOLVE.RES: * Se der *match*, o `state_counter` da peça sofre um *override* instantâneo para o valor de `anm_state` definido na tabela. Isso força a peça a pular visualmente para um novo `state_id` no ANM e começar uma nova física. * Em alguns casos (`anm_state` nulo na tabela), a peça assume que o estado de animação atual flui diretamente do gatilho (`self`). * Por fim, existem dezenas de "Estados Órfãos" no arquivo ANM que o SOLVE.RES nunca chama; eles atuam puramente como frames de transição em animações contínuas impulsionadas pelo avanço livre do `state_counter`.

Dinâmica do Ciclo de Vida: Início, Transições e Sprites

Para que uma peça ganhe vida de forma determinística, seu comportamento depende do seu estágio na simulação:

1. **Estado Inicial:** Quando o nível é carregado (ou a simulação inicia ao apertar *Play*), cada peça recebe um `state_counter` inicial. Na grande maioria das peças, o valor é cravado em **0** (o estado "Padrão/Desligado"). Algumas peças específicas herdam estados iniciais predefinidos gravados no cabeçalho do arquivo `.LEV` ou definidos pelo jogador no editor (ex: interruptores pré-ligados, canhões invertidos, gravidade do balão).
2. **O Que Ocasiona Cada Transição?** O `state_counter` avança por três gatilhos motores distintos:
3. **Colisões Físicas (SOLVE.RES):** O gatilho mestre. A sobreposição exata de duas caixas delimitadoras (*hitboxes*) faz a engine buscar as regras na tabela (ex: Laser toca no Balão $\rightarrow$ estado transita para "Estourando").
4. **Avanço Temporal (Tick Loop):** O motor a 60 Hz possui um loop incondicional que auto-incrementa o `state_counter` se a peça tiver limites superiores definidos. Isso causa ciclos de animações contínuas (ex: moinhos girando infinitamente) ou atuam como *timers* de atraso (ex: a chama sobe pelo pavo da dinamite *tick* até o limite de explosão).
5. **Estímulos Lógicos/Mecânicos:** Peças ligadas a redes elétricas (fios) ou mecânicas (cordas, correias, engrenagens) sofrem transições de estado remotamente quando o torque ou a energia do sistema muda (ex: a correia transmite força e altera o estado das polias atreladas).

6. **Os Sprites de Cada Estado:** No motor original, um sprite nunca é chamado "manualmente"; ele é um reflexo submisso do estado através das Seções do arquivo `.ANM`:
7. A **Seção C** do formato cruza a variável global com o design visual: ela diz "O estado `2` corresponde ao conjunto de frames `X`".
8. A **Seção B** define o tamanho, largura, altura e o deslocamento de tela desses frames.
9. A **Seção D** contém as cores em pixel art puras.
10. **No Clean Room (Godot):** A equipe de desenvolvimento traduzirá essa cadeia complexa para um modelo limpo. Cada peça utilizará um `AnimatedSprite2D` (ou `AnimationPlayer`) que conterá animações nomeadas exatamente como os IDs lógicos (ex: `"state_0"`, `"state_1"`, `"state_2"`). O único papel do script de renderização da Godot será escutar as alterações da variável física `state_counter` e dar um simples `.play("state_" + str(state_counter))`.

O Papel do Design Limpo (Clean Room)

Ao implementar cada peça, desenvolvedores da Godot devem consultar **exclusivamente** os dados da pasta `docs/parts/`, cruzando sempre a lógica com a especificação canônica em `tim2_specs/state-machine-specification.md`. Nós não portamos código de C sujo para GDScript. O correto é traduzir a semântica: programamos o loop de 60 Hz avançando o `state_counter` e deixamos as reações de colisão buscarem o novo estado através de uma cópia da tabela `SOLVE.RES` estática, mantendo o determinismo 100% idêntico ao jogo de 1993.

4. UI/UX e Controles

A interface de usuário de um simulador puzzle-sandbox é o elemento que determina se o jogador se sentirá no controle de uma máquina complexa ou frustrado brigando com caixas.

A implementação da UI em Godot tem o luxo de ser livre de amarras de motor antigo, desde que mantenha a coerência mecânica descrita no `tim2_specs/user-journey-specification.md`.

HUD do Modo Puzzle

A tela deve prover ao jogador visibilidade máxima da "Área do Puzzle" e rápido acesso às peças.

1. **Painel de Catálogo/Inventário:** Localizado nas laterais da tela (usualmente à direita ou inferior).
2. O catálogo de peças apresenta o inventário limitado disponível para aquela fase.
3. Ícones das peças em "caixas" renderizando seu primeiro frame (*idle*).
4. Indicação numérica das peças restantes.
5. **Barra de Controle de Simulação:**
6. **Play (Teste):** Congela o Modo Edição e inicia a simulação física.
7. **Stop/Reset:** Finaliza a simulação e devolve todas as peças para o exato estado em que estavam antes do clique em Play.
8. **Fast Forward:** Um botão *quality-of-life* moderno que acelera os 60Hz originais, útil para aguardar roldanas girarem até o topo de uma tela lenta.
9. **Display de Informações de Fase:**
10. Botão para rever o Objetivo e as Dicas (*Hints*).
11. Elementos de leitura da fase: Indicador global de Gravidade (Normal/Espaço) e Pressão do Ar.

Mapeamento de Controles Modernizados

1. Mouse & Teclado (Core PC)

O fluxo clássico validado em `tim2_editor/main.py`. - **Drag & Drop:** Clique esquerdo para segurar, arrastar e soltar (Snap na grade opcional, tipicamente de 16px). - **Conectar (Cabos e Eletricidade):** Modos de clique dedicados para

criar amarras ou ligar interruptores em tomadas elétricas. O *feedback* visual (cor da conexão) é fundamental aqui. - **Atalhos (Hotkeys):** Rotação rápida de peça (Espaço), deletar peça (Del), Pan pela área do puzzle (Middle-click drag).

2. Gamepad (Consoles e Steam Deck)

Adaptar a jogabilidade original "point-and-click" exige sistemas modernos: - **Cursor Magnético / Assistência:** O analógico esquerdo não apenas mimetiza o mouse, mas aplica uma atração gravitacional quando o cursor está perto de um conector de roldana ou um botão de UI. - **Menus Radiais:** Em vez de mover o cursor cruzando toda a tela até o inventário lateral, apertar `Y` ou `Triângulo` abre um menu radial rápido agrupando peças da fase pelo cursor. - **Direcionais (D-Pad):** Alterna as opções de conectores de uma engrenagem, por exemplo.

3. Touch/Mobile

- **Pinch-to-Zoom:** Essencial para fases imensas que passam o tamanho de uma tela.
- **Tap & Hold:** Exibe informações ("Tooltips") da peça, atributos de massa ou atrito.
- **Arrastar Flutuante:** Para evitar que o dedo tampe a peça ao ser arrastada, o ícone sob posse flutua ligeiramente acima do ponto de toque físico.

Fluxo de "Jornada do Usuário"

Toda transição de tela deve respeitar os 14 diagramas contidos em `user-journey-specification.md`, da tela de Assinatura inicial (*J01*) à passagem pelo Seletor de Níveis com separação de dificuldade e tela final de Vitória, certificando-se de reproduzir as corretas instâncias do fluxo clássico.

5. Design de Fases (Level Design)

A integridade do jogo como produto final está em apresentar os desafios lógicos de maneira equilibrada. A conversão limpa resgatou todos os **206 níveis originais**, que já contêm curva de dificuldade aprovada.

Progressão e Estrutura de Dificuldade

A organização de fases foi extraída da taxonomia do `game.py`.

1. **Tutoriais:** Micro-cenários onde apenas uma categoria é explorada de cada vez (ex: gravidade com esferas, como ligar polias).
2. **Easy:** Cenários lineares com soluções de um único caminho claro.
3. **Medium:** Níveis em que o jogador tem *red herrings* (peças disponíveis no inventário que não são necessárias para completar a fase) para induzir falso raciocínio.
4. **Hard e Really Hard:** Fases com peças que operam fora da tela ou que dependem do limite de cordas complexas e física interdependente fina de múltiplos alvos.
5. **Head-to-Head (Multiplayer):** Níveis balanceados que suportam competição entre dois jogadores de forma assimétrica.

O Formato de Fase (A Abstração YAML)

O projeto base descartou trabalhar o motor em cima dos dados binários crus (`.LEV`) como padrão e consolidou as fases num repositório YAML (`build/phase-5/yaml/`).

O Level Loader (`LevelLoader.gd` na arquitetura final) deve processar e carregar dados dessa forma:

- **Metadados:** Título da fase, Texto do Objetivo, Música a tocar, Dicas (*Hints*) ocultas.
- **Física da Fase:** `gravity` global e `pressure`.
- **fixed_parts :** Arrays contendo tipo da peça (`part_type`), posição (X/Y no plano pixel) e flags de orientação espacial. Elas são a base fixa onde a jogabilidade acontece. A simulação da colisão não pode empurrá-las.

- **moving_parts**: Array idêntico de peças que respondem à gravidade e física ou podem ser manipuladas livremente em modo *sandbox*.

O Avaliador de Vitória (Solution Condition)

Ao contrário dos quebra-cabeças modernos limitados ao toque em uma "bandeira final", o mecanismo avaliador reversado mapeia vitória por *Estados Disparadores*. O sistema opera através do `SolutionCondition` (referenciado em `docs/godot_game_spec/03_data_structures.md`).

Tipos de validação de vitória: 1. **Pelo Estado de Peça (state_match)**: Condição clássica "Acenda a vela". A verificação valida se o objeto da ID Específica (Vela) passou do seu estado de ID 0 para seu estado ID 1. 2. **Overlap Posicional (AABB overlap)**: Condição clássica "Faça a bola cair no cesto". Um retângulo virtual invisível (Retângulo Alvo) é instanciado na fase, atuando como um *trigger volume* moderno. O motor checa continuamente se o centro do `part_type` (ex: bola de basquete) se encontra na bounding box especificada. 3. **Condições Múltiplas**: Certos níveis Hard exigem que os estados acima sejam todos atingidos simultaneamente para acionar a janela de vitória.

6. Direção de Arte e Áudio

O projeto tem restrições técnicas vitais advindas de sua estratégia jurídica e histórica ("Clean Room"), forçando o departamento de arte e design de som a operarem em moldes diferentes do desenvolvimento tradicional.

Art Bible e Arte Clean Room

Existem duas rotas implementáveis na direção de arte. O *Game Design* base deve comportar ambas:

1. **A Abordagem "Port de Engine" (Modelo ScummVM)**: O jogo provê **zero arte visual original**. O jogador providencia o recurso legal `RESOURCE.001` do MS-DOS.
2. **Renderizador Opcodes (ANMPlayer.gd)**: O backend reconstrói os mais de 3.000 *sprites* extraídos por processamento cru, sendo essencial possuir um código que suporte 6 opcodes base (A, B, C, D) ditados por `tim2_specs/experiments/R-002-anm-format.md`. Nenhuma textura é modificada, respeitando paletas de 256 cores originais (`PARTTEMP.json`).
3. **A Abordagem "Remake HD" (Propriedade Intelectual Nova)**: Caso a rota escolhida seja evitar o requisito de arquivos originais, toda a arte precisa ser recriada.
4. **Restrição de Gameplay**: Apesar de poder re-imaginar um gato, os **hitboxes (raios de colisão) e offset (x/y center)** devem obrigatoriamente manter os tamanhos ditados na base do projeto (`build/phase-9/part-properties.json`) em ponto fixo (`AABB`). Se o raio de colisão da engrenagem nova no remake em HD for maior que a clássica em 1 pixel, **fases antigas vão quebrar**.
5. **Direção de Estilo**: Sugere-se uma arte limpa, vetorial ou 2.5D com shaders cartunescos, mantendo as animações expressivas e exageradas (*slapstick comedy*).

Audio Design Determinístico

O sistema de áudio difere da física porque a simulação física não pode jamais depender de um frame de áudio terminando.

- **Dispatcher Reativo**: O áudio é tratado exclusivamente como "Side-Effect" (efeito colateral) da mudança de estado e atrito de colisão do backend de mecânicas. Baseando-se no `R-016-sound-param-table.md`, cada peça (seja 14 FM Patches OPL2 ou os 113 sons Waveform de PCM cru 8-bit/11kHz) tem um trigger associado e despacha um som sem bloquear o fio da física.
- **Exemplo de Pipeline Musical**: Se o cenário determina *Laser Firing*, o código avança o state e aciona uma flag global; o *AudioDispatcher* "ouve" a flag no próximo tick visual de renderização de Godot e aciona o áudio local correspondente.

- **Remasterização:** Trilhas de fundo deverão ser repensadas caso se afaste o modelo ScummVM (Option A), recriando temas de synth-jazz adequados para ambientes sem distrair a lógica complexa do player construindo seus cenários.

7. Recursos Modernos e Online

Embora a espinha dorsal de simulação permaneça idêntica ao design de 1994, o remake de The Incredible Machine 2 adota padrões contemporâneos para garantir a retenção de público no ecossistema atual de PC e Consoles.

Sistema de Compartilhamento (Workshop)

O valor principal do jogo reside na expressão mecânica infinita e desafios da comunidade.

- **Fases Definidas pelo Usuário (UGC):** Graças à abstração de serialização (descrita no motor de fase `YAML` em `level.py`), uma fase de usuário gerada no "Editor de Nível Livre" exporta um arquivo `.yaml` limpo contendo as peças soltas, condições de vitória customizadas e física padrão.
- **Integração com Plataformas:** Conexão nativa com **Steam Workshop**. Jogadores podem dar *Upload* dos seus quebra-cabeças. O cliente de jogo embutido tem a seção *Community Browser* com filtros de "Melhor Avaliados", "Mais Desafiadores", "Engenhocas Cômicas".

Quality of Life (Melhorias de Interface)

A recriação introduz melhorias essenciais para evitar fadiga cognitiva: * **Undo / Redo (Desfazer / Refazer):** Pilha de histórico rastreando as manipulações de array no Node Tree, garantindo que o acidental esbarrão que apagou um conector complexo não seja um *game-over* manual. * **Magnetic Grid Dinâmico:** Ao arrastar conectores de polias (ropes/belts), o motor cria assistência (snap-in) com destaque colorido em torno da extremidade correta (verde para conectar, vermelho para impossibilitado), sanando o problema opaco da década de 90. * **Dicas Progressivas (Hint System):** Integrado às flags do YAML (`hints:` array). O jogador apertando no botão de lâmpada revela as dicas em camadas textuais sucessivas, auxiliando progressão sem usar *walkthroughs* de sites externos.

Multiplayer e Co-op

O formato possui o histórico modo **Head-to-Head**.

- **Modo Competitivo Paralelo:** Dois jogadores recebem a mesma fase exata. No modelo clássico local, a tela é dividida e um timer dita quem resolve mais rápido usando recursos mais escassos. A modernização trará este conceito para o *Online Matchmaking*.
- **Sandbox Co-op (Brainstorm):** Uma adição puramente moderna, permitindo a dois jogadores manipularem o cenário online via conexão P2P no modo *Freeplay*, colaborando para montar superestruturas de rube goldberg numa lousa gigantesca em sincronia com o motor determinístico validando ambos simultaneamente.

8. Catálogo Detalhado de Peças

Este documento lista detalhadamente todas as peças presentes no jogo, gerado automaticamente a partir da documentação original do projeto `docs/parts/`.

Peça 0 — Bowling Ball



Categoria: Balls

Fases em que aparece: 40 fases

Quantidade de Estados: 1 estado(s)

This bowling ball is very heavy and doesn't bounce much.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2832	mass/density (relative scale)
damping	200	drag/air-resistance or density
cor_q8	128	COR = 0.500
radius	16	collision radius in internal units
sprite_w	240	sprite width
sprite_h	240	sprite height
data_offset	32	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–7 (8 frames)	32×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART0_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART0_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART0_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART0_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART0_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART0_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART0_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART0_f007.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	8
END_FRAME	8
END_ANM	1

Notas de Comportamento:

Bowling Ball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** the part file `part-000-bowling-ball.md` describes part type `0`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 0`

TC-002: Mass value is positive

- **Given** Bowling Ball has `mass = 2832`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Bowling Ball has `cor_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Radius is non-negative

- **Given** Bowling Ball has `radius = 16`
- **When** physics engine initializes collision
- **Then** `radius >= 0`

TC-005: Initial state at simulation start

- **Given** Bowling Ball is placed in a level
- **When** simulation starts
- **Then** `state_counter` initializes to `0`

TC-006: Single default state — no named transitions

- **Given** Bowling Ball has no SOLVE.RES state data
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-007: ANM animation frames match state count

- **Given** Bowling Ball ANM has `Total Frames = 8`, `Total States = 1`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 1`

TC-008: Bytecode draws correct sprites

- **Given** Bowling Ball ANM frame `0`
- **When** bytecode is executed

- **Then** `DRAW_BMP` opcode references valid sprite in `build/phase-3/bitmaps/PART0_f*.png`

TC-009: Connection flags match category rules

- **Given** Bowling Ball (Balls category) declares no connections
- **When** checked against `category_rules`
- **Then** `Electrical=No, Belt=No, Rope=No, Fire=No, Laser=No, Projectile=No`

TC-010: Part used in real levels

- **Given** Bowling Ball (type 0)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 40 levels reference `part_num = 0`

TC-011: Snap-to-grid (16px) in edit mode

- **Given** Bowling Ball placed at position `(47, 23)`
- **When** editor snaps position
- **Then** result is `(48, 16)` — both coordinates multiple of 16

TC-012: Physics disabled in edit mode

- **Given** Bowling Ball in edit mode
- **Then** `velocity_x = velocity_y = 0` and `state_counter` frozen

TC-013: Created Part flag is No

- **Given** Bowling Ball Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-014: Interactive flag is No

- **Given** Bowling Ball Identity shows `Interactive: No`
- **When** `nt_part_info_t2_base.behavior` is checked
- **Then** part does not require user interaction to activate

TC-015: Namedtuple field count matches format

- **Given** `nt_part_info_t2_base` parsed from binary
- **Then** `len(nt_part_info_t2_base._fields) == 25` fields

TC-016: Flags_3 bitfield decodes correctly

- **Given** Bowling Ball `flags_3` from level data
- **When** decoded per `fformat.txt`
- **Then** gravity direction bit (bit 3) and initial state bit (bit 12) are valid

Peça 1 — Brick Wall



Categoria: Walls / Floors

Fases em que aparece: 37 fases

Quantidade de Estados: 2 estado(s)

This brick wall or floor can be stretched vertically or horizontally to any length you need. It's not as slippery as a caution wall. Explosives will blow it up.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
friction_q8	1024	friction = 4.000
unk_6	24	category-specific property
tile_w	16	base width in tile units
tile_h	16	base height in tile units
collision_w	576	AABB collision width
collision_h	416	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	4	4–7 (4 frames)	16×16	28
1	8	8–19 (12 frames)	40×16	56

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART1_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART1_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART1_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART1_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART1_f004.png :material-check:

Frame	Sprites	PNG References
5	0001_0005	build/phase-3/bitmaps/PART1_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART1_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART1_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART1_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART1_f009.png :material-check:
10	0001_0008	build/phase-3/bitmaps/PART1_f008.png :material-check:
11	0001_0009	build/phase-3/bitmaps/PART1_f009.png :material-check:
12	0001_0008	build/phase-3/bitmaps/PART1_f008.png :material-check:
13	0001_0009	build/phase-3/bitmaps/PART1_f009.png :material-check:
14	0001_0010	build/phase-3/bitmaps/PART1_f010.png :material-check:
15	0001_0010	build/phase-3/bitmaps/PART1_f010.png :material-check:
16	0001_0010	build/phase-3/bitmaps/PART1_f010.png :material-check:
17	0001_0011	build/phase-3/bitmaps/PART1_f011.png :material-check:
18	0001_0011	build/phase-3/bitmaps/PART1_f011.png :material-check:
19	0001_0011	build/phase-3/bitmaps/PART1_f011.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	20
END_FRAME	20
END_ANM	1

Notas de Comportamento:

Brick Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** the part file `part-001-brick-wall.md` describes part type `1`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 1`

TC-002: Mass value is positive

- **Given** Brick Wall has `mass = 4153`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Friction within valid range

- **Given** Brick Wall has `friction_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `friction = 4.000` (reasonable for rough surface)

TC-004: Destructible flag matches category rules

- **Given** Brick Wall shows `Destructible: Yes`
- **When** explosive part collides with it
- **Then** SOLVE.RES triggers destruction state

TC-005: Category Part Count is 15

- **Given** Brick Wall category rules show `Category Part Count: 15`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** 15 parts share this category

TC-006: No SOLVE.RES state transitions

- **Given** Brick Wall has "No SOLVE.RES state data"
- **When** collision occurs
- **Then** no programmatic state change occurs

TC-007: ANM has 2 states but single default behavior

- **Given** Brick Wall ANM shows `Total States = 2`
- **When** animation plays
- **Then** appearance determined by `appearance` field, not state machine

TC-008: Collision dimensions match bitmap

- **Given** Brick Wall has `collision_w = 576, collision_h = 416`
- **When** AABB collision check runs
- **Then** collision box is larger than bitmap (due to stretch gizmo)

TC-009: Connection flags all disabled

- **Given** Brick Wall declares no connections
- **When** checked against connections table
- **Then** `Electrical=No, Belt=No, Rope=No, Fire=No, Laser=No, Projectile=No`

TC-010: Part used in 37 levels

- **Given** Brick Wall (type 1)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 37 levels reference `part_num = 1`

TC-011: Tile dimensions are multiples of 16

- **Given** Brick Wall has `tile_w = 16, tile_h = 16`
- **When** gizmo stretches the wall
- **Then** dimensions snap to 16-pixel grid increments

TC-012: Created Part flag is No

- **Given** Brick Wall Identity shows `Created Part: No`
- **When** level is loaded
- **Then** part appears in Parts Bin

TC-013: Interactive flag is No

- **Given** Brick Wall Identity shows `Interactive: No`
- **When** player clicks on it during simulation
- **Then** no action triggered

TC-014: Scenery-part boundary respected

- **Given** Brick Wall has `part_num 1 (< 149)`
- **When** placed in level

- **Then** part is NOT scenery (scenery starts at 149)

TC-015: Wall stretch gizmo is single-axis

- **Given** Brick Wall is a Wall/Floor category
 - **When** gizmo type is determined
 - **Then** stretch is constrained to horizontal or vertical axis
-

Peça 2 — Wood incline



Categoria: Inclines
Fases em que aparece: 36 fases
Quantidade de Estados: 1 estado(s)

You can roll things up or down this wood incline, or use it to direct balloons. It can be stretched or shrunk, which changes the angle. Explosives won't hurt it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1510	mass/density (relative scale)
unk_2	1000	category-specific property
friction_q8	1024	friction = 4.000
unk_6	16	category-specific property
tile_w	16	base width in tile units
tile_h	240	base height in tile units
collision_w	64	AABB collision width
collision_h	0	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	6

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No

Capability	Supported
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	16×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART2_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART2_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART2_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART2_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Wood incline is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** the part file `part-002-wood-incline.md` describes part type `2`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 2`

TC-002: Mass value is positive

- **Given** Wood Incline has `mass = 1510`
- **When** validated
- **Then** `mass > 0`

TC-003: Friction value is valid

- **Given** Wood Incline has `friction_q8 = 1024`
- **When** converted
- **Then** `friction = 4.000` (wood surface)

TC-004: Destructible flag is No

- **Given** Wood Incline shows `Destructible: No`
- **When** explosive collides with it
- **Then** wall is NOT destroyed

TC-005: No SOLVE.RES state transitions

- **Given** Wood Incline has no SOLVE.RES data
- **When** any collision occurs
- **Then** no state change triggered

TC-006: Collision height is 0 (degenerate AABB)

- **Given** Wood Incline has `collision_h = 0`
- **When** AABB collision check runs
- **Then** this represents a 1D line collision (incline has no thickness)

TC-007: Collision width is 64

- **Given** Wood Incline has `collision_w = 64`
- **When** collision check runs
- **Then** collision extends 64 units along incline

TC-008: ANM single state with 4 frames

- **Given** Wood Incline ANM shows `Total Frames = 4, Total States = 1`
- **When** rendered
- **Then** only one visual appearance state

TC-009: No connections supported

- **Given** Wood Incline declares no connections
- **When** checked
- **Then** `Electrical=No, Belt=No, Rope=No, Fire=No, Laser=No, Projectile=No`

TC-010: Part used in 36 levels

- **Given** Wood Incline (type 2)
- **When** checked against YAML levels
- **Then** at least 36 levels reference it

TC-011: Incline angle determined by stretch gizmo

- **Given** Wood Incline can be stretched
- **When** gizmo changes `width_1` dimension
- **Then** collision box updates accordingly

TC-012: Bitmap height matches incline sprite

- **Given** Wood Incline has `bitmap_h = 32`
- **When** sprite is rendered
- **Then** frame height is 32 pixels

TC-013: Category is Inclines

- **Given** Wood Incline Category is "Inclines"
- **When** placed with other inclines
- **Then** they can stack/be adjacent

TC-014: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-015: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs

- **Then** no player interaction required

Peça 3 — Topsy Trailer



Categoria: Special Mechanics
Fases em que aparece: 38 fases
Quantidade de Estados: 1 estado(s)

This little trailer acts as a teeter-totter. Drop something heavy on the high end to catapult an object off the low end. Tie a rope to either end and use it to hoist and lower objects, pull the trigger of the phazer, or cause other reactions.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1888	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	80	bitmap frame width
bitmap_h	39	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
6	0	0–2 (3 frames)	80×39	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0003, 0001_0000	build/phase-3/bitmaps/PART3_f003.png :material-check: build/phase-3/bitmaps/PART3_f000.png :material-check:
1	0001_0002, 0001_0000	build/phase-3/bitmaps/PART3_f002.png :material-check: build/phase-3/bitmaps/PART3_f000.png :material-check:
2	0001_0001, 0001_0000	build/phase-3/bitmaps/PART3_f001.png :material-check: build/phase-3/bitmaps/PART3_f000.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	6
END_FRAME	3
END_ANM	1

Notas de Comportamento:

Tipsy Trailer is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has custom spawn behavior defined in SOLVE.RES.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-003-tipsy-trailer.md` describes part type `3`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 3`

TC-002: Mass value is positive

- **Given** Tipsy Trailer has `mass = 1888`
- **When** validated
- **Then** `mass > 0`

TC-003: property_q8 encodes mechanical property

- **Given** Tipsy Trailer has `property_q8 = 1024` (Q8.8 = 4.000)
- **When** behavior handler runs
- **Then** value represents pivot sensitivity or counterweight effect

TC-004: Category triggers include collision, electrical, proximity

- **Given** Tipsy Trailer shows `Triggers: collision, electrical, proximity`
- **When** any of these stimuli occur
- **Then** state transition may be triggered

TC-005: Spawns custom behavior

- **Given** Tipsy Trailer shows `Spawns: custom`

- **When** activated
- **Then** special behavior handler is invoked (teeter-totter pivot)

TC-006: Electrical connection supported

- **Given** Topsy Trailer shows `Electrical: Yes`
- **When** connected to power network
- **Then** can be activated via electrical signal

TC-007: ANM state ID is 6 (non-default initial)

- **Given** Topsy Trailer ANM shows `State ID: 6` for first frame
- **When** simulation starts
- **Then** `state_counter` may initialize to `6` (check `flags_3`)

TC-008: 3 animation frames for pivot motion

- **Given** Topsy Trailer ANM has `Total Frames = 3`
- **When** animated
- **Then** frames show trailer pivoting between positions

TC-009: Bytecode draws 6 sprites (2 per frame)

- **Given** Bytecode shows `DRAW_BMP: 6`
- **When** rendering
- **Then** each frame composes 2 sprite images

TC-010: Part used in 38 levels

- **Given** Topsy Trailer (type 3)
- **When** checked against YAML levels
- **Then** at least 38 levels reference it

TC-011: Rope connection anchor points

- **Given** Description mentions rope attachment at either end
- **When** rope connects to trailer
- **Then** force applied at attachment point creates pivot torque

TC-012: Teeter-totter pivot physics

- **Given** heavy object dropped on high end
- **When** physics tick runs
- **Then** angular momentum transfers to low end (catapult effect)

TC-013: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-014: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** player clicks during sim
- **Then** no direct interaction

TC-015: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
 - **When** validated
 - **Then** 15 parts in Special Mechanics category
-

Peça 4 — Balloon



Categoria: Balloons / Airships
Fases em que aparece: 68 fases
Quantidade de Estados: 8 estado(s)

This balloon can be programmed to have four different appearances, which all act exactly the same. It will float up into the air unless it's tied down with a rope or held back by another object. Use it to lift the low end of the teeter-totter, shoot the phazer, trigger the boxing glove, push the bellows, or bump various objects. Balloons will pop if they touch moving gears, hedge trimmers, tin snips, any flame, laser beams or tacks.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Popped	3→7 , 2→6 , 4→8 , 5→9	—	No
Popped	7→2 , 6→4 , 8→5 , 9→1	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	9	mass/density (relative scale)
unk_2	1	category-specific property
property_q8	64	Q8.8 = 0.250
unk_6	32	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	32	bitmap frame width
bitmap_h	48	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity
Destructible	Yes
Spawns	(none)
Category Part Count	3

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No

Capability	Supported
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0	40×51	0
7	1	1–5 (5 frames)	40×51	7
2	6	6	48×51	44
6	7	7–12 (6 frames)	48×51	51
4	13	13	56×51	95
8	14	14–17 (4 frames)	56×51	102
5	18	18	40×42	132
9	19	19–22 (4 frames)	40×42	139

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0010	build/phase-3/bitmaps/PART4_f010.png :material-check:
1	0001_0010	build/phase-3/bitmaps/PART4_f010.png :material-check:
2	0001_0011	build/phase-3/bitmaps/PART4_f011.png :material-check:
3	0001_0012	build/phase-3/bitmaps/PART4_f012.png :material-check:
4	0001_0013	build/phase-3/bitmaps/PART4_f013.png :material-check:
5	0001_0014	build/phase-3/bitmaps/PART4_f014.png :material-check:
6	0001_0000	build/phase-3/bitmaps/PART4_f000.png :material-check:
7	0001_0000	build/phase-3/bitmaps/PART4_f000.png :material-check:
8	0001_0001	build/phase-3/bitmaps/PART4_f001.png :material-check:
9	0001_0002	build/phase-3/bitmaps/PART4_f002.png :material-check:
10	0001_0003	build/phase-3/bitmaps/PART4_f003.png :material-check:
11	0001_0004	build/phase-3/bitmaps/PART4_f004.png :material-check:
12	0001_0005	build/phase-3/bitmaps/PART4_f005.png :material-check:
13	0003_0000	build/phase-3/bitmaps/PART4_3_f000.png :material-alert-outline: (not found)
14	0003_0000	build/phase-3/bitmaps/PART4_3_f000.png :material-alert-outline: (not found)
15	0003_0001	build/phase-3/bitmaps/PART4_3_f001.png :material-alert-outline: (not found)
16	0003_0002	build/phase-3/bitmaps/PART4_3_f002.png :material-alert-outline: (not found)
17	0003_0003	build/phase-3/bitmaps/PART4_3_f003.png :material-alert-outline: (not found)
18	0004_0000	build/phase-3/bitmaps/PART4_4_f000.png :material-alert-outline: (not found)
19	0004_0000	build/phase-3/bitmaps/PART4_4_f000.png :material-alert-outline: (not found)
20	0004_0001	build/phase-3/bitmaps/PART4_4_f001.png :material-alert-outline: (not found)
21	0004_0002	build/phase-3/bitmaps/PART4_4_f002.png :material-alert-outline: (not found)
22		

Frame	Sprites	PNG References
	0004_0003	build/phase-3/bitmaps/PART4_4_f003.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
2	3004	0	build/phase-6/raw-wav/SX_3004_11025.wav
8	3004	0	build/phase-6/raw-wav/SX_3004_11025.wav
15	3004	0	build/phase-6/raw-wav/SX_3004_11025.wav
20	3004	0	build/phase-6/raw-wav/SX_3004_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	23
END_FRAME	23
PLAY_SFX	4
END_ANM	1

Notas de Comportamento:

Balloon has 2 semantic states: Not Popped, Popped. Transitions are triggered by: collision, proximity.

Terminal state(s): Popped (part is removed from simulation).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-004-balloon.md` describes part type 4
- **When** parsed
- **Then** `part_num == 4`

TC-002: Very low mass (buoyant object)

- **Given** Balloon has `mass = 9`
- **When** gravity applied
- **Then** `mass` is lowest in category, enabling buoyancy

TC-003: property_q8 encodes lift force

- **Given** Balloon has `property_q8 = 64` ($Q8.8 = 0.250$)
- **When** buoyancy calculated
- **Then** `gravity_buoyancy = pressure * (1.0 - mass/density)`

TC-004: 8 ANM states (4 colors × 2 states each)

- **Given** Balloon ANM shows `Total States = 8`
- **When** rendered
- **Then** states map to 4 appearances × (Normal + Popped)

TC-005: SOLVE.RES transitions for popping

- **Given** Balloon has transitions `3→7, 2→6, 4→8, 5→9`
- **When** collision/proximity with sharp objects occurs

- **Then** state transitions to popped variant

TC-006: Terminal state is Popped (removed from sim)

- **Given** Balloon transitions to Popped state
- **When** `state_counter` reaches terminal value
- **Then** `part` is removed from simulation

TC-007: 4 PLAY_SFX opcodes (pop sound)

- **Given** Balloon ANM has `PLAY_SFX: 4`
- **When** pop animation plays
- **Then** sound `SX_3004_11025.wav` plays at each pop frame

TC-008: Destructible by explosives

- **Given** Balloon shows `Destructible: Yes`
- **When** explosive part collides
- **Then** balloon pops immediately

TC-009: Triggers include collision and proximity

- **Given** Balloon shows `Triggers: collision, proximity`
- **When** any object enters proximity radius
- **Then** SOLVE.RES lookup determines new state

TC-010: Part used in 68 levels

- **Given** Balloon (type 4)
- **When** checked against YAML levels
- **Then** at least 68 levels reference it

TC-011: Rope connection lifts balloon

- **Given** Balloon tied with rope to object
- **When** simulation runs
- **Then** rope constrains upward buoyancy

TC-012: Balloon pops near lasers, flames, gears

- **Given** sharp objects (laser beams, flames, hedge trimmers, tacks) nearby
- **When** proximity check runs
- **Then** balloon state transitions to Popped

TC-013: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-014: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** player clicks during simulation
- **Then** may trigger state change (placement orientation)

TC-015: 4 balloon color appearances

- **Given** Description mentions "four different appearances"
 - **When** `appearance` field varies
 - **Then** visual sprite set changes but physics unchanged
-

Peça 5 — Conveyor Belt



Categoria: Rotating Power Sources
Fases em que aparece: 61 fases
Quantidade de Estados: 10 estado(s)

Hitch this conveyor belt to a motor by adding a belt. Then use it to move objects. Add a gear to make it change directions.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	32	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	96	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	32	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, rope
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
	0	0	32×16	0

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2				
11	1	1–4 (4 frames)	32×16	7
4	5	5	48×16	43
12	6	6–9 (4 frames)	48×16	50
6	10	10	64×16	86
13	11	11–14 (4 frames)	64×16	93
8	15	15	80×16	129
14	16	16–19 (4 frames)	80×16	136
10	20	20	96×16	172
15	21	21–24 (4 frames)	96×16	179

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART5_f000.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART5_f000.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART5_f001.png :material-check:
3	0001_0002	build/phase-3/bitmaps/PART5_f002.png :material-check:
4	0001_0003	build/phase-3/bitmaps/PART5_f003.png :material-check:
5	0001_0004	build/phase-3/bitmaps/PART5_f004.png :material-check:
6	0001_0004	build/phase-3/bitmaps/PART5_f004.png :material-check:
7	0001_0005	build/phase-3/bitmaps/PART5_f005.png :material-check:
8	0001_0006	build/phase-3/bitmaps/PART5_f006.png :material-check:
9	0001_0007	build/phase-3/bitmaps/PART5_f007.png :material-check:
10	0001_0008	build/phase-3/bitmaps/PART5_f008.png :material-check:
11	0001_0008	build/phase-3/bitmaps/PART5_f008.png :material-check:
12	0001_0009	build/phase-3/bitmaps/PART5_f009.png :material-check:
13	0001_0010	build/phase-3/bitmaps/PART5_f010.png :material-check:
14	0001_0011	build/phase-3/bitmaps/PART5_f011.png :material-check:
15	0001_0012	build/phase-3/bitmaps/PART5_f012.png :material-check:
16	0001_0012	build/phase-3/bitmaps/PART5_f012.png :material-check:
17	0001_0013	build/phase-3/bitmaps/PART5_f013.png :material-check:
18	0001_0014	build/phase-3/bitmaps/PART5_f014.png :material-check:
19	0001_0015	build/phase-3/bitmaps/PART5_f015.png :material-check:
20	0001_0016	build/phase-3/bitmaps/PART5_f016.png :material-check:
21	0001_0016	build/phase-3/bitmaps/PART5_f016.png :material-check:
22	0001_0017	build/phase-3/bitmaps/PART5_f017.png :material-check:
23	0001_0018	build/phase-3/bitmaps/PART5_f018.png :material-check:
24	0001_0019	build/phase-3/bitmaps/PART5_f019.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	5	0	build/phase-6/raw-wav/SX_5_11025.wav
2	5	0	build/phase-6/raw-wav/SX_5_11025.wav
3	5	0	build/phase-6/raw-wav/SX_5_11025.wav
4	5	0	build/phase-6/raw-wav/SX_5_11025.wav
6	5	0	build/phase-6/raw-wav/SX_5_11025.wav
7	5	0	build/phase-6/raw-wav/SX_5_11025.wav
8	5	0	build/phase-6/raw-wav/SX_5_11025.wav
9	5	0	build/phase-6/raw-wav/SX_5_11025.wav
11	5	0	build/phase-6/raw-wav/SX_5_11025.wav
12	5	0	build/phase-6/raw-wav/SX_5_11025.wav
13	5	0	build/phase-6/raw-wav/SX_5_11025.wav
14	5	0	build/phase-6/raw-wav/SX_5_11025.wav
16	5	0	build/phase-6/raw-wav/SX_5_11025.wav
17	5	0	build/phase-6/raw-wav/SX_5_11025.wav
18	5	0	build/phase-6/raw-wav/SX_5_11025.wav
19	5	0	build/phase-6/raw-wav/SX_5_11025.wav
21	5	0	build/phase-6/raw-wav/SX_5_11025.wav
22	5	0	build/phase-6/raw-wav/SX_5_11025.wav
23	5	0	build/phase-6/raw-wav/SX_5_11025.wav
24	5	0	build/phase-6/raw-wav/SX_5_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	25
END_FRAME	25
PLAY_SFX	20
END_ANM	1

Notas de Comportamento:

Conveyor Belt is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-005-conveyor-belt.md` describes part type `5`
- **When** parsed
- **Then** `part_num == 5`

TC-002: Mass value is positive

- **Given** Conveyor Belt has `mass = 3776`
- **When** validated
- **Then** `mass > 0`

TC-003: 10 ANM states (5 widths × 2 for direction)

- **Given** Conveyor Belt ANM shows `Total States = 10`
- **When** rendered
- **Then** states map to 5 size variants × 2 animation directions

TC-004: Electrical connection required

- **Given** Conveyor Belt shows `Electrical: Yes`
- **When** motor starts
- **Then** conveyor belt activates

TC-005: Belt and Rope connections supported

- **Given** Conveyor Belt shows `Belt=Yes`, `Rope=Yes`
- **When** connected to motor
- **Then** torque is transmitted

TC-006: Triggers include collision, electrical, rope

- **Given** Conveyor Belt shows `Triggers: collision, electrical, rope`
- **When** any trigger activates
- **Then** belt motion state changes

TC-007: 20 PLAY_SFX opcodes (belt movement sound)

- **Given** Conveyor Belt ANM has `PLAY_SFX: 20`
- **When** belt is running
- **Then** mechanical sound plays

TC-008: Animation frames tied to state

- **Given** Belt has 5 width variants: 32, 48, 64, 80, 96 pixels
- **When** gizmo stretches belt
- **Then** appropriate ANM state is selected

TC-009: Part used in 61 levels

- **Given** Conveyor Belt (type 5)
- **When** checked against YAML levels
- **Then** at least 61 levels reference it

TC-010: Conveyor belt moves objects on collision

- **Given** object placed on active conveyor belt
- **When** physics tick runs
- **Then** object velocity += belt_speed (direction based on belt orientation)

TC-011: Belt speed standardized at ±60px/s

- **Given** Conveyor Belt is active
- **When** `belt_speed` is applied to object
- **Then** speed magnitude is `±60px/s` per `02_core_mechanics.md`

TC-012: Motor connection starts belt

- **Given** Conveyor Belt connected to Mouse Motor via belt
- **When** motor runs
- **Then** belt animation starts

TC-013: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated

- Then 7 parts in Rotating Power Sources category

TC-014: Created Part flag is No

- Given Identity shows Created Part: No
- When level loads
- Then part is in Parts Bin

TC-015: Interactive flag is No

- Given Identity shows Interactive: No
- When simulation runs
- Then no direct player interaction

Peça 6 — Mouse Motor



Categoria: Rotating Power Sources

Fases em que aparece: 44 fases

Quantidade de Estados: 2 estado(s)

This little cage is actually a Mouse Motor. Bump the cage to make the mouse run around on his wheel.
Add a belt and hitch it up to things like the conveyor belt or the Jack-in-the- box to make them work.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Running	3→6	—	No
Running	6→-1	6	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	49	bitmap frame width
bitmap_h	33	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, rope
Destructible	No

Rule	Value
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0–30 (31 frames)	48×32	0
6	31	31	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0002_0000, 0001_0000	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check:
1	0001_0001, 0002_0000, 0001_0000, 0003_0000	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f000.png :material-alert-outline: (not found)
2	0001_0001, 0002_0000, 0001_0000, 0003_0001	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f001.png :material-alert-outline: (not found)
3	0001_0001, 0002_0000, 0001_0000, 0003_0002, 0003_0000	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_3_f000.png :material-alert-outline: (not found)
4	0001_0001, 0002_0000, 0001_0000, 0003_0003, 0003_0001	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART6_3_f001.png :material-alert-outline: (not found)
5	0001_0001, 0002_0000, 0001_0000, 0003_0002	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f002.png :material-alert-outline: (not found)
6	0001_0001, 0002_0000, 0001_0000, 0003_0003	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f003.png :material-alert-outline: (not found)
7	0001_0001, 0002_0000, 0001_0000	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check:
8	0001_0001, 0002_0000, 0001_0000, 0003_0000	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f000.png :material-alert-outline: (not found)
9	0001_0001, 0002_0000, 0001_0000, 0003_0001	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f001.png :material-alert-outline: (not found)
10	0001_0001, 0002_0000, 0001_0000, 0003_0002	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f002.png :material-alert-outline: (not found)
11	0001_0001, 0002_0000, 0001_0000, 0003_0003	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_3_f003.png :material-alert-outline: (not found)
12	0001_0001, 0001_0000, 0002_0002	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_2_f002.png :material-alert-outline: (not found)
13	0001_0001, 0001_0000, 0002_0002	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_2_f002.png :material-alert-outline: (not found)
14	0001_0002, 0001_0000, 0001_0003, 0002_0004	build/phase-3/bitmaps/PART6_f002.png :material-check: build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_f003.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART6_2_f004.png :material-alert-outline: (not found)
15	0001_0001, 0001_0000, 0001_0004, 0002_0014	build/phase-3/bitmaps/PART6_f001.png :material-check: build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_f004.png :material-check: build/phase-3/bitmaps/PART6_2_f014.png :material-alert-outline: (not found)
16	0001_0002, 0001_0000, 0002_0013, 0001_0005	build/phase-3/bitmaps/PART6_f002.png :material-check: build/phase-3/bitmaps/PART6_f000.png :material-check: build/phase-3/bitmaps/PART6_2_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART6_f005.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
12	3006	0	build/phase-6/raw-wav/SX_3006_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	66
END_FRAME	17
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Mouse Motor has 2 semantic states: Not Running, Running. Transitions are triggered by: collision, electrical, rope.

Terminal state(s): Running (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-006-mouse-motor.md` describes part type `6`
- **When** parsed
- **Then** `part_num == 6`

TC-002: Mass value is positive

- **Given** Mouse Motor has `mass = 3776`
- **When** validated
- **Then** `mass > 0`

TC-003: property_q8 encodes rotation speed

- **Given** Mouse Motor has `property_q8 = 1024` (Q8.8 = 4.000)
- **When** motor runs
- **Then** angular velocity is set per property

TC-004: SOLVE.RES transition to Running state

- **Given** Mouse Motor transitions `3→6` (Not Running → Running)
- **When** collision/electrical/rope trigger occurs
- **Then** ANM state becomes `6`

TC-005: Terminal state removes part from sim

- **Given** Mouse Motor transitions to `6→-1`
- **When** Running animation completes
- **Then** `part is removed from simulation`

TC-006: Timed Section A animation at 60Hz

- **Given** Mouse Motor has `Animated (Section A): Yes` with 36 entries
- **When** simulation tick runs
- **Then** animation advances at 60 Hz rate

TC-007: PLAY_SFX at frame 12 (squeak sound)

- **Given** Mouse Motor ANM has `PLAY_SFX: 1` at frame 12
- **When** running animation reaches frame 12
- **Then** `SX_3006_11025.wav` plays

TC-008: Electrical connection supported

- **Given** Mouse Motor shows `Electrical: Yes`
- **When** connected to power network
- **Then** can be activated remotely

TC-009: Belt connection to conveyor/gear

- **Given** Mouse Motor shows `Belt: Yes`
- **When** belt connects to other rotating parts
- **Then** torque is transmitted

TC-010: Part used in 44 levels

- **Given** Mouse Motor (type 6)
- **When** checked against YAML levels
- **Then** at least 44 levels reference it

TC-011: Mouse in cage animation

- **Given** Description: "bump cage to make mouse run around on wheel"
- **When** activated
- **Then** 31-frame animation shows mouse running

TC-012: Belt connection starts motor

- **Given** Mouse Motor connected via belt to conveyor
- **When** motor runs
- **Then** connected parts receive angular velocity

TC-013: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Rotating Power Sources category

TC-014: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads

- Then part is in Parts Bin

TC-015: Interactive flag is Yes

- Given Identity shows Interactive: Yes
- When player bumps cage
- Then motor starts running

Peça 7 — Pulley



Categoria: Ropes / Belts / Pulleys

Fases em que aparece: 91 fases

Quantidade de Estados: 2 estado(s)

You can place this pulley between any two parts that may be connected by rope or cable. For example: tie one end of a rope to an object (a laundry basket for instance), then run the rope over as many pulleys as necessary (click on each pulley), and tie the other end of the rope to another part, such as the phazer. When you run the puzzle, the laundry basket will fall, pulling the rope, which will pull the trigger and fire the phazer.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	999	category-specific property
property_q8	0	Q8.8 = 0.000
unk_6	0	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	22	bitmap frame width
bitmap_h	24	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	4

Conexões e Capacidades:

Capability	Supported
Electrical	No

Capability	Supported
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
4	0	0	24x22	0
3	1	1–3 (3 frames)	24x19	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART7_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART7_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART7_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART7_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Pulley is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-007-pulley.md` describes part type `7`
- **When** parsed
- **Then** `part_num == 7`

TC-002: Mass value is positive

- **Given** Pulley has `mass = 3776`
- **When** validated
- **Then** `mass > 0`

TC-003: Extra data structure for pulley

- **Given** Pulley requires extra data per `d_item_extra_data_1_type_t2`
- **When** parsed
- **Then** `nt_part_extra_pulley` (8 bytes) is appended

TC-004: No SOLVE.RES state transitions

- **Given** Pulley has no SOLVE.RES state data
- **When** collision occurs
- **Then** no state change (static anchor point)

TC-005: Belt and Rope connections supported

- **Given** Pulley shows `Belt=Yes`, `Rope=Yes`
- **When** rope goes over pulley
- **Then** mechanical advantage is calculated

TC-006: 2 ANM states for pulley wheel rotation

- **Given** Pulley ANM shows `Total States = 2`
- **When** rope moves
- **Then** wheel rotates through 3 frames

TC-007: Rope constraint routing through pulley

- **Given** Rope connects object A to object B with pulley in between
- **When** constraint solver runs
- **Then** pulley redirects rope force vector

TC-008: Part used in 91 levels

- **Given** Pulley (type 7)
- **When** checked against YAML levels
- **Then** at least 91 levels reference it (highest of Batch 1)

TC-009: Pulley redirects force direction

- **Given** Description: "run the rope over as many pulleys"
- **When** object pulls rope down
- **Then** pulley redirects force to lift other object

TC-010: Category Part Count is 4

- **Given** Category Rules show `Category Part Count: 4`
- **When** validated
- **Then** 4 parts in Ropes/Belts/Pulleys category

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-012: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** no direct player interaction

TC-013: Bitmap dimensions match pulley wheel

- **Given** Pulley has `bitmap_w = 22`, `bitmap_h = 24`
- **When** rendered
- **Then** wheel sprite is approximately square

TC-014: Mechanical advantage calculation

- **Given** Pulley connects two rope segments
- **When** force applied

- **Then** force magnitude divided by segment count

TC-015: property_q8 is 0 (no motor)

- **Given** Pulley has `property_q8 = 0`
- **When** physics runs
- **Then** pulley is passive (no self-propulsion)

Peça 8 — Belt



Categoria: Ropes / Belts / Pulleys

Fases em que aparece: 83 fases

Quantidade de Estados: 1 estado(s)

Use this belt to hitch any two rotating parts together. Click on the first part you wish to connect (the Mandrill Motor, for instance), then stretch the belt over to the second part (such as the pinwheel). When the line turns from red to green the belt is in position. Click again to attach it. Belts can only be stretched a limited distance.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	0	Q8.8 = 0.000
unk_6	0	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	656	bitmap frame width
bitmap_h	4467	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	4

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	Yes

Capability	Supported
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	0x0	0

Opcodes do Bytecode:

Opcode	Count
END_ANM	1

Notas de Comportamento:

Belt is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-008-belt.md` describes part type `8`
- **When** parsed
- **Then** `part_num == 8`

TC-002: Mass value is positive

- **Given** Belt has `mass = 3776`
- **When** validated
- **Then** `mass > 0`

TC-003: Extra data structure for belt

- **Given** Belt requires extra data per `d_item_extra_data_1_type_t2`
- **When** parsed
- **Then** `nt_part_extra_belt` (4 bytes) is appended

TC-004: No ANM animation frames

- **Given** Belt ANM shows `Total Frames = 0`
- **When** rendered
- **Then** belt is drawn as line between two pulleys (procedural)

TC-005: Belt connects two rotating parts

- **Given** Description: "hitch any two rotating parts together"
- **When** belt placed between motor and conveyor
- **Then** torque transmits from motor to conveyor

TC-006: Belt has max stretch distance

- **Given** Description: "Belt can only be stretched a limited distance"
- **When** distance exceeds maximum
- **Then** placement is rejected (red line)

TC-007: Belt turns red then green when valid

- **Given** Belt being placed
- **When** distance is within range
- **Then** line color changes from red to green

TC-008: Belt and Rope connections supported

- **Given** Belt shows `Belt=Yes`, `Rope=Yes`
- **When** connected
- **Then** both belt and rope systems recognize connection

TC-009: Part used in 83 levels

- **Given** Belt (type 8)
- **When** checked against YAML levels
- **Then** at least 83 levels reference it

TC-010: Angular velocity sync between connected parts

- **Given** Motor (type 6) connected to Conveyor Belt (type 5) via Belt (type 8)
- **When** simulation runs
- **Then** all connected parts share same angular velocity

TC-011: Category Part Count is 4

- **Given** Category Rules show `Category Part Count: 4`
- **When** validated
- **Then** 4 parts in Ropes/Belts/Pulleys category

TC-012: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-013: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** belt auto-transmits when motor runs

TC-014: Bitmap dimensions are large (stretched line)

- **Given** Belt has `bitmap_w = 656`, `bitmap_h = 4467`
- **When** sprite rendered
- **Then** represents maximum stretched length

TC-015: Torque direction based on gear ratio

- **Given** Description: "place gears side by side to change direction of rotation"
- **When** odd number of gears in chain
- **Then** rotation direction inverts

Peça 9 — Basketball



Categoria: Balls

Fases em que aparece: 21 fases

Quantidade de Estados: 2 estado(s)

This basketball is medium in weight and very bouncy.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1322	mass/density (relative scale)
damping	20	drag/air-resistance or density
cor_q8	192	COR = 0.750
radius	16	collision radius in internal units
sprite_w	240	sprite width
sprite_h	240	sprite height
data_offset	32	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0	32×32	0
4	1	1–16 (16 frames)	32×32	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART9_f000.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART9_f000.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART9_f001.png :material-check:
3	0001_0002	build/phase-3/bitmaps/PART9_f002.png :material-check:
4	0001_0003	build/phase-3/bitmaps/PART9_f003.png :material-check:
5	0001_0004	build/phase-3/bitmaps/PART9_f004.png :material-check:
6	0001_0005	build/phase-3/bitmaps/PART9_f005.png :material-check:

Frame	Sprites	PNG References
7	0001_0006	build/phase-3/bitmaps/PART9_f006.png :material-check:
8	0001_0007	build/phase-3/bitmaps/PART9_f007.png :material-check:
9	0001_0008	build/phase-3/bitmaps/PART9_f008.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART9_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART9_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART9_f012.png :material-check:
13	0001_0013	build/phase-3/bitmaps/PART9_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART9_f014.png :material-check:
15	0001_0015	build/phase-3/bitmaps/PART9_f015.png :material-check:
16	0001_0009	build/phase-3/bitmaps/PART9_f009.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	17
END_FRAME	17
END_ANM	1

Notas de Comportamento:

Basketball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-009-basketball.md` describes part type `9`
- **When** parsed
- **Then** `part_num == 9`

TC-002: Mass value is positive

- **Given** Basketball has `mass = 1322`
- **When** validated
- **Then** `mass > 0`

TC-003: COR is high (bouncy)

- **Given** Basketball has `cor_q8 = 192`
- **When** converted
- **Then** `COR = 0.750` (75% restitution — much bouncier than bowling ball)

TC-004: Low damping (air resistance)

- **Given** Basketball has `damping = 20`
- **When** physics runs
- **Then** drag is minimal (lighter than bowling ball's 200)

TC-005: Radius is 16 (same as bowling ball)

- **Given** Basketball has `radius = 16`
- **When** collision check runs

- **Then** same collision sphere size as bowling ball

TC-006: Single default state (no transitions)

- **Given** Basketball has "Single default state — no named transitions"
- **When** collision occurs
- **Then** no state change (balls don't transform)

TC-007: 2 ANM states (normal + bounce frames)

- **Given** Basketball ANM shows `Total States = 2`
- **When** rendered
- **Then** state 3 = normal, state 4 = bounce sequence (16 frames)

TC-008: 17 animation frames for bounce

- **Given** Basketball ANM has `Total Frames = 17`
- **When** ball bounces
- **Then** 16-frame squash-and-stretch animation plays

TC-009: No connections supported

- **Given** Basketball declares no connections
- **When** checked
- **Then** `Electrical=No`, `Belt=No`, `Rope=No`, `Fire=No`, `Laser=No`, `Projectile=No`

TC-010: Part used in 21 levels

- **Given** Basketball (type 9)
- **When** checked against YAML levels
- **Then** at least 21 levels reference it

TC-011: Category triggers include collision and trap

- **Given** Basketball shows `Triggers: collision, trap`
- **When** hits ground or wall
- **Then** bounce physics applied with COR

TC-012: Ball category mass range

- **Given** Basketball (type 9) in Balls category (9 parts)
- **When** mass values compared
- **Then** bowling ball (2832) > basket (1322) > tennis (lower)

TC-013: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-014: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** ball physics are automatic

TC-015: Sprite dimensions are 32x32

- **Given** Basketball ANM frame 0 has dimensions `32x32`
 - **When** rendered
 - **Then** sprite matches collision radius
-

Peça 10 — Rope



Categoria: Ropes / Belts / Pulleys
Fases em que aparece: 117 fases
Quantidade de Estados: 1 estado(s)

Use this rope to tie objects together, hang things in the air, or hoist things off the ground with the help of a pulley. It attaches to teeter-totters, boat cleats, laundry baskets, buckets, phazers, balloons, the Mandrill Motor, and several other parts. To use rope: pull it out of the Parts Bin onto the screen. Click on the first object you want tied, then drag the cursor over the second object. When the line turns from red to green the rope is in position. Click again to attach it. You can cut the rope with hedge trimmers or tin snips.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1600	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	0	Q8.8 = 0.000
unk_6	0	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	13	bitmap frame width
bitmap_h	5	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	4

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	16×5	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART10.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Rope is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-010-rope.md` describes part type `10`
- **When** parsed
- **Then** `part_num == 10`

TC-002: Extra data structure for rope

- **Given** Rope requires extra data per `d_item_extra_data_1_type_t2`
- **When** parsed
- **Then** `nt_part_extra_rope` (6 bytes) is appended

TC-003: Rope connects two parts

- **Given** Description: "tie objects together", "tie one end to object, other end to second object"
- **When** rope placed between A and B
- **Then** `rope_info_1/2` stores connected part indices

TC-004: Rope turns red then green when valid

- **Given** rope being placed
- **When** both endpoints can connect
- **Then** line color changes red → green

TC-005: Rope cut by hedge trimmers or tin snips

- **Given** rope exists between two objects
- **When** hedge trimmers (type 37) or tin snips (type 77) touch rope
- **Then** rope connection is severed

TC-006: Max stretch distance enforced

- **Given** rope between two objects
- **When** `distance(A, B) > rope.max_length`
- **Then** constraint solver corrects position

TC-007: Single-pass rope constraint (per Q-018)

- **Given** rope constraint in simulation loop
- **When** constraint solver runs
- **Then** rope correction is single-pass (no iterations)

TC-008: Belt and Rope connections supported

- **Given** Rope shows `Belt=Yes, Rope=Yes`
- **When** connected
- **Then** both mechanical systems recognize

TC-009: Part used in 117 levels

- **Given** Rope (type 10)
- **When** checked against YAML levels
- **Then** at least 117 levels reference it (highest usage in all parts)

TC-010: Rope used with pulleys

- **Given** Description: "hang things in the air, or hoist things with pulley"
- **When** pulley redirects rope
- **Then** mechanical advantage applies

TC-011: Rope can attach to teeter-totter, boat cleat, laundry basket, bucket, phazer, balloon

- **Given** rope placement mode
- **When** click on valid anchor part
- **Then** connection established

TC-012: Category Part Count is 4

- **Given** Category Rules show `Category Part Count: 4`
- **When** validated
- **Then** 4 parts in Ropes/Belts/Pulleys category

TC-013: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-014: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** rope physics are automatic

TC-015: Bitmap dimensions small (13×5)

- **Given** Rope has `bitmap_w = 13, bitmap_h = 5`
- **When** rendered
- **Then** sprite represents rope segment texture

Peça 11 — Laundry Basket



Categoria: Containers

Fases em que aparece: 12 fases

Quantidade de Estados: 2 estado(s)

Use this laundry basket to trap Curie Cat, Newton Mouse, or Mel Schlemming. Tie one end of a rope to the laundry basket, and tie the other end to another part (like a bucket or teeter-totter).

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4000	mass/density (relative scale)
unk_2	70	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	64	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	64	bitmap frame width
bitmap_h	62	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
6	0	0	64×62	0
5	1	1	64×62	13

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001	build/phase-3/bitmaps/PART11_f000.png :material-check: build/phase-3/bitmaps/PART11_f001.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART11_f000.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	3
END_FRAME	2
END_ANM	1

Notas de Comportamento:

Laundry Basket is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-011-laundry-basket.md` describes part type `11`
- **When** parsed
- **Then** `part_num == 11`

TC-002: Mass value is positive

- **Given** Laundry Basket has `mass = 4000`
- **When** validated
- **Then** `mass > 0`

TC-003: property_q8 encodes container property

- **Given** Laundry Basket has `property_q8 = 128` ($Q8.8 = 0.500$)
- **When** containment check runs
- **Then** value represents hold strength

TC-004: Collision trigger only

- **Given** Laundry Basket shows `Triggers: collision`
- **When** object enters basket
- **Then** only collision response, no state change

TC-005: Single default state

- **Given** Laundry Basket has "Single default state"
- **When** rendered
- **Then** no SOLVE.RES transitions

TC-006: 2 ANM states (open/closed appearance)

- **Given** Laundry Basket ANM shows `Total States = 2`
- **When** rendered
- **Then** state 6 = default, state 5 = alternate (containment states)

TC-007: Rope connection anchor

- **Given** Description: "tie one end of rope to laundry basket"
- **When** rope attached
- **Then** basket can be lifted/lowered

TC-008: Container for Curie Cat, Newton Mouse, Mel Schlemming

- **Given** Description lists valid trapped characters
- **When** animal enters basket
- **Then** animal is contained

TC-009: Part used in 12 levels

- **Given** Laundry Basket (type 11)
- **When** checked against YAML levels
- **Then** at least 12 levels reference it

TC-010: Category Part Count is 5

- **Given** Category Rules show `Category Part Count: 5`
- **When** validated
- **Then** 5 parts in Containers category

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-012: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** container behavior is automatic

TC-013: Bitmap dimensions (64x62)

- **Given** Laundry Basket has `bitmap_w = 64, bitmap_h = 62`
- **When** rendered
- **Then** sprite is rectangular container shape

TC-014: Destructible flag is No

- **Given** Laundry Basket shows `Destructible: No`
- **When** collision with force occurs
- **Then** basket holds shape

TC-015: unk_2 = 70 (category-specific)

- **Given** Laundry Basket has `unk_2 = 70`
- **When** behavior handler processes
- **Then** value encodes containment capacity

Peça 12 — Curie Cat



Categoria: Characters
Fases em que aparece: 21 fases
Quantidade de Estados: 12 estado(s)

Curie Cat will head toward Newton Mouse or Bill the Goldfish whenever he can see them. He'll turn around if he's bumped or runs into something. He also likes the goo that comes out of the can when it falls from the can opener.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Hasn't Eaten	1→9	1	No

State Name	Triggers (self→other)	ANM State	Explosive
Has Eaten	9→-1	9	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
unk_2	120	category-specific property
property_q8	0	Q8.8 = 0.000
unk_6	64	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	39	bitmap frame width
bitmap_h	42	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity
Destructible	No
Spawns	(none)
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	45×42	0
8	1	1–74 (74 frames)	45×42	19
7	75	75–96 (22 frames)	72×57	1249
6	97	97–148 (52 frames)	45×42	1439
4	149	149–159 (11 frames)	48×33	2349
5	160	160–169 (10 frames)	48×33	2492
9	170	170	72×57	2576
	171	171–244 (74 frames)	72×57	2583

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
11				
12	245	245–261 (17 frames)	0×0	0
13	262	262–313 (52 frames)	0×0	0
14	314	314–323 (10 frames)	0×0	0
10	324	324	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0002, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f002.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
1	0001_0002, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f002.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
2	0001_0000, 0001_0004, 0001_0001	build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
3	0001_0000, 0001_0003, 0001_0001	build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
4	0001_0005, 0001_0001, 0001_0000	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check:
5	0001_0006, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
6	0001_0004, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f004.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
7	0001_0002, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f002.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
8	0001_0004, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f004.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
9	0001_0003, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
10	0001_0005, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
11	0001_0006, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
12	0001_0004, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f004.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
13	0001_0005, 0001_0000, 0001_0001	

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Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f022.png :material-check:
47	0001_0003, 0001_0017	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f017.png :material-check:
48	0001_0003, 0001_0018	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f018.png :material-check:
49	0001_0017, 0001_0004	build/phase-3/bitmaps/PART12_f017.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
50	0001_0018, 0001_0004	build/phase-3/bitmaps/PART12_f018.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
51	0001_0006, 0001_0017	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f017.png :material-check:
52	0001_0006, 0001_0018	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f018.png :material-check:
53	0001_0005, 0001_0019	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
54	0001_0005, 0001_0020	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f020.png :material-check:
55	0001_0006, 0001_0021	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f021.png :material-check:
56	0001_0006, 0001_0019	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
57	0001_0020, 0001_0004	build/phase-3/bitmaps/PART12_f020.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
58	0001_0021, 0001_0004	build/phase-3/bitmaps/PART12_f021.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
59	0001_0003, 0001_0019	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
60	0001_0003, 0001_0020	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f020.png :material-check:
61	0001_0005, 0001_0021	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f021.png :material-check:
62	0001_0005, 0001_0018	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f018.png :material-check:
63	0001_0006, 0001_0017	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f017.png :material-check:
64	0001_0006, 0001_0018	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f018.png :material-check:
65	0001_0004, 0001_0017	build/phase-3/bitmaps/PART12_f004.png :material-check: build/phase-3/bitmaps/PART12_f017.png :material-check:
66	0001_0018, 0001_0004	build/phase-3/bitmaps/PART12_f018.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
67	0001_0006, 0001_0019	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
68	0001_0006, 0001_0020	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f020.png :material-check:

Frame	Sprites	PNG References
69	0001_0003, 0001_0021	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f021.png :material-check:
70	0001_0003, 0001_0019	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
71	0001_0005, 0001_0020	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f020.png :material-check:
72	0001_0005, 0001_0021	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f021.png :material-check:
73	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
74	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
75	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
76	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
77	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
78	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
79	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
80	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
81	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
82	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
83	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
84	0001_0007	build/phase-3/bitmaps/PART12_f007.png :material-check:
85	0001_0008	build/phase-3/bitmaps/PART12_f008.png :material-check:
86	0001_0009	build/phase-3/bitmaps/PART12_f009.png :material-check:
87	0001_0010	build/phase-3/bitmaps/PART12_f010.png :material-check:
88	0001_0011	build/phase-3/bitmaps/PART12_f011.png :material-check:
89	0001_0012	build/phase-3/bitmaps/PART12_f012.png :material-check:
90	0001_0013	build/phase-3/bitmaps/PART12_f013.png :material-check:
91	0001_0014	build/phase-3/bitmaps/PART12_f014.png :material-check:
92	0001_0015	build/phase-3/bitmaps/PART12_f015.png :material-check:
93	0001_0016	build/phase-3/bitmaps/PART12_f016.png :material-check:
94	0001_0002, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f002.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
95	0001_0002, 0001_0000, 0001_0001	build/phase-3/bitmaps/PART12_f002.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
96	0001_0000, 0001_0004, 0001_0001	build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
97	0001_0000, 0001_0003, 0001_0001	build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f001.png :material-check:
98	0001_0005, 0001_0001, 0001_0000	

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Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f024.png :material-check:
132	0001_0003, 0001_0000, 0001_0023	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f023.png :material-check:
133	0001_0000, 0001_0023, 0001_0004	build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f023.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
134	0001_0000, 0001_0022, 0001_0004	build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f022.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
135	0001_0006, 0001_0000, 0001_0022	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f000.png :material-check: build/phase-3/bitmaps/PART12_f022.png :material-check:
136	0001_0003, 0001_0017	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f017.png :material-check:
137	0001_0003, 0001_0018	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f018.png :material-check:
138	0001_0017, 0001_0004	build/phase-3/bitmaps/PART12_f017.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
139	0001_0018, 0001_0004	build/phase-3/bitmaps/PART12_f018.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
140	0001_0006, 0001_0017	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f017.png :material-check:
141	0001_0006, 0001_0018	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f018.png :material-check:
142	0001_0005, 0001_0019	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
143	0001_0005, 0001_0020	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f020.png :material-check:
144	0001_0006, 0001_0021	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f021.png :material-check:
145	0001_0006, 0001_0019	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
146	0001_0020, 0001_0004	build/phase-3/bitmaps/PART12_f020.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
147	0001_0021, 0001_0004	build/phase-3/bitmaps/PART12_f021.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
148	0001_0003, 0001_0019	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
149	0001_0003, 0001_0020	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f020.png :material-check:
150	0001_0005, 0001_0021	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f021.png :material-check:
151	0001_0005, 0001_0018	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f018.png :material-check:
152	0001_0006, 0001_0017	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f017.png :material-check:
153	0001_0006, 0001_0018	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f018.png :material-check:
154	0001_0004, 0001_0017	build/phase-3/bitmaps/PART12_f004.png :material-check: build/phase-3/bitmaps/PART12_f017.png :material-check:
155	0001_0018, 0001_0004	build/phase-3/bitmaps/PART12_f018.png :material-check: build/phase-3/bitmaps/PART12_f004.png :material-check:
156	0001_0006, 0001_0019	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
157	0001_0006, 0001_0020	build/phase-3/bitmaps/PART12_f006.png :material-check: build/phase-3/bitmaps/PART12_f020.png :material-check:
158	0001_0003, 0001_0021	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f021.png :material-check:
159	0001_0003, 0001_0019	build/phase-3/bitmaps/PART12_f003.png :material-check: build/phase-3/bitmaps/PART12_f019.png :material-check:
160	0001_0005, 0001_0020	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f020.png :material-check:
161	0001_0005, 0001_0021	build/phase-3/bitmaps/PART12_f005.png :material-check: build/phase-3/bitmaps/PART12_f021.png :material-check:
162	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
163	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
164	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
165	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
166	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
167	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
168	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
169	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
170	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
171	0001_0026	build/phase-3/bitmaps/PART12_f026.png :material-check:
172	0001_0007	build/phase-3/bitmaps/PART12_f007.png :material-check:
173	0001_0008	build/phase-3/bitmaps/PART12_f008.png :material-check:
174	0001_0009	build/phase-3/bitmaps/PART12_f009.png :material-check:
175	0001_0010	build/phase-3/bitmaps/PART12_f010.png :material-check:
176	0001_0011	build/phase-3/bitmaps/PART12_f011.png :material-check:
177	0001_0012	build/phase-3/bitmaps/PART12_f012.png :material-check:
178	0001_0013	build/phase-3/bitmaps/PART12_f013.png :material-check:
179	0001_0014	build/phase-3/bitmaps/PART12_f014.png :material-check:
180	0001_0015	build/phase-3/bitmaps/PART12_f015.png :material-check:
181	0001_0016	build/phase-3/bitmaps/PART12_f016.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
30	3312	0	build/phase-6/raw-wav/SX_3312_11025.wav
73	3012	0	build/phase-6/raw-wav/SX_3012_11025.wav
162	3012	0	build/phase-6/raw-wav/SX_3012_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	412
END_FRAME	182
PLAY_SFX	3
END_ANM	1

Notas de Comportamento:

Curie Cat has 2 semantic states: Hasn't Eaten, Has Eaten. Transitions are triggered by: collision, proximity.

Terminal state(s): Has Eaten (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 13 — Jack-in-the-box



Categoria: Springs / Pneumatics

Fases em que aparece: 25 fases

Quantidade de Estados: 4 estado(s)

You can hitch this Jack-in-the-box to any rotating part by adding a belt. When the wheel turns, it will pop open. And when Jack pops out, anything on top of his box will be shot into the air.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Closed	3→4	—	No
Popped Out	5→-1	4	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
	1000	category-specific property

Property	Raw Value	Interpretation
unk_2		
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	33	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0	33×32	0
4	1	1–11 (11 frames)	33×32	25
5	12	12–28 (17 frames)	33×49	300
13	29	29	61×52	865

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0001_0000, 0001_0003, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:

Frame	Sprites	PNG References
1	0001_0001, 0001_0000, 0001_0003, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
2	0001_0001, 0001_0000, 0001_0004, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f004.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
3	0001_0001, 0001_0000, 0001_0005, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f005.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
4	0001_0001, 0001_0000, 0001_0004, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f004.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
5	0001_0001, 0001_0000, 0001_0003, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
6	0001_0001, 0001_0000, 0001_0006, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f006.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
7	0001_0001, 0001_0000, 0001_0003, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:

Frame	Sprites	PNG References
8	0001_0001, 0001_0000, 0001_0004, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f004.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
9	0001_0001, 0001_0000, 0001_0005, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f005.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
10	0001_0001, 0001_0000, 0001_0003, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
11	0001_0001, 0001_0000, 0001_0006, 0001_0002	build/phase-3/bitmaps/ PART13_f001.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f006.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
12	0001_0007, 0001_0000, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART13_f007.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check:
13	0001_0009, 0001_0015, 0001_0000, 0001_0003, 0001_0002	build/phase-3/bitmaps/ PART13_f009.png :material-check: build/phase-3/bitmaps/ PART13_f015.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
14	0001_0000, 0001_0003, 0001_0002, 0001_0008, 0001_0015	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART13_f008.png :material-check: build/phase-3/bitmaps/ PART13_f015.png :material-check:
15	0001_0000, 0001_0003, 0001_0002, 0001_0010, 0001_0016, 0001_0017	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f010.png :material-check: build/phase-3/bitmaps/ PART13_f016.png :material-check: build/phase-3/bitmaps/ PART13_f017.png :material-check:
16	0001_0000, 0001_0003, 0001_0002, 0001_0011, 0001_0020, 0001_0019	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f011.png :material-check: build/phase-3/bitmaps/ PART13_f020.png :material-check: build/phase-3/bitmaps/ PART13_f019.png :material-check:
17	0001_0000, 0001_0003, 0001_0002, 0001_0012, 0001_0020, 0001_0021	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f012.png :material-check: build/phase-3/bitmaps/ PART13_f020.png :material-check: build/phase-3/bitmaps/ PART13_f021.png :material-check:
18	0001_0000, 0001_0003, 0001_0013, 0001_0002, 0001_0022, 0001_0023	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f013.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f022.png :material-check: build/phase-3/bitmaps/ PART13_f023.png :material-check:
19	0001_0016, 0001_0000, 0001_0003, 0001_0014, 0001_0023, 0001_0002	build/phase-3/bitmaps/ PART13_f016.png :material-check: build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f023.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
20	0001_0000, 0001_0003, 0001_0014, 0001_0002, 0001_0024	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f024.png :material-check:
21	0001_0000, 0001_0003, 0001_0014, 0001_0002, 0001_0025	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f025.png :material-check:
22	0001_0000, 0001_0003, 0001_0014, 0001_0002, 0001_0016, 0001_0019	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f016.png :material-check: build/phase-3/bitmaps/ PART13_f019.png :material-check:
23	0001_0000, 0001_0003, 0001_0014, 0001_0002, 0001_0016, 0001_0021	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check: build/phase-3/bitmaps/ PART13_f016.png :material-check: build/phase-3/bitmaps/ PART13_f021.png :material-check:
24	0001_0000, 0001_0003, 0001_0014, 0001_0025, 0001_0002	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART13_f025.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
25	0001_0000, 0001_0003, 0001_0014, 0001_0026, 0001_0002	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f026.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
26	0001_0000, 0001_0003, 0001_0014, 0001_0027, 0001_0002	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f027.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
27	0001_0000, 0001_0003, 0001_0014, 0001_0028, 0001_0002	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f028.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
28	0001_0000, 0001_0003, 0001_0014, 0001_0029, 0001_0002	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f029.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:
29	0001_0000, 0001_0003, 0001_0014, 0001_0029, 0001_0002	build/phase-3/bitmaps/ PART13_f000.png :material-check: build/phase-3/bitmaps/ PART13_f003.png :material-check: build/phase-3/bitmaps/ PART13_f014.png :material-check: build/phase-3/bitmaps/ PART13_f029.png :material-check: build/phase-3/bitmaps/ PART13_f002.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
12	3013	0	build/phase-6/raw-wav/SX_3013_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	144
END_FRAME	30
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Jack-in-the-box has 2 semantic states: Closed, Popped Out. Transitions are triggered by: collision, electrical.

Terminal state(s): Popped Out (part is removed from simulation).

Compliance Test Cases

TC-001: Part type number matches filename

- Given part-013-jack-in-the-box.md describes part type 13
- When parsed
- Then part_num == 13

TC-002: Mass value is positive

- Given Jack-in-the-box has mass = 3776
- When validated
- Then mass > 0

TC-003: Electrical connection required

- Given Jack-in-the-box shows Electrical: Yes
- When belt connects to motor
- Then spring mechanism activates

TC-004: SOLVE.RES transition to Popped Out

- Given Jack-in-the-box transitions 3→4 (Closed → Popped Out)
- When belt/electrical trigger
- Then ANM state becomes 4

TC-005: Terminal state removes part

- Given Jack-in-the-box transitions 5→-1
- When popped out animation completes
- Then part is removed from simulation

TC-006: 4 ANM states

- Given Jack-in-the-box ANM shows Total States = 4
- When rendered
- Then states: closed, popping, popped, jack airborne

TC-007: Belt connection to motor

- Given Description: "hitch to any rotating part by adding a belt"
- When motor wheel turns

- **Then** jack-in-the-box activates

TC-008: Shoots object on top

- **Given** Description: "when Jack pops out, anything on top will be shot into air"
- **When** pop animation triggers
- **Then** upward force applied to objects above

TC-009: Part used in 25 levels

- **Given** Jack-in-the-box (type 13)
- **When** checked against YAML levels
- **Then** at least 25 levels reference it

TC-010: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Springs/Pneumatics category

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-012: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** motor activates
- **Then** spring-loaded pop is mechanical

TC-013: 30 total animation frames

- **Given** Jack-in-the-box ANM has `Total Frames = 30`
- **When** popping
- **Then** smooth jack extension animation

TC-014: PLAY_SFX at frame 12 (spring sound)

- **Given** Jack-in-the-box ANM has `PLAY_SFX: 1` at frame 12
- **When** popping animation
- **Then** `SX_3013_11025.wav` plays

TC-015: property_q8 encodes spring force

- **Given** Jack-in-the-box has `property_q8 = 1024`
- **When** popping
- **Then** upward velocity proportional to property

Peça 14 — Gear



Categoria: Rotating Power Sources

Fases em que aparece: 32 fases

Quantidade de Estados: 2 estado(s)

You can make this belt turn by adding a belt and hitching it to other rotating parts (like Mandrill Motors and generators). Place gears side by side or above/below each other to reach the distance necessary or to change the direction of rotation.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	48	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	35	bitmap frame width
bitmap_h	35	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, rope
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	40×35	0
4	1	1–4 (4 frames)	40×35	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART14_f000.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART14_f000.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART14_f001.png :material-check:

Frame	Sprites	PNG References
3	0001_0002	build/phase-3/bitmaps/PART14_f002.png :material-check:
4	0001_0003	build/phase-3/bitmaps/PART14_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	5
END_FRAME	5
END_ANM	1

Notas de Comportamento:

Gear is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-014-gear.md` describes part type `14`
- **When** parsed
- **Then** `part_num == 14`

TC-002: Mass value is positive

- **Given** Gear has `mass = 7552` (heaviest in Batch 1)
- **When** validated
- **Then** `mass > 0`

TC-003: Electrical, Belt, Rope connections all supported

- **Given** Gear shows `Electrical=Yes`, `Belt=Yes`, `Rope=Yes`
- **When** connected to system
- **Then** torque transmits through all connection types

TC-004: Gear ratio changes direction

- **Given** Description: "place gears side by side to change direction of rotation"
- **When** odd number of gears in chain
- **Then** rotation direction inverts

TC-005: No SOLVE.RES (static rotation state)

- **Given** Gear has no SOLVE.RES state data
- **When** belt drives it
- **Then** continuous rotation animation

TC-006: 2 ANM states (still/rotating)

- **Given** Gear ANM shows `Total States = 2`
- **When** rendered
- **Then** state 1 = still, state 4 = rotating (4 frames)

TC-007: Collision triggers gear

- **Given** Gear shows `Triggers: collision, electrical, rope`
- **When** collision occurs
- **Then** SOLVE.RES may trigger animation

TC-008: Part used in 32 levels

- **Given** Gear (type 14)
- **When** checked against YAML levels
- **Then** at least 32 levels reference it

TC-009: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Rotating Power Sources category

TC-010: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-011: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** gear rotation is automatic when driven

TC-012: Bitmap dimensions (35×35 square)

- **Given** Gear has `bitmap_w = 35, bitmap_h = 35`
- **When** rendered
- **Then** square gear sprite

TC-013: property_q8 encodes gear teeth count

- **Given** Gear has `property_q8 = 1024` ($Q8.8 = 4.000$)
- **When** behavior handler runs
- **Then** value affects mesh with adjacent gears

TC-014: unk_6 = 48 (teeth spacing)

- **Given** Gear has `unk_6 = 48`
- **When** collision with other gear
- **Then** tooth engagement distance

TC-015: Belt connects gears together

- **Given** Description: "belt and hitch to other rotating parts"
- **When** belt placed
- **Then** multiple gears can be driven from single motor

Peça 15 — Fish Tank



Categoria: Containers

Fases em que aparece: 11 fases

Quantidade de Estados: 3 estado(s)

This is Bill the Goldfish. Curie Cat will come after him if she's close enough and they're on the same level of flooring. If his tank breaks, she'll be attracted from a greater distance. Drop just about anything on the tank to break it.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Broken	15→9	15	No
Broken	9→-1	14	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	64	bitmap frame width
bitmap_h	51	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
15	0	0–254 (255 frames)	67×51	0
9	255	255–298 (44 frames)	0×0	0
14	299	299	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0007, 0001_0017, 0001_0001, 0001_0055, 0001_0060, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f017.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
1	0001_0000, 0001_0008, 0001_0018, 0001_0001, 0001_0056, 0001_0061, 0001_0062, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f018.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
2	0001_0000, 0001_0009, 0001_0019, 0001_0001, 0001_0057, 0001_0063, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f019.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
3	0001_0000, 0001_0010, 0001_0020, 0001_0001, 0001_0058, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f020.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
4	0001_0000, 0001_0011, 0001_0021, 0001_0001, 0001_0059, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f021.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
5	0001_0000, 0001_0012, 0001_0022, 0001_0001, 0001_0055, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f012.png :material-check: build/phase-3/bitmaps/ PART15_f022.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
6	0001_0000, 0001_0013, 0001_0023, 0001_0001, 0001_0056, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f013.png :material-check: build/phase-3/bitmaps/ PART15_f023.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f066.png :material-check:
7	0001_0000, 0001_0014, 0001_0024, 0001_0001, 0001_0057, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f024.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
8	0001_0000, 0001_0015, 0001_0025, 0001_0001, 0001_0058, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f015.png :material-check: build/phase-3/bitmaps/ PART15_f025.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
9	0001_0000, 0001_0016, 0001_0026, 0001_0001, 0001_0059, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f016.png :material-check: build/phase-3/bitmaps/ PART15_f026.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
10	0001_0000, 0001_0007, 0001_0027, 0001_0001, 0001_0055, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f027.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
11	0001_0000, 0001_0056, 0001_0008, 0001_0018, 0001_0001, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f018.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
12	0001_0000, 0001_0057, 0001_0009, 0001_0019, 0001_0001, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f019.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
13	0001_0000, 0001_0059, 0001_0010, 0001_0020, 0001_0001, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f020.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:

Frame	Sprites	PNG References
14	0001_0000, 0001_0055, 0001_0011, 0001_0021, 0001_0001, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f021.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
15	0001_0000, 0001_0056, 0001_0012, 0001_0022, 0001_0001, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f012.png :material-check: build/phase-3/bitmaps/ PART15_f022.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
16	0001_0000, 0001_0057, 0001_0013, 0001_0023, 0001_0001, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f013.png :material-check: build/phase-3/bitmaps/ PART15_f023.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
17	0001_0000, 0001_0058, 0001_0014, 0001_0024, 0001_0001, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f024.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
18	0001_0000, 0001_0059, 0001_0015, 0001_0025, 0001_0001, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f015.png :material-check: build/phase-3/bitmaps/ PART15_f025.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
19	0001_0000, 0001_0055, 0001_0016, 0001_0026, 0001_0001, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f016.png :material-check: build/phase-3/bitmaps/ PART15_f026.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
20	0001_0000, 0001_0056, 0001_0007, 0001_0017, 0001_0001, 0001_0061, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f017.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f066.png :material-check:
21	0001_0000, 0001_0057, 0001_0008, 0001_0018, 0001_0001, 0001_0062, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f018.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
22	0001_0000, 0001_0058, 0001_0009, 0001_0019, 0001_0001, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f019.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
23	0001_0000, 0001_0059, 0001_0010, 0001_0020, 0001_0001, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f020.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
24	0001_0000, 0001_0055, 0001_0011, 0001_0021, 0001_0001, 0001_0002, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f021.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
25	0001_0000, 0001_0056, 0001_0012, 0001_0022, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f012.png :material-check: build/phase-3/bitmaps/ PART15_f022.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
26	0001_0000, 0001_0057, 0001_0013, 0001_0023, 0001_0002, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f013.png :material-check: build/phase-3/bitmaps/ PART15_f023.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
27	0001_0000, 0001_0058, 0001_0014, 0001_0024, 0001_0002, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f024.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
28	0001_0000, 0001_0059, 0001_0015, 0001_0025, 0001_0002, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f015.png :material-check: build/phase-3/bitmaps/ PART15_f025.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
29	0001_0000, 0001_0057, 0001_0016, 0001_0026, 0001_0002, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f016.png :material-check: build/phase-3/bitmaps/ PART15_f026.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
30	0001_0000, 0001_0058, 0001_0007, 0001_0017, 0001_0002, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f017.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
31	0001_0000, 0001_0059, 0001_0008, 0001_0018, 0001_0002, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f018.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
32	0001_0000, 0001_0055, 0001_0009, 0001_0019, 0001_0002, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f019.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
33	0001_0000, 0001_0056, 0001_0010, 0001_0020, 0001_0002, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f020.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
34	0001_0000, 0001_0057, 0001_0011, 0001_0021, 0001_0002, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f021.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
35		

Frame	Sprites	PNG References
	0001_0000, 0001_0058, 0001_0012, 0001_0022, 0001_0002, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f012.png :material-check: build/phase-3/bitmaps/ PART15_f022.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
36	0001_0000, 0001_0059, 0001_0013, 0001_0023, 0001_0002, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f013.png :material-check: build/phase-3/bitmaps/ PART15_f023.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
37	0001_0000, 0001_0055, 0001_0014, 0001_0024, 0001_0002, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f024.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
38	0001_0000, 0001_0056, 0001_0015, 0001_0025, 0001_0002, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f015.png :material-check: build/phase-3/bitmaps/ PART15_f025.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
39	0001_0000, 0001_0057, 0001_0016, 0001_0026, 0001_0002, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f016.png :material-check: build/phase-3/bitmaps/ PART15_f026.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
40	0001_0000, 0001_0058, 0001_0017, 0001_0002, 0001_0007, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f017.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
41	0001_0000, 0001_0008, 0001_0018, 0001_0002, 0001_0059, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f018.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f066.png :material-check:
42	0001_0000, 0001_0009, 0001_0019, 0001_0002, 0001_0055, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f019.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
43	0001_0000, 0001_0010, 0001_0020, 0001_0002, 0001_0056, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f020.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
44	0001_0000, 0001_0011, 0001_0027, 0001_0057, 0001_0002, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f027.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
45	0001_0000, 0001_0012, 0001_0028, 0001_0002, 0001_0058, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f012.png :material-check: build/phase-3/bitmaps/ PART15_f028.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
46	0001_0000, 0001_0013, 0001_0029, 0001_0002, 0001_0059, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f013.png :material-check: build/phase-3/bitmaps/ PART15_f029.png :material-check: build/phase-3/bitmaps/ PART15_f002.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
47	0001_0000, 0001_0014, 0001_0030, 0001_0055, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f030.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
48	0001_0000, 0001_0015, 0001_0031, 0001_0056, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f015.png :material-check: build/phase-3/bitmaps/ PART15_f031.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
49	0001_0000, 0001_0016, 0001_0032, 0001_0057, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f016.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f032.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
50	0001_0000, 0001_0058, 0001_0007, 0001_0033, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f033.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
51	0001_0000, 0001_0059, 0001_0008, 0001_0034, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f034.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
52	0001_0000, 0001_0057, 0001_0009, 0001_0035, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f035.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
53	0001_0000, 0001_0058, 0001_0010, 0001_0036, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f036.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
54	0001_0000, 0001_0059, 0001_0011, 0001_0037, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f037.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
55	0001_0000, 0001_0055, 0001_0012, 0001_0038, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f012.png :material-check: build/phase-3/bitmaps/ PART15_f038.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
56	0001_0000, 0001_0056, 0001_0013, 0001_0039, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f013.png :material-check: build/phase-3/bitmaps/ PART15_f039.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:

Frame	Sprites	PNG References
57	0001_0000, 0001_0057, 0001_0014, 0001_0040, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f040.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
58	0001_0000, 0001_0058, 0001_0015, 0001_0041, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f015.png :material-check: build/phase-3/bitmaps/ PART15_f041.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
59	0001_0000, 0001_0059, 0001_0016, 0001_0042, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f016.png :material-check: build/phase-3/bitmaps/ PART15_f042.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
60	0001_0000, 0001_0055, 0001_0007, 0001_0033, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f033.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f065.png :material-check:
61	0001_0000, 0001_0056, 0001_0008, 0001_0034, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f034.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
62	0001_0000, 0001_0057, 0001_0009, 0001_0035, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f035.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
63	0001_0000, 0001_0058, 0001_0010, 0001_0036, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f036.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
64	0001_0000, 0001_0059, 0001_0011, 0001_0037, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f037.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f065.png :material-check:
65	0001_0000, 0001_0055, 0001_0012, 0001_0038, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f012.png :material-check: build/phase-3/bitmaps/ PART15_f038.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
66	0001_0000, 0001_0056, 0001_0013, 0001_0039, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f013.png :material-check: build/phase-3/bitmaps/ PART15_f039.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
67	0001_0000, 0001_0057, 0001_0014, 0001_0040, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f040.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
68	0001_0000, 0001_0058, 0001_0015, 0001_0041, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f015.png :material-check: build/phase-3/bitmaps/ PART15_f041.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f065.png :material-check:
69	0001_0000, 0001_0059, 0001_0016, 0001_0042, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f016.png :material-check: build/phase-3/bitmaps/ PART15_f042.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
70	0001_0000, 0001_0055, 0001_0033, 0001_0007, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f033.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
71	0001_0000, 0001_0056, 0001_0008, 0001_0034, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f034.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
72	0001_0000, 0001_0057, 0001_0009, 0001_0035, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f035.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
73	0001_0000, 0001_0058, 0001_0010, 0001_0036, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f036.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
74	0001_0000, 0001_0059, 0001_0011, 0001_0037, 0001_0060, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f037.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
75	0001_0000, 0001_0055, 0001_0012, 0001_0038, 0001_0061, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f012.png :material-check: build/phase-3/bitmaps/ PART15_f038.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
76	0001_0000, 0001_0056, 0001_0013, 0001_0039, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check: build/phase-3/bitmaps/ PART15_f013.png :material-check: build/phase-3/bitmaps/ PART15_f039.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f066.png :material-check:
77	0001_0000, 0001_0057, 0001_0014, 0001_0040, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f040.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
78	0001_0000, 0001_0058, 0001_0015, 0001_0041, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f015.png :material-check: build/phase-3/bitmaps/ PART15_f041.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
79	0001_0000, 0001_0016, 0001_0042, 0001_0059, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f016.png :material-check: build/phase-3/bitmaps/ PART15_f042.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
80	0001_0000, 0001_0007, 0001_0043, 0001_0057, 0001_0061, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f007.png :material-check: build/phase-3/bitmaps/ PART15_f043.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f061.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
81	0001_0000, 0001_0008, 0001_0044, 0001_0058, 0001_0062, 0001_0067, 0001_0066	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f008.png :material-check: build/phase-3/bitmaps/ PART15_f044.png :material-check: build/phase-3/bitmaps/ PART15_f058.png :material-check: build/phase-3/bitmaps/ PART15_f062.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check:
82	0001_0000, 0001_0009, 0001_0045, 0001_0001, 0001_0059, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f009.png :material-check: build/phase-3/bitmaps/ PART15_f045.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f059.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
83	0001_0000, 0001_0010, 0001_0046, 0001_0001, 0001_0055, 0001_0064, 0001_0067, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f010.png :material-check: build/phase-3/bitmaps/ PART15_f046.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f055.png :material-check: build/phase-3/bitmaps/ PART15_f064.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
84	0001_0000, 0001_0011, 0001_0017, 0001_0001, 0001_0056, 0001_0060, 0001_0067, 0001_0066, 0001_0065	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f011.png :material-check: build/phase-3/bitmaps/ PART15_f017.png :material-check: build/phase-3/bitmaps/ PART15_f001.png :material-check: build/phase-3/bitmaps/ PART15_f056.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f060.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check: build/phase-3/bitmaps/ PART15_f066.png :material-check: build/phase-3/bitmaps/ PART15_f065.png :material-check:
85	0001_0000, 0001_0057, 0001_0014, 0001_0040, 0001_0063, 0001_0067	build/phase-3/bitmaps/ PART15_f000.png :material-check: build/phase-3/bitmaps/ PART15_f057.png :material-check: build/phase-3/bitmaps/ PART15_f014.png :material-check: build/phase-3/bitmaps/ PART15_f040.png :material-check: build/phase-3/bitmaps/ PART15_f063.png :material-check: build/phase-3/bitmaps/ PART15_f067.png :material-check:
86	0001_0048, 0001_0003	build/phase-3/bitmaps/ PART15_f048.png :material-check: build/phase-3/bitmaps/ PART15_f003.png :material-check:
87	0001_0048, 0001_0003	build/phase-3/bitmaps/ PART15_f048.png :material-check: build/phase-3/bitmaps/ PART15_f003.png :material-check:
88	0001_0003, 0001_0047	build/phase-3/bitmaps/ PART15_f003.png :material-check: build/phase-3/bitmaps/ PART15_f047.png :material-check:
89	0001_0003, 0001_0047	build/phase-3/bitmaps/ PART15_f003.png :material-check: build/phase-3/bitmaps/ PART15_f047.png :material-check:
90	0001_0004, 0001_0049	build/phase-3/bitmaps/ PART15_f004.png :material-check: build/phase-3/bitmaps/ PART15_f049.png :material-check:
91	0001_0004, 0001_0050	build/phase-3/bitmaps/ PART15_f004.png :material-check: build/phase-3/bitmaps/ PART15_f050.png :material-check:
92	0001_0005, 0001_0052	build/phase-3/bitmaps/ PART15_f005.png :material-check: build/phase-3/bitmaps/ PART15_f052.png :material-check:
93	0001_0005, 0001_0051	build/phase-3/bitmaps/ PART15_f005.png :material-check: build/phase-3/bitmaps/ PART15_f051.png :material-check:
94	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f053.png :material-check:
95	0001_0006, 0001_0049	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f049.png :material-check:
96	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
97	0001_0006, 0001_0054	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f054.png :material-check:
98	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
99	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
100	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
101	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
102	0001_0006, 0001_0047	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f047.png :material-check:
103	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
104	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
105	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
106	0001_0006, 0001_0047	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f047.png :material-check:
107	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART15_f053.png :material-check:
108	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
109	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
110	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
111	0001_0006, 0001_0053	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f053.png :material-check:
112	0001_0006, 0001_0047	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f047.png :material-check:
113	0001_0006, 0001_0049	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f049.png :material-check:
114	0001_0006, 0001_0047	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f047.png :material-check:
115	0001_0006, 0001_0047	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f047.png :material-check:
116	0001_0006, 0001_0047	build/phase-3/bitmaps/ PART15_f006.png :material-check: build/phase-3/bitmaps/ PART15_f047.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
86	3015	0	build/phase-6/raw-wav/SX_3015_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	726
DRAW_LINE	24
END_FRAME	117
PLAY_SFX	1

Opcode	Count
END_ANM	1

Notas de Comportamento:

Fish Tank has 2 semantic states: Not Broken, Broken. Transitions are triggered by: collision.

Terminal state(s): Broken (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 16 — Bike Pump



Categoria: Springs / Pneumatics

Fases em que aparece: 27 fases

Quantidade de Estados: 3 estado(s)

Drop something on the top handle of this bike pump to make it blow air. It can be used to push away balloons and other objects, or to make pinwheels turn.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Pumped	2→4	2	No
Pumped	4→-1	5	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	46	bitmap frame width
bitmap_h	51	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	48x51	0
4	1	1	48x41	7
5	2	2	56x24	16

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART16_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART16_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART16_f002.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3016	0	build/phase-6/raw-wav/SX_3016_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	3
END_FRAME	3
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Bike Pump has 2 semantic states: Not Pumped, Pumped. Transitions are triggered by: collision, electrical.

Terminal state(s): Pumped (part is removed from simulation).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-016-bike-pump.md` describes part type `16`
- **When** parsed
- **Then** `part_num == 16`

TC-002: Mass value is positive

- **Given** Bike Pump has `mass = 3776`
- **When** validated
- **Then** `mass > 0`

TC-003: Electrical connection supported

- **Given** Bike Pump shows `Electrical: Yes`
- **When** triggered
- **Then** pump activates

TC-004: SOLVE.RES transition to Pumped

- **Given** Bike Pump transitions `2→4` (Not Pumped → Pumped)
- **When** collision or electrical trigger
- **Then** ANM state becomes `5`

TC-005: Terminal state removes part

- **Given** Bike Pump transitions `4↔--1`
- **When** pumping animation completes
- **Then** `part` is removed from simulation

TC-006: 3 ANM states

- **Given** Bike Pump ANM shows `Total States = 3`
- **When** rendered
- **Then** states: not pumped, pumping (1 frame), pumped

TC-007: Pushes balloons and objects

- **Given** Description: "blow air to push away balloons and other objects"
- **When** pumped
- **Then** air burst applies force to nearby objects

TC-008: Makes pinwheels turn

- **Given** Description: "to make pinwheels turn"
- **When** air burst hits pinwheel
- **Then** pinwheel angular velocity increases

TC-009: Part used in 27 levels

- **Given** Bike Pump (type 16)
- **When** checked against YAML levels
- **Then** at least 27 levels reference it

TC-010: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Springs/Pneumatics category

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-012: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** object drops on pump handle
- **Then** pump activates

TC-013: PLAY_SFX at frame 1 (pump sound)

- **Given** Bike Pump ANM has `PLAY_SFX: 1` at frame 1
- **When** pumping
- **Then** `SX_3016_11025.wav` plays

TC-014: property_q8 encodes air pressure

- **Given** Bike Pump has `property_q8 = 128` ($Q8.8 = 0.500$)
- **When** air burst calculated
- **Then** force magnitude proportional to property

TC-015: Collision trigger from dropped object

- **Given** Bike Pump shows `Triggers: collision, electrical`
- **When** object drops on top handle
- **Then** pump activates

Peça 17 — Bucket



Categoria: Containers
Fases em que aparece: 48 fases
Quantidade de Estados: 2 estado(s)

You can drop things inside this bucket. Tie a rope to it, and tie the other end to a second object. Then drop something heavy into the bucket to lift the other object. You can also lift teeter-totters, pull triggers, and affect other parts that may be attached to a rope.

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	7552	mass/density (relative scale)
<code>unk_2</code>	100	category-specific property
<code>property_q8</code>	92	$Q8.8 = 0.359$
<code>unk_6</code>	32	category-specific property
<code>dim_w1</code>	240	dimension 1 width
<code>dim_h1</code>	240	dimension 1 height
<code>dim_w2</code>	0	dimension 2 width
<code>dim_h2</code>	0	dimension 2 height
<code>bitmap_w</code>	37	bitmap frame width
<code>bitmap_h</code>	49	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	40×49	0
4	1	1	40×49	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART17_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART17_f001.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	2
END_FRAME	2
END_ANM	1

Notas de Comportamento:

Bucket is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-017-bucket.md` describes part type `17`
- **When** parsed
- **Then** `part_num == 17`

TC-002: Mass value is positive

- **Given** Bucket has `mass = 7552` (heavy for lifting)
- **When** validated
- **Then** `mass > 0`

TC-003: property_q8 encodes hold strength

- **Given** Bucket has `property_q8 = 92` (Q8.8 = 0.359)
- **When** containment check
- **Then** value affects how much weight it can hold

TC-004: Collision trigger only

- **Given** Bucket shows `Triggers: collision`
- **When** object enters bucket
- **Then** collision response applied

TC-005: Single default state

- **Given** Bucket has "Single default state"
- **When** rendered
- **Then** no state transitions

TC-006: 2 ANM states (empty/filled appearance)

- **Given** Bucket ANM shows `Total States = 2`
- **When** rendered
- **Then** state 1 and 4 represent visual variants

TC-007: Rope connection for lifting

- **Given** Description: "tie a rope to it, tie other end to second object"
- **When** rope attached
- **Then** bucket can lift other objects

TC-008: Heavy object in bucket lifts other end

- **Given** Description: "drop something heavy into bucket to lift other object"
- **When** bucket weighted
- **Then** rope pulls other attached object up

TC-009: Part used in 48 levels

- **Given** Bucket (type 17)
- **When** checked against YAML levels
- **Then** at least 48 levels reference it

TC-010: Category Part Count is 5

- **Given** Category Rules show `Category Part Count: 5`
- **When** validated
- **Then** 5 parts in Containers category

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** level loads
- **Then** part is in Parts Bin

TC-012: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs

- **Then** bucket physics automatic

TC-013: Bitmap dimensions (37×49)

- **Given** Bucket has `bitmap_w = 37`, `bitmap_h = 49`
- **When** rendered
- **Then** rectangular container shape

TC-014: unk_2 = 100 (structural)

- **Given** Bucket has `unk_2 = 100`
- **When** force applied
- **Then** determines break threshold

TC-015: Rope mechanical advantage

- **Given** Bucket tied to teeter-totter or pulley
- **When** bucket descends
- **Then** rope pulls connected object up

Peça 18 — Cannon



Categoria: Explosives / Projectiles
Fases em que aparece: 21 fases
Quantidade de Estados: 37 estado(s)

Light the fuse of this cannon with a laser, a flaming part (like a candle or rocket), or by using a magnifying glass and light source. It fires cannon balls which can be used to break and bump things. It can be rotated to aim in six different angles.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Fired	1→23 , 4→25 , 7→27 , 34→35 , 41→42 , 47→48	47	No
Fired	19→4 , 20→7 , 31→34 , 38→41 , 45→47 , 50→-1	48	No

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	14726	mass/density (relative scale)
<code>unk_2</code>	1000	category-specific property
<code>property_q8</code>	192	Q8.8 = 0.750
<code>unk_6</code>	12	category-specific property
<code>dim_w1</code>	240	dimension 1 width
<code>dim_h1</code>	240	dimension 1 height
<code>dim_w2</code>	0	dimension 2 width
<code>dim_h2</code>	0	dimension 2 height
<code>bitmap_w</code>	101	bitmap frame width
<code>bitmap_h</code>	78	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	80×61	0
23	1	1–11 (11 frames)	91×82	19
24	12	12–21 (10 frames)	80×59	296
19	22	22–26 (5 frames)	80×59	566
29	27	27–28 (2 frames)	90×80	675
3	29	29	88×93	725
4	30	30	98×96	750
25	31	31–41 (11 frames)	99×78	775
26	42	42–51 (10 frames)	107×43	1082
20	52	52–56 (5 frames)	86×78	1304
30	57	57–58 (2 frames)	79×78	1429
6	59	59	56×69	1467
7	60	60	64×63	1480
27	61	61–71 (11 frames)	132×170	1493
28	72	72–81 (10 frames)	94×80	1808
22	82	82–85 (4 frames)	72×58	2046
31	86	86	194×129	2160
15	87	87–89 (3 frames)	184×129	2203
9	90	90	88×61	2290
34	91	91	80×59	2303
35	92	92–101 (10 frames)	88×65	2316
36	102	102–111 (10 frames)	98×67	2562
38	112	112–116 (5 frames)	96×46	2844

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
39	117	117–118 (2 frames)	91×78	2941
40	119	119	100×78	2991
41	120	120	103×78	3016
42	121	121–130 (10 frames)	90×82	3041
43	131	131–140 (10 frames)	128×176	3311
45	141	141–145 (5 frames)	0×0	0
44	146	146–147 (2 frames)	0×0	0
46	148	148	0×0	0
47	149	149	0×0	0
48	150	150–159 (10 frames)	0×0	0
49	160	160–169 (10 frames)	0×0	0
50	170	170–174 (5 frames)	0×0	0
51	175	175–176 (2 frames)	0×0	0
52	177	177	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0002_0000, 0002_0004	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found)
1	0001_0000, 0002_0000, 0002_0004, 0002_0009	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
2	0001_0000, 0002_0000, 0002_0004, 0002_0009	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
3	0001_0000, 0002_0000, 0002_0004, 0002_0010	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
4		

Frame	Sprites	PNG References
	0001_0000, 0002_0000, 0002_0005, 0002_0011	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
5	0001_0000, 0002_0000, 0002_0005, 0002_0009	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
6	0001_0000, 0002_0000, 0002_0006, 0002_0010	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
7	0001_0000, 0002_0000, 0002_0006, 0002_0011	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
8	0001_0000, 0002_0000, 0002_0007, 0002_0009	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
9	0001_0000, 0002_0000, 0002_0007, 0002_0010	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
10	0001_0000, 0002_0000, 0002_0008, 0002_0011	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
11	0001_0000, 0002_0000, 0002_0008, 0002_0009	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
12	0001_0000, 0002_0000	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
13	0001_0001, 0002_0001	build/phase-3/bitmaps/PART18_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f001.png :material-alert-outline: (not found)
14	0001_0002, 0002_0002	build/phase-3/bitmaps/PART18_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f002.png :material-alert-outline: (not found)
15	0003_0000, 0003_0001, 0003_0002, 0001_0005, 0002_0000, 0001_0003, 0001_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found)
16	0003_0000, 0003_0001, 0003_0002, 0001_0005, 0002_0000, 0001_0003, 0001_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found)
17	0003_0000, 0003_0001, 0001_0005, 0001_0004, 0002_0003, 0001_0006, 0001_0003	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f003.png :material-alert-outline: (not found)
18	0003_0000, 0001_0005, 0001_0004, 0002_0003, 0001_0007, 0001_0003	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f003.png :material-alert-outline: (not found)
19	0001_0005, 0001_0004, 0002_0003, 0001_0008	build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f008.png :material-alert-outline: (not found)
20	0001_0005, 0001_0004, 0002_0003, 0001_0009	build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f009.png :material-alert-outline: (not found)
21	0001_0010, 0002_0000	build/phase-3/bitmaps/PART18_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
22	0001_0000, 0002_0000	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
23	0004_0000, 0002_0000, 0002_0004	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found)
24	0004_0000, 0002_0000, 0002_0004, 0002_0009	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
25	0004_0000, 0002_0000, 0002_0004, 0002_0009	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
26	0004_0000, 0002_0000, 0002_0004, 0002_0010	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
27	0004_0000, 0002_0000, 0002_0005, 0002_0011	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
28	0004_0000, 0002_0000, 0002_0005, 0002_0009	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
29	0004_0000, 0002_0000, 0002_0006, 0002_0010	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
30	0004_0000, 0002_0000, 0002_0006, 0002_0011	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
31	0004_0000, 0002_0000, 0002_0007, 0002_0009	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f007.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
32	0004_0000, 0002_0000, 0002_0007, 0002_0010	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
33	0004_0000, 0002_0000, 0002_0008, 0002_0011	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
34	0004_0000, 0002_0000, 0002_0008, 0002_0009	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
35	0004_0000, 0002_0000	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
36	0004_0001, 0004_0011	build/phase-3/bitmaps/PART18_4_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f011.png :material-alert-outline: (not found)
37	0004_0002, 0004_0012	build/phase-3/bitmaps/PART18_4_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f012.png :material-alert-outline: (not found)
38	0003_0000, 0003_0001, 0003_0002, 0004_0005, 0002_0001, 0004_0003, 0004_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f004.png :material-alert-outline: (not found)
39	0003_0000, 0003_0001, 0003_0002, 0004_0005, 0002_0001, 0004_0003, 0004_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f004.png :material-alert-outline: (not found)
40	0003_0000, 0003_0001, 0004_0005, 0002_0003, 0004_0003, 0004_0004, 0004_0006	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f006.png :material-alert-outline: (not found)
41	0003_0000, 0004_0005, 0002_0003, 0004_0007, 0004_0003, 0004_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f004.png :material-alert-outline: (not found)
42	0004_0005, 0002_0003, 0004_0003, 0004_0008	build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f008.png :material-alert-outline: (not found)
43	0004_0005, 0002_0003, 0004_0003, 0004_0009	build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f009.png :material-alert-outline: (not found)
44	0004_0010, 0002_0003	build/phase-3/bitmaps/PART18_4_f010.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found)
45	0004_0000, 0002_0000	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
46	0005_0000, 0002_0000, 0005_0013	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f013.png :material-alert-outline: (not found)
47	0005_0000, 0002_0000, 0005_0013, 0002_0009	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
48	0005_0000, 0002_0000, 0005_0013, 0002_0009	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
49	0005_0000, 0002_0000, 0005_0013, 0002_0010	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
50	0005_0000, 0002_0000, 0005_0014, 0002_0011	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
51	0005_0000, 0002_0000, 0005_0014, 0002_0009	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
52	0005_0000, 0002_0000, 0005_0015, 0002_0010	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
53	0005_0000, 0002_0000, 0005_0015, 0002_0011	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
54	0005_0000, 0002_0000, 0005_0016, 0002_0009	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f016.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
55	0005_0000, 0002_0000, 0005_0016, 0002_0011	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f016.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
56	0005_0000, 0002_0000, 0005_0017, 0002_0009	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f017.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
57	0005_0000, 0002_0000, 0005_0017, 0002_0010	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f017.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found)
58	0005_0000, 0002_0000	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
59	0005_0001, 0002_0001	build/phase-3/bitmaps/PART18_5_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f001.png :material-alert-outline: (not found)
60	0005_0002, 0002_0002	build/phase-3/bitmaps/PART18_5_f002.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f002.png :material-alert-outline: (not found)
61	0003_0000, 0003_0001, 0003_0002, 0005_0005, 0005_0003, 0002_0000, 0005_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found)
62	0003_0000, 0003_0001, 0003_0002, 0005_0005, 0005_0003, 0002_0000, 0005_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found)
63	0003_0000, 0003_0001, 0005_0005, 0005_0003, 0005_0011, 0005_0004, 0005_0006	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f006.png :material-alert-outline: (not found)
64	0003_0000, 0005_0005, 0005_0003, 0005_0011, 0005_0004, 0005_0007	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f007.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
65	0005_0005, 0005_0003, 0005_0011, 0005_0008	build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f008.png :material-alert-outline: (not found)
66	0005_0005, 0005_0003, 0005_0011, 0005_0009	build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f009.png :material-alert-outline: (not found)
67	0005_0005, 0005_0003, 0005_0011, 0005_0004	build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found)
68	0005_0010, 0005_0012	build/phase-3/bitmaps/PART18_5_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f012.png :material-alert-outline: (not found)
69	0005_0000, 0002_0000	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
70	0002_0004, 0001_0000, 0002_0000	build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
71	0002_0004, 0001_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
72	0002_0004, 0001_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
73		

Frame	Sprites	PNG References
	0002_0004, 0001_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
74	0002_0005, 0001_0000, 0002_0000, 0002_0011	build/phase-3/bitmaps/PART18_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)
75	0002_0005, 0001_0000, 0002_0000, 0002_0009	build/phase-3/bitmaps/PART18_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)
76	0002_0006, 0001_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
77	0002_0006, 0001_0000, 0002_0011, 0002_0000	build/phase-3/bitmaps/PART18_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
78	0002_0007, 0001_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
79	0002_0007, 0001_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
80	0002_0008, 0001_0000, 0002_0011, 0002_0000	build/phase-3/bitmaps/PART18_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
81	0001_0000, 0002_0000	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
82	0001_0001, 0002_0001	build/phase-3/bitmaps/PART18_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f001.png :material-alert-outline: (not found)
83	0001_0002, 0002_0002	build/phase-3/bitmaps/PART18_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f002.png :material-alert-outline: (not found)
84	0003_0000, 0003_0001, 0003_0002, 0001_0004, 0001_0005, 0002_0000, 0001_0003	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f003.png :material-alert-outline: (not found)
85	0003_0000, 0003_0001, 0003_0002, 0001_0004, 0001_0005, 0002_0000, 0001_0003	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f003.png :material-alert-outline: (not found)
86	0003_0000, 0003_0001, 0001_0005, 0001_0004, 0002_0003, 0001_0003, 0001_0006	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f006.png :material-alert-outline: (not found)
87	0003_0000, 0001_0005, 0001_0004, 0002_0003, 0001_0003, 0001_0007	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f007.png :material-alert-outline: (not found)
88	0001_0005, 0001_0004, 0002_0003, 0001_0008	build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f008.png :material-alert-outline: (not found)
89	0001_0005, 0001_0004, 0002_0003, 0001_0009	build/phase-3/bitmaps/PART18_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_f009.png :material-alert-outline: (not found)
90	0001_0010, 0002_0000	build/phase-3/bitmaps/PART18_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
91	0001_0000, 0002_0000	build/phase-3/bitmaps/PART18_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
92	0002_0004, 0004_0000, 0002_0000	build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
93	0002_0004, 0004_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
94	0002_0004, 0004_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
95	0002_0004, 0004_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
96	0002_0005, 0004_0000, 0002_0011, 0002_0000	build/phase-3/bitmaps/PART18_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
97	0002_0005, 0004_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
98	0002_0006, 0004_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
99	0002_0006, 0004_0000, 0002_0011, 0002_0000	build/phase-3/bitmaps/PART18_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
100	0002_0007, 0004_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
101	0002_0007, 0004_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
102	0002_0008, 0004_0000, 0002_0011, 0002_0000	build/phase-3/bitmaps/PART18_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
103	0004_0000, 0002_0000	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
104	0004_0001, 0004_0011	build/phase-3/bitmaps/PART18_4_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f011.png :material-alert-outline: (not found)
105	0004_0002, 0004_0012	build/phase-3/bitmaps/PART18_4_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f012.png :material-alert-outline: (not found)
106	0003_0000, 0003_0001, 0003_0002, 0004_0005, 0004_0004, 0004_0003, 0002_0000	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
107	0003_0000, 0003_0001, 0003_0002, 0004_0005, 0004_0004, 0004_0003, 0002_0000	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
108	0003_0000, 0003_0001, 0004_0005, 0002_0003, 0004_0004, 0004_0003, 0004_0006	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f006.png :material-alert-outline: (not found)
109	0003_0000, 0004_0005, 0002_0003, 0004_0004, 0004_0003, 0004_0007	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f007.png :material-alert-outline: (not found)
110	0004_0005, 0002_0003, 0004_0003, 0004_0008	build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f008.png :material-alert-outline: (not found)
111	0004_0005, 0002_0003, 0004_0003, 0004_0009	build/phase-3/bitmaps/PART18_4_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_4_f009.png :material-alert-outline: (not found)
112	0004_0010, 0002_0000	build/phase-3/bitmaps/PART18_4_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
113	0004_0000, 0002_0000	build/phase-3/bitmaps/PART18_4_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
114	0005_0013, 0005_0000, 0002_0000	build/phase-3/bitmaps/PART18_5_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
115	0005_0013, 0005_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_5_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
116	0005_0013, 0005_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_5_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
117	0005_0013, 0005_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_5_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
118	0005_0014, 0005_0000, 0002_0011, 0002_0000	build/phase-3/bitmaps/PART18_5_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
119	0005_0014, 0005_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_5_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
120	0005_0015, 0005_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_5_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
121	0005_0015, 0005_0000, 0002_0011, 0002_0000	build/phase-3/bitmaps/PART18_5_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
122	0005_0016, 0005_0000, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART18_5_f016.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
123	0005_0016, 0005_0000, 0002_0010, 0002_0000	build/phase-3/bitmaps/PART18_5_f016.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
124	0005_0017, 0005_0000, 0002_0011, 0002_0000	build/phase-3/bitmaps/PART18_5_f017.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
125	0005_0000, 0002_0000	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
126	0005_0001, 0002_0001	build/phase-3/bitmaps/PART18_5_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f001.png :material-alert-outline: (not found)
127	0005_0002, 0002_0002	build/phase-3/bitmaps/PART18_5_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f002.png :material-alert-outline: (not found)
128	0003_0000, 0003_0001, 0003_0002, 0005_0003, 0005_0005, 0005_0012, 0005_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f012.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found)
129	0003_0000, 0003_0001, 0003_0002, 0005_0003, 0005_0005, 0005_0012, 0005_0004	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f012.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found)
130	0003_0000, 0003_0001, 0005_0003, 0005_0005, 0005_0011, 0005_0004, 0005_0006	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f006.png :material-alert-outline: (not found)
131	0003_0000, 0005_0003, 0005_0005, 0005_0011, 0005_0004, 0005_0007	build/phase-3/bitmaps/PART18_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f007.png :material-alert-outline: (not found)
132	0005_0003, 0005_0005, 0005_0011, 0005_0008	build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f008.png :material-alert-outline: (not found)
133	0005_0003, 0005_0005, 0005_0011, 0005_0009	build/phase-3/bitmaps/PART18_5_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_5_f009.png :material-alert-outline: (not found)
134	0005_0010, 0002_0000	build/phase-3/bitmaps/PART18_5_f010.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)
135	0005_0000, 0002_0000	build/phase-3/bitmaps/PART18_5_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART18_2_f000.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
15	3018	0	build/phase-6/raw-wav/SX_3018_11025.wav
24	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
38	3018	0	build/phase-6/raw-wav/SX_3018_11025.wav
47	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
61	3018	0	build/phase-6/raw-wav/SX_3018_11025.wav
71	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
84	3018	0	build/phase-6/raw-wav/SX_3018_11025.wav
93	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
106	3018	0	build/phase-6/raw-wav/SX_3018_11025.wav
115	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
128	3018	0	build/phase-6/raw-wav/SX_3018_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	544
END_FRAME	136
PLAY_SFX	12
END_ANM	1

Notas de Comportamento:

Cannon has 2 semantic states: Not Fired, Fired. Transitions are triggered by: collision, proximity, timer.

Terminal state(s): Fired (part is removed from simulation).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
 - **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
 - **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
-

Peça 19 — Dynamite



Categoria: Explosives / Projectiles
Fases em que aparece: 28 fases
Quantidade de Estados: 3 estado(s)

Light the fuse of this dynamite with a laser, a flaming part (like a candle or rocket), or by using a magnifying glass and light source. It will blow up all kinds of things, including some walls.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Blown up	2→3	—	No
Blown up	3→-1	3	Yes :material-bomb:

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1132	mass/density (relative scale)
unk_2	90	category-specific property
property_q8	64	Q8.8 = 0.250
unk_6	32	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	53	bitmap frame width
bitmap_h	15	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No

Capability	Supported
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	53×15	0
3	1	1–16 (16 frames)	92×45	13
4	17	17	40×15	319

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f001.png :material-check:
1	0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f001.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
2	0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f001.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
3	0001_0000, 0001_0001, 0001_0007	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f001.png :material-check: build/phase-3/bitmaps/PART19_f007.png :material-check:
4	0001_0000, 0001_0002, 0001_0008	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f002.png :material-check: build/phase-3/bitmaps/PART19_f008.png :material-check:
5	0001_0000, 0001_0002, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f002.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
6	0001_0000, 0001_0002, 0001_0007	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f002.png :material-check: build/phase-3/bitmaps/PART19_f007.png :material-check:
7	0001_0000, 0001_0002, 0001_0008	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f002.png :material-check: build/phase-3/bitmaps/PART19_f008.png :material-check:
8	0001_0000, 0001_0003, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f003.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
9	0001_0000, 0001_0004, 0001_0004, 0001_0007	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f004.png :material-check: build/phase-3/bitmaps/PART19_f004.png :material-check: build/phase-3/bitmaps/PART19_f007.png :material-check:
10	0001_0000, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
11	0001_0000, 0001_0003, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f003.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
12	0001_0000, 0001_0004, 0001_0004, 0001_0007	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f004.png :material-check: build/phase-3/bitmaps/PART19_f004.png :material-check: build/phase-3/bitmaps/PART19_f007.png :material-check:
13	0001_0000, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
14	0001_0000, 0001_0003, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f003.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
15	0001_0000, 0001_0004, 0001_0004, 0001_0007	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f004.png :material-check: build/phase-3/bitmaps/PART19_f004.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART19_f007.png :material-check:
16	0001_0000, 0001_0006	build/phase-3/bitmaps/PART19_f000.png :material-check: build/phase-3/bitmaps/PART19_f006.png :material-check:
17	0001_0000	build/phase-3/bitmaps/PART19_f000.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	51
END_FRAME	18
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Dynamite has 2 semantic states: Not Blown up, Blown up. Transitions are triggered by: collision, proximity, timer.

State(s) Blown up are explosive — reaching these destroys the part.

Terminal state(s): Blown up (part is removed from simulation).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-019-dynamite.md` describes part type `19`
- **When** parsed
- **Then** `part_num == 19`

TC-002: Mass value is positive

- **Given** Dynamite has `mass = 1132`
- **When** validated
- **Then** `mass > 0`

TC-003: property_q8 encodes fuse timer

- **Given** Dynamite has `property_q8 = 64` (Q8.8 = 0.250)
- **When** lit
- **Then** explosion after `0.250 * 255` ticks (fuse duration)

TC-004: SOLVE.RES transition to Blown up

- **Given** Dynamite transitions 2→3 (Not Blown up → Blown up)
- **When** fuse timer expires OR collision/proximity trigger
- **Then** ANM state becomes 3

TC-005: Terminal explosive state removes part

- **Given** Dynamite transitions 3→-1
- **When** explosion animation completes
- **Then** part is removed from simulation and destroys nearby parts

TC-006: Explosive state destroys walls

- **Given** Dynamite has Explosive: Yes flag for state 3
- **When** in blown up state
- **Then** any wall with Destructible: Yes within blast radius is destroyed

TC-007: Fire via laser, flame, or magnifying glass

- **Given** Description: "light with laser, flaming part, or magnifying glass"
- **When** fire/proximity trigger
- **Then** fuse timer starts

TC-008: 3 ANM states

- **Given** Dynamite ANM shows Total States = 3
- **When** rendered
- **Then** states: fuse (idle), burning (16 frames), explosion

TC-009: 18 animation frames

- **Given** Dynamite ANM has Total Frames = 18
- **When** lit
- **Then** fuse burning animation plays

TC-010: PLAY_SFX at frame 1 (sizzle)

- **Given** Dynamite ANM has PLAY_SFX: 1 at frame 1
- **When** fuse burning
- **Then** SX_3019_11025.wav plays

TC-011: Part used in 28 levels

- **Given** Dynamite (type 19)
- **When** checked against YAML levels
- **Then** at least 28 levels reference it

TC-012: Category Part Count is 10

- **Given** Category Rules show Category Part Count: 10
- **When** validated
- **Then** 10 parts in Explosives/Projectiles category

TC-013: Spawns projectile

- **Given** Dynamite shows Spawns: projectile
- **When** explosion occurs
- **Then** debris projectile created

TC-014: Destructible by other explosives

- **Given** Dynamite shows Destructible: Yes
- **When** other explosion nearby

- **Then** chain reaction triggers

TC-015: Triggers include collision, proximity, timer

- **Given** Dynamite shows Triggers: collision, proximity, timer
- **When** any trigger activates
- **Then** fuse burns or explosion triggers

Peça 20 — Phazer pulse



Categoria: Explosives / Projectiles

Fases em que aparece: 0 fases

Quantidade de Estados: 1 estado(s)

[CREATED PART]

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	20000	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	16	bitmap frame width
bitmap_h	4	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–6 (7 frames)	24×12	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART20_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART20_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART20_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART20_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART20_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART20_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART20_f006.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	7
END_FRAME	7
END_ANM	1

Notas de Comportamento:

Phazer pulse is a dynamically created (phantom) part. It is not placed by the player but spawned at runtime by another part's behavior (e.g., explosion debris, laser beam, rope end phantom).

Phazer pulse is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-020-phazer-pulse.md` describes part type `20`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 20`

TC-002: Created Part flag is Yes

- **Given** Phazer Pulse Identity shows `Created Part: Yes`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == True`

TC-003: Not placed by player — no level usage

- **Given** Phazer Pulse is a dynamically created phantom part
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 20`

TC-004: Spawns projectile (category rule)

- **Given** Category Rules show `Spawns: projectile`
- **When** behavior handler executes
- **Then** a separate part instance is spawned on trigger

TC-005: Destructible flag is Yes

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** phazer pulse is removed from simulation

TC-006: Triggers include collision, proximity, timer

- **Given** Category Rules show `Triggers: collision, proximity, timer`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-007: No SOLVE.RES state data

- **Given** Phazer Pulse has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-008: ANM animation frames match state count

- **Given** Phazer Pulse ANM has `Total Frames = 7, Total States = 1`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 1`

TC-009: All 7 frames have DRAW_BMP bytecode

- **Given** Phazer Pulse ANM has 7 frames
- **When** bytecode is executed
- **Then** each frame contains a `DRAW_BMP` opcode referencing valid sprite

TC-010: No electrical/belt/rope/laser connections

- **Given** Connections & Capabilities shows `Electrical=No, Belt=No, Rope=No, Laser=No`
- **When** checked against `category_rules`
- **Then** only `Projectile=Yes`

TC-011: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** phazer pulse does not require user interaction

TC-012: 7-frame static animation (no Section A durations)

- **Given** Phazer Pulse ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty and frames are static

TC-013: Bitmap dimensions 16×4

- **Given** Physics Properties show `bitmap_w = 16, bitmap_h = 4`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: No sound effects

- **Given** Phazer Pulse ANM has no Sound Effects table
- **When** frame renders
- **Then** no `PLAY_SFX` opcode is present

TC-015: Extra data for phantom projectile

- **Given** Phazer Pulse is a created projectile part

- **When** `nt_part_info_t2_base` is parsed
 - **Then** extra data fields are initialized for spawn parameters
-

Peça 21 — Electric Switch & Outlet



Categoria: Electrical
Fases em que aparece: 54 fases
Quantidade de Estados: 8 estado(s)

Plug any of the electrical parts (like the toaster, can opener, or fan) into this electric outlet, then drop something on the switch to turn on the power. Or bump up on the switch if it's upside-down.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	32	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
6	0	0	32×32	0
7	1	1	32×32	13
8	2	2	32×32	32
9	3	3	32×32	51
10	4	4	32×32	76
11	5	5	32×32	89
12	6	6	32×32	108
13	7	7	32×32	127

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001	build/phase-3/bitmaps/PART21_f000.png :material-check: build/phase-3/bitmaps/PART21_f001.png :material-check:
1	0001_0000, 0001_0001, 0001_0003	build/phase-3/bitmaps/PART21_f000.png :material-check: build/phase-3/bitmaps/PART21_f001.png :material-check: build/phase-3/bitmaps/PART21_f003.png :material-check:
2	0001_0000, 0001_0001, 0001_0003	build/phase-3/bitmaps/PART21_f000.png :material-check: build/phase-3/bitmaps/PART21_f001.png :material-check: build/phase-3/bitmaps/PART21_f003.png :material-check:
3	0001_0000, 0001_0001, 0001_0003, 0001_0003	build/phase-3/bitmaps/PART21_f000.png :material-check: build/phase-3/bitmaps/PART21_f001.png :material-check: build/phase-3/bitmaps/PART21_f003.png :material-check: build/phase-3/bitmaps/PART21_f003.png :material-check:
4	0001_0000, 0001_0002	build/phase-3/bitmaps/PART21_f000.png :material-check: build/phase-3/bitmaps/PART21_f002.png :material-check:
5	0001_0000, 0001_0002, 0001_0003	build/phase-3/bitmaps/PART21_f000.png :material-check: build/phase-3/bitmaps/PART21_f002.png :material-check: build/phase-3/bitmaps/PART21_f003.png :material-check:
6	0001_0000, 0001_0002, 0001_0003	build/phase-3/bitmaps/PART21_f000.png :material-check: build/phase-3/bitmaps/PART21_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART21_f003.png :material-check:
7	0001_0000, 0001_0002, 0001_0003, 0001_0003	build/phase-3/bitmaps/PART21_f000.png :material-check: build/phase-3/bitmaps/PART21_f002.png :material-check: build/phase-3/bitmaps/PART21_f003.png :material-check: build/phase-3/bitmaps/PART21_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	24
END_FRAME	8
END_ANM	1

Notas de Comportamento:

Electric Switch & Outlet is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-021-electric-switch-&-outlet.md` describes part type `21`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 21`

TC-002: Mass value is positive

- **Given** Electric Switch & Outlet has `mass = 3776`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Electric Switch & Outlet has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical = Yes`
- **When** checked against `category_rules`
- **Then** electrical parts can be plugged in

TC-005: 8 ANM states (state IDs 6-13)

- **Given** Electric Switch & Outlet ANM has `Total States = 8`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 8`

TC-006: No SOLVE.RES state transitions

- **Given** Electric Switch & Outlet has no programmatic state transitions

- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-007: Part used in 54 levels

- **Given** Electric Switch & Outlet (type 21)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 54 levels reference `part_num = 21`

TC-008: No belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows `Belt=No, Rope=No, Fire=No, Laser=No, Projectile=No`
- **When** checked against `category_rules`
- **Then** only `Electrical=Yes`

TC-009: Triggers are electrical and collision

- **Given** Category Rules show `Triggers: electrical, collision`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-011: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** switch does not require user interaction to activate

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** switch is not destroyed

TC-013: Bitmap dimensions 32×32

- **Given** Physics Properties show `bitmap_w = 32, bitmap_h = 32`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: 24 DRAW_BMP opcodes across 8 states

- **Given** Bytecode Opcodes show `DRAW_BMP = 24` across 8 states
- **When** bytecode is executed
- **Then** each frame renders correct sprites for that state



Categoria: Explosives / Projectiles
Fases em que aparece: 57 fases
Quantidade de Estados: 2 estado(s)

As soon as you set this remote control down on the screen, a second part made up of explosives appears. Drag the explosives to the area or object you want to blow up. Drop something on top of the remote control button to set off the explosion. You can also tie one end of a rope to the button and hitch the other end to a teeter-totter, or another object that will allow you to pull the button down.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	32	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	30	bitmap frame width
bitmap_h	37	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0	35×37	0
4	1	1–13 (13 frames)	35×37	25

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0002, 0001_0001, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f001.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check:
1	0001_0002, 0001_0001, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f001.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check:
2	0001_0001, 0001_0002, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART22_f001.png :material-check: build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check:
3	0001_0001, 0001_0002, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART22_f001.png :material-check: build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check:
4	0001_0001, 0001_0002, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART22_f001.png :material-check: build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check:
5	0001_0002, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check:
6	0001_0002, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f004.png :material-check:
7	0001_0002, 0001_0000, 0001_0003	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check:
8	0001_0002, 0001_0000, 0001_0005, 0001_0009	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f005.png :material-check: build/phase-3/bitmaps/PART22_f009.png :material-check:
9	0001_0002, 0001_0000, 0001_0003, 0001_0010	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check: build/phase-3/bitmaps/PART22_f010.png :material-check:
10	0001_0002, 0001_0000, 0001_0004, 0001_0011	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f004.png :material-check: build/phase-3/bitmaps/PART22_f011.png :material-check:
11	0001_0002, 0001_0000, 0001_0003, 0001_0012	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check: build/phase-3/bitmaps/PART22_f012.png :material-check:
12	0001_0002, 0001_0000, 0001_0003, 0001_0013	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check: build/phase-3/bitmaps/PART22_f013.png :material-check:
13	0001_0002, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART22_f002.png :material-check: build/phase-3/bitmaps/PART22_f000.png :material-check: build/phase-3/bitmaps/PART22_f003.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
3	3022	0	build/phase-6/raw-wav/SX_3022_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	52
END_FRAME	14
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Remote Control is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-022-remote-control.md` describes part type `22`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 22`

TC-002: Mass value is positive

- **Given** Remote Control has `mass = 4153`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Remote Control has `property_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (1024 >> 8) <= 1.0` i.e. `COR = 4.000`

TC-004: 2 ANM states (state IDs 3-4)

- **Given** Remote Control ANM has `Total States = 2`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 2`

TC-005: No SOLVE.RES state transitions

- **Given** Remote Control has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-006: Part used in 57 levels

- **Given** Remote Control (type 22)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 57 levels reference `part_num = 22`

TC-007: PLAY_SFX opcode triggers sound ID 3022

- **Given** Bytecode Opcodes show `PLAY_SFX = 1` at frame 3
- **When** animation reaches frame 3

- **Then** `PLAY_SFX` opcode dispatches sound ID `3022`

TC-008: 14 frames across 2 states

- **Given** Remote Control ANM has `Total Frames = 14`, `Total States = 2`
- **When** bytecode is executed
- **Then** state 3 has frame 0, state 4 has frames 1-13

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** remote control does not require user interaction

TC-011: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows `Electrical=No`, `Belt=No`, `Rope=No`, `Fire=No`, `Laser=No`, `Projectile=No`
- **When** checked against `category_rules`
- **Then** only default connections

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** remote control is not destroyed

TC-013: Bitmap dimensions 30×37

- **Given** Physics Properties show `bitmap_w = 30`, `bitmap_h = 37`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: 52 DRAW_BMP opcodes across 14 frames

- **Given** Bytecode Opcodes show `DRAW_BMP = 52`
- **When** bytecode is executed
- **Then** each frame renders correct sprites

TC-015: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

Peça 23 — Boat Cleat



Categoria: Special Mechanics

Fases em que aparece: 75 fases

Quantidade de Estados: 2 estado(s)

Any object that can be tied with a rope may be hitched to this boat cleat. Use it to hang things from the air, or to keep balloons from floating away.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	998	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	30	bitmap frame width
bitmap_h	15	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	32×15	0
2	1	1–3 (3 frames)	32×18	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART23.png :material-check:
1	0001_0000, 0002_0000	build/phase-3/bitmaps/PART23.png :material-check: build/phase-3/bitmaps/PART23_2_f000.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
2	0001_0000, 0002_0001	build/phase-3/bitmaps/PART23.png :material-check: build/phase-3/bitmaps/PART23_2_f001.png :material-alert-outline: (not found)
3	0001_0000, 0002_0002	build/phase-3/bitmaps/PART23.png :material-check: build/phase-3/bitmaps/PART23_2_f002.png :material-alert-outline: (not found)

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	7
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Boat Cleat is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has custom spawn behavior defined in SOLVE.RES.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-023-boat-cleat.md` describes part type `23`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 23`

TC-002: Mass value is positive

- **Given** Boat Cleat has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Boat Cleat has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Rope connection supported

- **Given** Description: "Any object that can be tied with a rope may be hitched to this boat cleat"
- **When** rope is attached
- **Then** `rope_info_1/2` stores connected part indices

TC-005: No SOLVE.RES state transitions

- **Given** Boat Cleat has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-006: Part used in 75 levels

- **Given** Boat Cleat (type 23)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 75 levels reference `part_num = 23`

TC-007: Triggers include collision, electrical, proximity

- **Given** Category Rules show `Triggers: collision, electrical, proximity`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-008: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-009: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** boat cleat does not require user interaction

TC-010: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** boat cleat is not destroyed

TC-011: Bitmap dimensions 30×15

- **Given** Physics Properties show `bitmap_w = 30, bitmap_h = 15`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-012: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-013: Anchor point for rope connections

- **Given** Boat Cleat is used to "hang things from the air, or to keep balloons from floating away"
- **When** rope placement connects
- **Then** cleat acts as fixed anchor point

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: No belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows only default connections
- **When** checked against `category_rules`
- **Then** rope is a logical connector, not a connection flag

Peça 24 — Electric Fan



Categoria: Electrical

Fases em que aparece: 23 fases

Quantidade de Estados: 2 estado(s)

Plug this electric fan into an outlet to make it blow air. Flip it to change wind direction. Use it to blow objects away or to turn the pinwheel.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	1→2	—	No
On	2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	27	bitmap frame width
bitmap_h	46	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	32×46	0
2	1	1–68 (68 frames)	32×46	25

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
1	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
2	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
3	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
4	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
5	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
6	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f013.png :material-check:
7	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
8	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
9	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
10	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
11	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
12	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
13	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f013.png :material-check:
14	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
15	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
16	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
17	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
18	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
19	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
20	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f013.png :material-check:
21	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
22	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
23	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
24	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
25	0001_0000, 0001_0001, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f006.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
26	0001_0000, 0001_0001, 0001_0005, 0001_0013	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f001.png :material-check: build/phase-3/bitmaps/PART24_f005.png :material-check: build/phase-3/bitmaps/PART24_f013.png :material-check:
27	0001_0000, 0001_0002, 0001_0008, 0001_0014	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f002.png :material-check: build/phase-3/bitmaps/PART24_f008.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f014.png :material-check:
28	0001_0000, 0001_0002, 0001_0007, 0001_0014	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f002.png :material-check: build/phase-3/bitmaps/PART24_f007.png :material-check: build/phase-3/bitmaps/PART24_f014.png :material-check:
29	0001_0000, 0001_0002, 0001_0008, 0001_0014	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f002.png :material-check: build/phase-3/bitmaps/PART24_f008.png :material-check: build/phase-3/bitmaps/PART24_f014.png :material-check:
30	0001_0000, 0001_0002, 0001_0007, 0001_0014	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f002.png :material-check: build/phase-3/bitmaps/PART24_f007.png :material-check: build/phase-3/bitmaps/PART24_f014.png :material-check:
31	0001_0000, 0001_0003, 0001_0009, 0001_0015	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f003.png :material-check: build/phase-3/bitmaps/PART24_f009.png :material-check: build/phase-3/bitmaps/PART24_f015.png :material-check:
32	0001_0000, 0001_0003, 0001_0010, 0001_0015	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f003.png :material-check: build/phase-3/bitmaps/PART24_f010.png :material-check: build/phase-3/bitmaps/PART24_f015.png :material-check:
33	0001_0000, 0001_0003, 0001_0009, 0001_0015	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f003.png :material-check: build/phase-3/bitmaps/PART24_f009.png :material-check: build/phase-3/bitmaps/PART24_f015.png :material-check:
34	0001_0000, 0001_0003, 0001_0010, 0001_0015	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f003.png :material-check: build/phase-3/bitmaps/PART24_f010.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f015.png :material-check:
35	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
36	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
37	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
38	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
39	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
40	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
41	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f004.png :material-check:
42	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
43	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
44	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
45	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
46	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
47	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
48	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f004.png :material-check:
49	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
50	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
51	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
52	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
53	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
54	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
55	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f004.png :material-check:
56	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
57	0001_0011, 0001_0016, 0001_0000, 0001_0004	build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
58	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
59	0001_0000, 0001_0011, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f011.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
60	0001_0000, 0001_0012, 0001_0016, 0001_0004	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f012.png :material-check: build/phase-3/bitmaps/PART24_f016.png :material-check: build/phase-3/bitmaps/PART24_f004.png :material-check:
61	0001_0000, 0001_0003, 0001_0009, 0001_0015	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f003.png :material-check: build/phase-3/bitmaps/PART24_f009.png :material-check: build/phase-3/bitmaps/PART24_f015.png :material-check:
62	0001_0000, 0001_0003, 0001_0010, 0001_0015	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f003.png :material-check: build/phase-3/bitmaps/PART24_f010.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART24_f015.png :material-check:
63	0001_0000, 0001_0003, 0001_0009, 0001_0015	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f003.png :material-check: build/phase-3/bitmaps/PART24_f009.png :material-check: build/phase-3/bitmaps/PART24_f015.png :material-check:
64	0001_0000, 0001_0003, 0001_0010, 0001_0015	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f003.png :material-check: build/phase-3/bitmaps/PART24_f010.png :material-check: build/phase-3/bitmaps/PART24_f015.png :material-check:
65	0001_0000, 0001_0002, 0001_0008, 0001_0014	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f002.png :material-check: build/phase-3/bitmaps/PART24_f008.png :material-check: build/phase-3/bitmaps/PART24_f014.png :material-check:
66	0001_0000, 0001_0002, 0001_0007, 0001_0014	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f002.png :material-check: build/phase-3/bitmaps/PART24_f007.png :material-check: build/phase-3/bitmaps/PART24_f014.png :material-check:
67	0001_0000, 0001_0002, 0001_0008, 0001_0014	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f002.png :material-check: build/phase-3/bitmaps/PART24_f008.png :material-check: build/phase-3/bitmaps/PART24_f014.png :material-check:
68	0001_0000, 0001_0002, 0001_0007, 0001_0014	build/phase-3/bitmaps/PART24_f000.png :material-check: build/phase-3/bitmaps/PART24_f002.png :material-check: build/phase-3/bitmaps/PART24_f007.png :material-check: build/phase-3/bitmaps/PART24_f014.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	24	0	build/phase-6/raw-wav/SX_24_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
2	24	0	build/phase-6/raw-wav/SX_24_11025.wav
3	24	0	build/phase-6/raw-wav/SX_24_11025.wav
4	24	0	build/phase-6/raw-wav/SX_24_11025.wav
5	24	0	build/phase-6/raw-wav/SX_24_11025.wav
6	24	0	build/phase-6/raw-wav/SX_24_11025.wav
7	24	0	build/phase-6/raw-wav/SX_24_11025.wav
8	24	0	build/phase-6/raw-wav/SX_24_11025.wav
9	24	0	build/phase-6/raw-wav/SX_24_11025.wav
10	24	0	build/phase-6/raw-wav/SX_24_11025.wav
11	24	0	build/phase-6/raw-wav/SX_24_11025.wav
12	24	0	build/phase-6/raw-wav/SX_24_11025.wav
13	24	0	build/phase-6/raw-wav/SX_24_11025.wav
14	24	0	build/phase-6/raw-wav/SX_24_11025.wav
15	24	0	build/phase-6/raw-wav/SX_24_11025.wav
16	24	0	build/phase-6/raw-wav/SX_24_11025.wav
17	24	0	build/phase-6/raw-wav/SX_24_11025.wav
18	24	0	build/phase-6/raw-wav/SX_24_11025.wav
19	24	0	build/phase-6/raw-wav/SX_24_11025.wav
20	24	0	build/phase-6/raw-wav/SX_24_11025.wav
21	24	0	build/phase-6/raw-wav/SX_24_11025.wav
22	24	0	build/phase-6/raw-wav/SX_24_11025.wav
23	24	0	build/phase-6/raw-wav/SX_24_11025.wav
24	24	0	build/phase-6/raw-wav/SX_24_11025.wav
25	24	0	build/phase-6/raw-wav/SX_24_11025.wav
26	24	0	build/phase-6/raw-wav/SX_24_11025.wav
27	24	0	build/phase-6/raw-wav/SX_24_11025.wav
28	24	0	build/phase-6/raw-wav/SX_24_11025.wav
29	24	0	build/phase-6/raw-wav/SX_24_11025.wav
30	24	0	build/phase-6/raw-wav/SX_24_11025.wav
31	24	0	build/phase-6/raw-wav/SX_24_11025.wav
32	24	0	build/phase-6/raw-wav/SX_24_11025.wav
33	24	0	build/phase-6/raw-wav/SX_24_11025.wav
34	24	0	build/phase-6/raw-wav/SX_24_11025.wav
35	24	0	build/phase-6/raw-wav/SX_24_11025.wav
36	24	0	build/phase-6/raw-wav/SX_24_11025.wav
37	24	0	build/phase-6/raw-wav/SX_24_11025.wav
38	24	0	build/phase-6/raw-wav/SX_24_11025.wav
39	24	0	build/phase-6/raw-wav/SX_24_11025.wav
40	24	0	build/phase-6/raw-wav/SX_24_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
41	24	0	build/phase-6/raw-wav/SX_24_11025.wav
42	24	0	build/phase-6/raw-wav/SX_24_11025.wav
43	24	0	build/phase-6/raw-wav/SX_24_11025.wav
44	24	0	build/phase-6/raw-wav/SX_24_11025.wav
45	24	0	build/phase-6/raw-wav/SX_24_11025.wav
46	24	0	build/phase-6/raw-wav/SX_24_11025.wav
47	24	0	build/phase-6/raw-wav/SX_24_11025.wav
48	24	0	build/phase-6/raw-wav/SX_24_11025.wav
49	24	0	build/phase-6/raw-wav/SX_24_11025.wav
50	24	0	build/phase-6/raw-wav/SX_24_11025.wav
51	24	0	build/phase-6/raw-wav/SX_24_11025.wav
52	24	0	build/phase-6/raw-wav/SX_24_11025.wav
53	24	0	build/phase-6/raw-wav/SX_24_11025.wav
54	24	0	build/phase-6/raw-wav/SX_24_11025.wav
55	24	0	build/phase-6/raw-wav/SX_24_11025.wav
56	24	0	build/phase-6/raw-wav/SX_24_11025.wav
57	24	0	build/phase-6/raw-wav/SX_24_11025.wav
58	24	0	build/phase-6/raw-wav/SX_24_11025.wav
59	24	0	build/phase-6/raw-wav/SX_24_11025.wav
60	24	0	build/phase-6/raw-wav/SX_24_11025.wav
61	24	0	build/phase-6/raw-wav/SX_24_11025.wav
62	24	0	build/phase-6/raw-wav/SX_24_11025.wav
63	24	0	build/phase-6/raw-wav/SX_24_11025.wav
64	24	0	build/phase-6/raw-wav/SX_24_11025.wav
65	24	0	build/phase-6/raw-wav/SX_24_11025.wav
66	24	0	build/phase-6/raw-wav/SX_24_11025.wav
67	24	0	build/phase-6/raw-wav/SX_24_11025.wav
68	24	0	build/phase-6/raw-wav/SX_24_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	276
END_FRAME	69
PLAY_SFX	68
END_ANM	1

Notas de Comportamento:

Electric Fan has 2 semantic states: Off, On. Transitions are triggered by: electrical, collision.

Terminal state(s): On (part is removed from simulation).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 25 — Flashlight



Categoria: Light / Flame Sources

Fases em que aparece: 55 fases

Quantidade de Estados: 3 estado(s)

Drop something on the button of this flashlight to turn it on. Use it to power solar panels. Or put a magnifying glass right in front of it and use it to light fuses and candles.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	3→4	3	No
On	4→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	46	bitmap frame width
bitmap_h	21	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	proximity, electrical
Destructible	No
Spawns	fire

Rule	Value
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	Yes
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0	48×21	0
4	1	1	69×34	19
2	2	2	69×34	40

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001, 0001_0002	build/phase-3/bitmaps/PART25_f000.png :material-check: build/phase-3/bitmaps/PART25_f001.png :material-check: build/phase-3/bitmaps/PART25_f002.png :material-check:
1	0001_0002, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART25_f002.png :material-check: build/phase-3/bitmaps/PART25_f000.png :material-check: build/phase-3/bitmaps/PART25_f003.png :material-check:
2	0001_0002, 0001_0000, 0001_0003	build/phase-3/bitmaps/PART25_f002.png :material-check: build/phase-3/bitmaps/PART25_f000.png :material-check: build/phase-3/bitmaps/PART25_f003.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3025	0	build/phase-6/raw-wav/SX_3025_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	9
END_FRAME	3
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Flashlight has 2 semantic states: Off, On. Transitions are triggered by: proximity, electrical.

Terminal state(s): On (part is removed from simulation).

This part can ignite nearby combustible parts (balloons, explosives). Uses proximity trigger.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-025-flashlight.md` describes part type `25`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 25`

TC-002: Mass value is positive

- **Given** Flashlight has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Flashlight has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: SOLVE.RES state transitions (Off → On)

- **Given** Flashlight has 2 states: Off and On
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Terminal state On (part removed)

- **Given** Flashlight State Machine shows `On: 2→-1`
- **When** state reaches On
- **Then** part is removed from simulation (terminal)

TC-006: Spawns fire (light source)

- **Given** Category Rules show `Spawns: fire`
- **When** flashlight is activated
- **Then** fire/light beam is emitted

TC-007: Triggers are proximity and electrical

- **Given** Category Rules show `Triggers: proximity, electrical`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-008: Part used in 55 levels

- **Given** Flashlight (type 25)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 55 levels reference `part_num = 25`

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs

- **Then** flashlight can be manually activated

TC-011: Powers solar panels (electrical output)

- **Given** Description: "Use it to power solar panels"
- **When** flashlight is On
- **Then** electrical power is supplied to connected parts

TC-012: Can light fuses with magnifying glass

- **Given** Description: "put a magnifying glass right in front of it and use it to light fuses and candles"
- **When** light beam intersects magnifying glass
- **Then** focused beam ignites fuse/candle

TC-013: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** flashlight is not destroyed

TC-014: Bitmap dimensions 46x21

- **Given** Physics Properties show `bitmap_w = 46, bitmap_h = 21`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Category Part Count is 5

- **Given** Category Rules show `Category Part Count: 5`
- **When** validated
- **Then** 5 parts in Light / Flame Sources category

Peça 26 — Generator



Categoria: Electrical
Fases em que aparece: 15 fases
Quantidade de Estados: 8 estado(s)

This generator comes with its own outlet. Use it to supply electricity to power parts by connecting the generator's wheel to a rotational motor (like the Mandrill Motor, the Mouse Motor, or the Electric Motor) by adding a belt.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	9→10	—	No
On	2→3	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500

Property	Raw Value	Interpretation
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	81	bitmap frame width
bitmap_h	34	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
9	0	0	81×34	0
10	1	1	81×34	13
11	2	2	81×34	32
12	3	3	81×34	51
2	4	4–11 (8 frames)	81×34	76
3	12	12–19 (8 frames)	81×34	244
4	20	20–27 (8 frames)	81×34	460
5	28	28–35 (8 frames)	81×34	676

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0002_0000	build/phase-3/bitmaps/PART26_f000.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found)
1	0001_0000, 0002_0000, 0002_0001	build/phase-3/bitmaps/PART26_f000.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material- alert-outline: (not found)
2	0001_0000, 0002_0000, 0002_0001	build/phase-3/bitmaps/PART26_f000.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material- alert-outline: (not found)
3	0001_0000, 0002_0000, 0002_0001, 0002_0001	build/phase-3/bitmaps/PART26_f000.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material- alert-outline: (not found)
4	0002_0000, 0001_0001, 0003_0000	build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_f001.png :material-check: build/phase-3/bitmaps/PART26_3_f000.png :material- alert-outline: (not found)
5	0001_0002, 0002_0000, 0003_0001	build/phase-3/bitmaps/PART26_f002.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f001.png :material- alert-outline: (not found)
6	0001_0003, 0002_0000, 0003_0002	build/phase-3/bitmaps/PART26_f003.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f002.png :material- alert-outline: (not found)
7	0001_0004, 0002_0000, 0003_0003	build/phase-3/bitmaps/PART26_f004.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f003.png :material- alert-outline: (not found)
8	0002_0000, 0001_0001, 0003_0004	build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_f001.png :material-check: build/phase-3/bitmaps/PART26_3_f004.png :material- alert-outline: (not found)
9	0001_0002, 0002_0000, 0003_0005	build/phase-3/bitmaps/PART26_f002.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f005.png :material- alert-outline: (not found)
10	0001_0003, 0002_0000, 0003_0006	build/phase-3/bitmaps/PART26_f003.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f006.png :material- alert-outline: (not found)
11	0001_0004, 0002_0000, 0003_0007	build/phase-3/bitmaps/PART26_f004.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material- alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART26_3_f007.png :material-alert-outline: (not found)
12	0002_0000, 0001_0001, 0002_0001, 0003_0000	build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_f001.png :material-check: build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f000.png :material-alert-outline: (not found)
13	0001_0002, 0002_0000, 0002_0001, 0003_0001	build/phase-3/bitmaps/PART26_f002.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f001.png :material-alert-outline: (not found)
14	0001_0003, 0002_0000, 0002_0001, 0003_0002	build/phase-3/bitmaps/PART26_f003.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f002.png :material-alert-outline: (not found)
15	0001_0004, 0002_0000, 0002_0001, 0003_0003	build/phase-3/bitmaps/PART26_f004.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f003.png :material-alert-outline: (not found)
16	0002_0000, 0001_0001, 0002_0001, 0003_0004	build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_f001.png :material-check: build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f004.png :material-alert-outline: (not found)
17	0001_0002, 0002_0000, 0002_0001, 0003_0005	build/phase-3/bitmaps/PART26_f002.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f005.png :material-alert-outline: (not found)
18	0001_0003, 0002_0000, 0002_0001, 0003_0006	build/phase-3/bitmaps/PART26_f003.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f006.png :material-alert-outline: (not found)
19	0001_0004, 0002_0000, 0002_0001, 0003_0007	build/phase-3/bitmaps/PART26_f004.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f007.png :material-alert-outline: (not found)
20	0002_0000, 0001_0001, 0003_0000, 0002_0001	build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_f001.png :material-check: build/phase-3/bitmaps/PART26_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
21	0001_0002, 0002_0000, 0003_0001, 0002_0001	build/phase-3/bitmaps/PART26_f002.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
22	0001_0003, 0002_0000, 0003_0002, 0002_0001	build/phase-3/bitmaps/PART26_f003.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
23	0001_0004, 0002_0000, 0003_0003, 0002_0001	build/phase-3/bitmaps/PART26_f004.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
24	0002_0000, 0001_0001, 0003_0004, 0002_0001	build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_f001.png :material-check: build/phase-3/bitmaps/PART26_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
25	0001_0002, 0002_0000, 0003_0005, 0002_0001	build/phase-3/bitmaps/PART26_f002.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
26	0001_0003, 0002_0000, 0003_0006, 0002_0001	build/phase-3/bitmaps/PART26_f003.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
27	0001_0004, 0002_0000, 0003_0007, 0002_0001	build/phase-3/bitmaps/PART26_f004.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART26_3_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
28	0002_0000, 0001_0001, 0003_0000, 0002_0001, 0002_0001	build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_f001.png :material-check: build/phase-3/bitmaps/PART26_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
29	0001_0002, 0002_0000, 0002_0001, 0003_0001, 0002_0001	build/phase-3/bitmaps/PART26_f002.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
30	0001_0003, 0002_0000, 0002_0001, 0003_0002, 0002_0001	build/phase-3/bitmaps/PART26_f003.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
31	0001_0004, 0002_0000, 0002_0001, 0003_0003, 0002_0001	build/phase-3/bitmaps/PART26_f004.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
32	0002_0000, 0001_0001, 0002_0001, 0003_0004, 0002_0001	build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_f001.png :material-check: build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
33	0001_0002, 0002_0000, 0002_0001, 0003_0005, 0002_0001	build/phase-3/bitmaps/PART26_f002.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f005.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
34	0001_0003, 0002_0000, 0002_0001, 0003_0006, 0002_0001	build/phase-3/bitmaps/PART26_f003.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)
35	0001_0004, 0002_0000, 0002_0001, 0003_0007, 0002_0001	build/phase-3/bitmaps/PART26_f004.png :material-check: build/phase-3/bitmaps/PART26_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_3_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART26_2_f001.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
4	26	0	build/phase-6/raw-wav/SX_26_11025.wav
5	26	0	build/phase-6/raw-wav/SX_26_11025.wav
6	26	0	build/phase-6/raw-wav/SX_26_11025.wav
7	26	0	build/phase-6/raw-wav/SX_26_11025.wav
8	26	0	build/phase-6/raw-wav/SX_26_11025.wav
9	26	0	build/phase-6/raw-wav/SX_26_11025.wav
10	26	0	build/phase-6/raw-wav/SX_26_11025.wav
11	26	0	build/phase-6/raw-wav/SX_26_11025.wav
12	26	0	build/phase-6/raw-wav/SX_26_11025.wav
13	26	0	build/phase-6/raw-wav/SX_26_11025.wav
14	26	0	build/phase-6/raw-wav/SX_26_11025.wav
15	26	0	build/phase-6/raw-wav/SX_26_11025.wav
16	26	0	build/phase-6/raw-wav/SX_26_11025.wav
17	26	0	build/phase-6/raw-wav/SX_26_11025.wav
18	26	0	build/phase-6/raw-wav/SX_26_11025.wav
19	26	0	build/phase-6/raw-wav/SX_26_11025.wav
20	26	0	build/phase-6/raw-wav/SX_26_11025.wav
21	26	0	build/phase-6/raw-wav/SX_26_11025.wav
22	26	0	build/phase-6/raw-wav/SX_26_11025.wav
23	26	0	build/phase-6/raw-wav/SX_26_11025.wav
24	26	0	build/phase-6/raw-wav/SX_26_11025.wav
25	26	0	build/phase-6/raw-wav/SX_26_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
26	26	0	build/phase-6/raw-wav/SX_26_11025.wav
27	26	0	build/phase-6/raw-wav/SX_26_11025.wav
28	26	0	build/phase-6/raw-wav/SX_26_11025.wav
29	26	0	build/phase-6/raw-wav/SX_26_11025.wav
30	26	0	build/phase-6/raw-wav/SX_26_11025.wav
31	26	0	build/phase-6/raw-wav/SX_26_11025.wav
32	26	0	build/phase-6/raw-wav/SX_26_11025.wav
33	26	0	build/phase-6/raw-wav/SX_26_11025.wav
34	26	0	build/phase-6/raw-wav/SX_26_11025.wav
35	26	0	build/phase-6/raw-wav/SX_26_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	140
END_FRAME	36
PLAY_SFX	32
END_ANM	1

Notas de Comportamento:

Generator has 2 semantic states: Off, On. Transitions are triggered by: electrical, collision.

Compliance Test Cases

TC-001: Part type number matches filename

- Given part-026-generator.md describes part type 26
- When parsed by decoder/d_puzzle.py → nt_part_info_t2_base
- Then part_num == 26

TC-002: Mass value is positive

- Given Generator has mass = 3776
- When validated against build/phase-9/part-properties.json
- Then mass > 0

TC-003: COR within valid range [0.0, 1.0]

- Given Generator has property_q8 = 128
- When converted from Q8.8 fixed-point
- Then 0.0 <= (128 >> 8) <= 1.0 i.e. COR = 0.500

TC-004: SOLVE.RES state transitions (Off → On)

- Given Generator has 2 states: Off (9→10) and On (2→3)
- When behavior handler evaluates trigger
- Then state transitions follow SOLVE.RES rules

TC-005: Belt connection to power rotating parts

- Given Description: "Attach a belt to the treadmill to power anything that is driven with a belt"
- When belt connects generator to motor

- **Then** rotational power is transferred

TC-006: 8 ANM states (state IDs span multiple values)

- **Given** Generator ANM has `Total States = 8`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 8`

TC-007: Part used in 15 levels

- **Given** Generator (type 26)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 15 levels reference `part_num = 26`

TC-008: Triggers are electrical and collision

- **Given** Category Rules show `Triggers: electrical, collision`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** generator can be manually activated

TC-011: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical = Yes`
- **When** checked against `category_rules`
- **Then** generator has its own outlet for electrical parts

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** generator is not destroyed

TC-013: Belt Connection flag is Yes

- **Given** Connections & Capabilities shows `Belt Connection = Yes`
- **When** belt is attached
- **Then** rotational power is transferred to connected part

TC-014: Bitmap dimensions 81×34

- **Given** Physics Properties show `bitmap_w = 81, bitmap_h = 34`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Category Part Count is 9

- **Given** Category Rules show `Category Part Count: 9`
 - **When** validated
 - **Then** 9 parts in Electrical category
-

Peça 27 — Captain Z Super Phazer



Categoria: Explosives / Projectiles

Fases em que aparece: 22 fases

Quantidade de Estados: 4 estado(s)

This spiffy toy phazer shoots pulses of energy. Tie one end of a rope to the trigger and run it through a pulley (placed behind the phazer), then tie the other end to a balloon or something heavy. You can program the number of energy pulses you want to fire, but the gun will only shoot as long as the rope is pulling on the trigger. Use phazer pulses to bump things, pop balloons and blimps, light candles and fuses, and blow up explosives.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Fired	1→6	—	No
Fired	6→-1	6	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	192	Q8.8 = 0.750
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	65	bitmap frame width
bitmap_h	44	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No

Capability	Supported
Fire/Flame	No
Laser	No
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	70×44	0
6	1	1–7 (7 frames)	70×44	19
5	8	8	70×44	196
4	9	9	70×44	221

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0001_0020, 0001_0000	build/phase-3/bitmaps/PART27_f001.png :material-check: build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check:
1	0001_0020, 0001_0000, 0001_0002, 0001_0011	build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f002.png :material-check: build/phase-3/bitmaps/PART27_f011.png :material-check:
2	0001_0020, 0001_0000, 0001_0003, 0001_0012	build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f003.png :material-check: build/phase-3/bitmaps/PART27_f012.png :material-check:
3	0001_0020, 0001_0000, 0001_0004, 0001_0013	build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f004.png :material-check: build/phase-3/bitmaps/PART27_f013.png :material-check:
4	0001_0020, 0001_0000, 0001_0005, 0001_0014	build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f005.png :material-check: build/phase-3/bitmaps/PART27_f014.png :material-check:

Frame	Sprites	PNG References
5	0001_0020, 0001_0000, 0001_0006, 0001_0015	build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f006.png :material-check: build/phase-3/bitmaps/PART27_f015.png :material-check:
6	0001_0020, 0001_0000, 0001_0007, 0001_0016	build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f007.png :material-check: build/phase-3/bitmaps/PART27_f016.png :material-check:
7	0001_0020, 0001_0000, 0001_0008, 0001_0016	build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f008.png :material-check: build/phase-3/bitmaps/PART27_f016.png :material-check:
8	0001_0020, 0001_0000, 0001_0009, 0001_0018	build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f009.png :material-check: build/phase-3/bitmaps/PART27_f018.png :material-check:
9	0001_0001, 0001_0020, 0001_0000, 0001_0000	build/phase-3/bitmaps/PART27_f001.png :material-check: build/phase-3/bitmaps/PART27_f020.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check: build/phase-3/bitmaps/PART27_f000.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3027	0	build/phase-6/raw-wav/SX_3027_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	39
END_FRAME	10
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Captain Z Super Phazer has 2 semantic states: Not Fired, Fired. Transitions are triggered by: collision, proximity, timer.

Terminal state(s): Fired (part is removed from simulation).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-027-captain-z-super-phazer.md` describes part type `27`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 27`

TC-002: Mass value is positive

- **Given** Captain Z Super Phazer has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Captain Z Super Phazer has `property_q8 = 192`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (192 >> 8) <= 1.0` i.e. `COR = 0.750`

TC-004: SOLVE.RES state transitions (Not Fired → Fired)

- **Given** Captain Z has 2 states: Not Fired (`1→6`) and Fired (`6→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Terminal state Fired (part removed)

- **Given** State Machine shows `Fired: 6→-1`
- **When** state reaches Fired
- **Then** part is removed from simulation (terminal)

TC-006: Spawns projectile (phazer pulse)

- **Given** Category Rules show `Spawns: projectile`
- **When** rope pulls trigger
- **Then** phazer pulse (type 20) is spawned

TC-007: Destructible flag is Yes

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** Captain Z is destroyed

TC-008: Triggers include collision, proximity, timer

- **Given** Category Rules show `Triggers: collision, proximity, timer`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-009: Part used in 22 levels

- **Given** Captain Z Super Phazer (type 27)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 22 levels reference `part_num = 27`

TC-010: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-011: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** phazer requires rope-trigger mechanism to fire

TC-012: Rope-activated trigger mechanism

- **Given** Description: "Tie one end of a rope to the trigger and run it through a pulley"
- **When** rope pulls trigger
- **Then** phazer fires programmed number of pulses

TC-013: Can pop balloons and blimps

- **Given** Description: "use phazer pulses to bump things, pop balloons and blimps"
- **When** projectile hits balloon/blimp
- **Then** target is destroyed

TC-014: 4 ANM states (state IDs 1, 6, and others)

- **Given** Captain Z ANM has `Total States = 4`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 4`

TC-015: Projectile connection flag is Yes

- **Given** Connections & Capabilities shows `Projectile = Yes`
- **When** checked against `category_rules`
- **Then** part can spawn projectiles

Peça 28 — Baseball



Categoria: Balls
Fases em que aparece: 47 fases
Quantidade de Estados: 1 estado(s)

This baseball is pretty light and not very bouncy.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
damping	9	drag/air-resistance or density
cor_q8	112	COR = 0.438
radius	24	collision radius in internal units

Property	Raw Value	Interpretation
sprite_w	240	sprite width
sprite_h	240	sprite height
data_offset	18	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–7 (8 frames)	24×18	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART28_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART28_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART28_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART28_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART28_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART28_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART28_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART28_f007.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	8
END_FRAME	8
END_ANM	1

Notas de Comportamento:

Baseball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-028-baseball.md` describes part type `28`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 28`

TC-002: Mass value is positive

- **Given** Baseball has `mass = 2000`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Baseball has `cor_q8 = 112`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (112 >> 8) <= 1.0` i.e. `COR = 0.438`

TC-004: Radius is non-negative

- **Given** Baseball has `radius = 24`
- **When** physics engine initializes collision
- **Then** `radius >= 0`

TC-005: Damping value affects air resistance

- **Given** Baseball has `damping = 9`
- **When** physics simulation runs
- **Then** velocity decay matches expected air resistance

TC-006: Single default state — no named transitions

- **Given** Baseball has no SOLVE.RES state data
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-007: Part used in 47 levels

- **Given** Baseball (type 28)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 47 levels reference `part_num = 28`

TC-008: Category Part Count is 9 (Balls category)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Balls category

TC-009: Triggers are collision and trap

- **Given** Category Rules show `Triggers: collision, trap`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Created Part flag is No

- **Given** Identity shows `Created Part: No`

- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-011: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** baseball physics are automatic

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** baseball is not destroyed

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Sprite dimensions 240×240

- **Given** Physics Properties show `sprite_w = 240, sprite_h = 240`
- **When** sprite is loaded
- **Then** sprite dimensions match

TC-015: Lower COR than bowling ball (0.438 vs 0.500)

- **Given** Baseball COR = 0.438, Bowling Ball COR = 0.500
- **When** both collide with surface
- **Then** baseball bounces less than bowling ball

Peça 29 — Lava Lamp



Categoria: Light / Flame Sources

Fases em que aparece: 19 fases

Quantidade de Estados: 3 estado(s)

This groovy lava lamp isn't just a cool piece of retro-decor taken from Professor Tim's attic. It's also an excellent light source. Tie a rope to the chain and give it a tug to turn on the lamp. Use it to power up the solar panel, or shine it through a magnifying glass to light fuses and candles.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	1→2	1	No
On	2→-1	3	No

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	1300	mass/density (relative scale)
<code>unk_2</code>	1000	category-specific property

Property	Raw Value	Interpretation
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	24	bitmap frame width
bitmap_h	65	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	proximity, electrical
Destructible	No
Spawns	fire
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	Yes
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	28×65	0
2	1	1–6 (6 frames)	27×79	25
3	7	7	28×65	265

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART29_f000.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check:
1	0001_0000, 0001_0001, 0001_0002, 0001_0002, 0001_0003, 0001_0001, 0001_0004	build/phase-3/bitmaps/ PART29_f000.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f004.png :material-check:
2	0001_0000, 0001_0001, 0001_0002, 0001_0002, 0001_0003, 0001_0001, 0001_0005	build/phase-3/bitmaps/ PART29_f000.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f005.png :material-check:
3	0001_0000, 0001_0001, 0001_0002, 0001_0003, 0001_0006	build/phase-3/bitmaps/ PART29_f000.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check: build/phase-3/bitmaps/ PART29_f006.png :material-check:
4	0001_0000, 0001_0001, 0001_0002, 0001_0003, 0001_0007	build/phase-3/bitmaps/ PART29_f000.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check: build/phase-3/bitmaps/ PART29_f007.png :material-check:
5	0001_0000, 0001_0001, 0001_0002, 0001_0003, 0001_0008	build/phase-3/bitmaps/ PART29_f000.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check: build/phase-3/bitmaps/ PART29_f008.png :material-check:
6		build/phase-3/bitmaps/ PART29_f000.png :material-check:

Frame	Sprites	PNG References
	0001_0000, 0001_0001, 0001_0002, 0001_0003, 0001_0004, 0001_0000, 0001_0001, 0001_0002, 0001_0003, 0001_0004	build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check: build/phase-3/bitmaps/ PART29_f004.png :material-check: build/phase-3/bitmaps/ PART29_f000.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check: build/phase-3/bitmaps/ PART29_f004.png :material-check:
7	0001_0000, 0001_0001, 0001_0002, 0001_0003, 0001_0005	build/phase-3/bitmaps/ PART29_f000.png :material-check: build/phase-3/bitmaps/ PART29_f001.png :material-check: build/phase-3/bitmaps/ PART29_f002.png :material-check: build/phase-3/bitmaps/ PART29_f003.png :material-check: build/phase-3/bitmaps/ PART29_f005.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	48
END_FRAME	8
END_ANM	1

Notas de Comportamento:

Lava Lamp has 2 semantic states: Off, On. Transitions are triggered by: proximity, electrical.

Terminal state(s): On (part is removed from simulation).

This part can ignite nearby combustible parts (balloons, explosives). Uses proximity trigger.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-029-lava-lamp.md` describes part type `29`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 29`

TC-002: Mass value is positive

- **Given** Lava Lamp has `mass = 1300`
- **When** validated against `build/phase-9/part-properties.json`

- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Lava Lamp has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: SOLVE.RES state transitions (Off → On)

- **Given** Lava Lamp has 2 states: Off (`1→2`) and On (`2→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Terminal state On (part removed)

- **Given** State Machine shows `On: 2→-1`
- **When** state reaches On
- **Then** part is removed from simulation (terminal)

TC-006: Spawns fire (light source)

- **Given** Category Rules show `Spawns: fire`
- **When** lava lamp is activated
- **Then** fire/light beam is emitted

TC-007: Triggers are proximity and electrical

- **Given** Category Rules show `Triggers: proximity, electrical`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-008: Part used in 19 levels

- **Given** Lava Lamp (type 29)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 19 levels reference `part_num = 29`

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** lava lamp can be rope-activated

TC-011: Rope chain pull activation

- **Given** Description: "Tie a rope to the chain and give it a tug to turn on the lamp"
- **When** rope pulls chain
- **Then** lamp state changes from Off to On

TC-012: Powers solar panels (light energy)

- **Given** Description: "Use it to power up the solar panel"
- **When** lamp is On and solar panel is nearby
- **Then** electrical power is generated

TC-013: Can light fuses through magnifying glass

- **Given** Description: "shine it through a magnifying glass to light fuses and candles"
- **When** light beam intersects magnifying glass
- **Then** focused beam ignites fuse/candle

TC-014: Animated Section A (22 duration entries)

- **Given** ANM shows `Animated (Section A): Yes`, `Section A Durations: 22 entries`
- **When** animation renderer processes frame timing
- **Then** durations control playback speed at 60 Hz

TC-015: 3 ANM states (state IDs 1, 2, 3)

- **Given** Lava Lamp ANM has `Total States = 3`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 3`

Peça 30 — Magnifying Glass



Categoria: Lasers / Optics
Fases em que aparece: 52 fases
Quantidade de Estados: 1 estado(s)

Place this magnifying glass in front of any light source to ignite fuses or candles. Make sure it's close enough to both the light source and the fuse you're trying to light. Flip it if necessary.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	18	bitmap frame width
bitmap_h	40	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, proximity, collision
Destructible	No
Spawns	laser_beam

Rule	Value
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	Yes
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×40	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART30.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Magnifying Glass is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part emits a **laser beam**. The beam interacts with mirrors, mixers, and laser-activated plugs.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-030-magnifying-glass.md` describes part type `30`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 30`

TC-002: Mass value is positive

- **Given** Magnifying Glass has `mass = 3776`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Magnifying Glass has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point

- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Spawns laser beam

- **Given** Category Rules show `Spawns: laser_beam`
- **When** light source shines through glass
- **Then** focused laser beam is emitted

TC-005: No SOLVE.RES state transitions

- **Given** Magnifying Glass has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-006: 1 ANM state (single frame)

- **Given** Magnifying Glass ANM has `Total States = 1, Total Frames = 1`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 1`

TC-007: Part used in 52 levels

- **Given** Magnifying Glass (type 30)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 52 levels reference `part_num = 30`

TC-008: Triggers include electrical, proximity, collision

- **Given** Category Rules show `Triggers: electrical, proximity, collision`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-009: Laser connection supported

- **Given** Connections & Capabilities shows `Laser = Yes`
- **When** checked against `category_rules`
- **Then** part emits laser beam

TC-010: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical = Yes`
- **When** checked against `category_rules`
- **Then** part can interact with electrical systems

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-012: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** magnifying glass activates passively with light

TC-013: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** magnifying glass is not destroyed

TC-014: Can ignite fuses and candles

- **Given** Description: "Place this magnifying glass in front of any light source to ignite fuses or candles"
- **When** laser beam hits fuse/candle
- **Then** target is ignited

TC-015: Category Part Count is 8

- **Given** Category Rules show `Category Part Count: 8`
- **When** validated
- **Then** 8 parts in Lasers / Optics category

Peça 31 — Mandrill Motor



Categoria: Rotating Power Sources

Fases em que aparece: 9 fases

Quantidade de Estados: 7 estado(s)

To start up this Mandrill Motor, tie a rope to the shade and give it a tug. When Pavlov Mandrill sees the banana he pedals like crazy. Attach a belt to the treadmill to power anything that is driven with a belt.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Sitting	1→4	1	No
Treading	4→7	5	No
Bonked	7→-1	9	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	101	bitmap frame width
bitmap_h	79	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, rope
Destructible	No

Rule	Value
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	106×79	0
3	1	1–127 (127 frames)	106×79	61
4	128	128–136 (9 frames)	0×0	0
9	137	137–158 (22 frames)	0×0	0
5	159	159–172 (14 frames)	0×0	0
7	173	173–178 (6 frames)	0×0	0
8	179	179	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0008	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f008.png :material-check:
1	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check:
2	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
3	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f003.png :material-check:
4	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
5	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check:
6	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
7	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
8	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
9	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check:
10	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0004	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f004.png :material-check:
11	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0005	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f005.png :material-check:
12	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f006.png :material-check:
13	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f006.png :material-check:
14	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0007	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f007.png :material-check:
15		build/phase-3/bitmaps/ PART31_f039.png :material-check:

Frame	Sprites	PNG References
	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f006.png :material-check:
16	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0005	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f005.png :material-check:
17	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0004	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f004.png :material-check:

Frame	Sprites	PNG References
18	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check:
19	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
20	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
21	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
22	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check:
23	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
24	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002, 0001_0003	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check: build/phase-3/bitmaps/ PART31_f003.png :material-check:
25	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f002.png :material-check:
26	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0001, 0001_0008	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f008.png :material-check:
27	0001_0039, 0001_0031, 0001_0021, 0001_0033, 0001_0041, 0001_0000, 0001_0001, 0001_0008	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f033.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f008.png :material-check:
28	0001_0039, 0001_0031, 0001_0021, 0001_0033, 0001_0041, 0001_0000, 0001_0001, 0001_0008	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f033.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f008.png :material-check:
29	0001_0039, 0001_0031, 0001_0021, 0001_0034, 0001_0041, 0001_0000, 0001_0001, 0001_0008	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f034.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f008.png :material-check:
30	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0000, 0001_0001, 0001_0008	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f008.png :material-check:
31	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0000, 0001_0001, 0001_0008, 0001_0036	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f008.png :material-check: build/phase-3/bitmaps/ PART31_f036.png :material-check:
32	0001_0039, 0001_0031, 0001_0021, 0001_0038, 0001_0041, 0001_0000, 0001_0001, 0001_0008	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f038.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f001.png :material-check: build/phase-3/bitmaps/ PART31_f008.png :material-check:
33		

Frame	Sprites	PNG References
	0001_0039, 0001_0031, 0001_0021, 0001_0033, 0001_0041, 0001_0009	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f033.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f009.png :material-check:
34	0001_0039, 0001_0031, 0001_0021, 0001_0034, 0001_0041, 0001_0009	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f034.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f009.png :material-check:
35	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0022, 0001_0010	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f022.png :material-check: build/phase-3/bitmaps/ PART31_f010.png :material-check:
36	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0036, 0001_0022, 0001_0010	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f036.png :material-check: build/phase-3/bitmaps/ PART31_f022.png :material-check: build/phase-3/bitmaps/ PART31_f010.png :material-check:
37	0001_0039, 0001_0031, 0001_0021, 0001_0038, 0001_0041, 0001_0023, 0001_0011	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f038.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f023.png :material-check: build/phase-3/bitmaps/ PART31_f011.png :material-check:
38	0001_0039, 0001_0031, 0001_0021, 0001_0033, 0001_0041, 0001_0023, 0001_0011	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f033.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f023.png :material-check: build/phase-3/bitmaps/ PART31_f011.png :material-check:
39	0001_0039, 0001_0031, 0001_0021, 0001_0034, 0001_0041, 0001_0024, 0001_0012	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f034.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f024.png :material-check: build/phase-3/bitmaps/ PART31_f012.png :material-check:
40	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0024, 0001_0012	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f024.png :material-check: build/phase-3/bitmaps/ PART31_f012.png :material-check:
41	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0036, 0001_0013	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f036.png :material-check: build/phase-3/bitmaps/ PART31_f013.png :material-check:
42	0001_0039, 0001_0031, 0001_0021, 0001_0038, 0001_0041, 0001_0013	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f038.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f013.png :material-check:
43	0001_0039, 0001_0031, 0001_0021, 0001_0033, 0001_0041, 0001_0022, 0001_0014	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f033.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f022.png :material-check: build/phase-3/bitmaps/ PART31_f014.png :material-check:
44	0001_0039, 0001_0031, 0001_0021, 0001_0034, 0001_0041, 0001_0022, 0001_0014	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f034.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f022.png :material-check: build/phase-3/bitmaps/ PART31_f014.png :material-check:
45	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0023, 0001_0015	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f023.png :material-check: build/phase-3/bitmaps/ PART31_f015.png :material-check:
46	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0036, 0001_0023, 0001_0015	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f036.png :material-check: build/phase-3/bitmaps/ PART31_f023.png :material-check: build/phase-3/bitmaps/ PART31_f015.png :material-check:
47	0001_0039, 0001_0031, 0001_0021, 0001_0038, 0001_0041, 0001_0024, 0001_0016	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f038.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f024.png :material-check: build/phase-3/bitmaps/ PART31_f016.png :material-check:
48	0001_0039, 0001_0031, 0001_0021, 0001_0033, 0001_0041, 0001_0024, 0001_0016	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f033.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f024.png :material-check: build/phase-3/bitmaps/ PART31_f016.png :material-check:
49	0001_0039, 0001_0031, 0001_0021, 0001_0034, 0001_0041, 0001_0010	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f034.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f010.png :material-check:
50	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0010	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f010.png :material-check:
51	0001_0039, 0001_0031, 0001_0021, 0001_0035, 0001_0041, 0001_0036, 0001_0022, 0001_0011	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f036.png :material-check: build/phase-3/bitmaps/ PART31_f022.png :material-check: build/phase-3/bitmaps/ PART31_f011.png :material-check:
52	0001_0039, 0001_0031, 0001_0021, 0001_0038, 0001_0041, 0001_0022, 0001_0011	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f038.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f022.png :material-check: build/phase-3/bitmaps/ PART31_f011.png :material-check:
53	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0023, 0001_0013	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f023.png :material-check: build/phase-3/bitmaps/ PART31_f013.png :material-check:
54	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0023, 0001_0013	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f023.png :material-check: build/phase-3/bitmaps/ PART31_f013.png :material-check:
55	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0024, 0001_0014	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f024.png :material-check: build/phase-3/bitmaps/ PART31_f014.png :material-check:
56	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0024, 0001_0014	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f024.png :material-check: build/phase-3/bitmaps/ PART31_f014.png :material-check:
57	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0015	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART31_f015.png :material-check:
58	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0022, 0001_0016	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f022.png :material-check: build/phase-3/bitmaps/ PART31_f016.png :material-check:
59	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0023, 0001_0010	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f023.png :material-check: build/phase-3/bitmaps/ PART31_f010.png :material-check:
60	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0024, 0001_0011	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f024.png :material-check: build/phase-3/bitmaps/ PART31_f011.png :material-check:
61	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0012	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f012.png :material-check:

Frame	Sprites	PNG References
62	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0022, 0001_0013	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f022.png :material-check: build/phase-3/bitmaps/ PART31_f013.png :material-check:
63	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0023, 0001_0014	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f023.png :material-check: build/phase-3/bitmaps/ PART31_f014.png :material-check:
64	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0000, 0001_0019, 0001_0017	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f019.png :material-check: build/phase-3/bitmaps/ PART31_f017.png :material-check:
65	0001_0039, 0001_0031, 0001_0021, 0001_0041, 0001_0035, 0001_0000, 0001_0020, 0001_0018	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f041.png :material-check: build/phase-3/bitmaps/ PART31_f035.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART31_f020.png :material-check: build/phase-3/bitmaps/ PART31_f018.png :material-check:
66	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0020, 0001_0018	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f020.png :material-check: build/phase-3/bitmaps/ PART31_f018.png :material-check:
67	0001_0039, 0001_0031, 0001_0021, 0001_0032, 0001_0037, 0001_0040, 0001_0042, 0001_0000, 0001_0019, 0001_0017	build/phase-3/bitmaps/ PART31_f039.png :material-check: build/phase-3/bitmaps/ PART31_f031.png :material-check: build/phase-3/bitmaps/ PART31_f021.png :material-check: build/phase-3/bitmaps/ PART31_f032.png :material-check: build/phase-3/bitmaps/ PART31_f037.png :material-check: build/phase-3/bitmaps/ PART31_f040.png :material-check: build/phase-3/bitmaps/ PART31_f042.png :material-check: build/phase-3/bitmaps/ PART31_f000.png :material-check: build/phase-3/bitmaps/ PART31_f019.png :material-check: build/phase-3/bitmaps/ PART31_f017.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
2	3206	0	build/phase-6/raw-wav/SX_3206_11025.wav
6	3206	0	build/phase-6/raw-wav/SX_3206_11025.wav
12	3031	0	build/phase-6/raw-wav/SX_3031_11025.wav
19	3206	0	build/phase-6/raw-wav/SX_3206_11025.wav
23	3206	0	build/phase-6/raw-wav/SX_3206_11025.wav
27	3231	0	build/phase-6/raw-wav/SX_3231_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	579
END_FRAME	68
PLAY_SFX	6
END_ANM	1

Notas de Comportamento:

Mandrill Motor has 3 semantic states: Sitting, Treading, Bonked. Transitions are triggered by: collision, electrical, rope.

Terminal state(s): Bonked (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-031-mandrill-motor.md` describes part type `31`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 31`

TC-002: Mass value is positive

- **Given** Mandrill Motor has `mass = 3776`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Mandrill Motor has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: SOLVE.RES state transitions (Sitting → Treading → Bonked)

- **Given** Mandrill Motor has 3 states: Sitting (`1→4`), Treading (`4→7`), Bonked (`7→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Terminal state Bonked (part removed)

- **Given** State Machine shows `Bonked: 7→-1`
- **When** state reaches Bonked
- **Then** part is removed from simulation (terminal)

TC-006: Belt connection to power other parts

- **Given** Description: "Attach a belt to the treadmill to power anything that is driven with a belt"
- **When** belt connects motor to generator/fan
- **Then** rotational power is transferred

TC-007: 7 ANM states

- **Given** Mandrill Motor ANM has `Total States = 7`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 7`

TC-008: Animated Section A (185 duration entries)

- **Given** ANM shows `Animated (Section A): Yes, Section A Durations: 185 entries`
- **When** animation renderer processes frame timing
- **Then** durations control playback speed at 60 Hz

TC-009: Part used in 9 levels

- **Given** Mandrill Motor (type 31)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 9 levels reference `part_num = 31`

TC-010: Triggers are collision, electrical, rope

- **Given** Category Rules show `Triggers: collision, electrical, rope`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-012: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** motor requires rope activation by player

TC-013: Belt Connection flag is Yes

- **Given** Connections & Capabilities shows `Belt Connection = Yes`
- **When** belt is attached
- **Then** rotational power is transferred to connected part

TC-014: Rope-activated shade trigger

- **Given** Description: "tie a rope to the shade and give it a tug"
- **When** rope pulls shade
- **Then** motor starts pedaling

TC-015: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
 - **When** validated
 - **Then** 7 parts in Rotating Power Sources category
-

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-031-mandrill-motor.md` describes part type 31
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 31`

TC-002: Mass value is positive

- **Given** Mandrill Motor has `mass = 3776`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Mandrill Motor has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: 3 SOLVE.RES states (Sitting → Treading → Bonked)

- **Given** Mandrill Motor has states: Sitting (`1→4`), Treading (`4→7`), Bonked (`7→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Belt connection to power devices

- **Given** Description: "Attach a belt to the treadmill to power anything that is driven with a belt"
- **When** belt connects motor to device
- **Then** rotational power is transferred

TC-006: Rope trigger for activation

- **Given** Description: "tie a rope to the shade and give it a tug"
- **When** rope pulls shade
- **Then** motor starts pedaling

TC-007: 7 ANM states with timed animation (185 duration entries)

- **Given** Mandrill Motor ANM has `Total States = 7`, `Section A Durations: 185 entries`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 7`

TC-008: Animated Section A controls playback

- **Given** ANM shows `Animated (Section A): Yes`, `Section A Durations: 185 entries`
- **When** animation renderer processes frame timing
- **Then** durations control playback speed at 60 Hz

TC-009: Part used in 9 levels

- **Given** Mandrill Motor (type 31)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 9 levels reference `part_num = 31`

TC-010: Triggers include collision, electrical, rope

- **Given** Category Rules show `Triggers: collision, electrical, rope`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-012: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** motor can be rope-activated

TC-013: Belt Connection flag is Yes

- **Given** Connections & Capabilities shows `Belt Connection = Yes`

- **When** belt is attached
- **Then** rotational power is transferred to connected part

TC-014: Terminal state Bonked (part removed)

- **Given** State Machine shows `Bonked: 7→-1`
- **When** state reaches Bonked
- **Then** part is removed from simulation (terminal)

TC-015: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Rotating Power Sources category

Peça 32 — Boris the Bat



Categoria: Characters
Fases em que aparece: 0 fases
Quantidade de Estados: 2 estado(s)

Happy Halloween! Meet Boris the Bat. He hangs in mid-air until he's bumped. Then he flies around acting batty.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2400	mass/density (relative scale)
unk_2	40	category-specific property
property_q8	64	Q8.8 = 0.250
unk_6	32	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	22	bitmap frame width
bitmap_h	54	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	<code>collision, proximity</code>
Destructible	No
Spawns	<i>(none)</i>
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×54	0
2	1	1–3 (3 frames)	106×40	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART32_f000.png :material-check:
1	0001_0002, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART32_f002.png :material-check: build/phase-3/bitmaps/PART32_f002.png :material-check: build/phase-3/bitmaps/PART32_f001.png :material-check:
2	0001_0001, 0001_0003	build/phase-3/bitmaps/PART32_f001.png :material-check: build/phase-3/bitmaps/PART32_f003.png :material-check:
3	0001_0001, 0001_0003	build/phase-3/bitmaps/PART32_f001.png :material-check: build/phase-3/bitmaps/PART32_f003.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
2	3032	0	build/phase-6/raw-wav/SX_3032_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	8
END_FRAME	4
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Boris the Bat is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-032-boris-the-bat.md` describes part type `32`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`

- **Then** `part_num == 32`

TC-002: Mass value is positive

- **Given** Boris the Bat has `mass = 2400`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Boris the Bat has `property_q8 = 64`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (64 >> 8) <= 1.0` i.e. `COR = 0.250`

TC-004: No SOLVE.RES state transitions

- **Given** Boris the Bat has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-005: 2 ANM states (state IDs 1-2)

- **Given** Boris the Bat ANM has `Total States = 2`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 2`

TC-006: Animated Section A (6 duration entries)

- **Given** ANM shows `Animated (Section A): Yes, Section A Durations: 6 entries`
- **When** animation renderer processes frame timing
- **Then** durations control playback speed at 60 Hz

TC-007: Sound effect on flight activation

- **Given** Sound Effects table shows frame 2 triggers sound ID `3032`
- **When** animation reaches frame 2
- **Then** `PLAY_SFX` opcode dispatches sound ID `3032`

TC-008: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** bat activates passively on bump

TC-011: Not used in any built-in level

- **Given** Boris the Bat has no level references
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 32`

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** bat is not destroyed

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Hangs in mid-air until bumped

- **Given** Description: "He hangs in mid-air until he's bumped. Then he flies around acting batty"
- **When** collision or proximity trigger fires
- **Then** bat enters flying state

TC-015: Category Part Count is 8

- **Given** Category Rules show `Category Part Count: 8`
 - **When** validated
 - **Then** 8 parts in Characters category
-

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-032-boris-the-bat.md` describes part type `32`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 32`

TC-002: Mass value is positive

- **Given** Boris the Bat has `mass = 2400`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Boris the Bat has `property_q8 = 64`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (64 >> 8) <= 1.0` i.e. `COR = 0.250`

TC-004: Not used in any built-in level

- **Given** Boris the Bat is a Character type
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 32`

TC-005: 2 ANM states with animated flight

- **Given** Boris the Bat ANM has `Total States = 2`, `Section A Durations: 6 entries`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 2`

TC-006: Animated Section A for bat movement

- **Given** ANM shows `Animated (Section A): Yes`, `Section A Durations: 6 entries`
- **When** animation renderer processes frame timing
- **Then** durations control playback speed at 60 Hz

TC-007: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-008: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-009: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** bat behavior is automatic (AI-driven)

TC-010: Character AI activates on bump

- **Given** Description: "He hangs in mid-air until he's bumped. Then he flies around acting batty"
- **When** collision or proximity trigger fires
- **Then** bat enters flying state

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** bat is not destroyed

TC-012: Sound effect on flight activation

- **Given** Sound Effects show `Frame 2: Sound ID 3032`
- **When** bat enters flying state
- **Then** `PLAY_SFX` opcode dispatches sound ID `3032`

TC-013: Bitmap dimensions 22x54

- **Given** Physics Properties show `bitmap_w = 22, bitmap_h = 54`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: Category Part Count is 8

- **Given** Category Rules show `Category Part Count: 8`
- **When** validated
- **Then** 8 parts in Characters category

TC-015: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

Peça 33 — Cupid



Categoria: Characters

Fases em que aparece: 0 fases

Quantidade de Estados: 2 estado(s)

Well, well, if it isn't Cupid, dropping in for Valentine's Day! Bump him to make him fly around. Any balloons that touch his arrow will pop.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	11	mass/density (relative scale)
unk_2	4	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	8	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	67	bitmap frame width
bitmap_h	50	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity
Destructible	No
Spawns	(none)
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	72×50	0
2	1	1–4 (4 frames)	72×50	19

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0002, 0001_0001, 0001_0000	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART33_f002.png :material-check: build/phase-3/bitmaps/PART33_f001.png :material-check: build/phase-3/bitmaps/PART33_f000.png :material-check:
1	0001_0002, 0001_0001, 0001_0000	build/phase-3/bitmaps/PART33_f002.png :material-check: build/phase-3/bitmaps/PART33_f001.png :material-check: build/phase-3/bitmaps/PART33_f000.png :material-check:
2	0001_0004, 0001_0003, 0001_0000	build/phase-3/bitmaps/PART33_f004.png :material-check: build/phase-3/bitmaps/PART33_f003.png :material-check: build/phase-3/bitmaps/PART33_f000.png :material-check:
3	0001_0002, 0001_0001, 0001_0000	build/phase-3/bitmaps/PART33_f002.png :material-check: build/phase-3/bitmaps/PART33_f001.png :material-check: build/phase-3/bitmaps/PART33_f000.png :material-check:
4	0001_0005, 0001_0006, 0001_0000	build/phase-3/bitmaps/PART33_f005.png :material-check: build/phase-3/bitmaps/PART33_f006.png :material-check: build/phase-3/bitmaps/PART33_f000.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	15
END_FRAME	5
END_ANM	1

Notas de Comportamento:

Cupid is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-033-cupid.md` describes part type `33`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 33`

TC-002: Mass value is positive

- **Given** Cupid has `mass = 11`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Cupid has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: No SOLVE.RES state transitions

- **Given** Cupid has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-005: 2 ANM states (state IDs 1-2)

- **Given** Cupid ANM has `Total States = 2`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 2`

TC-006: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** cupid activates passively on bump

TC-009: Not used in any built-in level

- **Given** Cupid has no level references
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 33`

TC-010: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** cupid is not destroyed

TC-011: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-012: Pops balloons on arrow contact

- **Given** Description: "Any balloons that touch his arrow will pop"
- **When** cupid arrow intersects balloon
- **Then** balloon is destroyed

TC-013: Flies around on bump

- **Given** Description: "Bump him to make him fly around"
- **When** collision or proximity trigger fires
- **Then** cupid enters flying state

TC-014: 5 frames with static animation

- **Given** Cupid ANM has `Total Frames = 5, Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: Category Part Count is 8

- **Given** Category Rules show `Category Part Count: 8`

- **When** validated
 - **Then** 8 parts in Characters category
-

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-033-cupid.md` describes part type `33`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 33`

TC-002: Mass value is positive

- **Given** Cupid has `mass = 11` (very light)
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Cupid has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Not used in any built-in level

- **Given** Cupid is a Character type
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 33`

TC-005: 2 ANM states (static animation)

- **Given** Cupid ANM has `Total States = 2, Animated (Section A): No`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 2`

TC-006: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** cupid behavior is automatic (AI-driven)

TC-009: Character AI activates on bump

- **Given** Description: "Bump him to make him fly around"
- **When** collision or proximity trigger fires
- **Then** cupid enters flying state

TC-010: Pops balloons on contact

- **Given** Description: "Any balloons that touch his arrow will pop"

- **When** cupid arrow collides with balloon
- **Then** balloon is destroyed

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** cupid is not destroyed

TC-012: Bitmap dimensions 67x50

- **Given** Physics Properties show `bitmap_w = 67, bitmap_h = 50`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-013: Category Part Count is 8

- **Given** Category Rules show `Category Part Count: 8`
- **When** validated
- **Then** 8 parts in Characters category

TC-014: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-015: 5 frames across 2 states

- **Given** Cupid ANM has `Total Frames = 5, Total States = 2`
- **When** bytecode is executed
- **Then** state 1 has frame 0, state 2 has frames 1-4

Peça 34 — Santa Claus



Categoria: Characters

Fases em que aparece: 0 fases

Quantidade de Estados: 2 estado(s)

Ho, ho, ho! Happy Holidays! This cheesy plastic Santa Claus lamp will only light up when you place it next to an electric socket.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	1→2	1	No
On	2→-1	2	No
!Part	(none)	—	No
Boxing Glove	(none)	—	No
Cocked	2→3	—	No
Punched	3→-1	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	53	bitmap frame width
bitmap_h	62	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity
Destructible	No
Spawns	(none)
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	56×62	0
2	1	1–12 (12 frames)	56×62	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART34_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART34_f001.png :material-check:
2	0001_0000	build/phase-3/bitmaps/PART34_f000.png :material-check:
3	0001_0001	build/phase-3/bitmaps/PART34_f001.png :material-check:
4	0001_0000	build/phase-3/bitmaps/PART34_f000.png :material-check:
5	0001_0001	build/phase-3/bitmaps/PART34_f001.png :material-check:
6		

Frame	Sprites	PNG References
	0001_0000	build/phase-3/bitmaps/PART34_f000.png :material-check:
7	0001_0001	build/phase-3/bitmaps/PART34_f001.png :material-check:
8	0001_0000	build/phase-3/bitmaps/PART34_f000.png :material-check:
9	0001_0001	build/phase-3/bitmaps/PART34_f001.png :material-check:
10	0001_0000	build/phase-3/bitmaps/PART34_f000.png :material-check:
11	0001_0001	build/phase-3/bitmaps/PART34_f001.png :material-check:
12	0001_0000	build/phase-3/bitmaps/PART34_f000.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	13
END_FRAME	13
END_ANM	1

Notas de Comportamento:

Santa Claus has 6 semantic states: Off, On, !Part, Boxing Glove, Cocked, Punched. Transitions are triggered by: collision, proximity.

Terminal state(s): On, Punched (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-034-santa-claus.md` describes part type `34`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 34`

TC-002: Mass value is positive

- **Given** Santa Claus has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Santa Claus has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: SOLVE.RES state transitions (Off → On)

- **Given** Santa Claus has 2 primary states: Off (`1→2`) and On (`2→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Terminal state On (part removed)

- **Given** State Machine shows `On: 2→-1`
- **When** state reaches On
- **Then** part is removed from simulation (terminal)

TC-006: 2 ANM states (state IDs 1-2)

- **Given** Santa Claus ANM has `Total States = 2`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 2`

TC-007: Animated Section A (56 duration entries)

- **Given** ANM shows `Animated (Section A): Yes, Section A Durations: 56 entries`
- **When** animation renderer processes frame timing
- **Then** durations control playback speed at 60 Hz

TC-008: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** santa can be activated by player

TC-011: Not used in any built-in level

- **Given** Santa Claus has no level references
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 34`

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** santa is not destroyed

TC-013: Electrical activation via socket

- **Given** Description: "This cheesy plastic Santa Claus lamp will only light up when you place it next to an electric socket"
- **When** connected to electrical outlet
- **Then** lamp state changes from Off to On

TC-014: Category Part Count is 8

- **Given** Category Rules show `Category Part Count: 8`
- **When** validated
- **Then** 8 parts in Characters category

TC-015: 13 frames alternating between 2 sprites

- **Given** Santa Claus ANM has `Total Frames = 13`
 - **When** bytecode is executed
 - **Then** frames alternate between sprite 0 and 1
-

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-034-santa-claus.md` describes part type `34`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 34`

TC-002: Mass value is positive

- **Given** Santa Claus has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Santa Claus has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: 6 SOLVE.RES states (Off, On, !Part, Boxing Glove, Cocked, Punched)

- **Given** Santa Claus has states: Off (`1→2`), On (`2→1`), !Part, Boxing Glove, Cocked (`2→3`), Punched (`3→1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Not used in any built-in level

- **Given** Santa Claus is a Character type
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 34`

TC-006: Animated Section A (56 duration entries)

- **Given** Santa Claus ANM has `Animated (Section A): Yes`, `Section A Durations: 56 entries`
- **When** animation renderer processes frame timing
- **Then** durations control playback speed at 60 Hz

TC-007: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-008: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-009: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** santa can be activated by electrical connection

TC-010: Electrical activation

- **Given** Description: "This cheesy plastic Santa Claus lamp will only light up when you place it next to an electric socket"
- **When** electrical connection is made
- **Then** santa lamp turns on

TC-011: Terminal states On and Punched (part removed)

- **Given** State Machine shows On: 2→-1, Punched: 3→-1
- **When** state reaches terminal
- **Then** part is removed from simulation

TC-012: Destructible flag is No

- **Given** Category Rules show Destructible: No
- **When** explosive part collides
- **Then** santa is not destroyed

TC-013: 2 ANM states (states 1-2)

- **Given** Santa Claus ANM has Total States = 2
- **When** decoder/d_t2anm.py parses Section C
- **Then** len(Section C entries) == 2

TC-014: Category Part Count is 8

- **Given** Category Rules show Category Part Count: 8
- **When** validated
- **Then** 8 parts in Characters category

TC-015: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against category_rules
- **Then** only default connections

Peça 35 — Boxing Glove



Categoria: Springs / Pneumatics
Fases em que aparece: 44 fases
Quantidade de Estados: 3 estado(s)

Bump the button on the back of this boxing glove to make it punch things.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	32	Q8.8 = 0.125
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	53	bitmap frame width

Property	Raw Value	Interpretation
bitmap_h	39	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	56×39	0
3	1	1–12 (12 frames)	56×39	7
4	13	13	88×79	93

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART35_f000.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART35_f000.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART35_f001.png :material-check:
3	0001_0002	build/phase-3/bitmaps/PART35_f002.png :material-check:
4	0001_0002	build/phase-3/bitmaps/PART35_f002.png :material-check:
5	0001_0003	build/phase-3/bitmaps/PART35_f003.png :material-check:
6	0001_0004	build/phase-3/bitmaps/PART35_f004.png :material-check:
7	0001_0005	build/phase-3/bitmaps/PART35_f005.png :material-check:
8	0001_0006	build/phase-3/bitmaps/PART35_f006.png :material-check:
9	0001_0007	build/phase-3/bitmaps/PART35_f007.png :material-check:
10	0001_0008	build/phase-3/bitmaps/PART35_f008.png :material-check:
11	0001_0009	build/phase-3/bitmaps/PART35_f009.png :material-check:
12	0001_0010	build/phase-3/bitmaps/PART35_f010.png :material-check:
13	0001_0010	build/phase-3/bitmaps/PART35_f010.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
3	3035	0	build/phase-6/raw-wav/SX_3035_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	14
END_FRAME	14
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Boxing Glove is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-035-boxing-glove.md` describes part type `35`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 35`

TC-002: Mass value is positive

- **Given** Boxing Glove has `mass = 3776`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Boxing Glove has `property_q8 = 32`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (32 >> 8) <= 1.0` i.e. `COR = 0.125`

TC-004: No SOLVE.RES state transitions

- **Given** Boxing Glove has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-005: Part used in 44 levels

- **Given** Boxing Glove (type 35)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 44 levels reference `part_num = 35`

TC-006: No triggers defined

- **Given** Category Rules show `Triggers: (none)`
- **When** behavior handler evaluates
- **Then** part activates purely from collision

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`

- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** glove punches on button activation

TC-009: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** glove is not destroyed

TC-010: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-011: Punch force applies to impacted part

- **Given** Description: "Bump the button on the back of this boxing glove to make it punch things"
- **When** button is activated
- **Then** glove extends and applies force to target

TC-012: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Springs / Pneumatics category

TC-013: Bitmap dimensions 53×39

- **Given** Physics Properties show `bitmap_w = 53, bitmap_h = 39`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: Low COR (0.125) indicates soft surface

- **Given** Boxing Glove COR = 0.125 (lowest in springs category)
- **When** collision occurs
- **Then** energy absorption is high

TC-015: Collision trigger activates punch

- **Given** Glove button is depressed by falling object
- **When** collision force exceeds threshold
- **Then** punch animation triggers

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-035-boxing-glove.md` describes part type `35`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 35`

TC-002: Mass value is positive

- **Given** Boxing Glove has `mass = 3776`

- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Boxing Glove has `property_q8 = 32`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (32 >> 8) <= 1.0` i.e. `COR = 0.125`

TC-004: No SOLVE.RES state transitions

- **Given** Boxing Glove has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-005: Part used in 44 levels

- **Given** Boxing Glove (type 35)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 44 levels reference `part_num = 35`

TC-006: Triggers are collision-based activation

- **Given** Description: "Bump the button on the back of this boxing glove to make it punch things"
- **When** collision activates button
- **Then** glove punches in direction of collision

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** glove activates on collision without player input

TC-009: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** glove is not destroyed

TC-010: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Springs / Pneumatics category

TC-011: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-012: Bitmap dimensions 53×39

- **Given** Physics Properties show `bitmap_w = 53, bitmap_h = 39`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-013: Punch force transfers momentum

- **Given** Boxing Glove activates on collision
- **When** glove strikes target
- **Then** target receives momentum transfer from punch

TC-014: Low COR (0.125) — minimal bounce

- **Given** Boxing Glove COR = 0.125
- **When** glove collides with surface
- **Then** minimal bounce response

TC-015: Spring/pneumatic category behavior

- **Given** Boxing Glove is in Springs / Pneumatics category
- **When** behavior handler processes
- **Then** pneumatic punch force is applied

Peça 36 — Rocket



Categoria: Explosives / Projectiles

Fases em que aparece: 24 fases

Quantidade de Estados: 15 estado(s)

You can light the fuse of this rocket with a candle (or other flaming part), a laser beam, a phazer, or by using a magnifying glass and light source. It can be flipped to fly straight up, left, or right.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Fired	2→17 , 12→18 , 25→21	25	No
Fired	8→12 , 15→25 , 24→-1	21	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	18000	mass/density (relative scale)
unk_2	1800	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	32	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	84	bitmap frame width
bitmap_h	84	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	32×85	0
10	1	1–3 (3 frames)	32×87	13
17	4	4–30 (27 frames)	64×110	70
9	31	31–34 (4 frames)	86×29	557
8	35	35–40 (6 frames)	114×62	633
12	41	41	111×50	749
16	42	42–44 (3 frames)	105×62	768
18	45	45–65 (21 frames)	102×62	825
14	66	66–71 (6 frames)	88×29	1178
15	72	72–77 (6 frames)	86×51	1274
25	78	78	110×49	1390
20	79	79–82 (4 frames)	112×50	1409
21	83	83–103 (21 frames)	102×62	1485
23	104	104–110 (7 frames)	84×29	1838
24	111	111	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0002_0000, 0001_0000	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f000.png :material-check:
1	0002_0000, 0001_0000, 0003_0000	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART36_3_f000.png :material-alert-outline: (not found)
2	0002_0000, 0001_0000, 0003_0001	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f000.png :material-check: build/phase-3/bitmaps/PART36_3_f001.png :material-alert-outline: (not found)
3	0002_0000, 0001_0000, 0003_0002	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f000.png :material-check: build/phase-3/bitmaps/PART36_3_f002.png :material-alert-outline: (not found)
4	0002_0000, 0001_0000, 0001_0005	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f000.png :material-check: build/phase-3/bitmaps/PART36_f005.png :material-check:
5	0002_0000, 0001_0000, 0001_0005	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f000.png :material-check: build/phase-3/bitmaps/PART36_f005.png :material-check:
6	0002_0000, 0001_0000, 0001_0006	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f000.png :material-check: build/phase-3/bitmaps/PART36_f006.png :material-alert-outline: (not found)
7	0002_0000, 0001_0001, 0001_0007	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f001.png :material-check: build/phase-3/bitmaps/PART36_f007.png :material-alert-outline: (not found)
8	0002_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f001.png :material-check: build/phase-3/bitmaps/PART36_f006.png :material-alert-outline: (not found)
9	0002_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f001.png :material-check: build/phase-3/bitmaps/PART36_f006.png :material-alert-outline: (not found)
10	0002_0000, 0001_0002, 0001_0007	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART36_f007.png :material-alert-outline: (not found)
11	0002_0000, 0001_0002, 0001_0007	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f002.png :material-check: build/phase-3/bitmaps/PART36_f007.png :material-alert-outline: (not found)
12	0002_0000, 0001_0002, 0001_0006	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f002.png :material-check: build/phase-3/bitmaps/PART36_f006.png :material-alert-outline: (not found)
13	0002_0000, 0001_0003, 0001_0007	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f003.png :material-check: build/phase-3/bitmaps/PART36_f007.png :material-alert-outline: (not found)
14	0002_0000, 0001_0003, 0001_0007	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f003.png :material-check: build/phase-3/bitmaps/PART36_f007.png :material-alert-outline: (not found)
15	0002_0000, 0001_0003, 0001_0006	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f003.png :material-check: build/phase-3/bitmaps/PART36_f006.png :material-alert-outline: (not found)
16	0002_0000, 0001_0003, 0001_0007	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f003.png :material-check: build/phase-3/bitmaps/PART36_f007.png :material-alert-outline: (not found)
17	0002_0000, 0001_0004, 0001_0007	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f004.png :material-check: build/phase-3/bitmaps/PART36_f007.png :material-alert-outline: (not found)
18	0001_0004, 0001_0006, 0002_0000	build/phase-3/bitmaps/PART36_f004.png :material-check: build/phase-3/bitmaps/PART36_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)
19	0002_0000	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)
20	0002_0001	build/phase-3/bitmaps/PART36_2_f001.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
21	0001_0008, 0001_0012, 0002_0002	build/phase-3/bitmaps/PART36_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f012.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f002.png :material-alert-outline: (not found)
22	0001_0008, 0001_0012, 0002_0002	build/phase-3/bitmaps/PART36_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f012.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f002.png :material-alert-outline: (not found)
23	0001_0009, 0001_0013, 0001_0014, 0001_0015, 0002_0000	build/phase-3/bitmaps/PART36_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)
24	0001_0014, 0001_0017, 0001_0015, 0001_0016, 0001_0011, 0002_0000	build/phase-3/bitmaps/PART36_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f017.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f016.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)
25	0001_0010, 0002_0000	build/phase-3/bitmaps/PART36_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)
26	0002_0000, 0001_0009	build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f009.png :material-alert-outline: (not found)
27	0001_0008, 0002_0000	build/phase-3/bitmaps/PART36_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)
28	0001_0009, 0002_0000	build/phase-3/bitmaps/PART36_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)
29	0001_0010, 0002_0000	build/phase-3/bitmaps/PART36_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
30	0001_0011, 0002_0000	build/phase-3/bitmaps/PART36_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f000.png :material-alert-outline: (not found)
31	0002_0003, 0001_0018	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found)
32	0002_0003, 0002_0003, 0001_0018, 0003_0000	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_3_f000.png :material-alert-outline: (not found)
33	0002_0003, 0001_0018, 0003_0001	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_3_f001.png :material-alert-outline: (not found)
34	0002_0003, 0001_0018, 0003_0002	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_3_f002.png :material-alert-outline: (not found)
35	0002_0003, 0001_0018, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
36	0002_0003, 0001_0018, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
37	0002_0003, 0001_0018, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
38	0002_0003, 0001_0018, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
39	0002_0003, 0001_0019, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART36_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
40	0002_0003, 0001_0019, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
41	0002_0003, 0001_0019, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
42	0002_0003, 0001_0020, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
43	0002_0003, 0001_0020, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
44	0002_0003, 0001_0020, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
45	0002_0003, 0001_0021, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
46	0002_0003, 0001_0021, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
47	0002_0003, 0001_0021, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
48	0002_0003, 0001_0022, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART36_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
49	0002_0003, 0001_0022, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
50	0002_0003, 0001_0022, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
51	0002_0003, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
52	0002_0003, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
53	0002_0003, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
54	0002_0003	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found)
55	0002_0004	build/phase-3/bitmaps/PART36_2_f004.png :material-alert-outline: (not found)
56	0002_0005, 0001_0026, 0001_0030	build/phase-3/bitmaps/PART36_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f026.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f030.png :material-alert-outline: (not found)
57	0002_0005, 0001_0026, 0001_0030	build/phase-3/bitmaps/PART36_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f026.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f030.png :material-alert-outline: (not found)
58	0002_0003, 0001_0026, 0001_0031, 0001_0032, 0001_0034	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f026.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f031.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f032.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f034.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
59	0002_0003, 0001_0027, 0001_0032, 0001_0034, 0001_0035, 0001_0032	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f032.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f034.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f035.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f032.png :material-alert-outline: (not found)
60	0002_0003, 0001_0028	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f028.png :material-alert-outline: (not found)
61	0002_0003, 0001_0029	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f029.png :material-alert-outline: (not found)
62	0002_0003, 0001_0028	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f028.png :material-alert-outline: (not found)
63	0002_0003, 0001_0027	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f027.png :material-alert-outline: (not found)
64	0002_0003, 0001_0026	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f026.png :material-alert-outline: (not found)
65	0002_0003, 0001_0027	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f027.png :material-alert-outline: (not found)
66	0002_0003, 0001_0028	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f028.png :material-alert-outline: (not found)
67	0002_0003, 0001_0029	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f029.png :material-alert-outline: (not found)
68	0002_0003, 0001_0018	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found)
69	0002_0003, 0001_0018, 0003_0000	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART36_3_f000.png :material-alert-outline: (not found)
70	0002_0003, 0001_0018, 0003_0001	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_3_f001.png :material-alert-outline: (not found)
71	0002_0003, 0001_0018, 0003_0002	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_3_f002.png :material-alert-outline: (not found)
72	0002_0003, 0001_0018, 0003_0003	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_3_f003.png :material-alert-outline: (not found)
73	0002_0003, 0001_0018, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
74	0002_0003, 0001_0018, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
75	0002_0003, 0001_0018, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
76	0002_0003, 0001_0018, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
77	0002_0003, 0001_0019, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
78	0002_0003, 0001_0019, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f019.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
79	0002_0003, 0001_0019, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
80	0002_0003, 0001_0020, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
81	0002_0003, 0001_0020, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
82	0002_0003, 0001_0020, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
83	0002_0003, 0001_0021, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
84	0002_0003, 0001_0021, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
85	0002_0003, 0001_0021, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
86	0002_0003, 0001_0022, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
87	0002_0003, 0001_0022, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f022.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
88	0002_0003, 0001_0022, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
89	0002_0003, 0001_0023	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f023.png :material-alert-outline: (not found)
90	0002_0003, 0001_0024	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f024.png :material-alert-outline: (not found)
91	0002_0003, 0001_0025	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f025.png :material-alert-outline: (not found)
92	0002_0003	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found)
93	0002_0004	build/phase-3/bitmaps/PART36_2_f004.png :material-alert-outline: (not found)
94	0002_0005, 0001_0026, 0001_0030	build/phase-3/bitmaps/PART36_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f026.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f030.png :material-alert-outline: (not found)
95	0002_0005, 0001_0026, 0001_0030	build/phase-3/bitmaps/PART36_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f026.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f030.png :material-alert-outline: (not found)
96	0002_0003, 0001_0027, 0001_0031, 0001_0032, 0001_0035	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f031.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f032.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f035.png :material-alert-outline: (not found)
97	0002_0003, 0001_0028, 0001_0035, 0001_0032, 0001_0034, 0001_0033	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f028.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f035.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f032.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART36_f034.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f033.png :material-alert-outline: (not found)
98	0002_0003, 0001_0029	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f029.png :material-alert-outline: (not found)
99	0002_0003, 0001_0028	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f028.png :material-alert-outline: (not found)
100	0002_0003, 0001_0027	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f027.png :material-alert-outline: (not found)
101	0002_0003, 0001_0026	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f026.png :material-alert-outline: (not found)
102	0002_0003, 0001_0027	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f027.png :material-alert-outline: (not found)
103	0002_0003, 0001_0028	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f028.png :material-alert-outline: (not found)
104	0002_0003, 0001_0029	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f029.png :material-alert-outline: (not found)
105	0002_0003, 0001_0028	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f028.png :material-alert-outline: (not found)
106	0002_0003, 0001_0027	build/phase-3/bitmaps/PART36_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART36_f027.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
4	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
21	3036	0	build/phase-6/raw-wav/SX_3036_11025.wav
35	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
56	3036	0	build/phase-6/raw-wav/SX_3036_11025.wav
73	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
94	3036	0	build/phase-6/raw-wav/SX_3036_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	293
END_FRAME	107
PLAY_SFX	6
END_ANM	1

Notas de Comportamento:

Rocket has 2 semantic states: Not Fired, Fired. Transitions are triggered by: collision, proximity, timer.

Terminal state(s): Fired (part is removed from simulation).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 37 — Hedge Trimmers



Categoria: Cutting / Popping
Fases em que aparece: 51 fases
Quantidade de Estados: 6 estado(s)

You can cut ropes with these hedge trimmers by bumping the handles with another object. Balloons and blimps pop against the tips.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Open	7→11	7	No
Closed	3→-1	4	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property

Property	Raw Value	Interpretation
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	55	bitmap frame width
bitmap_h	36	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	3

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
7	0	0	61×36	0
8	1	1–3 (3 frames)	61×36	13
11	4	4	32×8	70
3	5	5	65×36	77
4	6	6	64×17	122
9	7	7–9 (3 frames)	70×17	129

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001	build/phase-3/bitmaps/PART37_f000.png :material-check: build/phase-3/bitmaps/PART37_f001.png :material-check:
1	0001_0000, 0001_0001, 0002_0000	build/phase-3/bitmaps/PART37_f000.png :material-check: build/phase-3/bitmaps/PART37_f001.png :material-check: build/phase-3/bitmaps/PART37_2_f000.png :material-alert-outline: (not found)
2	0001_0000, 0001_0001, 0002_0001	build/phase-3/bitmaps/PART37_f000.png :material-check: build/phase-3/bitmaps/PART37_f001.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART37_2_f001.png :material-alert-outline: (not found)
3	0001_0000, 0001_0001, 0002_0002	build/phase-3/bitmaps/PART37_f000.png :material-check: build/phase-3/bitmaps/PART37_f001.png :material-check: build/phase-3/bitmaps/PART37_2_f002.png :material-alert-outline: (not found)
4	0001_0001	build/phase-3/bitmaps/PART37_f001.png :material-check:
5	0001_0002	build/phase-3/bitmaps/PART37_f002.png :material-check:
6	0001_0002	build/phase-3/bitmaps/PART37_f002.png :material-check:
7	0001_0002, 0002_0000	build/phase-3/bitmaps/PART37_f002.png :material-check: build/phase-3/bitmaps/PART37_2_f000.png :material-alert-outline: (not found)
8	0001_0002, 0002_0001	build/phase-3/bitmaps/PART37_f002.png :material-check: build/phase-3/bitmaps/PART37_2_f001.png :material-alert-outline: (not found)
9	0001_0002, 0002_0002	build/phase-3/bitmaps/PART37_f002.png :material-check: build/phase-3/bitmaps/PART37_2_f002.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
5	3037	0	build/phase-6/raw-wav/SX_3037_11025.wav

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	20
END_FRAME	10
DRAW_LINE	6
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Hedge Trimmers has 2 semantic states: Open, Closed. Transitions are triggered by: collision.

Terminal state(s): Closed (part is removed from simulation).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-037-hedge-trimmers.md` describes part type `37`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 37`

TC-002: Mass value is positive

- **Given** Hedge Trimmers has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Hedge Trimmers has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: SOLVE.RES state transitions

- **Given** Hedge Trimmers has named state transitions
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Cuts rope connections

- **Given** Description: "You can cut ropes with these hedge trimmers"
- **When** trimmer blades contact rope
- **Then** rope connection is severed

TC-006: Pops balloons and blimps

- **Given** Description: "Balloons and blimps pop against the tips"
- **When** balloon/blimp contacts trimmer tips
- **Then** target is destroyed

TC-007: Collision trigger activates

- **Given** Category Rules show `Triggers: collision`
- **When** object bumps trimmers
- **Then** cutting animation plays

TC-008: Part used in 51 levels

- **Given** Hedge Trimmers (type 37)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 51 levels reference `part_num = 37`

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** trimmers respond to object collision

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** trimmers are not destroyed

TC-012: No electrical/belt/rope/fire/laser connections

- **Given** Connections & Capabilities shows all mechanical connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-013: Category Part Count is 3

- **Given** Category Rules show `Category Part Count: 3`
- **When** validated

- **Then** 3 parts in Cutting / Popping category

TC-014: Bitmap dimensions 55×36

- **Given** Physics Properties show `bitmap_w = 55, bitmap_h = 36`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Rope can be cut by mechanical interaction

- **Given** Rope exists between two objects
 - **When** Hedge Trimmers (type 37) contact rope
 - **Then** rope connection is severed (per Q-018 rope constraint)
-

Peça 38 — Solar Panel



Categoria: Electrical
Fases em que aparece: 16 fases
Quantidade de Estados: 2 estado(s)

This solar panel comes with its own electrical outlet. Shine a light on the panel, then plug in any electric part you want to operate.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	71	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0–3 (4 frames)	72×32	0
2	4	4–7 (4 frames)	72×32	52

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART38_f000.png :material-check:
1	0001_0000, 0001_0002	build/phase-3/bitmaps/PART38_f000.png :material-check: build/phase-3/bitmaps/PART38_f002.png :material-check:
2	0001_0000, 0001_0002	build/phase-3/bitmaps/PART38_f000.png :material-check: build/phase-3/bitmaps/PART38_f002.png :material-check:
3	0001_0000, 0001_0002, 0001_0002	build/phase-3/bitmaps/PART38_f000.png :material-check: build/phase-3/bitmaps/PART38_f002.png :material-check: build/phase-3/bitmaps/PART38_f002.png :material-check:
4	0001_0001	build/phase-3/bitmaps/PART38_f001.png :material-check:
5	0001_0001, 0001_0002	build/phase-3/bitmaps/PART38_f001.png :material-check: build/phase-3/bitmaps/PART38_f002.png :material-check:
6	0001_0001, 0001_0002	build/phase-3/bitmaps/PART38_f001.png :material-check: build/phase-3/bitmaps/PART38_f002.png :material-check:
7	0001_0001, 0001_0002, 0001_0002	build/phase-3/bitmaps/PART38_f001.png :material-check: build/phase-3/bitmaps/PART38_f002.png :material-check: build/phase-3/bitmaps/PART38_f002.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	16
END_FRAME	8
END_ANM	1

Notas de Comportamento:

Solar Panel is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- Given `part-038-solar-panel.md` describes part type 38

- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 38`

TC-002: Mass value is positive

- **Given** Solar Panel has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Solar Panel has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: No SOLVE.RES state transitions

- **Given** Solar Panel has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-005: 2 ANM states (state IDs 2-3)

- **Given** Solar Panel ANM has `Total States = 2`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 2`

TC-006: Electrical power generation from light

- **Given** Description: "Shine a light on the panel, then plug in any electric part you want to operate"
- **When** light source (flashlight, lava lamp) shines on panel
- **Then** panel generates electrical power

TC-007: Part used in 16 levels

- **Given** Solar Panel (type 38)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 16 levels reference `part_num = 38`

TC-008: Triggers are electrical and collision

- **Given** Category Rules show `Triggers: electrical, collision`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** panel activates passively with light

TC-011: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical = Yes`
- **When** checked against `category_rules`
- **Then** panel has outlet for electrical parts

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** panel is not destroyed

TC-013: Category Part Count is 9

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Electrical category

TC-014: 4 frames per state (8 total)

- **Given** Solar Panel ANM has frames 0-3 (state 3) and frames 4-7 (state 2)
- **When** bytecode is executed
- **Then** each state renders correct sprites

TC-015: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

Peça 39 — Springboard



Categoria: Springs / Pneumatics

Fases em que aparece: 54 fases

Quantidade de Estados: 3 estado(s)

Anything you drop on this springboard will go higher with each bounce.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	32	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	49	bitmap frame width
bitmap_h	27	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–1 (2 frames)	56×27	0
5	2	2–7 (6 frames)	56×43	26
4	8	8	56×27	106

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0005, 0001_0000	build/phase-3/bitmaps/PART39_f005.png :material-check: build/phase-3/bitmaps/PART39_f000.png :material-check:
1	0001_0003, 0001_0000	build/phase-3/bitmaps/PART39_f003.png :material-check: build/phase-3/bitmaps/PART39_f000.png :material-check:
2	0001_0004, 0001_0002	build/phase-3/bitmaps/PART39_f004.png :material-check: build/phase-3/bitmaps/PART39_f002.png :material-check:
3	0001_0004, 0001_0000	build/phase-3/bitmaps/PART39_f004.png :material-check: build/phase-3/bitmaps/PART39_f000.png :material-check:
4	0001_0004, 0001_0001	build/phase-3/bitmaps/PART39_f004.png :material-check: build/phase-3/bitmaps/PART39_f001.png :material-check:
5	0001_0006, 0001_0001	build/phase-3/bitmaps/PART39_f006.png :material-check: build/phase-3/bitmaps/PART39_f001.png :material-check:
6	0001_0003, 0001_0001	build/phase-3/bitmaps/PART39_f003.png :material-check: build/phase-3/bitmaps/PART39_f001.png :material-check:
7	0001_0006, 0001_0002	build/phase-3/bitmaps/PART39_f006.png :material-check: build/phase-3/bitmaps/PART39_f002.png :material-check:
8	0001_0005, 0001_0000	build/phase-3/bitmaps/PART39_f005.png :material-check: build/phase-3/bitmaps/PART39_f000.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
2	3039	0	build/phase-6/raw-wav/SX_3039_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	18
END_FRAME	9
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Springboard is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-039-springboard.md` describes part type `39`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 39`

TC-002: Mass value is positive

- **Given** Springboard has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Springboard has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: No SOLVE.RES state transitions

- **Given** Springboard has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-005: Triggers are collision and electrical

- **Given** Category Rules show `Triggers: collision, electrical`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-006: Part used in 54 levels

- **Given** Springboard (type 39)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 54 levels reference `part_num = 39`

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`

- **When** simulation runs
- **Then** springboard activates passively on contact

TC-009: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** springboard is not destroyed

TC-010: Progressive bounce height increase

- **Given** Description: "Anything you drop on this springboard will go higher with each bounce"
- **When** object contacts springboard
- **Then** bounce velocity increases with each contact

TC-011: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Springs / Pneumatics category

TC-012: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-013: Bitmap dimensions 49x27

- **Given** Physics Properties show `bitmap_w = 49, bitmap_h = 27`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: Spring physics with increasing energy

- **Given** Springboard collision response
- **When** object lands on springboard
- **Then** upward impulse is applied with energy multiplier

TC-015: Multi-bounce constraint enforcement

- **Given** Object bounces on springboard multiple times
- **When** physics engine resolves collision
- **Then** each bounce applies progressively larger impulse

Peça 40 — Pinwheel



Categoria: Rotating Power Sources

Fases em que aparece: 16 fases

Quantidade de Estados: 2 estado(s)

You can make this pinwheel spin by blowing air on it (from parts like the fan or the bike pump). Attach a belt and use it to turn other rotating parts.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Spinning	1→2	1	No
Spinning	2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	50	bitmap frame width
bitmap_h	55	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, rope
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	56×55	0
2	1	1–10 (10 frames)	56×55	13

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0010, 0001_0000	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f000.png :material-check:
1	0001_0010, 0001_0000	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f000.png :material-check:
2	0001_0010, 0001_0001	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f001.png :material-check:
3	0001_0010, 0001_0002	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f002.png :material-check:
4	0001_0010, 0001_0003	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f003.png :material-check:
5	0001_0010, 0001_0004	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f004.png :material-check:
6	0001_0010, 0001_0005	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f005.png :material-check:
7	0001_0010, 0001_0006	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f006.png :material-check:
8	0001_0010, 0001_0007	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f007.png :material-check:
9	0001_0010, 0001_0008	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f008.png :material-check:
10	0001_0010, 0001_0009	build/phase-3/bitmaps/PART40_f010.png :material-check: build/phase-3/bitmaps/PART40_f009.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	22
END_FRAME	11
END_ANM	1

Notas de Comportamento:

Pinwheel has 2 semantic states: Not Spinning, Spinning. Transitions are triggered by: collision, electrical, rope.

Terminal state(s): Spinning (part is removed from simulation).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-040-pinwheel.md` describes part type `40`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 40`

TC-002: Mass value is positive

- **Given** Pinwheel has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Pinwheel has `property_q8 = 128`

- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: SOLVE.RES state transitions (Not Spinning → Spinning)

- **Given** Pinwheel has 2 states: Not Spinning (`1→2`) and Spinning (`2→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Terminal state Spinning (part removed)

- **Given** State Machine shows `Spinning: 2→-1`
- **When** state reaches Spinning
- **Then** part is removed from simulation (terminal)

TC-006: Belt connection to power devices

- **Given** Description: "Attach a belt and use it to turn other rotating parts"
- **When** belt connects pinwheel to device
- **Then** rotational power is transferred

TC-007: Wind-activated rotation

- **Given** Description: "You can make this pinwheel spin by blowing air on it (from parts like the fan or the bike pump)"
- **When** wind from fan/bike pump contacts pinwheel
- **Then** pinwheel begins spinning

TC-008: 2 ANM states (state IDs 1-2)

- **Given** Pinwheel ANM has `Total States = 2`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 2`

TC-009: Part used in 16 levels

- **Given** Pinwheel (type 40)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 16 levels reference `part_num = 40`

TC-010: Triggers include collision, electrical, rope

- **Given** Category Rules show `Triggers: collision, electrical, rope`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-012: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** pinwheel can be manually positioned/activated

TC-013: Belt Connection flag is Yes

- **Given** Connections & Capabilities shows `Belt Connection = Yes`
- **When** belt is attached
- **Then** rotational power is transferred to connected part

TC-014: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Rotating Power Sources category

TC-015: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

Peça 41 — explosion



Categoria: Created / Phantom
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

[CREATED PART]

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	1	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	70	bitmap frame width
bitmap_h	66	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	1

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0–2 (3 frames)	72×66	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0002	build/phase-3/bitmaps/PART41_f002.png :material-check:
1	0001_0001, 0001_0000	build/phase-3/bitmaps/PART41_f001.png :material-check: build/phase-3/bitmaps/PART41_f000.png :material-check:
2	0001_0001, 0001_0002	build/phase-3/bitmaps/PART41_f001.png :material-check: build/phase-3/bitmaps/PART41_f002.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
0	3211	0	build/phase-6/raw-wav/SX_3211_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	5
PLAY_SFX	1
END_FRAME	3
END_ANM	1

Notas de Comportamento:

explosion is a dynamically created (phantom) part. It is not placed by the player but spawned at runtime by another part's behavior (e.g., explosion debris, laser beam, rope end phantom).

explosion is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-041-explosion.md` describes part type `41`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 41`

TC-002: Created Part flag is Yes

- **Given** Explosion Identity shows `Created Part: Yes`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == True`

TC-003: Not placed by player — no level usage

- **Given** Explosion is a dynamically spawned effect
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 41`

TC-004: 3-frame explosion animation

- **Given** Explosion ANM has `Total Frames = 3, Total States = 1`

- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 1`

TC-005: Sound effect at frame 0

- **Given** Sound Effects table shows frame 0 triggers sound ID `3211`
- **When** explosion spawns
- **Then** `PLAY_SFX` opcode dispatches sound ID `3211`

TC-006: 5 DRAW_BMP opcodes across 3 frames

- **Given** Bytecode Opcodes show `DRAW_BMP = 5` across 3 frames
- **When** bytecode is executed
- **Then** each frame renders correct explosion sprites

TC-007: PLAY_SFX opcode present

- **Given** Bytecode Opcodes show `PLAY_SFX = 1`
- **When** animation reaches trigger frame
- **Then** sound effect is dispatched

TC-008: No SOLVE.RES state transitions

- **Given** Explosion has no programmatic state transitions
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-009: No triggers defined

- **Given** Category Rules show `Triggers: (none)`
- **When** behavior handler evaluates
- **Then** explosion is purely visual effect

TC-010: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** checked
- **Then** explosion is not subject to destruction

TC-011: Created / Phantom category

- **Given** Category shows `Created / Phantom`
- **When** part is spawned
- **Then** it is treated as temporary visual effect

TC-012: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** explosion requires no player interaction

TC-013: Bitmap dimensions 70×66

- **Given** Physics Properties show `bitmap_w = 70, bitmap_h = 66`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

Peça 42 — Newton Mouse



Categoria: Characters
Fases em que aparece: 37 fases
Quantidade de Estados: 9 estado(s)

This is Newton Mouse. He'll go after any cheese he can see. He'll also run away if Curie Cat comes after him. He'll run inside a mouse hole if you place a hunk of cheese on the other side of it, or if Curie chases him toward one. Newton also has to watch out for alligators.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Hasn't Eaten	1→10	1	No
Has Eaten	10→-1	10	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
unk_2	1	category-specific property
property_q8	0	Q8.8 = 0.000
unk_6	256	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	20	bitmap frame width
bitmap_h	20	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity
Destructible	No
Spawns	(none)
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×20	0
9	1	1–22 (22 frames)	32×26	7
7	23	23–26 (4 frames)	32×24	163
8	27	27–29 (3 frames)	32×24	191
10	30	30–49 (20 frames)	32×23	214
11	50	50	48×14	354
14	51	51–72 (22 frames)	32×13	361
12	73	73–76 (4 frames)	0×0	0
13	77	77	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART42_f000.png :material-check:
1	0001_0007	build/phase-3/bitmaps/PART42_f007.png :material-check:
2	0001_0008	build/phase-3/bitmaps/PART42_f008.png :material-check:
3	0001_0009	build/phase-3/bitmaps/PART42_f009.png :material-check:
4	0001_0008	build/phase-3/bitmaps/PART42_f008.png :material-check:
5	0001_0009	build/phase-3/bitmaps/PART42_f009.png :material-check:
6	0001_0008	build/phase-3/bitmaps/PART42_f008.png :material-check:
7	0001_0009	build/phase-3/bitmaps/PART42_f009.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART42_f008.png :material-check:
9	0001_0007	build/phase-3/bitmaps/PART42_f007.png :material-check:
10	0001_0001	build/phase-3/bitmaps/PART42_f001.png :material-check:
11	0001_0002	build/phase-3/bitmaps/PART42_f002.png :material-check:
12	0001_0003	build/phase-3/bitmaps/PART42_f003.png :material-check:
13	0001_0005	build/phase-3/bitmaps/PART42_f005.png :material-check:
14	0001_0002	build/phase-3/bitmaps/PART42_f002.png :material-check:
15	0001_0003	build/phase-3/bitmaps/PART42_f003.png :material-check:
16	0001_0005	build/phase-3/bitmaps/PART42_f005.png :material-check:
17	0001_0010	build/phase-3/bitmaps/PART42_f010.png :material-check:
18	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:

Frame	Sprites	PNG References
19	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
20	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:
21	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
22	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:
23	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
24	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:
25	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
26	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:
27	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
28	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:
29	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
30	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:
31	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
32	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:
33	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
34	0001_0011	build/phase-3/bitmaps/PART42_f011.png :material-check:
35	0001_0012	build/phase-3/bitmaps/PART42_f012.png :material-check:
36	0001_0010	build/phase-3/bitmaps/PART42_f010.png :material-check:
37	0001_0000	build/phase-3/bitmaps/PART42_f000.png :material-check:
38	0001_0007	build/phase-3/bitmaps/PART42_f007.png :material-check:
39	0001_0008	build/phase-3/bitmaps/PART42_f008.png :material-check:
40	0001_0009	build/phase-3/bitmaps/PART42_f009.png :material-check:
41	0001_0008	build/phase-3/bitmaps/PART42_f008.png :material-check:
42	0001_0009	build/phase-3/bitmaps/PART42_f009.png :material-check:
43	0001_0008	build/phase-3/bitmaps/PART42_f008.png :material-check:
44	0001_0009	build/phase-3/bitmaps/PART42_f009.png :material-check:
45	0001_0008	build/phase-3/bitmaps/PART42_f008.png :material-check:
46	0001_0007	build/phase-3/bitmaps/PART42_f007.png :material-check:
47	0001_0001	build/phase-3/bitmaps/PART42_f001.png :material-check:
48	0001_0002	build/phase-3/bitmaps/PART42_f002.png :material-check:
49	0001_0003	build/phase-3/bitmaps/PART42_f003.png :material-check:
50	0001_0005	build/phase-3/bitmaps/PART42_f005.png :material-check:
51	0001_0002	build/phase-3/bitmaps/PART42_f002.png :material-check:
52	0001_0003	build/phase-3/bitmaps/PART42_f003.png :material-check:
53	0001_0005	build/phase-3/bitmaps/PART42_f005.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
17	3042	0	build/phase-6/raw-wav/SX_3042_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
27	3042	0	build/phase-6/raw-wav/SX_3042_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	54
END_FRAME	54
PLAY_SFX	2
END_ANM	1

Notas de Comportamento:

Newton Mouse has 2 semantic states: Hasn't Eaten, Has Eaten. Transitions are triggered by: collision, proximity.

Terminal state(s): Has Eaten (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-042-newton-mouse.md` describes part type `42`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 42`

TC-002: Mass value is positive

- **Given** Newton Mouse has `mass = 2000`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR is 0.000 (fixed-point property)

- **Given** Newton Mouse has `property_q8 = 0`
- **When** converted from Q8.8 fixed-point
- **Then** `COR = 0.000`

TC-004: Character AI with cheese-seeking behavior

- **Given** Description: "He'll go after any cheese he can see"
- **When** cheese (part 52) is placed in level
- **Then** mouse pathfinds toward cheese

TC-005: Fear response to Curie Cat

- **Given** Description: "He'll also run away if Curie Cat comes after him"
- **When** Curie Cat (part 12) enters proximity
- **Then** mouse flees in opposite direction

TC-006: Mouse hole destination behavior

- **Given** Description: "He'll run inside a mouse hole if you place a hunk of cheese on the other side of it"
- **When** cheese placed beyond mouse hole
- **Then** mouse enters hole and disappears

TC-007: SOLVE.RES state transitions (complex character AI)

- **Given** Newton Mouse has named states in SOLVE.RES

- **When** behavior handler evaluates triggers
- **Then** state transitions follow SOLVE.RES rules

TC-008: Part used in 37 levels

- **Given** Newton Mouse (type 42)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 37 levels reference `part_num = 42`

TC-009: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-011: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** mouse AI responds to environment

TC-012: Character category (Category Part Count = 8)

- **Given** Category Rules show `Category Part Count: 8`
- **When** validated
- **Then** 8 parts in Characters category

TC-013: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** mouse is not destroyed

TC-014: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-015: Alligator avoidance behavior

- **Given** Description mentions alligators as threat
- **When** alligator enters proximity
- **Then** mouse pathfinds to avoid

Peça 43 — Pinball



Categoria: Balls

Fases em que aparece: 78 fases

Quantidade de Estados: 2 estado(s)

This pinball is very hard and heavy, and doesn't bounce much.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	21428	mass/density (relative scale)
damping	28000	drag/air-resistance or density
cor_q8	96	COR = 0.375
radius	16	collision radius in internal units
sprite_w	240	sprite width
sprite_h	240	sprite height
data_offset	23	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×23	0
2	1	1–3 (3 frames)	26×24	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART43.png :material-check:
1	0001_0000, 0002_0000	build/phase-3/bitmaps/PART43.png :material-check: build/phase-3/bitmaps/PART43_2_f000.png :material-alert-outline: (not found)
2	0001_0000, 0002_0001	build/phase-3/bitmaps/PART43.png :material-check: build/phase-3/bitmaps/PART43_2_f001.png :material-alert-outline: (not found)
3	0001_0000, 0002_0002	build/phase-3/bitmaps/PART43.png :material-check: build/phase-3/bitmaps/PART43_2_f002.png :material-alert-outline: (not found)

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	7
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Pinball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-043-pinball.md` describes part type `43`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 43`

TC-002: Mass value is positive

- **Given** Pinball has `mass = 21428` (heaviest ball so far)
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Pinball has `cor_q8 = 96`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (96 >> 8) <= 1.0` i.e. `COR = 0.375`

TC-004: Radius is non-negative

- **Given** Pinball has `radius = 16`
- **When** physics engine initializes collision
- **Then** `radius >= 0`

TC-005: Damping value affects behavior

- **Given** Pinball has `damping = 28000` (very high, near immovable)
- **When** physics simulation runs
- **Then** velocity decay is extremely high

TC-006: Single default state — no named transitions

- **Given** Pinball has no SOLVE.RES state data
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-007: Part used in 78 levels (highest usage in batch 3)

- **Given** Pinball (type 43)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 78 levels reference `part_num = 43`

TC-008: Category Part Count is 9 (Balls category)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Balls category

TC-009: Triggers are collision and trap

- **Given** Category Rules show `Triggers: collision, trap`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-011: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** pinball physics are automatic

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** pinball is not destroyed

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Sprite dimensions 240×240

- **Given** Physics Properties show `sprite_w = 240, sprite_h = 240`
- **When** sprite is loaded
- **Then** sprite dimensions match

TC-015: Highest mass of all balls (21428 vs bowling ball 2832)

- **Given** Pinball mass = 21428, Bowling Ball mass = 2832
- **When** collision occurs
- **Then** pinball transfers significantly more momentum

Peça 44 — Tennis Ball



Categoria: Balls
Fases em que aparece: 44 fases
Quantidade de Estados: 1 estado(s)

This tennis ball is very light and bouncy.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1322	mass/density (relative scale)
damping	5	drag/air-resistance or density
cor_q8	192	COR = 0.750
radius	16	collision radius in internal units

Property	Raw Value	Interpretation
sprite_w	240	sprite width
sprite_h	240	sprite height
data_offset	15	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–7 (8 frames)	16×15	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART44_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART44_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART44_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART44_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART44_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART44_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART44_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART44_f007.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	8
END_FRAME	8
END_ANM	1

Notas de Comportamento:

Tennis Ball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-044-tennis-ball.md` describes part type `44`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 44`

TC-002: Mass value is positive

- **Given** Tennis Ball has `mass = 1322`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Tennis Ball has `cor_q8 = 192`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (192 >> 8) <= 1.0` i.e. `COR = 0.750`

TC-004: Radius is non-negative

- **Given** Tennis Ball has `radius = 16`
- **When** physics engine initializes collision
- **Then** `radius >= 0`

TC-005: Damping value affects air resistance

- **Given** Tennis Ball has `damping = 5`
- **When** physics simulation runs
- **Then** velocity decay is minimal (light and bouncy)

TC-006: Single default state — no named transitions

- **Given** Tennis Ball has no SOLVE.RES state data
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-007: Part used in 44 levels

- **Given** Tennis Ball (type 44)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 44 levels reference `part_num = 44`

TC-008: Category Part Count is 9 (Balls category)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Balls category

TC-009: Triggers are collision and trap

- **Given** Category Rules show `Triggers: collision, trap`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Created Part flag is No

- **Given** Identity shows `Created Part: No`

- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-011: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** tennis ball physics are automatic

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** tennis ball is not destroyed

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Sprite dimensions 240×240

- **Given** Physics Properties show `sprite_w = 240, sprite_h = 240`
- **When** sprite is loaded
- **Then** sprite dimensions match

TC-015: Highest COR of all balls (0.750 vs bowling ball 0.500)

- **Given** Tennis Ball COR = 0.750, Bowling Ball COR = 0.500
- **When** bouncing
- **Then** tennis ball bounces significantly higher than bowling ball

Peça 45 — Aladdin's Lamp



Categoria: Light / Flame Sources
Fases em que aparece: 8 fases
Quantidade de Estados: 3 estado(s)

Light this oil lamp with a laser, a flaming part (like flint rocks or a match-on-a-spring), or a light source and a magnifying glass. Once it's burning, you can use it to light candles and fuses, heat up coffee pots, or to pop blimps and balloons.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Out	1→2	1	No
Lit	2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
unk_2	50	category-specific property

Property	Raw Value	Interpretation
property_q8	84	Q8.8 = 0.328
unk_6	32	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	51	bitmap frame width
bitmap_h	29	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	proximity, electrical
Destructible	No
Spawns	fire
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	Yes
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	56×29	0
3	1	1–3 (3 frames)	57×29	7
2	4	4–8 (5 frames)	57×46	46

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART45.png :material-check:
1	0001_0000, 0003_0000	build/phase-3/bitmaps/PART45.png :material-check: build/phase-3/bitmaps/PART45_3_f000.png :material-alert-outline: (not found)
2	0001_0000, 0003_0001	build/phase-3/bitmaps/PART45.png :material-check: build/phase-3/bitmaps/PART45_3_f001.png :material-alert-outline: (not found)
3	0001_0000, 0003_0002	build/phase-3/bitmaps/PART45.png :material-check: build/phase-3/bitmaps/PART45_3_f002.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
4	0001_0000, 0002_0000	build/phase-3/bitmaps/PART45.png :material-check: build/phase-3/bitmaps/PART45_2_f000.png :material-alert-outline: (not found)
5	0002_0001, 0001_0000	build/phase-3/bitmaps/PART45_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART45.png :material-check:
6	0001_0000, 0002_0002	build/phase-3/bitmaps/PART45.png :material-check: build/phase-3/bitmaps/PART45_2_f002.png :material-alert-outline: (not found)
7	0001_0000, 0002_0003	build/phase-3/bitmaps/PART45.png :material-check: build/phase-3/bitmaps/PART45_2_f003.png :material-alert-outline: (not found)
8	0001_0000, 0002_0004	build/phase-3/bitmaps/PART45.png :material-check: build/phase-3/bitmaps/PART45_2_f004.png :material-alert-outline: (not found)

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	17
END_FRAME	9
END_ANM	1

Notas de Comportamento:

Aladdin's Lamp has 2 semantic states: Out, Lit. Transitions are triggered by: proximity, electrical.

Terminal state(s): Lit (part is removed from simulation).

This part can ignite nearby combustible parts (balloons, explosives). Uses proximity trigger.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-045-aladdins-lamp.md` describes part type `45`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 45`

TC-002: Mass value is positive

- **Given** Aladdin's Lamp has `mass = 2000`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Aladdin's Lamp has `property_q8 = 84`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (84 >> 8) <= 1.0` i.e. `COR = 0.328`

TC-004: SOLVE.RES state transitions (Out → Lit)

- **Given** Aladdin's Lamp has 2 primary states: Out (`1→2`) and Lit (`2→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Terminal state Lit (part removed)

- **Given** State Machine shows `Lit: 2→-1`
- **When** state reaches Lit
- **Then** part is removed from simulation (terminal)

TC-006: Spawns fire (light source)

- **Given** Category Rules show `Spawns: fire`
- **When** lamp is lit
- **Then** fire/light beam is emitted

TC-007: Triggers are proximity and electrical

- **Given** Category Rules show `Triggers: proximity, electrical`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-008: Part used in 8 levels

- **Given** Aladdin's Lamp (type 45)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 8 levels reference `part_num = 45`

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** lamp can be manually activated

TC-011: Multiple ignition sources

- **Given** Description: "Light this oil lamp with a laser, a flaming part, or a light source and a magnifying glass"
- **When** any valid ignition source contacts lamp
- **Then** lamp transitions to Lit state

TC-012: Can pop blimps and balloons

- **Given** Description: "Once it's burning, you can use it to... pop blimps and balloons"
- **When** fire beam contacts balloon/blimp
- **Then** target is destroyed

TC-013: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** lamp is not destroyed

TC-014: 3 ANM states (state IDs 1, 2, 3)

- **Given** Aladdin's Lamp ANM has `Total States = 3`
- **When** `decoder/d_t2anm.py` parses Section C
- **Then** `len(Section C entries) == 3`

TC-015: Category Part Count is 5

- **Given** Category Rules show `Category Part Count: 5`

- **When** validated
 - **Then** 5 parts in Light / Flame Sources category
-

Peça 46 — Pipe Wall



Categoria: Walls / Floors
Fases em que aparece: 27 fases
Quantidade de Estados: 2 estado(s)

This is a pipe wall or floor. Stretch it vertically or horizontally to any length you need. It has a slippery surface. Explosions won't affect it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	1024	friction = 4.000
unk_6	16	category-specific property
tile_w	32	base width in tile units
tile_h	32	base height in tile units
collision_w	576	AABB collision width
collision_h	416	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	4	4–7 (4 frames)	16×16	28
1	8	8–19 (12 frames)	32×25	56

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART46_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART46_f001.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART46_f001.png :material-check:
3	0001_0002	build/phase-3/bitmaps/PART46_f002.png :material-check:
4	0001_0003	build/phase-3/bitmaps/PART46_f003.png :material-check:
5	0001_0004	build/phase-3/bitmaps/PART46_f004.png :material-check:
6	0001_0004	build/phase-3/bitmaps/PART46_f004.png :material-check:
7	0001_0005	build/phase-3/bitmaps/PART46_f005.png :material-check:
8	0001_0006	build/phase-3/bitmaps/PART46_f006.png :material-check:
9	0001_0007	build/phase-3/bitmaps/PART46_f007.png :material-check:
10	0001_0008	build/phase-3/bitmaps/PART46_f008.png :material-check:
11	0001_0009	build/phase-3/bitmaps/PART46_f009.png :material-check:
12	0001_0010	build/phase-3/bitmaps/PART46_f010.png :material-check:
13	0001_0011	build/phase-3/bitmaps/PART46_f011.png :material-check:
14	0001_0012	build/phase-3/bitmaps/PART46_f012.png :material-check:
15	0001_0013	build/phase-3/bitmaps/PART46_f013.png :material-check:
16	0001_0014	build/phase-3/bitmaps/PART46_f014.png :material-check:
17	0001_0015	build/phase-3/bitmaps/PART46_f015.png :material-check:
18	0001_0016	build/phase-3/bitmaps/PART46_f016.png :material-check:
19	0001_0017	build/phase-3/bitmaps/PART46_f017.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	20
END_FRAME	20
END_ANM	1

Notas de Comportamento:

Pipe Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-046-pipe-wall.md` describes part type `46`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 46`

TC-002: Mass value is positive

- **Given** Pipe Wall has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Pipe Wall has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Slippery surface (low friction)

- **Given** Description: "It has a slippery surface"
- **When** objects slide on pipe wall
- **Then** friction is reduced

TC-005: Part used in 27 levels

- **Given** Pipe Wall (type 46)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 27 levels reference `part_num = 46`

TC-006: No triggers defined

- **Given** Category Rules show `Triggers: (none)`
- **When** behavior handler evaluates
- **Then** wall is purely static geometry

TC-007: Destructible flag is No (explosion-immune)

- **Given** Description: "Explosions won't affect it"
- **When** explosive part collides
- **Then** pipe wall is not destroyed

TC-008: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-009: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** pipe wall is static geometry

TC-010: Walls / Floors category

- **Given** Category shows `Walls / Floors`
- **When** physics collision detected
- **Then** wall provides collision surface

TC-011: Vertically/horizontally stretchable

- **Given** Description: "Stretch it vertically or horizontally to any length you need"
- **When** editor resizes part
- **Then** collision geometry updates accordingly

TC-012: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Bitmap dimensions variable

- **Given** Pipe Wall has variable length
- **When** sprite is rendered
- **Then** bitmap dimensions reflect actual size

TC-015: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

Peça 47 — Curved Pipe Wall



Categoria: Walls / Floors
Fases em que aparece: 7 fases
Quantidade de Estados: 1 estado(s)

This curved pipe section can be connected to a pipe wall or floor. Flip it to curve in the direction needed.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	7552	mass/density (relative scale)
<code>unk_2</code>	0	category-specific property
<code>friction_q8</code>	1024	friction = 4.000
<code>unk_6</code>	16	category-specific property
<code>tile_w</code>	240	base width in tile units
<code>tile_h</code>	240	base height in tile units
<code>collision_w</code>	0	AABB collision width
<code>collision_h</code>	0	AABB collision height
<code>bitmap_w</code>	32	bitmap frame width

Property	Raw Value	Interpretation
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	32×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART47_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART47_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART47_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART47_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Curved Pipe Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-047-curved-pipe-wall.md` describes part type `47`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 47`

TC-002: Mass value is positive

- **Given** Curved Pipe Wall has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Friction value within range

- **Given** Curved Pipe Wall has `friction_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `friction = 4.000` (high friction surface)

TC-004: Destructible flag is Yes

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** curved pipe wall can be destroyed

TC-005: Part used in 7 levels

- **Given** Curved Pipe Wall (type 47)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 7 levels reference `part_num = 47`

TC-006: No triggers defined

- **Given** Category Rules show `Triggers: (none)`
- **When** behavior handler evaluates
- **Then** wall is purely static geometry

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** curved pipe wall is static geometry

TC-009: Flip capability changes curve direction

- **Given** Description: "Flip it to curve in the direction needed"
- **When** player flips part
- **Then** pipe curve direction changes

TC-010: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-011: Connects to pipe wall or floor

- **Given** Description: "This curved pipe section can be connected to a pipe wall or floor"
- **When** placed adjacent to pipe wall
- **Then** connection is formed

TC-012: Tile dimensions 240×240 base

- **Given** Physics Properties show `tile_w = 240`, `tile_h = 240`
- **When** sprite is loaded
- **Then** base tile dimensions match

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Bitmap dimensions 32×32

- **Given** Physics Properties show `bitmap_w = 32`, `bitmap_h = 32`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
 - **When** animation renderer processes frame timing
 - **Then** Section A durations is empty
-

Peça 48 — Wood Wall



Categoria: Walls / Floors

Fases em que aparece: 39 fases

Quantidade de Estados: 3 estado(s)

This is a wooden wall or floor. Stretch it vertically or horizontally to any length you need. Its surface isn't very slippery. Explosions will blast holes through it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	7552	mass/density (relative scale)
<code>unk_2</code>	0	category-specific property
<code>friction_q8</code>	1024	friction = 4.000
<code>unk_6</code>	12	category-specific property
<code>tile_w</code>	32	base width in tile units
<code>tile_h</code>	32	base height in tile units
<code>collision_w</code>	576	AABB collision width
<code>collision_h</code>	416	AABB collision height

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0–12 (13 frames)	16×16	0
3	13	13–25 (13 frames)	16×16	91
4	26	26–37 (12 frames)	16×16	182

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART48_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART48_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART48_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART48_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART48_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART48_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART48_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART48_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART48_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART48_f009.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART48_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART48_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART48_f012.png :material-check:
13		

Frame	Sprites	PNG References
	0001_0013	build/phase-3/bitmaps/PART48_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART48_f014.png :material-check:
15	0001_0015	build/phase-3/bitmaps/PART48_f015.png :material-check:
16	0001_0016	build/phase-3/bitmaps/PART48_f016.png :material-check:
17	0001_0017	build/phase-3/bitmaps/PART48_f017.png :material-check:
18	0001_0018	build/phase-3/bitmaps/PART48_f018.png :material-check:
19	0001_0019	build/phase-3/bitmaps/PART48_f019.png :material-check:
20	0001_0020	build/phase-3/bitmaps/PART48_f020.png :material-check:
21	0001_0021	build/phase-3/bitmaps/PART48_f021.png :material-check:
22	0001_0022	build/phase-3/bitmaps/PART48_f022.png :material-check:
23	0001_0023	build/phase-3/bitmaps/PART48_f023.png :material-check:
24	0001_0024	build/phase-3/bitmaps/PART48_f024.png :material-check:
25	0001_0025	build/phase-3/bitmaps/PART48_f025.png :material-check:
26	0001_0026	build/phase-3/bitmaps/PART48_f026.png :material-check:
27	0001_0027	build/phase-3/bitmaps/PART48_f027.png :material-check:
28	0001_0028	build/phase-3/bitmaps/PART48_f028.png :material-check:
29	0001_0029	build/phase-3/bitmaps/PART48_f029.png :material-check:
30	0001_0030	build/phase-3/bitmaps/PART48_f030.png :material-check:
31	0001_0031	build/phase-3/bitmaps/PART48_f031.png :material-check:
32	0001_0032	build/phase-3/bitmaps/PART48_f032.png :material-check:
33	0001_0033	build/phase-3/bitmaps/PART48_f033.png :material-check:
34	0001_0034	build/phase-3/bitmaps/PART48_f034.png :material-check:
35	0001_0035	build/phase-3/bitmaps/PART48_f035.png :material-check:
36	0001_0036	build/phase-3/bitmaps/PART48_f036.png :material-check:
37	0001_0037	build/phase-3/bitmaps/PART48_f037.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	38
END_FRAME	38
END_ANM	1

Notas de Comportamento:

Wood Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-048-wood-wall.md` describes part type `48`

- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 48`

TC-002: Mass value is positive

- **Given** Wood Wall has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Wood Wall has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Destructible by explosives

- **Given** Description: "Explosions will blast holes through it"
- **When** explosive part collides
- **Then** wood wall is destroyed (hole created)

TC-005: Part used in 39 levels

- **Given** Wood Wall (type 48)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 39 levels reference `part_num = 48`

TC-006: No triggers defined

- **Given** Category Rules show `Triggers: (none)`
- **When** behavior handler evaluates
- **Then** wall is purely static geometry

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** wood wall is static geometry

TC-009: Walls / Floors category

- **Given** Category shows `Walls / Floors`
- **When** physics collision detected
- **Then** wall provides collision surface

TC-010: Vertically/horizontally stretchable

- **Given** Description: "Stretch it vertically or horizontally to any length you need"
- **When** editor resizes part
- **Then** collision geometry updates accordingly

TC-011: Non-slippery surface (contrasts with pipe wall)

- **Given** Description: "Its surface isn't very slippery"
- **When** objects slide on wood wall
- **Then** friction is higher than pipe wall

TC-012: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: Different from pipe wall (destructible vs explosion-immune)

- **Given** Wood Wall (destructible) vs Pipe Wall (explosion-immune)
- **When** explosive collision occurs
- **Then** wood wall is destroyed, pipe wall persists

Peça 50 — Electric Motor



Categoria: Rotating Power Sources

Fases em que aparece: 32 fases

Quantidade de Estados: 3 estado(s)

Plug this electric motor into an outlet and flick on the switch. Then use a belt to attach it to gears, conveyor belts, and other rotating parts. This motor can also be flipped.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	<code>1→4</code>	—	No
Running	<code>4→-1</code>	3	No

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	7552	mass/density (relative scale)
<code>unk_2</code>	0	category-specific property
<code>property_q8</code>	1024	Q8.8 = 4.000
<code>unk_6</code>	16	category-specific property
<code>dim_w1</code>	240	dimension 1 width
<code>dim_h1</code>	240	dimension 1 height
<code>dim_w2</code>	0	dimension 2 width
<code>dim_h2</code>	0	dimension 2 height

Property	Raw Value	Interpretation
bitmap_w	66	bitmap frame width
bitmap_h	54	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, rope
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	70×54	0
4	1	1–9 (9 frames)	70×54	19
3	10	10–37 (28 frames)	70×50	244

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0005, 0001_0008	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check:
1	0001_0000, 0001_0005, 0001_0009, 0001_0022	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f022.png :material-check:
2	0001_0001, 0001_0005, 0001_0006, 0001_0021	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f021.png :material-check:
3	0001_0002, 0001_0005, 0001_0007, 0001_0020	build/phase-3/bitmaps/ PART50_f002.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f020.png :material-check:
4	0001_0003, 0001_0005, 0001_0008, 0001_0019	build/phase-3/bitmaps/ PART50_f003.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f019.png :material-check:
5	0001_0004, 0001_0005, 0001_0009, 0001_0018	build/phase-3/bitmaps/ PART50_f004.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f018.png :material-check:
6	0001_0003, 0001_0005, 0001_0006, 0001_0025	build/phase-3/bitmaps/ PART50_f003.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f025.png :material-check:
7	0001_0002, 0001_0005, 0001_0007, 0001_0024	build/phase-3/bitmaps/ PART50_f002.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f024.png :material-check:
8	0001_0001, 0001_0005, 0001_0008, 0001_0023	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f023.png :material-check:
9	0001_0000, 0001_0005, 0001_0009, 0001_0022	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f022.png :material-check:
10	0001_0001, 0001_0005, 0001_0006, 0001_0010	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f010.png :material-check:
11	0001_0000, 0001_0005, 0001_0007, 0001_0014	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f014.png :material-check:
12	0001_0001, 0001_0005, 0001_0008, 0001_0011	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f011.png :material-check:
13	0001_0000, 0001_0005, 0001_0009, 0001_0015	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f015.png :material-check:
14	0001_0001, 0001_0005, 0001_0006, 0001_0012	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f012.png :material-check:
15	0001_0000, 0001_0005, 0001_0007, 0001_0016	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f016.png :material-check:
16	0001_0001, 0001_0005, 0001_0008, 0001_0013	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f013.png :material-check:
17	0001_0000, 0001_0005, 0001_0009, 0001_0014	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f014.png :material-check:
18	0001_0001, 0001_0005, 0001_0006, 0001_0017	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f017.png :material-check:
19	0001_0000, 0001_0005, 0001_0007, 0001_0022, 0001_0026	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f022.png :material-check: build/phase-3/bitmaps/ PART50_f026.png :material-check:
20	0001_0001, 0001_0005, 0001_0008, 0001_0027	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f027.png :material-check:
21	0001_0000, 0001_0005, 0001_0009, 0001_0013	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f013.png :material-check:
22	0001_0001, 0001_0005, 0001_0006, 0001_0028	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f028.png :material-check:
23	0001_0000, 0001_0005, 0001_0007, 0001_0015	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f015.png :material-check:
24	0001_0001, 0001_0005, 0001_0008, 0001_0029	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f029.png :material-check:
25	0001_0000, 0001_0005, 0001_0009, 0001_0016	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f016.png :material-check:
26	0001_0001, 0001_0005, 0001_0006, 0001_0030	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f030.png :material-check:
27	0001_0000, 0001_0005, 0001_0007, 0001_0014	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f014.png :material-check:
28	0001_0001, 0001_0005, 0001_0008, 0001_0031	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f031.png :material-check:
29	0001_0000, 0001_0005, 0001_0009, 0001_0014	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f014.png :material-check:
30	0001_0001, 0001_0005, 0001_0006, 0001_0032	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f032.png :material-check:
31	0001_0000, 0001_0005, 0001_0007, 0001_0015	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f015.png :material-check:
32	0001_0001, 0001_0005, 0001_0008, 0001_0033	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f033.png :material-check:
33	0001_0000, 0001_0005, 0001_0009, 0001_0013	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f013.png :material-check:
34	0001_0001, 0001_0005, 0001_0006, 0001_0010	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f006.png :material-check: build/phase-3/bitmaps/ PART50_f010.png :material-check:
35	0001_0000, 0001_0005, 0001_0007, 0001_0015	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f007.png :material-check: build/phase-3/bitmaps/ PART50_f015.png :material-check:
36	0001_0001, 0001_0005, 0001_0008, 0001_0010	build/phase-3/bitmaps/ PART50_f001.png :material-check: build/phase-3/bitmaps/ PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f008.png :material-check: build/phase-3/bitmaps/ PART50_f010.png :material-check:
37	0001_0000, 0001_0005, 0001_0009, 0001_0014	build/phase-3/bitmaps/ PART50_f000.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART50_f005.png :material-check: build/phase-3/bitmaps/ PART50_f009.png :material-check: build/phase-3/bitmaps/ PART50_f014.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
10	50	0	build/phase-6/raw-wav/SX_50_11025.wav
11	50	0	build/phase-6/raw-wav/SX_50_11025.wav
12	50	0	build/phase-6/raw-wav/SX_50_11025.wav
13	50	0	build/phase-6/raw-wav/SX_50_11025.wav
14	50	0	build/phase-6/raw-wav/SX_50_11025.wav
15	50	0	build/phase-6/raw-wav/SX_50_11025.wav
16	50	0	build/phase-6/raw-wav/SX_50_11025.wav
17	50	0	build/phase-6/raw-wav/SX_50_11025.wav
18	50	0	build/phase-6/raw-wav/SX_50_11025.wav
19	50	0	build/phase-6/raw-wav/SX_50_11025.wav
20	50	0	build/phase-6/raw-wav/SX_50_11025.wav
21	50	0	build/phase-6/raw-wav/SX_50_11025.wav
22	50	0	build/phase-6/raw-wav/SX_50_11025.wav
23	50	0	build/phase-6/raw-wav/SX_50_11025.wav
24	50	0	build/phase-6/raw-wav/SX_50_11025.wav
25	50	0	build/phase-6/raw-wav/SX_50_11025.wav
26	50	0	build/phase-6/raw-wav/SX_50_11025.wav
27	50	0	build/phase-6/raw-wav/SX_50_11025.wav
28	50	0	build/phase-6/raw-wav/SX_50_11025.wav
29	50	0	build/phase-6/raw-wav/SX_50_11025.wav
30	50	0	build/phase-6/raw-wav/SX_50_11025.wav
31	50	0	build/phase-6/raw-wav/SX_50_11025.wav
32	50	0	build/phase-6/raw-wav/SX_50_11025.wav
33	50	0	build/phase-6/raw-wav/SX_50_11025.wav
34	50	0	build/phase-6/raw-wav/SX_50_11025.wav
35	50	0	build/phase-6/raw-wav/SX_50_11025.wav
36	50	0	build/phase-6/raw-wav/SX_50_11025.wav
37	50	0	build/phase-6/raw-wav/SX_50_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	152
END_FRAME	38

Opcode	Count
PLAY_SFX	28
END_ANM	1

Notas de Comportamento:

Electric Motor has 2 semantic states: Off, Running. Transitions are triggered by: collision, electrical, rope.

Terminal state(s): Running (part is removed from simulation).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-050-electric-motor.md` describes part type `50`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 50`

TC-002: Mass value is positive

- **Given** Electric Motor has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Electric Motor has `property_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (1024 >> 8) <= 1.0` i.e. `COR = 4.000`

TC-004: SOLVE.RES state transitions

- **Given** Electric Motor has named state transitions
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Belt connection to power devices

- **Given** Description: "use a belt to attach it to gears, conveyor belts, and other rotating parts"
- **When** belt connects motor to device
- **Then** rotational power is transferred

TC-006: Electrical activation via outlet

- **Given** Description: "Plug this electric motor into an outlet and flick on the switch"
- **When** connected to electrical outlet
- **Then** motor activates

TC-007: Part used in 32 levels

- **Given** Electric Motor (type 50)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 32 levels reference `part_num = 50`

TC-008: Triggers include collision, electrical, rope

- **Given** Category Rules show `Triggers: collision, electrical, rope`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** motor can be manually activated

TC-011: Belt Connection flag is Yes

- **Given** Connections & Capabilities shows `Belt Connection = Yes`
- **When** belt is attached
- **Then** rotational power is transferred to connected part

TC-012: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical = Yes`
- **When** checked against `category_rules`
- **Then** motor can be plugged into outlet

TC-013: Category Part Count is 7

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Rotating Power Sources category

TC-014: Flip capability changes motor orientation

- **Given** Description: "This motor can also be flipped"
- **When** player flips motor
- **Then** belt connection orientation changes

TC-015: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** motor is not destroyed

Peça 51 — Vacuum



Categoria: Electrical
Fases em que aparece: 7 fases
Quantidade de Estados: 2 estado(s)

Plug this vacuum into an electrical outlet and use it to suck up any object that's affected by gravity.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	9→10	—	No
Vacuuming	10→-1	10	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	59	bitmap frame width
bitmap_h	33	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
9	0	0	65×33	0
10	1	1–59 (59 frames)	65×33	13

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0001_0000	build/phase-3/bitmaps/PART51_f001.png :material-check: build/phase-3/bitmaps/PART51_f000.png :material-check:
1	0001_0001, 0001_0000, 0001_0006	build/phase-3/bitmaps/PART51_f001.png :material-check: build/phase-3/bitmaps/PART51_f000.png :material-check: build/phase-3/bitmaps/PART51_f006.png :material-check:
2	0001_0002, 0001_0003, 0001_0011	

[illegible]

[illegible]

[illegible]

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART51_f005.png :material-check: build/phase-3/bitmaps/PART51_f004.png :material-check: build/phase-3/bitmaps/PART51_f008.png :material-check:
53	0001_0002, 0001_0003, 0001_0009	build/phase-3/bitmaps/PART51_f002.png :material-check: build/phase-3/bitmaps/PART51_f003.png :material-check: build/phase-3/bitmaps/PART51_f009.png :material-check:
54	0001_0004, 0001_0001, 0001_0008	build/phase-3/bitmaps/PART51_f004.png :material-check: build/phase-3/bitmaps/PART51_f001.png :material-check: build/phase-3/bitmaps/PART51_f008.png :material-check:
55	0001_0002, 0001_0003, 0001_0009	build/phase-3/bitmaps/PART51_f002.png :material-check: build/phase-3/bitmaps/PART51_f003.png :material-check: build/phase-3/bitmaps/PART51_f009.png :material-check:
56	0001_0004, 0001_0001, 0001_0008	build/phase-3/bitmaps/PART51_f004.png :material-check: build/phase-3/bitmaps/PART51_f001.png :material-check: build/phase-3/bitmaps/PART51_f008.png :material-check:
57	0001_0005, 0001_0002, 0001_0009	build/phase-3/bitmaps/PART51_f005.png :material-check: build/phase-3/bitmaps/PART51_f002.png :material-check: build/phase-3/bitmaps/PART51_f009.png :material-check:
58	0001_0004, 0001_0001, 0001_0008	build/phase-3/bitmaps/PART51_f004.png :material-check: build/phase-3/bitmaps/PART51_f001.png :material-check: build/phase-3/bitmaps/PART51_f008.png :material-check:
59	0001_0002, 0001_0003, 0001_0009	build/phase-3/bitmaps/PART51_f002.png :material-check: build/phase-3/bitmaps/PART51_f003.png :material-check: build/phase-3/bitmaps/PART51_f009.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	51	0	build/phase-6/raw-wav/SX_51_11025.wav
2	51	0	build/phase-6/raw-wav/SX_51_11025.wav
3	51	0	build/phase-6/raw-wav/SX_51_11025.wav
4	51	0	build/phase-6/raw-wav/SX_51_11025.wav
5	51	0	build/phase-6/raw-wav/SX_51_11025.wav
6	51	0	build/phase-6/raw-wav/SX_51_11025.wav
7	51	0	build/phase-6/raw-wav/SX_51_11025.wav
8	51	0	build/phase-6/raw-wav/SX_51_11025.wav
9	51	0	build/phase-6/raw-wav/SX_51_11025.wav
10	51	0	build/phase-6/raw-wav/SX_51_11025.wav
11	51	0	build/phase-6/raw-wav/SX_51_11025.wav
12	51	0	build/phase-6/raw-wav/SX_51_11025.wav
13	51	0	build/phase-6/raw-wav/SX_51_11025.wav
14	51	0	build/phase-6/raw-wav/SX_51_11025.wav
15	51	0	build/phase-6/raw-wav/SX_51_11025.wav
16	51	0	build/phase-6/raw-wav/SX_51_11025.wav
17	51	0	build/phase-6/raw-wav/SX_51_11025.wav
18	51	0	build/phase-6/raw-wav/SX_51_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
19	51	0	build/phase-6/raw-wav/SX_51_11025.wav
20	51	0	build/phase-6/raw-wav/SX_51_11025.wav
21	51	0	build/phase-6/raw-wav/SX_51_11025.wav
22	51	0	build/phase-6/raw-wav/SX_51_11025.wav
23	51	0	build/phase-6/raw-wav/SX_51_11025.wav
24	51	0	build/phase-6/raw-wav/SX_51_11025.wav
25	51	0	build/phase-6/raw-wav/SX_51_11025.wav
26	51	0	build/phase-6/raw-wav/SX_51_11025.wav
27	51	0	build/phase-6/raw-wav/SX_51_11025.wav
28	51	0	build/phase-6/raw-wav/SX_51_11025.wav
29	51	0	build/phase-6/raw-wav/SX_51_11025.wav
30	51	0	build/phase-6/raw-wav/SX_51_11025.wav
31	51	0	build/phase-6/raw-wav/SX_51_11025.wav
32	51	0	build/phase-6/raw-wav/SX_51_11025.wav
33	51	0	build/phase-6/raw-wav/SX_51_11025.wav
34	51	0	build/phase-6/raw-wav/SX_51_11025.wav
35	51	0	build/phase-6/raw-wav/SX_51_11025.wav
36	51	0	build/phase-6/raw-wav/SX_51_11025.wav
37	51	0	build/phase-6/raw-wav/SX_51_11025.wav
38	51	0	build/phase-6/raw-wav/SX_51_11025.wav
39	51	0	build/phase-6/raw-wav/SX_51_11025.wav
40	51	0	build/phase-6/raw-wav/SX_51_11025.wav
41	51	0	build/phase-6/raw-wav/SX_51_11025.wav
42	51	0	build/phase-6/raw-wav/SX_51_11025.wav
43	51	0	build/phase-6/raw-wav/SX_51_11025.wav
44	51	0	build/phase-6/raw-wav/SX_51_11025.wav
45	51	0	build/phase-6/raw-wav/SX_51_11025.wav
46	51	0	build/phase-6/raw-wav/SX_51_11025.wav
47	51	0	build/phase-6/raw-wav/SX_51_11025.wav
48	51	0	build/phase-6/raw-wav/SX_51_11025.wav
49	51	0	build/phase-6/raw-wav/SX_51_11025.wav
50	51	0	build/phase-6/raw-wav/SX_51_11025.wav
51	51	0	build/phase-6/raw-wav/SX_51_11025.wav
52	51	0	build/phase-6/raw-wav/SX_51_11025.wav
53	51	0	build/phase-6/raw-wav/SX_51_11025.wav
54	51	0	build/phase-6/raw-wav/SX_51_11025.wav
55	51	0	build/phase-6/raw-wav/SX_51_11025.wav
56	51	0	build/phase-6/raw-wav/SX_51_11025.wav
57	51	0	build/phase-6/raw-wav/SX_51_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
58	51	0	build/phase-6/raw-wav/SX_51_11025.wav
59	51	0	build/phase-6/raw-wav/SX_51_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	177
END_FRAME	60
PLAY_SFX	59
END_ANM	1

Notas de Comportamento:

Vacuum has 2 semantic states: Off, Vacuuming. Transitions are triggered by: electrical, collision.

Terminal state(s): Vacuuming (part is removed from simulation).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 52 — Cheese



Categoria: Special Mechanics
Fases em que aparece: 29 fases
Quantidade de Estados: 1 estado(s)

Newton Mouse will come after this cheese whenever he's close enough and on the same level of flooring.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1500	mass/density (relative scale)
unk_2	40	category-specific property
property_q8	64	Q8.8 = 0.250
unk_6	64	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	31	bitmap frame width

Property	Raw Value	Interpretation
bitmap_h	18	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	32×18	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART52.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Cheese is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has custom spawn behavior defined in SOLVE.RES.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-052-cheese.md` describes part type `52`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 52`

TC-002: Newton Mouse attraction target

- **Given** Description: "Newton Mouse will come after this cheese whenever he's close enough and on the same level of flooring"
- **When** cheese placed in level with Newton Mouse
- **Then** mouse pathfinds toward cheese

TC-003: Same-level flooring requirement

- **Given** Cheese attraction requires same floor level
- **When** cheese is on different elevation than mouse
- **Then** mouse does not pathfind to cheese

TC-004: Part used in 29 levels

- **Given** Cheese (type 52)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 29 levels reference `part_num = 52`

TC-005: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-006: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** cheese is static bait

TC-007: Special Mechanics category

- **Given** Category shows `Special Mechanics`
- **When** behavior handler evaluates
- **Then** cheese acts as mouse attractor

TC-008: No triggers defined

- **Given** Category Rules show `Triggers: (none)` or passive triggers
- **When** behavior handler evaluates
- **Then** cheese behavior is purely attractor-based

TC-009: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** cheese is not destroyed (or may be eaten)

TC-010: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-011: Mouse hole destination pairing

- **Given** Description: "He'll run inside a mouse hole if you place a hunk of cheese on the other side of it"
- **When** cheese placed beyond mouse hole
- **Then** mouse enters hole when reaching cheese

TC-012: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`

- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-013: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-014: Sprite existence validation

- **Given** Cheese ANM references sprite files
- **When** `decoder/d_t2anm.py` parses Section D
- **Then** all referenced PNGs exist in `build/phase-3/bitmaps/`

TC-015: Level YAML cross-reference

- **Given** Cheese appears in level YAML files
- **When** `validate_spec.py` loads level
- **Then** cheese part is correctly instantiated

Peça 53 — Thumb Tack



Categoria: Cutting / Popping
Fases em que aparece: 34 fases
Quantidade de Estados: 8 estado(s)

This thumb tack is handy for popping blimps and balloons. It can be flipped so that the point is facing up, down, or either side.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	7552	mass/density (relative scale)
<code>unk_2</code>	20	category-specific property
<code>property_q8</code>	1024	Q8.8 = 4.000
<code>unk_6</code>	2	category-specific property
<code>dim_w1</code>	240	dimension 1 width
<code>dim_h1</code>	240	dimension 1 height
<code>dim_w2</code>	0	dimension 2 width
<code>dim_h2</code>	0	dimension 2 height
<code>bitmap_w</code>	24	bitmap frame width
<code>bitmap_h</code>	24	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	<code>collision</code>

Rule	Value
Destructible	No
Spawns	(none)
Category Part Count	3

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
6	0	0	24×24	0
7	1	1–3 (3 frames)	24×25	7
8	4	4	24×24	46
9	5	5–7 (3 frames)	24×25	53
11	8	8	24×17	92
10	9	9–11 (3 frames)	25×17	99
13	12	12	24×17	138
12	13	13–15 (3 frames)	78×30	145

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART53_f000.png :material-check:
1	0001_0000, 0002_0000	build/phase-3/bitmaps/PART53_f000.png :material-check: build/phase-3/bitmaps/PART53_2_f000.png :material-alert-outline: (not found)
2	0001_0000, 0002_0001	build/phase-3/bitmaps/PART53_f000.png :material-check: build/phase-3/bitmaps/PART53_2_f001.png :material-alert-outline: (not found)
3	0001_0000, 0002_0002	build/phase-3/bitmaps/PART53_f000.png :material-check: build/phase-3/bitmaps/PART53_2_f002.png :material-alert-outline: (not found)
4	0001_0003	build/phase-3/bitmaps/PART53_f003.png :material-check:
5	0001_0003, 0002_0000	build/phase-3/bitmaps/PART53_f003.png :material-check: build/phase-3/bitmaps/PART53_2_f000.png :material-alert-outline: (not found)
6	0001_0003, 0002_0001	build/phase-3/bitmaps/PART53_f003.png :material-check: build/phase-3/bitmaps/PART53_2_f001.png :material-alert-outline: (not found)
7	0001_0003, 0002_0002	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART53_f003.png :material-check: build/phase-3/bitmaps/PART53_2_f002.png :material-alert-outline: (not found)
8	0001_0002	build/phase-3/bitmaps/PART53_f002.png :material-check:
9	0001_0002, 0002_0000	build/phase-3/bitmaps/PART53_f002.png :material-check: build/phase-3/bitmaps/PART53_2_f000.png :material-alert-outline: (not found)
10	0001_0002, 0002_0001	build/phase-3/bitmaps/PART53_f002.png :material-check: build/phase-3/bitmaps/PART53_2_f001.png :material-alert-outline: (not found)
11	0001_0002, 0002_0002	build/phase-3/bitmaps/PART53_f002.png :material-check: build/phase-3/bitmaps/PART53_2_f002.png :material-alert-outline: (not found)
12	0001_0001	build/phase-3/bitmaps/PART53_f001.png :material-check:
13	0001_0001, 0002_0000, 0002_0000	build/phase-3/bitmaps/PART53_f001.png :material-check: build/phase-3/bitmaps/PART53_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART53_2_f000.png :material-alert-outline: (not found)
14	0001_0001, 0002_0001	build/phase-3/bitmaps/PART53_f001.png :material-check: build/phase-3/bitmaps/PART53_2_f001.png :material-alert-outline: (not found)
15	0001_0001, 0002_0002	build/phase-3/bitmaps/PART53_f001.png :material-check: build/phase-3/bitmaps/PART53_2_f002.png :material-alert-outline: (not found)

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	29
END_FRAME	16
END_ANM	1

Notas de Comportamento:

Thumb Tack is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-053-thumb-tack.md` describes part type `53`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 53`

TC-002: Popping capability for blimps and balloons

- **Given** Description: "This thumb tack is handy for popping blimps and balloons"
- **When** balloon/blimp contacts thumb tack point
- **Then** target is destroyed

TC-003: Part used in 34 levels

- **Given** Thumb Tack (type 53)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 34 levels reference `part_num = 53`

TC-004: No triggers defined (passive popping)

- **Given** Category Rules show `Triggers: (none)` or collision-only
- **When** behavior handler evaluates
- **Then** popping occurs on contact

TC-005: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-006: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** thumb tack is passive obstacle

TC-007: Cutting / Popping category

- **Given** Category shows `Cutting / Popping`
- **When** behavior handler processes
- **Then** popping mechanic is applied

TC-008: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** thumb tack is not destroyed

TC-009: No electrical/belt/rope/fire/laser connections

- **Given** Connections & Capabilities shows most flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-010: Flip capability changes point direction

- **Given** Description: "It can be flipped so that the point is facing up, down, or either side"
- **When** player flips thumb tack
- **Then** pop direction changes

TC-011: Category Part Count is 3

- **Given** Category Rules show `Category Part Count: 3`
- **When** validated
- **Then** 3 parts in Cutting / Popping category

TC-012: Collision trigger activates popping

- **Given** Thumb tack is stationary
- **When** balloon/blimp contacts point
- **Then** target is popped

TC-013: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing

- **Then** Section A durations is empty

TC-014: Sprite existence validation

- **Given** Thumb Tack ANM references sprite files
- **When** `decoder/d_t2anm.py` parses Section D
- **Then** all referenced PNGs exist in `build/phase-3/bitmaps/`

TC-015: Level YAML cross-reference

- **Given** Thumb Tack appears in level YAML files
- **When** `validate_spec.py` loads level
- **Then** thumb tack part is correctly instantiated

Peça 54 — Mel Schlemming



Categoria: Characters
Fases em que aparece: 32 fases
Quantidade de Estados: 14 estado(s)

Meet Mel Schlemming. Mel walks mindlessly forward until he bumps into something. Then he turns around and walks mindlessly in the other direction. You can program him to walk, run, or stand still until he's bumped. If he falls too far, he drops to the floor and takes a snooze. He also has to watch out for alligators.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Normal	1→8	1	No
Sleeping	51→-1	11	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2400	mass/density (relative scale)
unk_2	40	category-specific property
property_q8	0	Q8.8 = 0.000
unk_6	128	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	20	bitmap frame width
bitmap_h	26	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	

Rule	Value
	collision, proximity
Destructible	No
Spawns	(none)
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×26	0
18	1	1–23 (23 frames)	16×28	7
19	24	24–46 (23 frames)	16×28	446
20	47	47–69 (23 frames)	24×26	707
21	70	70–94 (25 frames)	24×26	872
23	95	95–133 (39 frames)	40×11	1129
25	134	134	0×0	0
3	135	135–150 (16 frames)	0×0	0
15	151	151	0×0	0
16	152	152–159 (8 frames)	0×0	0
8	160	160–167 (8 frames)	0×0	0
9	168	168–197 (30 frames)	0×0	0
11	198	198–234 (37 frames)	0×0	0
51	235	235	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART54_f000.png :material-check:
1	0001_0042	build/phase-3/bitmaps/PART54_f042.png :material-check:
2	0001_0043, 0001_0044, 0001_0046	build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check: build/phase-3/bitmaps/PART54_f046.png :material-check:

Frame	Sprites	PNG References
3	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
4	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
5	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
6	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
7	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
8	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
9	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
10	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
11	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:

Frame	Sprites	PNG References
12	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
13	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
14	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
15	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
16	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
17	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
18	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
19	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
20	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:

Frame	Sprites	PNG References
21	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
22	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
23	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
24	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
25	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
26	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
27	0001_0047, 0001_0043, 0001_0044	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f044.png :material-check:
28	0001_0047, 0001_0043, 0001_0045	build/phase-3/bitmaps/PART54_f047.png :material-check: build/phase-3/bitmaps/PART54_f043.png :material-check: build/phase-3/bitmaps/PART54_f045.png :material-check:
29	0001_0041, 0001_0050	build/phase-3/bitmaps/PART54_f041.png :material-check: build/phase-3/bitmaps/PART54_f050.png :material-check:
30	0001_0041, 0001_0050	build/phase-3/bitmaps/PART54_f041.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART54_f050.png :material-check:
31	0001_0041, 0001_0049	build/phase-3/bitmaps/PART54_f041.png :material-check: build/phase-3/bitmaps/PART54_f049.png :material-check:
32	0001_0041, 0001_0048	build/phase-3/bitmaps/PART54_f041.png :material-check: build/phase-3/bitmaps/PART54_f048.png :material-check:
33	0001_0041, 0001_0049	build/phase-3/bitmaps/PART54_f041.png :material-check: build/phase-3/bitmaps/PART54_f049.png :material-check:
34	0001_0041, 0001_0050	build/phase-3/bitmaps/PART54_f041.png :material-check: build/phase-3/bitmaps/PART54_f050.png :material-check:
35	0001_0000	build/phase-3/bitmaps/PART54_f000.png :material-check:
36	0001_0003	build/phase-3/bitmaps/PART54_f003.png :material-check:
37	0001_0004	build/phase-3/bitmaps/PART54_f004.png :material-check:
38	0001_0005	build/phase-3/bitmaps/PART54_f005.png :material-check:
39	0001_0006	build/phase-3/bitmaps/PART54_f006.png :material-check:
40	0001_0007	build/phase-3/bitmaps/PART54_f007.png :material-check:
41	0001_0008	build/phase-3/bitmaps/PART54_f008.png :material-check:
42	0001_0001	build/phase-3/bitmaps/PART54_f001.png :material-check:
43	0001_0002	build/phase-3/bitmaps/PART54_f002.png :material-check:
44	0001_0000	build/phase-3/bitmaps/PART54_f000.png :material-check:
45	0001_0010	build/phase-3/bitmaps/PART54_f010.png :material-check:
46	0001_0011	build/phase-3/bitmaps/PART54_f011.png :material-check:
47	0001_0012	build/phase-3/bitmaps/PART54_f012.png :material-check:
48	0001_0013	build/phase-3/bitmaps/PART54_f013.png :material-check:
49	0001_0014	build/phase-3/bitmaps/PART54_f014.png :material-check:

Frame	Sprites	PNG References
50	0001_0015	build/phase-3/bitmaps/PART54_f015.png :material-check:
51	0001_0016	build/phase-3/bitmaps/PART54_f016.png :material-check:
52	0001_0009	build/phase-3/bitmaps/PART54_f009.png :material-check:
53	0001_0017	build/phase-3/bitmaps/PART54_f017.png :material-check:
54	0001_0018	build/phase-3/bitmaps/PART54_f018.png :material-check:
55	0001_0019	build/phase-3/bitmaps/PART54_f019.png :material-check:
56	0001_0020	build/phase-3/bitmaps/PART54_f020.png :material-check:
57	0001_0021	build/phase-3/bitmaps/PART54_f021.png :material-check:
58	0001_0022	build/phase-3/bitmaps/PART54_f022.png :material-check:
59	0001_0023	build/phase-3/bitmaps/PART54_f023.png :material-check:
60	0001_0024	build/phase-3/bitmaps/PART54_f024.png :material-check:
61	0001_0025	build/phase-3/bitmaps/PART54_f025.png :material-check:
62	0001_0026	build/phase-3/bitmaps/PART54_f026.png :material-check:
63	0001_0027	build/phase-3/bitmaps/PART54_f027.png :material-check:
64	0001_0028	build/phase-3/bitmaps/PART54_f028.png :material-check:
65	0001_0027	build/phase-3/bitmaps/PART54_f027.png :material-check:
66	0001_0029	build/phase-3/bitmaps/PART54_f029.png :material-check:
67	0001_0029	build/phase-3/bitmaps/PART54_f029.png :material-check:
68	0001_0030	build/phase-3/bitmaps/PART54_f030.png :material-check:
69	0001_0031	build/phase-3/bitmaps/PART54_f031.png :material-check:
70	0001_0032	build/phase-3/bitmaps/PART54_f032.png :material-check:
71	0001_0029	build/phase-3/bitmaps/PART54_f029.png :material-check:
72	0001_0030	build/phase-3/bitmaps/PART54_f030.png :material-check:
73	0001_0031	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART54_f031.png :material-check:
74	0001_0032	build/phase-3/bitmaps/PART54_f032.png :material-check:
75	0001_0029	build/phase-3/bitmaps/PART54_f029.png :material-check:
76	0001_0030	build/phase-3/bitmaps/PART54_f030.png :material-check:
77	0001_0031	build/phase-3/bitmaps/PART54_f031.png :material-check:
78	0001_0032	build/phase-3/bitmaps/PART54_f032.png :material-check:
79	0001_0033	build/phase-3/bitmaps/PART54_f033.png :material-check:
80	0001_0034	build/phase-3/bitmaps/PART54_f034.png :material-check:
81	0001_0035	build/phase-3/bitmaps/PART54_f035.png :material-check:
82	0001_0035	build/phase-3/bitmaps/PART54_f035.png :material-check:
83	0001_0036	build/phase-3/bitmaps/PART54_f036.png :material-check:
84	0001_0037	build/phase-3/bitmaps/PART54_f037.png :material-check:
85	0001_0038	build/phase-3/bitmaps/PART54_f038.png :material-check:
86	0001_0039	build/phase-3/bitmaps/PART54_f039.png :material-check:
87	0001_0040	build/phase-3/bitmaps/PART54_f040.png :material-check:
88	0001_0040, 0001_0055	build/phase-3/bitmaps/PART54_f040.png :material-check: build/phase-3/bitmaps/PART54_f055.png :material-check:
89	0001_0040, 0001_0055	build/phase-3/bitmaps/PART54_f040.png :material-check: build/phase-3/bitmaps/PART54_f055.png :material-check:
90	0001_0052, 0001_0055, 0001_0056	build/phase-3/bitmaps/PART54_f052.png :material-check: build/phase-3/bitmaps/PART54_f055.png :material-check: build/phase-3/bitmaps/PART54_f056.png :material-check:
91	0001_0052, 0001_0055, 0001_0056, 0001_0057	build/phase-3/bitmaps/PART54_f052.png :material-check: build/phase-3/bitmaps/PART54_f055.png :material-check: build/phase-3/bitmaps/PART54_f056.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART54_f057.png :material-check:
92	0001_0052, 0001_0055, 0001_0056, 0001_0057	build/phase-3/bitmaps/PART54_f052.png :material-check: build/phase-3/bitmaps/PART54_f055.png :material-check: build/phase-3/bitmaps/PART54_f056.png :material-check: build/phase-3/bitmaps/PART54_f057.png :material-check:
93	0001_0052, 0001_0056, 0001_0057	build/phase-3/bitmaps/PART54_f052.png :material-check: build/phase-3/bitmaps/PART54_f056.png :material-check: build/phase-3/bitmaps/PART54_f057.png :material-check:
94	0001_0040, 0001_0057	build/phase-3/bitmaps/PART54_f040.png :material-check: build/phase-3/bitmaps/PART54_f057.png :material-check:
95	0001_0053	build/phase-3/bitmaps/PART54_f053.png :material-check:
96	0001_0053	build/phase-3/bitmaps/PART54_f053.png :material-check:
97	0001_0054	build/phase-3/bitmaps/PART54_f054.png :material-check:
98	0001_0054	build/phase-3/bitmaps/PART54_f054.png :material-check:
99	0001_0054	build/phase-3/bitmaps/PART54_f054.png :material-check:
100	0001_0054	build/phase-3/bitmaps/PART54_f054.png :material-check:
101	0001_0054	build/phase-3/bitmaps/PART54_f054.png :material-check:
102	0001_0053	build/phase-3/bitmaps/PART54_f053.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
5	3454	0	build/phase-6/raw-wav/SX_3454_11025.wav
8	3454	0	build/phase-6/raw-wav/SX_3454_11025.wav
11	3454	0	build/phase-6/raw-wav/SX_3454_11025.wav
14	3454	0	build/phase-6/raw-wav/SX_3454_11025.wav
17	3454	0	build/phase-6/raw-wav/SX_3454_11025.wav
20	3454	0	build/phase-6/raw-wav/SX_3454_11025.wav
23	3454	0	build/phase-6/raw-wav/SX_3454_11025.wav
26	3454	0	build/phase-6/raw-wav/SX_3454_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
29	3054	0	build/phase-6/raw-wav/SX_3054_11025.wav
53	3554	0	build/phase-6/raw-wav/SX_3554_11025.wav
66	3254	0	build/phase-6/raw-wav/SX_3254_11025.wav
81	3354	0	build/phase-6/raw-wav/SX_3354_11025.wav
89	3654	0	build/phase-6/raw-wav/SX_3654_11025.wav
95	3754	0	build/phase-6/raw-wav/SX_3754_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	176
END_FRAME	103
PLAY_SFX	14
END_ANM	1

Notas de Comportamento:

Mel Schlemming has 2 semantic states: Normal, Sleeping. Transitions are triggered by: collision, proximity.

Terminal state(s): Sleeping (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 55 — Remote Control Explosives



Categoria: Explosives / Projectiles

Fases em que aparece: 57 fases

Quantidade de Estados: 2 estado(s)

These explosives come with the remote control part. They'll blow up all kinds of things, including most walls. Drop something on top of the remote control button to set them off. You can also tie one end of a rope to the button and hitch the other end to a teeter-totter, or another object that will allow you to pull the button down.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Exploded	2→5	—	No
Exploded	5→-1	—	Yes :material-bomb:

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	32	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	60	bitmap frame width
bitmap_h	48	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	62×48	0
5	1	1–8 (8 frames)	62×48	13

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0006, 0001_0008	build/phase-3/bitmaps/PART55_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found)
1	0001_0006, 0001_0008	build/phase-3/bitmaps/PART55_f006.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found)
2	0001_0006, 0001_0008, 0001_0009	build/phase-3/bitmaps/PART55_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f009.png :material-alert-outline: (not found)
3	0001_0006, 0001_0008, 0001_0010	build/phase-3/bitmaps/PART55_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f010.png :material-alert-outline: (not found)
4	0001_0006, 0001_0008, 0001_0010	build/phase-3/bitmaps/PART55_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f010.png :material-alert-outline: (not found)
5	0001_0006, 0001_0008, 0001_0011	build/phase-3/bitmaps/PART55_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f011.png :material-alert-outline: (not found)
6	0001_0006, 0001_0008, 0001_0012	build/phase-3/bitmaps/PART55_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f012.png :material-alert-outline: (not found)
7	0001_0006, 0001_0008, 0001_0013	build/phase-3/bitmaps/PART55_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f013.png :material-alert-outline: (not found)
8	0001_0007, 0001_0008	build/phase-3/bitmaps/PART55_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART55_f008.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
3	3055	0	build/phase-6/raw-wav/SX_3055_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	24
END_FRAME	9

Opcode	Count
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Remote Control Explosives has 2 semantic states: Not Exploded, Exploded. Transitions are triggered by: collision, proximity, timer.

State(s) Exploded are explosive — reaching these destroys the part.

Terminal state(s): Exploded (part is removed from simulation).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-055-remote-control-explosives.md` describes part type `55`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 55`

TC-002: Mass value is positive

- **Given** Remote Control Explosives has `mass = 4153`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Remote Control Explosives has `property_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (1024 >> 8) <= 1.0` i.e. `COR = 4.000`

TC-004: SOLVE.RES state transitions

- **Given** Remote Control Explosives has named state transitions
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Destructible flag is Yes

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** explosives are destroyed

TC-006: Spawns projectile on detonation

- **Given** Category Rules show `Spawns: projectile`
- **When** remote triggers explosion
- **Then** explosive force propagates

TC-007: Part used in 57 levels

- **Given** Remote Control Explosives (type 55)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 57 levels reference `part_num = 55`

TC-008: Triggers include collision, proximity, timer

- **Given** Category Rules show `Triggers: collision, proximity, timer`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** explosives require remote trigger activation

TC-011: Rope-activated trigger mechanism

- **Given** Description: "tie one end of a rope to the button and hitch the other end to a teeter-totter"
- **When** rope pulls button
- **Then** explosion is triggered

TC-012: Destroys walls including most types

- **Given** Description: "They'll blow up all kinds of things, including most walls"
- **When** explosion occurs near wall
- **Then** wall is destroyed (except explosion-immune types)

TC-013: Category Part Count is 10

- **Given** Category Rules show `Category Part Count: 10`
- **When** validated
- **Then** 10 parts in Explosives / Projectiles category

TC-014: No electrical/belt/rope/fire/laser connections

- **Given** Connections & Capabilities shows most flags as No
- **When** checked against `category_rules`
- **Then** only default connections and Projectile=Yes

TC-015: Terminal state on detonation (part removed)

- **Given** State Machine shows terminal state on explosion
- **When** explosion triggers
- **Then** part is removed from simulation

Peça 56 — Caution Wall



Categoria: Walls / Floors

Fases em que aparece: 48 fases

Quantidade de Estados: 3 estado(s)

This is a caution wall or floor. Stretch it vertically or horizontally to any length you need. It's very slippery, and isn't affected by explosives.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	100	mass/density (relative scale)
unk_2	140	category-specific property
friction_q8	512	friction = 2.000
unk_6	32	category-specific property
tile_w	16	base width in tile units
tile_h	16	base height in tile units
collision_w	576	AABB collision width
collision_h	416	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–2 (3 frames)	16×16	0
2	3	3–5 (3 frames)	16×16	21
3	6	6–17 (12 frames)	24×9	42

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART56_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART56_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART56_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART56_f003.png :material-check:

Frame	Sprites	PNG References
4	0001_0004	build/phase-3/bitmaps/PART56_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART56_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART56_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART56_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART56_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART56_f009.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART56_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART56_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART56_f012.png :material-check:
13	0001_0013	build/phase-3/bitmaps/PART56_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART56_f014.png :material-check:
15	0001_0011	build/phase-3/bitmaps/PART56_f011.png :material-check:
16	0001_0015	build/phase-3/bitmaps/PART56_f015.png :material-check:
17	0001_0016	build/phase-3/bitmaps/PART56_f016.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	18
END_FRAME	18
END_ANM	1

Notas de Comportamento:

Caution Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-056-caution-wall.md` describes part type `56`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 56`

TC-002: Mass value is positive

- **Given** Caution Wall has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Caution Wall has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Very slippery surface (high friction reduction)

- **Given** Description: "It's very slippery"
- **When** objects slide on caution wall
- **Then** friction is minimized

TC-005: Explosion-immune

- **Given** Description: "isn't affected by explosives"
- **When** explosive part collides
- **Then** caution wall is not destroyed

TC-006: Part used in 48 levels

- **Given** Caution Wall (type 56)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 48 levels reference `part_num = 56`

TC-007: No triggers defined

- **Given** Category Rules show `Triggers: (none)`
- **When** behavior handler evaluates
- **Then** wall is purely static geometry

TC-008: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-009: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** caution wall is static geometry

TC-010: Walls / Floors category

- **Given** Category shows `Walls / Floors`
- **When** physics collision detected
- **Then** wall provides collision surface

TC-011: Vertically/horizontally stretchable

- **Given** Description: "Stretch it vertically or horizontally to any length you need"
- **When** editor resizes part
- **Then** collision geometry updates accordingly

TC-012: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Destructible flag is No (explosion-immune)

- **Given** Category Rules and description confirm immunity
- **When** explosive collision occurs

- **Then** caution wall persists

TC-015: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

Peça 57 — Large Curved Pipe



Categoria: Pipe Systems
Fases em que aparece: 43 fases
Quantidade de Estados: 1 estado(s)

Drop balls and other objects into this large curved pipe to make them come out the other end. It can be attached to straight sections of large pipe and t-connectors. You can also attach an accelerator tube to speed up or reverse the direction of objects moving through the pipes.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	70	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	<code>collision, electrical</code>
Destructible	No
Spawns	<i>(none)</i>
Category Part Count	4

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No

Capability	Supported
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	72×64	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART57_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART57_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART57_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART57_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Large Curved Pipe is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-057-large-curved-pipe.md` describes part type `57`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 57`

TC-002: Ball guidance through pipe system

- **Given** Description: "Drop balls and other objects into this large curved pipe to make them come out the other end"
- **When** ball enters pipe
- **Then** ball exits at pipe output

TC-003: Pipe System category

- **Given** Category shows `Pipe Systems`
- **When** physics simulation runs
- **Then** ball guidance through curve is calculated

TC-004: Part used in 43 levels

- **Given** Large Curved Pipe (type 57)
- **When** checked against `build/phase-5/yaml/*.yaml`

- **Then** at least 43 levels reference `part_num = 57`

TC-005: Connects to straight pipe sections and t-connectors

- **Given** Description: "It can be attached to straight sections of large pipe and t-connectors"
- **When** pipe placed adjacent to compatible pipe
- **Then** connection is formed

TC-006: Accelerator tube attachment

- **Given** Description: "You can also attach an accelerator tube to speed up or reverse the direction of objects"
- **When** accelerator attached to pipe
- **Then** ball velocity is modified

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** pipe is static geometry

TC-009: No triggers defined

- **Given** Category Rules show `Triggers: (none)` or passive
- **When** behavior handler evaluates
- **Then** pipe guides balls passively

TC-010: Destructible flag is No (pipe walls are indestructible)

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** pipe is not destroyed

TC-011: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-012: Category Part Count includes this pipe type

- **Given** Large Curved Pipe is in Pipe Systems category
- **When** validated
- **Then** category count reflects pipe system parts

TC-013: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-014: Sprite existence validation

- **Given** Large Curved Pipe ANM references sprite files
- **When** `decoder/d_t2anm.py` parses Section D
- **Then** all referenced PNGs exist in `build/phase-3/bitmaps/`

TC-015: Level YAML cross-reference

- **Given** Large Curved Pipe appears in level YAML files
- **When** `validate_spec.py` loads level
- **Then** pipe part is correctly instantiated

Peça 58 — Mel's House



Categoria: Special Mechanics
Fases em que aparece: 18 fases
Quantidade de Estados: 8 estado(s)

Here's Mel Schlemming's cozy suburban duplex. If he sees it, he'll head home. It can also be programmed to be a rustic log cabin.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Vacant	5→6, 4→15	4	No
Occupied	6→4, 15→-1	15	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	48	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
5	0	0	48×64	0
6	1	1–69 (69 frames)	48×64	13
9	70	70–107 (38 frames)	72×97	2298
8	108	108–110 (3 frames)	0×0	0
4	111	111	0×0	0
15	112	112–121 (10 frames)	0×0	0
13	122	122–130 (9 frames)	0×0	0
14	131	131	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0008, 0001_0009	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check:
1	0001_0008	build/phase-3/bitmaps/ PART58_f008.png :material-check:
2	0001_0008, 0001_0009, 0001_0010, 0001_0011	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check:
3	0001_0008, 0001_0009, 0001_0010, 0001_0011	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check:
4	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
5	0001_0008, 0001_0009, 0001_0010, 0001_0012, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f012.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
6	0001_0008, 0001_0009, 0001_0010, 0001_0012, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f012.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
7	0001_0008, 0001_0009, 0001_0010, 0001_0012, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f012.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
8	0001_0008, 0001_0009, 0001_0010, 0001_0012, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f012.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
9	0001_0008, 0001_0009, 0001_0010, 0001_0013, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f013.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
10	0001_0008, 0001_0009, 0001_0010, 0001_0013, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f013.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
11	0001_0008, 0001_0009, 0001_0010, 0001_0013, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f013.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
12	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
13	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
14	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
15	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART58_f021.png :material-check:
16	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
17	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
18	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
19	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
20	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
21	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
22	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
23	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
24	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
25	0001_0008, 0001_0009, 0001_0010, 0001_0014, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f014.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
26	0001_0008, 0001_0009, 0001_0010, 0001_0013, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f013.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:

Frame	Sprites	PNG References
27	0001_0008, 0001_0009, 0001_0010, 0001_0013, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f013.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
28	0001_0008, 0001_0009, 0001_0010, 0001_0013, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f013.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
29	0001_0008, 0001_0009, 0001_0010, 0001_0012, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f012.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
30	0001_0008, 0001_0009, 0001_0010, 0001_0012, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f012.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
31	0001_0008, 0001_0009, 0001_0010, 0001_0012, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f012.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
32	0001_0008, 0001_0009, 0001_0010, 0001_0012, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART58_f012.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
33	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
34	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
35	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
36	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
37	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
38	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
39	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
40	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0019, 0001_0015	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check: build/phase-3/bitmaps/ PART58_f015.png :material-check:
41	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0020, 0001_0015	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check: build/phase-3/bitmaps/ PART58_f015.png :material-check:
42	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0021, 0001_0015	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check: build/phase-3/bitmaps/ PART58_f015.png :material-check:
43	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0019, 0001_0015	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check: build/phase-3/bitmaps/ PART58_f015.png :material-check:
44	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0020, 0001_0015	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check: build/phase-3/bitmaps/ PART58_f015.png :material-check:
45	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0021, 0001_0015	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check: build/phase-3/bitmaps/ PART58_f015.png :material-check:
46	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0016, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f016.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
47	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0016, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f016.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:

Frame	Sprites	PNG References
48	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0016, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f016.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
49	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0017, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f017.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
50	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0017, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f017.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
51	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0017, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f017.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
52	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
53	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
54	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
55	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
56	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
57	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
58	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
59	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
60	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
61	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
62	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
63	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
64	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
65	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0018, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f018.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
66	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0017, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f017.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:

Frame	Sprites	PNG References
67	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0017, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f017.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
68	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0017, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f017.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
69	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0016, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f016.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
70	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0016, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f016.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
71	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0015, 0001_0019	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART58_f015.png :material-check: build/phase-3/bitmaps/ PART58_f019.png :material-check:
72	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0015, 0001_0020	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f015.png :material-check: build/phase-3/bitmaps/ PART58_f020.png :material-check:
73	0001_0008, 0001_0009, 0001_0010, 0001_0011, 0001_0015, 0001_0021	build/phase-3/bitmaps/ PART58_f008.png :material-check: build/phase-3/bitmaps/ PART58_f009.png :material-check: build/phase-3/bitmaps/ PART58_f010.png :material-check: build/phase-3/bitmaps/ PART58_f011.png :material-check: build/phase-3/bitmaps/ PART58_f015.png :material-check: build/phase-3/bitmaps/ PART58_f021.png :material-check:
74	0001_0000	build/phase-3/bitmaps/ PART58_f000.png :material-check:
75	0001_0000, 0001_0001	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f001.png :material-check:
76	0001_0000, 0001_0002	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f002.png :material-check:
77	0001_0000, 0001_0003	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f003.png :material-check:
78	0001_0000	build/phase-3/bitmaps/ PART58_f000.png :material-check:
79	0001_0000, 0001_0004, 0001_0004, 0001_0005	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f005.png :material-check:
80	0001_0000, 0001_0004, 0001_0004, 0001_0006	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f006.png :material-check:
81	0001_0000, 0001_0004, 0001_0004, 0001_0007	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f007.png :material-check:
82	0001_0000, 0001_0004, 0001_0004, 0001_0005, 0001_0001	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f005.png :material-check: build/phase-3/bitmaps/ PART58_f001.png :material-check:
83	0001_0000, 0001_0004, 0001_0004, 0001_0006, 0001_0002	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f006.png :material-check: build/phase-3/bitmaps/ PART58_f002.png :material-check:
84	0001_0000, 0001_0004, 0001_0004, 0001_0007, 0001_0003	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f007.png :material-check: build/phase-3/bitmaps/ PART58_f003.png :material-check:
85	0001_0000, 0001_0004, 0001_0004, 0001_0007	build/phase-3/bitmaps/ PART58_f000.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f004.png :material-check: build/phase-3/bitmaps/ PART58_f007.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
2	3058	0	build/phase-6/raw-wav/SX_3058_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
78	3058	0	build/phase-6/raw-wav/SX_3058_11025.wav
85	3058	0	build/phase-6/raw-wav/SX_3058_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	434
END_FRAME	86
PLAY_SFX	3
END_ANM	1

Notas de Comportamento:

Mel's House has 2 semantic states: Vacant, Occupied. Transitions are triggered by: collision, electrical, proximity.

Terminal state(s): Occupied (part is removed from simulation).

This part has custom spawn behavior defined in SOLVE.RES.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-058-mels-house.md` describes part type `58`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 58`

TC-002: Mel attraction destination

- **Given** Description: "Here's Mel Schlemming's cozy suburban duplex. If he sees it, he'll head home"
- **When** Mel (type 54) sees house
- **Then** Mel pathfinds toward house

TC-003: Programmable appearance (suburban vs log cabin)

- **Given** Description: "It can also be programmed to be a rustic log cabin"
- **When** player selects appearance
- **Then** house sprite changes

TC-004: Part used in 18 levels

- **Given** Mel's House (type 58)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 18 levels reference `part_num = 58`

TC-005: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-006: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** house appearance can be programmed

TC-007: Special Mechanics category

- **Given** Category shows `Special Mechanics`
- **When** behavior handler evaluates
- **Then** house acts as Mel's destination

TC-008: No triggers defined (passive destination)

- **Given** Category Rules show passive or no triggers
- **When** behavior handler evaluates
- **Then** house is passive destination marker

TC-009: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** house is not destroyed

TC-010: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-011: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-012: SOLVE.RES state transitions for programming

- **Given** Mel's House has SOLVE.RES states for appearance modes
- **When** behavior handler evaluates
- **Then** state reflects selected appearance

TC-013: Animated Section A for house details

- **Given** ANM shows `Animated (Section A): Yes`
- **When** simulation runs
- **Then** house animation plays (smoke, lights, etc.)

TC-014: Sprite existence validation

- **Given** Mel's House ANM references sprite files
- **When** `decoder/d_t2anm.py` parses Section D
- **Then** all referenced PNGs exist in `build/phase-3/bitmaps/`

TC-015: Level YAML cross-reference

- **Given** Mel's House appears in level YAML files
- **When** `validate_spec.py` loads level
- **Then** house part is correctly instantiated

Peça 59 — Super Ball



Categoria: Balls

Fases em que aparece: 7 fases

Quantidade de Estados: 3 estado(s)

This super ball gains height with each bounce.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1800	mass/density (relative scale)
damping	14	drag/air-resistance or density
cor_q8	512	COR = 2.000
radius	16	collision radius in internal units
sprite_w	240	sprite width
sprite_h	240	sprite height
data_offset	23	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×23	0
3	1	1	24×27	7
4	2	2	32×22	14

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART59_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART59_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART59_f002.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	3
END_FRAME	3
END_ANM	1

Notas de Comportamento:

Super Ball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-059-super-ball.md` describes part type `59`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 59`

TC-002: Mass value is positive

- **Given** Super Ball has `mass = 1800`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR exceeds 1.0 (bounce amplification)

- **Given** Super Ball has `cor_q8 = 512`
- **When** converted from Q8.8 fixed-point
- **Then** `COR = 2.000` (exceeds normal range, ball gains height)

TC-004: Radius is non-negative

- **Given** Super Ball has `radius = 16`
- **When** physics engine initializes collision
- **Then** `radius >= 0`

TC-005: Damping value affects air resistance

- **Given** Super Ball has `damping = 14`
- **When** physics simulation runs
- **Then** velocity decay reflects light mass

TC-006: Single default state — no named transitions

- **Given** Super Ball has no SOLVE.RES state data
- **When** the part is loaded
- **Then** `state_machine` section shows "Single default state"

TC-007: Part used in 7 levels

- **Given** Super Ball (type 59)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 7 levels reference `part_num = 59`

TC-008: Category Part Count is 9 (Balls category)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Balls category

TC-009: Triggers are collision and trap

- **Given** Category Rules show `Triggers: collision, trap`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-011: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** super ball physics are automatic

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** super ball is not destroyed

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Sprite dimensions 240×240

- **Given** Physics Properties show `sprite_w = 240, sprite_h = 240`
- **When** sprite is loaded
- **Then** sprite dimensions match

TC-015: COR > 1.0 validated (unique among all balls)

- **Given** Super Ball COR = 2.000 vs normal range [0.0, 1.0]
- **When** collision response calculated
- **Then** bounce height exceeds fall height (energy gain mechanic)

Peça 60 — Grass Floor



Categoria: Walls / Floors

Fases em que aparece: 10 fases

Quantidade de Estados: 3 estado(s)

This is a grass floor or vine wall. Stretch it vertically or horizontally to any length you need. It's not very slippery. Explosions will blast holes through it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	7552	mass/density (relative scale)

Property	Raw Value	Interpretation
unk_2	0	category-specific property
friction_q8	128	friction = 0.500
unk_6	128	category-specific property
tile_w	16	base width in tile units
tile_h	16	base height in tile units
collision_w	576	AABB collision width
collision_h	416	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–7 (8 frames)	16×16	0
2	8	8–17 (10 frames)	16×16	56
3	18	18–29 (12 frames)	40×20	126

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART60_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART60_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART60_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART60_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART60_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART60_f005.png :material-check:
6		

Frame	Sprites	PNG References
	0001_0006	build/phase-3/bitmaps/PART60_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART60_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART60_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART60_f009.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART60_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART60_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART60_f012.png :material-check:
13	0001_0013	build/phase-3/bitmaps/PART60_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART60_f014.png :material-check:
15	0001_0015	build/phase-3/bitmaps/PART60_f015.png :material-check:
16	0001_0016	build/phase-3/bitmaps/PART60_f016.png :material-check:
17	0001_0017	build/phase-3/bitmaps/PART60_f017.png :material-check:
18	0001_0018	build/phase-3/bitmaps/PART60_f018.png :material-check:
19	0001_0019	build/phase-3/bitmaps/PART60_f019.png :material-check:
20	0001_0020	build/phase-3/bitmaps/PART60_f020.png :material-check:
21	0001_0021	build/phase-3/bitmaps/PART60_f021.png :material-check:
22	0001_0022	build/phase-3/bitmaps/PART60_f022.png :material-check:
23	0001_0023	build/phase-3/bitmaps/PART60_f023.png :material-check:
24	0001_0024	build/phase-3/bitmaps/PART60_f024.png :material-check:
25	0001_0025	build/phase-3/bitmaps/PART60_f025.png :material-check:
26	0001_0026	build/phase-3/bitmaps/PART60_f026.png :material-check:
27	0001_0027	build/phase-3/bitmaps/PART60_f027.png :material-check:
28	0001_0028	build/phase-3/bitmaps/PART60_f028.png :material-check:
29	0001_0029	build/phase-3/bitmaps/PART60_f029.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	30
END_FRAME	30
END_ANM	1

Notas de Comportamento:

Grass Floor is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-060-grass-floor.md` describes part type `60`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`

- **Then** `part_num == 60`

TC-002: Mass value is positive

- **Given** Grass Floor has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Friction value within range

- **Given** Grass Floor has `friction_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `friction = 0.500` (not very slippery)

TC-004: Destructible by explosives

- **Given** Description: "Explosions will blast holes through it"
- **When** explosive part collides
- **Then** grass floor is destroyed (hole created)

TC-005: Part used in 10 levels

- **Given** Grass Floor (type 60)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 10 levels reference `part_num = 60`

TC-006: No triggers defined

- **Given** Category Rules show `Triggers: (none)`
- **When** behavior handler evaluates
- **Then** wall is purely static geometry

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** grass floor is static geometry

TC-009: Walls / Floors category

- **Given** Category shows `Walls / Floors`
- **When** physics collision detected
- **Then** wall provides collision surface

TC-010: Vertically/horizontally stretchable

- **Given** Description: "Stretch it vertically or horizontally to any length you need"
- **When** editor resizes part
- **Then** collision geometry updates accordingly

TC-011: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-012: Tile dimensions 16×16 base

- **Given** Physics Properties show `tile_w = 16`, `tile_h = 16`
- **When** sprite is loaded
- **Then** base tile dimensions match

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: Collision AABB dimensions 576×416

- **Given** Physics Properties show `collision_w = 576`, `collision_h = 416`
- **When** physics engine resolves collisions
- **Then** AABB matches grass floor geometry

Peça 61 — Alligator



Categoria: Characters
Fases em que aparece: 14 fases
Quantidade de Estados: 11 estado(s)

Meet Edison Alligator. He'll chow down Mel Schlemming or Newton Mouse if they get too close. He also flips things into the air with his snout.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Hasn't Eaten	1→5	1	No
Has Eaten	5→-1	5	No

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	2400	mass/density (relative scale)
<code>unk_2</code>	40	category-specific property
<code>property_q8</code>	192	Q8.8 = 0.750
<code>unk_6</code>	48	category-specific property
<code>dim_w1</code>	240	dimension 1 width
<code>dim_h1</code>	240	dimension 1 height
<code>dim_w2</code>	0	dimension 2 width
<code>dim_h2</code>	0	dimension 2 height

Property	Raw Value	Interpretation
bitmap_w	94	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity
Destructible	No
Spawns	(none)
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	100×16	0
7	1	1–10 (10 frames)	95×18	19
8	11	11–20 (10 frames)	100×26	209
4	21	21–23 (3 frames)	96×16	403
12	24	24–25 (2 frames)	96×17	460
5	26	26–64 (39 frames)	96×17	498
6	65	65–66 (2 frames)	0×0	0
9	67	67–76 (10 frames)	0×0	0
10	77	77–81 (5 frames)	0×0	0
11	82	82–83 (2 frames)	0×0	0
13	84	84	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0001_0000, 0001_0002	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f000.png :material-check: build/phase-3/bitmaps/PART61_f002.png :material-check:
1	0001_0001, 0001_0008, 0001_0000	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f008.png :material-check: build/phase-3/bitmaps/PART61_f000.png :material-check:
2	0001_0001, 0001_0009, 0001_0000	

[illegible]

[illegible]

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f010.png :material-check: build/phase-3/bitmaps/PART61_f000.png :material-check:
36	0001_0001, 0001_0012, 0001_0000	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f012.png :material-check: build/phase-3/bitmaps/PART61_f000.png :material-check:
37	0001_0001, 0001_0013, 0001_0000	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f013.png :material-check: build/phase-3/bitmaps/PART61_f000.png :material-check:
38	0001_0001, 0001_0008, 0001_0003	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f008.png :material-check: build/phase-3/bitmaps/PART61_f003.png :material-check:
39	0001_0001, 0001_0009, 0001_0004	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f009.png :material-check: build/phase-3/bitmaps/PART61_f004.png :material-check:
40	0001_0001, 0001_0010, 0001_0005	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f010.png :material-check: build/phase-3/bitmaps/PART61_f005.png :material-check:
41	0001_0001, 0001_0012, 0001_0006	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f012.png :material-check: build/phase-3/bitmaps/PART61_f006.png :material-check:
42	0001_0001, 0001_0013, 0001_0007	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f013.png :material-check: build/phase-3/bitmaps/PART61_f007.png :material-check:
43	0001_0001, 0001_0014, 0001_0002	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f014.png :material-check: build/phase-3/bitmaps/PART61_f002.png :material-check:
44	0001_0001, 0001_0015, 0001_0002	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f015.png :material-check: build/phase-3/bitmaps/PART61_f002.png :material-check:
45	0001_0001, 0001_0014, 0001_0002	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f014.png :material-check: build/phase-3/bitmaps/PART61_f002.png :material-check:
46	0001_0001, 0001_0015, 0001_0002	build/phase-3/bitmaps/PART61_f001.png :material-check: build/phase-3/bitmaps/PART61_f015.png :material-check: build/phase-3/bitmaps/PART61_f002.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
11	3061	0	build/phase-6/raw-wav/SX_3061_11025.wav
17	3261	0	build/phase-6/raw-wav/SX_3261_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	141
END_FRAME	47
PLAY_SFX	2
END_ANM	1

Notas de Comportamento:

Alligator has 2 semantic states: Hasn't Eaten, Has Eaten. Transitions are triggered by: collision, proximity.

Terminal state(s): Has Eaten (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-061-alligator.md` describes part type `61`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 61`

TC-002: Mass value is positive

- **Given** Alligator has `mass = 2400`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Alligator has `property_q8 = 192`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (192 >> 8) <= 1.0` i.e. `COR = 0.750`

TC-004: Predator of Mel and Newton Mouse

- **Given** Description: "He'll chow down Mel Schlemming or Newton Mouse if they get too close"
- **When** alligator proximity detects character
- **Then** character is eaten (removed from simulation)

TC-005: Snout flip mechanic

- **Given** Description: "He also flips things into the air with his snout"
- **When** alligator collides with object
- **Then** object is launched upward

TC-006: SOLVE.RES state transitions (character AI)

- **Given** Alligator has named states in SOLVE.RES
- **When** behavior handler evaluates triggers
- **Then** state transitions follow SOLVE.RES rules

TC-007: Part used in 14 levels

- **Given** Alligator (type 61)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 14 levels reference `part_num = 61`

TC-008: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-009: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-010: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** alligator AI responds to environment

TC-011: Character category (Category Part Count = 8)

- **Given** Category Rules show `Category Part Count: 8`
- **When** validated
- **Then** 8 parts in Characters category

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** alligator is not destroyed

TC-013: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-014: Bitmap dimensions 94x16

- **Given** Physics Properties show `bitmap_w = 94, bitmap_h = 16`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Threat to mouse and Mel AI systems

- **Given** Alligator is predator for two character types
- **When** simulation runs
- **Then** both Newton Mouse and Mel Schlemming pathfind to avoid

Peça 62 — Coffee Pot



Categoria: Special Mechanics
Fases em que aparece: 17 fases
Quantidade de Estados: 3 estado(s)

Make this coffee pot percolate by heating it with a candle, Aladdin's lamp, match-on-a-spring, or flint & tinder. Then use the steam to push things.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	1→3	1	No
Percolating	3→-1	4	No

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	2400	mass/density (relative scale)

Property	Raw Value	Interpretation
unk_2	40	category-specific property
property_q8	64	Q8.8 = 0.250
unk_6	64	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	38	bitmap frame width
bitmap_h	41	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	40×41	0
3	1	1–17 (17 frames)	40×41	7
4	18	18–23 (6 frames)	73×69	332

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART62_f000.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART62_f000.png :material-check:
2	0001_0000	build/phase-3/bitmaps/PART62_f000.png :material-check:
3	0001_0000, 0001_0001	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f001.png :material-check:
4	0001_0000, 0001_0003	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f003.png :material-check:

Frame	Sprites	PNG References
5	0001_0000, 0001_0002	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f002.png :material-check:
6	0001_0000, 0001_0004	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f004.png :material-check:
7	0001_0005, 0001_0008	build/phase-3/bitmaps/PART62_f005.png :material-check: build/phase-3/bitmaps/PART62_f008.png :material-check:
8	0001_0005, 0001_0006	build/phase-3/bitmaps/PART62_f005.png :material-check: build/phase-3/bitmaps/PART62_f006.png :material-check:
9	0001_0000, 0001_0002	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f002.png :material-check:
10	0001_0010, 0001_0014	build/phase-3/bitmaps/PART62_f010.png :material-check: build/phase-3/bitmaps/PART62_f014.png :material-check:
11	0001_0010, 0001_0012	build/phase-3/bitmaps/PART62_f010.png :material-check: build/phase-3/bitmaps/PART62_f012.png :material-check:
12	0001_0005, 0001_0008	build/phase-3/bitmaps/PART62_f005.png :material-check: build/phase-3/bitmaps/PART62_f008.png :material-check:
13	0001_0000, 0001_0004	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f004.png :material-check:
14	0001_0005, 0001_0007	build/phase-3/bitmaps/PART62_f005.png :material-check: build/phase-3/bitmaps/PART62_f007.png :material-check:
15	0001_0005, 0001_0009	build/phase-3/bitmaps/PART62_f005.png :material-check: build/phase-3/bitmaps/PART62_f009.png :material-check:
16	0001_0010, 0001_0013	build/phase-3/bitmaps/PART62_f010.png :material-check: build/phase-3/bitmaps/PART62_f013.png :material-check:
17	0001_0010, 0001_0011	build/phase-3/bitmaps/PART62_f010.png :material-check: build/phase-3/bitmaps/PART62_f011.png :material-check:
18	0001_0000, 0001_0015	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f015.png :material-check:
19	0001_0000, 0001_0016	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f016.png :material-check:
20	0001_0000, 0001_0017	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f017.png :material-check:
21	0001_0000, 0001_0015	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f015.png :material-check:
22	0001_0000, 0001_0016	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f016.png :material-check:
23	0001_0000, 0001_0017	build/phase-3/bitmaps/PART62_f000.png :material-check: build/phase-3/bitmaps/PART62_f017.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	45
END_FRAME	24
DRAW_LINE	23
DRAW_RECT	3
	1

Opcode	Count
END_ANM	

Notas de Comportamento:

Coffee Pot has 2 semantic states: Off, Percolating. Transitions are triggered by: collision, electrical, proximity.

Terminal state(s): Percolating (part is removed from simulation).

This part has custom spawn behavior defined in SOLVE.RES.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-062-coffee-pot.md` describes part type `62`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 62`

TC-002: Mass value is positive

- **Given** Coffee Pot has `mass = 2400`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Coffee Pot has `property_q8 = 64`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (64 >> 8) <= 1.0` i.e. `COR = 0.250`

TC-004: Steam propulsion mechanic

- **Given** Description: "Then use the steam to push things"
- **When** coffee pot is heated and percolating
- **Then** steam applies force to nearby objects

TC-005: Fire activation sources

- **Given** Description: "Make this coffee pot percolate by heating it with a candle, Aladdin's lamp, match-on-a-spring, or flint & tinder"
- **When** any valid heat source contacts pot
- **Then** percolation begins

TC-006: Part used in 17 levels

- **Given** Coffee Pot (type 62)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 17 levels reference `part_num = 62`

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** pot can be heated by player-placed fire sources

TC-009: Special Mechanics category

- **Given** Category shows `Special Mechanics`
- **When** behavior handler evaluates
- **Then** steam propulsion is applied

TC-010: Triggers include fire/heat activation

- **Given** Category Rules show `Triggers: fire` or passive heat detection
- **When** heat source contacts pot
- **Then** percolation state activates

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** pot is not destroyed

TC-012: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-013: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-014: Bitmap dimensions 38×41

- **Given** Physics Properties show `bitmap_w = 38, bitmap_h = 41`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

Peça 63 — Pool Ball



Categoria: Balls

Fases em que aparece: 21 fases

Quantidade de Estados: 16 estado(s)

This pool ball won't move until it's hit. The harder it's hit, the farther it will roll. It isn't affected by gravity. You can program it to show any number on its surface.

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	2400	mass/density (relative scale)
<code>damping</code>	40	drag/air-resistance or density
<code>cor_q8</code>	224	COR = 0.875

Property	Raw Value	Interpretation
radius	1	collision radius in internal units
sprite_w	240	sprite width
sprite_h	240	sprite height
data_offset	23	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	24×23	0
3	1	1	24×23	7
4	2	2	24×23	14
5	3	3	24×23	21
6	4	4	24×23	28
7	5	5	24×23	35
8	6	6	24×23	42
9	7	7	24×23	49
10	8	8	24×23	56
11	9	9	24×23	63
12	10	10	24×23	70
13	11	11	24×23	77
14	12	12	24×23	84
15	13	13	24×23	91
16	14	14	24×23	98
17	15	15	24×23	105

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART63_f000.png :material-check:
1	0001_0002	build/phase-3/bitmaps/PART63_f002.png :material-check:
2	0001_0003	build/phase-3/bitmaps/PART63_f003.png :material-check:
3	0001_0004	build/phase-3/bitmaps/PART63_f004.png :material-check:
4	0001_0005	build/phase-3/bitmaps/PART63_f005.png :material-check:
5	0001_0006	build/phase-3/bitmaps/PART63_f006.png :material-check:
6	0001_0007	build/phase-3/bitmaps/PART63_f007.png :material-check:
7	0001_0008	build/phase-3/bitmaps/PART63_f008.png :material-check:
8	0001_0009	build/phase-3/bitmaps/PART63_f009.png :material-check:
9	0001_0010	build/phase-3/bitmaps/PART63_f010.png :material-check:
10	0001_0011	build/phase-3/bitmaps/PART63_f011.png :material-check:
11	0001_0012	build/phase-3/bitmaps/PART63_f012.png :material-check:
12	0001_0013	build/phase-3/bitmaps/PART63_f013.png :material-check:
13	0001_0014	build/phase-3/bitmaps/PART63_f014.png :material-check:
14	0001_0015	build/phase-3/bitmaps/PART63_f015.png :material-check:
15	0001_0001	build/phase-3/bitmaps/PART63_f001.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	16
END_FRAME	16
END_ANM	1

Notas de Comportamento:

Pool Ball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-063-pool-ball.md` describes part type `63`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 63`

TC-002: Mass value is positive

- **Given** Pool Ball has `mass = 2400`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Pool Ball has `cor_q8 = 224`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (224 >> 8) <= 1.0` i.e. `COR = 0.875`

TC-004: Radius is non-negative

- **Given** Pool Ball has `radius = 1`
- **When** physics engine initializes collision
- **Then** `radius >= 0`

TC-005: No gravity behavior

- **Given** Description: "It isn't affected by gravity"
- **When** simulation runs
- **Then** pool ball maintains constant vertical position unless acted upon

TC-006: Programmable number display

- **Given** Description: "You can program it to show any number on its surface"
- **When** player sets ball number
- **Then** ANM sprite reflects programmed number

TC-007: Only moves when hit

- **Given** Description: "This pool ball won't move until it's hit"
- **When** collision transfers momentum
- **Then** pool ball velocity changes proportional to impact

TC-008: Part used in 21 levels

- **Given** Pool Ball (type 63)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 21 levels reference `part_num = 63`

TC-009: Category Part Count is 9 (Balls category)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Balls category

TC-010: Triggers are collision and trap

- **Given** Category Rules show `Triggers: collision, trap`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-011: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-012: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** pool ball physics are automatic

TC-013: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** pool ball is not destroyed

TC-014: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`

- **Then** only default connections

TC-015: Sprite dimensions 240×240

- **Given** Physics Properties show `sprite_w = 240, sprite_h = 240`
- **When** sprite is loaded
- **Then** sprite dimensions match

Peça 64 — Pinball Bumper



Categoria: Special Mechanics
Fases em que aparece: 20 fases
Quantidade de Estados: 2 estado(s)

This pinball bumper can be placed so that moving objects will bounce off in any direction.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	2400	mass/density (relative scale)
<code>unk_2</code>	40	category-specific property
<code>property_q8</code>	256	Q8.8 = 1.000
<code>unk_6</code>	16	category-specific property
<code>dim_w1</code>	240	dimension 1 width
<code>dim_h1</code>	240	dimension 1 height
<code>dim_w2</code>	0	dimension 2 width
<code>dim_h2</code>	0	dimension 2 height
<code>bitmap_w</code>	41	bitmap frame width
<code>bitmap_h</code>	41	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	<code>collision, electrical, proximity</code>
Destructible	No
Spawns	<code>custom</code>
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No

Capability	Supported
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
4	0	0	48×41	0
3	1	1–2 (2 frames)	48×41	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001	build/phase-3/bitmaps/PART64_f001.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART64_f000.png :material-check:
2	0001_0000	build/phase-3/bitmaps/PART64_f000.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3064	0	build/phase-6/raw-wav/SX_3064_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	3
END_FRAME	3
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Pinball Bumper is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has custom spawn behavior defined in SOLVE.RES.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-064-pinball-bumper.md` describes part type `64`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 64`

TC-002: Mass value is positive

- **Given** Pinball Bumper has `mass = 2400`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Pinball Bumper has `property_q8 = 256`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (256 >> 8) <= 1.0` i.e. `COR = 1.000`

TC-004: Bounce redirection mechanic

- **Given** Description: "This pinball bumper can be placed so that moving objects will bounce off in any direction"
- **When** object contacts bumper
- **Then** object bounces in direction determined by bumper orientation

TC-005: COR = 1.0 (perfect elastic collision)

- **Given** Pinball Bumper COR = 1.000
- **When** collision occurs
- **Then** no energy is lost (perfect bounce)

TC-006: Part used in 20 levels

- **Given** Pinball Bumper (type 64)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** at least 20 levels reference `part_num = 64`

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** bumper is passive collision device

TC-009: Special Mechanics category

- **Given** Category shows `Special Mechanics`
- **When** behavior handler evaluates
- **Then** bounce redirection is applied

TC-010: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** bumper is not destroyed

TC-011: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-012: Category Part Count is 15

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-013: Flip changes bounce direction

- **Given** Bumper can be flipped
- **When** player flips bumper

- **Then** bounce direction changes accordingly

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: Collision response redirects momentum

- **Given** Object contacts bumper at angle θ
 - **When** physics engine resolves collision
 - **Then** object velocity is redirected while magnitude is preserved (COR=1.0)
-

Peça 65 — Leprechaun



Categoria: Characters
Fases em que aparece: 0 fases
Quantidade de Estados: 3 estado(s)

Happy Saint Patrick's Day! Say hi to Blarney O'Reilly, the leprechaun who lives in Professor Tim's garden.
Give him a nudge and he'll dance a jig for you.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Resting	1→2	—	No
Dancing	2→-1	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
unk_2	800	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	64	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	39	bitmap frame width
bitmap_h	62	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	<code>collision, proximity</code>
Destructible	No

Rule	Value
Spawns	(none)
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	40×62	0
2	1	1–7 (7 frames)	40×62	13
3	8	8–17 (10 frames)	48×65	118

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0001_0000	build/phase-3/bitmaps/PART65_f001.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
1	0001_0001, 0001_0000	build/phase-3/bitmaps/PART65_f001.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
2	0001_0001, 0001_0000	build/phase-3/bitmaps/PART65_f001.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
3	0001_0002, 0001_0000	build/phase-3/bitmaps/PART65_f002.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
4	0001_0003, 0001_0000	build/phase-3/bitmaps/PART65_f003.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
5	0001_0004, 0001_0000	build/phase-3/bitmaps/PART65_f004.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
6	0001_0005, 0001_0000	build/phase-3/bitmaps/PART65_f005.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
7	0001_0004, 0001_0000	build/phase-3/bitmaps/PART65_f004.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
8	0001_0003, 0001_0000	build/phase-3/bitmaps/PART65_f003.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
9	0001_0002, 0001_0000	build/phase-3/bitmaps/PART65_f002.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
10	0001_0001, 0001_0000	build/phase-3/bitmaps/PART65_f001.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
11	0001_0002, 0001_0000	build/phase-3/bitmaps/PART65_f002.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:

Frame	Sprites	PNG References
12	0001_0003, 0001_0000	build/phase-3/bitmaps/PART65_f003.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
13	0001_0004, 0001_0000	build/phase-3/bitmaps/PART65_f004.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
14	0001_0005, 0001_0000	build/phase-3/bitmaps/PART65_f005.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
15	0001_0004, 0001_0000	build/phase-3/bitmaps/PART65_f004.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
16	0001_0003, 0001_0000	build/phase-3/bitmaps/PART65_f003.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:
17	0001_0002, 0001_0000	build/phase-3/bitmaps/PART65_f002.png :material-check: build/phase-3/bitmaps/PART65_f000.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	65	0	build/phase-6/raw-wav/SX_65_11025.wav
2	65	0	build/phase-6/raw-wav/SX_65_11025.wav
3	65	0	build/phase-6/raw-wav/SX_65_11025.wav
4	65	0	build/phase-6/raw-wav/SX_65_11025.wav
5	65	0	build/phase-6/raw-wav/SX_65_11025.wav
6	65	0	build/phase-6/raw-wav/SX_65_11025.wav
7	65	0	build/phase-6/raw-wav/SX_65_11025.wav
8	65	0	build/phase-6/raw-wav/SX_65_11025.wav
9	65	0	build/phase-6/raw-wav/SX_65_11025.wav
10	65	0	build/phase-6/raw-wav/SX_65_11025.wav
11	65	0	build/phase-6/raw-wav/SX_65_11025.wav
12	65	0	build/phase-6/raw-wav/SX_65_11025.wav
13	65	0	build/phase-6/raw-wav/SX_65_11025.wav
14	65	0	build/phase-6/raw-wav/SX_65_11025.wav
15	65	0	build/phase-6/raw-wav/SX_65_11025.wav
16	65	0	build/phase-6/raw-wav/SX_65_11025.wav
17	65	0	build/phase-6/raw-wav/SX_65_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	36
END_FRAME	18
PLAY_SFX	17
END_ANM	1

Notas de Comportamento:

Leprechaun has 2 semantic states: Resting, Dancing. Transitions are triggered by: collision, proximity.

Terminal state(s): Dancing (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-065-leprechaun.md` describes part type `65`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 65`

TC-002: Mass value is positive

- **Given** Leprechaun has `mass = 2000`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Leprechaun has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: SOLVE.RES state transitions (Resting → Dancing)

- **Given** Leprechaun has states: Resting (`1→2`) and Dancing (`2→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-005: Not used in any built-in level

- **Given** Leprechaun is a Character type
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no built-in level references `part_num = 65`

TC-006: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-007: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-008: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** leprechaun dances when nudged

TC-009: Character category (Category Part Count = 8)

- **Given** Category Rules show `Category Part Count: 8`
- **When** validated
- **Then** 8 parts in Characters category

TC-010: Destructible flag is No

- **Given** Category Rules show `Destructible: No`

- **When** explosive part collides
- **Then** leprechaun is not destroyed

TC-011: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-012: Dancing jig on nudge

- **Given** Description: "Give him a nudge and he'll dance a jig for you"
- **When** collision or proximity trigger fires
- **Then** leprechaun enters Dancing state

TC-013: Terminal state Dancing (part removed)

- **Given** State Machine shows `Dancing: 2→-1`
- **When** state reaches Dancing
- **Then** part is removed from simulation (terminal)

TC-014: Bitmap dimensions 39×62

- **Given** Physics Properties show `bitmap_w = 39, bitmap_h = 62`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Saint Patrick's Day themed character

- **Given** Description: "Happy Saint Patrick's Day! Say hi to Blarney O'Reilly, the leprechaun who lives in Professor Tim's garden"
- **When** character activates
- **Then** dance animation plays

Peça 66 — Mouse Hole



Categoria: Special Mechanics

Fases em que aparece: 6 fases

Quantidade de Estados: 3 estado(s)

Newton Mouse likes to hide in this mouse hole when he's chased by Curie Cat. Or you can lure him inside by placing some cheese on the far side of the hole.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Vacant	<code>1→3</code>	1	No
Occupied	<code>3→-1</code>	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>unk_2</code>	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
<code>bitmap_w</code>	19	bitmap frame width

Property	Raw Value	Interpretation
bitmap_h	21	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24x21	0
3	1	1–5 (5 frames)	24x21	7
2	6	6	24x21	74

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0004	build/phase-3/bitmaps/PART66_f004.png :material-check:
1	0001_0004, 0001_0000	build/phase-3/bitmaps/PART66_f004.png :material-check: build/phase-3/bitmaps/PART66_f000.png :material-check:
2	0001_0004, 0001_0000	build/phase-3/bitmaps/PART66_f004.png :material-check: build/phase-3/bitmaps/PART66_f000.png :material-check:
3	0001_0004, 0001_0001	build/phase-3/bitmaps/PART66_f004.png :material-check: build/phase-3/bitmaps/PART66_f001.png :material-check:
4	0001_0004, 0001_0002	build/phase-3/bitmaps/PART66_f004.png :material-check: build/phase-3/bitmaps/PART66_f002.png :material-check:
5	0001_0004, 0001_0003	build/phase-3/bitmaps/PART66_f004.png :material-check: build/phase-3/bitmaps/PART66_f003.png :material-check:
6	0001_0004, 0001_0003	build/phase-3/bitmaps/PART66_f004.png :material-check: build/phase-3/bitmaps/PART66_f003.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3066	0	build/phase-6/raw-wav/SX_3066_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	13
END_FRAME	7
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Mouse Hole has 2 semantic states: Vacant, Occupied. Transitions are triggered by: collision, electrical, proximity.

Terminal state(s): Occupied (part is removed from simulation).

This part has custom spawn behavior defined in SOLVE.RES.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-066-mouse-hole.md` describes part type `66`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 66`

TC-002: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-003: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** mouse hole responds to Newton Mouse proximity

TC-004: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-005: SOLVE.RES state transitions (Vacant → Occupied)

- **Given** Mouse Hole has states: Vacant (`1→3`) and Occupied (`3→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-006: Terminal state Occupied (part removed)

- **Given** State Machine shows `Occupied: 3→-1`
- **When** state reaches Occupied
- **Then** part is removed from simulation (terminal)

TC-007: Triggers are collision, electrical, proximity

- **Given** Category Rules show `Triggers: collision, electrical, proximity`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-008: Part used in 6 levels

- **Given** Mouse Hole (type 66)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 6 levels reference `part_num = 66`

TC-009: Custom spawn behavior

- **Given** Category Rules show `Spawns: custom`
- **When** behavior handler processes
- **Then** custom spawn logic is applied

TC-010: Special Mechanics category (Category Part Count = 15)

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** mouse hole is not destroyed

TC-012: Bitmap dimensions 19x21

- **Given** Physics Properties show `bitmap_w = 19, bitmap_h = 21`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-013: 7 total animation frames across 3 states

- **Given** ANM shows `Total Frames: 7, Total States: 3`
- **When** animation renderer processes
- **Then** all 7 frames map to correct states

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: Newton Mouse hides inside when chased

- **Given** Description: "Newton Mouse likes to hide in this mouse hole when he's chased by Curie Cat"
- **When** mouse is chased by cat and reaches hole
- **Then** mouse enters Occupied state and is hidden

Peça 67 — Can Opener



Categoria: Electrical

Fases em que aparece: 11 fases

Quantidade de Estados: 3 estado(s)

Plug this can opener into an electrical outlet to make it open the can. When the sauce spills out of the can, Curie Cat will come lap it up if she's within range.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Unopened	1→3	1	No
Opened	2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	37	bitmap frame width
bitmap_h	51	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	43×51	0
3	1	1–24 (24 frames)	43×49	31
2	25	25	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001, 0001_0001, 0001_0004, 0001_0002	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f004.png :material-check: build/phase-3/bitmaps/ PART67_f002.png :material-check:
1	0001_0001, 0001_0001, 0001_0000, 0001_0005, 0001_0003	build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f005.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check:
2	0001_0000, 0001_0001, 0001_0001, 0001_0006, 0001_0003	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f006.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check:
3	0001_0001, 0001_0001, 0001_0000, 0001_0003, 0001_0007	build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check: build/phase-3/bitmaps/ PART67_f007.png :material-check:
4	0001_0000, 0001_0001, 0001_0001, 0001_0003, 0001_0008	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check: build/phase-3/bitmaps/ PART67_f008.png :material-check:
5	0001_0001, 0001_0001, 0001_0000, 0001_0009, 0001_0003	build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART67_f009.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check:
6	0001_0000, 0001_0001, 0001_0001, 0001_0004, 0001_0003	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f004.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check:
7	0001_0001, 0001_0001, 0001_0000, 0001_0005, 0001_0003	build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f005.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check:
8	0001_0000, 0001_0001, 0001_0001, 0001_0006, 0001_0003	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f006.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check:
9	0001_0001, 0001_0001, 0001_0000, 0001_0003, 0001_0007	build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check: build/phase-3/bitmaps/ PART67_f007.png :material-check:
10	0001_0000, 0001_0001, 0001_0001, 0001_0003, 0001_0008	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check: build/phase-3/bitmaps/ PART67_f008.png :material-check:
11	0001_0001, 0001_0001, 0001_0000, 0001_0009, 0001_0003	build/phase-3/bitmaps/ PART67_f001.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f009.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check:
12	0001_0000, 0001_0001, 0001_0001, 0001_0004, 0001_0003	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f004.png :material-check: build/phase-3/bitmaps/ PART67_f003.png :material-check:
13	0001_0000, 0001_0001, 0001_0001, 0001_0002, 0001_0010	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f002.png :material-check: build/phase-3/bitmaps/ PART67_f010.png :material-check:
14	0001_0000, 0001_0001, 0001_0001, 0001_0002, 0001_0011	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f002.png :material-check: build/phase-3/bitmaps/ PART67_f011.png :material-check:
15	0001_0000, 0001_0001, 0001_0001, 0001_0002, 0001_0012, 0001_0015	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f002.png :material-check: build/phase-3/bitmaps/ PART67_f012.png :material-check: build/phase-3/bitmaps/ PART67_f015.png :material-check:
16	0001_0000, 0001_0001, 0001_0001, 0001_0002, 0001_0012, 0001_0015	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART67_f002.png :material-check: build/phase-3/bitmaps/ PART67_f012.png :material-check: build/phase-3/bitmaps/ PART67_f015.png :material-check:
17	0001_0000, 0001_0001, 0001_0001, 0001_0002, 0001_0012, 0001_0016	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f002.png :material-check: build/phase-3/bitmaps/ PART67_f012.png :material-check: build/phase-3/bitmaps/ PART67_f016.png :material-check:
18	0001_0000, 0001_0001, 0001_0001, 0001_0002, 0001_0012, 0001_0017	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f002.png :material-check: build/phase-3/bitmaps/ PART67_f012.png :material-check: build/phase-3/bitmaps/ PART67_f017.png :material-check:
19	0001_0000, 0001_0001, 0001_0001, 0001_0002, 0001_0012, 0001_0018	build/phase-3/bitmaps/ PART67_f000.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f001.png :material-check: build/phase-3/bitmaps/ PART67_f002.png :material-check: build/phase-3/bitmaps/ PART67_f012.png :material-check: build/phase-3/bitmaps/ PART67_f018.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	67	0	build/phase-6/raw-wav/SX_67_11025.wav
2	67	0	build/phase-6/raw-wav/SX_67_11025.wav
3	67	0	build/phase-6/raw-wav/SX_67_11025.wav
4	67	0	build/phase-6/raw-wav/SX_67_11025.wav
5	67	0	build/phase-6/raw-wav/SX_67_11025.wav
6	67	0	build/phase-6/raw-wav/SX_67_11025.wav
7	67	0	build/phase-6/raw-wav/SX_67_11025.wav
8	67	0	build/phase-6/raw-wav/SX_67_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
9	67	0	build/phase-6/raw-wav/SX_67_11025.wav
10	67	0	build/phase-6/raw-wav/SX_67_11025.wav
11	67	0	build/phase-6/raw-wav/SX_67_11025.wav
12	67	0	build/phase-6/raw-wav/SX_67_11025.wav
15	3067	0	build/phase-6/raw-wav/SX_3067_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	105
END_FRAME	20
PLAY_SFX	13
END_ANM	1

Notas de Comportamento:

Can Opener has 2 semantic states: Unopened, Opened. Transitions are triggered by: electrical, collision.

Terminal state(s): Opened (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-067-can-opener.md` describes part type `67`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 67`

TC-002: Mass value is positive

- **Given** Can Opener has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Can Opener has `property_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 4.000`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** can opener responds to electrical activation

TC-006: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-007: SOLVE.RES state transitions (Unopened → Opened)

- **Given** Can Opener has states: Unopened (`1→3`) and Opened (`2→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-008: Terminal state Opened (part removed)

- **Given** State Machine shows `Opened: 2→-1`
- **When** state reaches Opened
- **Then** part is removed from simulation (terminal)

TC-009: Triggers are electrical and collision

- **Given** Category Rules show `Triggers: electrical, collision`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Part used in 11 levels

- **Given** Can Opener (type 67)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 11 levels reference `part_num = 67`

TC-011: Electrical category (Category Part Count = 9)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Electrical category

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** can opener is not destroyed

TC-013: Animated (Section A) — 36 duration entries

- **Given** ANM shows `Animated (Section A): Yes, Section A Durations: 36 entries`
- **When** animation renderer processes frame timing
- **Then** Section A durations control playback at 60 Hz

TC-014: Bitmap dimensions 37×51

- **Given** Physics Properties show `bitmap_w = 37, bitmap_h = 51`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Curie Cat attracted by spilled sauce

- **Given** Description: "When the sauce spills out of the can, Curie Cat will come lap it up if she's within range"
 - **When** can opener opens and sauce spills
 - **Then** Curie Cat pathfinds toward sauce
-

Peça 68 — Soccer Ball



Categoria: Balls
Fases em que aparece: 20 fases
Quantidade de Estados: 1 estado(s)

This soccer ball is medium in weight and quite bouncy.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1522	mass/density (relative scale)
damping	28	drag/air-resistance or density
cor_q8	160	COR = 0.625
radius	24	collision radius in internal units
sprite_w	0	sprite width
sprite_h	0	sprite height
data_offset	32	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–7 (8 frames)	32×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART68_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART68_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART68_f002.png :material-check:

Frame	Sprites	PNG References
3	0001_0003	build/phase-3/bitmaps/PART68_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART68_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART68_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART68_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART68_f007.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	8
END_FRAME	8
END_ANM	1

Notas de Comportamento:

Soccer Ball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-068-soccer-ball.md` describes part type `68`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 68`

TC-002: Mass value is positive

- **Given** Soccer Ball has `mass = 1522`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Soccer Ball has `cor_q8 = 160`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (160 >> 8) <= 1.0` i.e. `COR = 0.625`

TC-004: Radius is non-negative

- **Given** Soccer Ball has `radius = 24`
- **When** physics engine initializes collision
- **Then** `radius >= 0`

TC-005: Damping value is non-negative

- **Given** Soccer Ball has `damping = 28`
- **When** physics engine applies drag
- **Then** `damping >= 0`

TC-006: Part used in 20 levels

- **Given** Soccer Ball (type 68)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 20 levels reference `part_num = 68`

TC-007: Triggers are collision and trap

- **Given** Category Rules show `Triggers: collision, trap`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-008: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-009: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** soccer ball physics are automatic

TC-010: Balls category (Category Part Count = 9)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Balls category

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** soccer ball is not destroyed

TC-012: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-013: Single default state (no SOLVE.RES transitions)

- **Given** State Machine shows *Single default state — no named transitions*
- **When** behavior handler evaluates
- **Then** no state transitions are triggered

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: 8 animation frames (rotating appearance)

- **Given** ANM shows `Total Frames: 8, Total States: 1`
- **When** animation renderer processes
- **Then** all 8 frames render in sequence for rolling effect

Peça 69 — Anti-Gravity Pad



Categoria: Special Mechanics

Fases em que aparece: 76 fases

Quantidade de Estados: 2 estado(s)

This anti-gravity pad reverses the gravity field for anything on top of it. Without gravity, most things will float up into the air. But balloons will drop.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Unactivated	1→2	1	No
Operating	2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	32	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	48	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	48×16	0
2	1	1–4 (4 frames)	48×16	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART69_f000.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART69_f000.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART69_f001.png :material-check:
3	0001_0002	build/phase-3/bitmaps/PART69_f002.png :material-check:
4	0001_0003	build/phase-3/bitmaps/PART69_f003.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	69	0	build/phase-6/raw-wav/SX_69_11025.wav
2	69	0	build/phase-6/raw-wav/SX_69_11025.wav
3	69	0	build/phase-6/raw-wav/SX_69_11025.wav
4	69	0	build/phase-6/raw-wav/SX_69_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	5
END_FRAME	5
PLAY_SFX	4
END_ANM	1

Notas de Comportamento:

Anti-Gravity Pad has 2 semantic states: Unactivated, Operating. Transitions are triggered by: collision, electrical, proximity.

Terminal state(s): Operating (part is removed from simulation).

This part has custom spawn behavior defined in SOLVE.RES.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-069-anti-gravity-pad.md` describes part type `69`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 69`

TC-002: Mass value is positive

- **Given** Anti-Gravity Pad has `mass = 3776`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Anti-Gravity Pad has `property_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 4.000`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** anti-gravity pad responds to triggers

TC-006: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-007: SOLVE.RES state transitions (Unactivated → Operating)

- **Given** Anti-Gravity Pad has states: Unactivated (`1→2`) and Operating (`2→1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-008: Terminal state Operating (part removed)

- **Given** State Machine shows `Operating: 2→1`
- **When** state reaches Operating
- **Then** part is removed from simulation (terminal)

TC-009: Triggers are collision, electrical, proximity

- **Given** Category Rules show `Triggers: collision, electrical, proximity`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-010: Part used in 76 levels (highest usage)

- **Given** Anti-Gravity Pad (type 69)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 76 levels reference `part_num = 69`

TC-011: Custom spawn behavior

- **Given** Category Rules show `Spawns: custom`
- **When** behavior handler processes
- **Then** custom spawn logic is applied

TC-012: Special Mechanics category (Category Part Count = 15)

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-013: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** anti-gravity pad is not destroyed

TC-014: Bitmap dimensions 48×16

- **Given** Physics Properties show `bitmap_w = 48, bitmap_h = 16`

- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Gravity reversal affects balloons inversely

- **Given** Description: "Without gravity, most things will float up into the air. But balloons will drop"
- **When** object lands on anti-gravity pad
- **Then** gravity field is reversed for normal objects but inverted for balloons

Peça 70 — Missile



Categoria: Explosives / Projectiles
Fases em que aparece: 23 fases
Quantidade de Estados: 15 estado(s)

This missile is handy for blowing up all sorts of things, including most walls. Light the fuse with a laser, a phazer, a candle (or other flaming part), or by using a magnifying glass and light source. You can flip it to fly straight up, right, or left.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Launched	2→8 , 13→14 , 18→19	18	No
Launched	3→12 , 16→17 , 21→22	19	No
Exploded	12→13 , 17→18 , 22→-1	—	Yes :material-bomb:

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	22000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	32	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	96	bitmap frame width
bitmap_h	96	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile

Rule	Value
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	32×95	0
8	1	1–27 (27 frames)	64×120	13
4	28	28–31 (4 frames)	8×2	506
3	32	32–37 (6 frames)	124×49	566
12	38	38	114×49	680
13	39	39	115×50	699
14	40	40–55 (16 frames)	109×62	718
15	56	56–63 (8 frames)	109×25	1018
16	64	64–69 (6 frames)	121×49	1130
17	70	70	116×49	1244
18	71	71	119×50	1263
19	72	72–87 (16 frames)	116×62	1282
20	88	88–95 (8 frames)	109×25	1576
21	96	96–101 (6 frames)	0×0	0
22	102	102	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0002_0000, 0001_0000	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f000.png :material-check:
1	0002_0000, 0001_0000, 0001_0005	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f000.png :material-check: build/phase-3/bitmaps/PART70_f005.png :material-check:
2		

Frame	Sprites	PNG References
	0002_0000, 0001_0000, 0001_0005	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f000.png :material-check: build/phase-3/bitmaps/PART70_f005.png :material-check:
3	0002_0000, 0001_0000, 0001_0006	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f000.png :material-check: build/phase-3/bitmaps/PART70_f006.png :material-alert-outline: (not found)
4	0002_0000, 0001_0001, 0001_0007	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f001.png :material-check: build/phase-3/bitmaps/PART70_f007.png :material-alert-outline: (not found)
5	0002_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f001.png :material-check: build/phase-3/bitmaps/PART70_f006.png :material-alert-outline: (not found)
6	0002_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f001.png :material-check: build/phase-3/bitmaps/PART70_f006.png :material-alert-outline: (not found)
7	0002_0000, 0001_0002, 0001_0007	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f002.png :material-check: build/phase-3/bitmaps/PART70_f007.png :material-alert-outline: (not found)
8	0002_0000, 0001_0002, 0001_0007	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f002.png :material-check: build/phase-3/bitmaps/PART70_f007.png :material-alert-outline: (not found)
9	0002_0000, 0001_0002, 0001_0006	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART70_f006.png :material-alert-outline: (not found)
10	0002_0000, 0001_0003, 0001_0007	build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f003.png :material-check: build/phase-3/bitmaps/PART70_f007.png :material-alert-outline: (not found)
11	0002_0000, 0001_0003, 0001_0007	build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f003.png :material-check: build/phase-3/bitmaps/PART70_f007.png :material-alert-outline: (not found)
12	0002_0000, 0001_0003, 0001_0006	build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f003.png :material-check: build/phase-3/bitmaps/PART70_f006.png :material-alert-outline: (not found)
13	0002_0000, 0001_0003, 0001_0007	build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f003.png :material-check: build/phase-3/bitmaps/PART70_f007.png :material-alert-outline: (not found)
14	0002_0000, 0001_0004, 0001_0007	build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f004.png :material-check: build/phase-3/bitmaps/PART70_f007.png :material-alert-outline: (not found)
15	0002_0000, 0001_0004, 0001_0006, 0002_0000	build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f004.png :material-check: build/phase-3/bitmaps/PART70_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found)
16	0002_0000	build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found)
17	0002_0001	build/phase-3/bitmaps/PART70_2_f001.png :material-alert-outline: (not found)
18	0001_0008, 0001_0012, 0002_0002	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART70_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f012.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f002.png :material-alert-outline: (not found)
19	0001_0008, 0001_0012, 0002_0002	build/phase-3/bitmaps/PART70_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f012.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f002.png :material-alert-outline: (not found)
20	0001_0009, 0001_0013, 0001_0014, 0001_0015, 0002_0000	build/phase-3/bitmaps/PART70_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found)
21	0001_0014, 0001_0017, 0001_0015, 0001_0016, 0001_0011, 0002_0000	build/phase-3/bitmaps/PART70_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f017.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f016.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found)
22	0001_0010, 0002_0000	build/phase-3/bitmaps/PART70_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found)
23	0002_0000, 0001_0009	build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f009.png :material-alert-outline: (not found)
24	0001_0008, 0002_0000	build/phase-3/bitmaps/PART70_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f000.png :material-alert-outline: (not found)
25	0001_0009, 0002_0000	build/phase-3/bitmaps/PART70_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART70_2_f000.png :material-alert-outline: (not found)
26	0001_0010, 0002_0000	build/phase-3/bitmaps/PART70_f010.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found)
27	0001_0011, 0002_0000	build/phase-3/bitmaps/PART70_f011.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_2_f000.png :material-alert-outline: (not found)
28	0003_0000	build/phase-3/bitmaps/PART70_3_f000.png :material-alert-outline: (not found)
29	0002_0003, 0001_0018	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found)
30	0002_0003, 0001_0018, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
31	0002_0003, 0001_0018, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
32	0002_0003, 0001_0018, 0001_0024	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
33	0002_0003, 0001_0018, 0001_0025	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
34	0002_0003, 0001_0019, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f019.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
35	0002_0003, 0001_0019, 0001_0024	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
36	0002_0003, 0001_0019, 0001_0025	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
37	0002_0003, 0001_0020, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
38	0002_0003, 0001_0020, 0001_0024	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
39	0002_0003, 0001_0020, 0001_0025	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
40	0002_0003, 0001_0021, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
41	0002_0003, 0001_0021, 0001_0024	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
42	0002_0003, 0001_0021, 0001_0025	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f021.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
43	0002_0003, 0001_0022, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
44	0002_0003, 0001_0022, 0001_0024	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
45	0002_0003, 0001_0022, 0001_0025	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
46	0002_0003	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found)
47	0002_0004	build/phase-3/bitmaps/PART70_2_f004.png :material-alert-outline: (not found)
48	0002_0005, 0001_0027, 0001_0030	build/phase-3/bitmaps/PART70_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f030.png :material-alert-outline: (not found)
49	0002_0005, 0001_0027, 0001_0030	build/phase-3/bitmaps/PART70_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f030.png :material-alert-outline: (not found)
50	0001_0026, 0002_0005, 0001_0031, 0001_0032, 0001_0034	build/phase-3/bitmaps/PART70_f026.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f031.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f032.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f034.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
51	0001_0027, 0002_0005, 0001_0035, 0001_0033, 0001_0034, 0001_0033, 0001_0034, 0001_0034	build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f035.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f033.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f034.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f033.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f034.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f034.png :material-alert-outline: (not found)
52	0001_0028, 0002_0003	build/phase-3/bitmaps/PART70_f028.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found)
53	0001_0029, 0002_0003	build/phase-3/bitmaps/PART70_f029.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found)
54	0001_0028, 0002_0003	build/phase-3/bitmaps/PART70_f028.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found)
55	0001_0027, 0002_0003	build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found)
56	0001_0026, 0002_0003	build/phase-3/bitmaps/PART70_f026.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found)
57	0001_0027, 0002_0003	build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found)
58	0002_0003, 0001_0028	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f028.png :material-alert-outline: (not found)
59	0002_0003, 0001_0029	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART70_f029.png :material-alert-outline: (not found)
60	0003_0000	build/phase-3/bitmaps/PART70_3_f000.png :material-alert-outline: (not found)
61	0002_0003, 0001_0018	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found)
62	0002_0003, 0001_0018, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
63	0002_0003, 0001_0018, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
64	0002_0003, 0001_0018, 0001_0024	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
65	0002_0003, 0001_0018, 0001_0025	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f018.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
66	0002_0003, 0001_0019, 0001_0023	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
67	0002_0003, 0001_0019, 0001_0024	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
68	0002_0003, 0001_0019, 0001_0025	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		found) build/phase-3/bitmaps/PART70_f019.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
69	0002_0003, 0001_0020, 0001_0023	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
70	0002_0003, 0001_0020, 0001_0024	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
71	0002_0003, 0001_0020, 0001_0025	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f020.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
72	0002_0003, 0001_0021, 0001_0023	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
73	0002_0003, 0001_0021, 0001_0024	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
74	0002_0003, 0001_0021, 0001_0025	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f021.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
75	0002_0003, 0001_0022, 0001_0023	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f023.png :material-alert-outline: (not found)
76	0002_0003, 0001_0022, 0001_0024	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f024.png :material-alert-outline: (not found)
77	0002_0003, 0001_0022, 0001_0025	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f022.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f025.png :material-alert-outline: (not found)
78	0002_0003	build/phase-3/bitmaps/ PART70_2_f003.png :material-alert-outline: (not found)
79	0002_0004	build/phase-3/bitmaps/ PART70_2_f004.png :material-alert-outline: (not found)
80	0002_0005, 0001_0027, 0001_0030	build/phase-3/bitmaps/ PART70_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f030.png :material-alert-outline: (not found)
81	0002_0005, 0001_0027, 0001_0030	build/phase-3/bitmaps/ PART70_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f030.png :material-alert-outline: (not found)
82	0002_0005, 0001_0026, 0001_0031, 0001_0032, 0001_0034	build/phase-3/bitmaps/ PART70_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f026.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f031.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f032.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f034.png :material-alert-outline: (not found)
83	0002_0005, 0001_0027, 0001_0033, 0001_0033, 0001_0034, 0001_0035, 0001_0034	build/phase-3/bitmaps/ PART70_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f033.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f033.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f034.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART70_f035.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f034.png :material-alert-outline: (not found)
84	0002_0003, 0001_0028	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f028.png :material-alert-outline: (not found)
85	0002_0003, 0001_0029	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f029.png :material-alert-outline: (not found)
86	0002_0003, 0001_0028	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f028.png :material-alert-outline: (not found)
87	0002_0003, 0001_0027	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found)
88	0002_0003, 0001_0026	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f026.png :material-alert-outline: (not found)
89	0002_0003, 0001_0027	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f027.png :material-alert-outline: (not found)
90	0002_0003, 0001_0028	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f028.png :material-alert-outline: (not found)
91	0002_0003, 0001_0029	build/phase-3/bitmaps/PART70_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART70_f029.png :material-alert-outline: (not found)
92	0003_0000	build/phase-3/bitmaps/PART70_3_f000.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
18	3070	0	build/phase-6/raw-wav/SX_3070_11025.wav

Frame	Sound ID	Bytecode Offset	WAV Reference
30	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
48	3070	0	build/phase-6/raw-wav/SX_3070_11025.wav
62	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
80	3070	0	build/phase-6/raw-wav/SX_3070_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	255
END_FRAME	93
PLAY_SFX	6
END_ANM	1

Notas de Comportamento:

Missile has 3 semantic states: Not Launched, Launched, Exploded. Transitions are triggered by: collision, proximity, timer.

State(s) Exploded are explosive — reaching these destroys the part.

Terminal state(s): Exploded (part is removed from simulation).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 71 — Boxes



Categoria: Containers
Fases em que aparece: 43 fases
Quantidade de Estados: 9 estado(s)

This box is really five boxes in one. Program it to be glass, wooden, wicker, metal, or cardboard, which are all different sizes. Drop things inside the box, or use it to catch falling objects.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	1000	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	48	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	50×32	0
3	1	1	64×32	13
4	2	2	48×48	26
5	3	3	64×48	39
6	4	4	80×64	52
7	5	5	48×32	59
8	6	6	64×32	66
9	7	7	48×48	73
10	8	8	64×48	80

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0005, 0001_0000	build/phase-3/bitmaps/PART71_f005.png :material-check: build/phase-3/bitmaps/PART71_f000.png :material-check:
1	0001_0001, 0001_0006	build/phase-3/bitmaps/PART71_f001.png :material-check: build/phase-3/bitmaps/PART71_f006.png :material-check:
2	0001_0007, 0001_0002	build/phase-3/bitmaps/PART71_f007.png :material-check: build/phase-3/bitmaps/PART71_f002.png :material-check:
3	0001_0008, 0001_0003	build/phase-3/bitmaps/PART71_f008.png :material-check: build/phase-3/bitmaps/PART71_f003.png :material-check:

Frame	Sprites	PNG References
4	0001_0004	build/phase-3/bitmaps/PART71_f004.png :material-check:
5	0001_0000	build/phase-3/bitmaps/PART71_f000.png :material-check:
6	0001_0001	build/phase-3/bitmaps/PART71_f001.png :material-check:
7	0001_0002	build/phase-3/bitmaps/PART71_f002.png :material-check:
8	0001_0003	build/phase-3/bitmaps/PART71_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	13
END_FRAME	9
END_ANM	1

Notas de Comportamento:

Boxes is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-071-boxes.md` describes part type `71`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 71`

TC-002: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-003: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** boxes are static containers

TC-004: Container category (Category Part Count = 5)

- **Given** Category Rules show `Category Part Count: 5`
- **When** validated
- **Then** 5 parts in Containers category

TC-005: Triggers are collision

- **Given** Category Rules show `Triggers: collision`
- **When** behavior handler evaluates
- **Then** collision trigger is registered

TC-006: Part used in 43 levels

- **Given** Boxes (type 71)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 43 levels reference `part_num = 71`

TC-007: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** boxes are not destroyed

TC-008: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-009: 9 states (5 box types + 4 material variants)

- **Given** ANM shows `Total States: 9`
- **When** box type is programmed
- **Then** correct ANM state is selected

TC-010: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-011: Bitmap dimensions 48×32

- **Given** Physics Properties show `bitmap_w = 48, bitmap_h = 32`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-012: 5 box material types (glass, wooden, wicker, metal, cardboard)

- **Given** Description: "Program it to be glass, wooden, wicker, metal, or cardboard, which are all different sizes"
- **When** player programs box
- **Then** selected material determines box size and appearance

TC-013: No SOLVE.RES state transitions (static part)

- **Given** State Machine shows *No SOLVE.RES state data — this part has no programmatic state transitions*
- **When** behavior handler evaluates
- **Then** no state transitions are triggered

TC-014: Container accepts dropped objects

- **Given** Description: "Drop things inside the box, or use it to catch falling objects"
- **When** object falls into box
- **Then** object is contained within box bounds

TC-015: State determined by appearance field at placement

- **Given** Behavior Notes: "Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file)"
- **When** level is loaded
- **Then** box renders with programmed appearance state

Peça 72 — Peg



Categoria: Special Mechanics

Fases em que aparece: 0 fases

Quantidade de Estados: 1 estado(s)

[CREATED PART]

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4000	mass/density (relative scale)
unk_2	8	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	64	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	15	bitmap frame width
bitmap_h	10	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	1	1–11 (11 frames)	24×7	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001	build/phase-3/bitmaps/PART72_f001.png :material-alert-outline: (not found)
1	0001_0000	build/phase-3/bitmaps/PART72_f000.png :material-alert-outline: (not found)
2	0001_0003	build/phase-3/bitmaps/PART72_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
3	0001_0004	build/phase-3/bitmaps/PART72_f004.png :material-alert-outline: (not found)
4	0001_0006	build/phase-3/bitmaps/PART72_f006.png :material-alert-outline: (not found)
5	0001_0005	build/phase-3/bitmaps/PART72_f005.png :material-alert-outline: (not found)
6	0001_0006	build/phase-3/bitmaps/PART72_f006.png :material-alert-outline: (not found)
7	0001_0007	build/phase-3/bitmaps/PART72_f007.png :material-alert-outline: (not found)
8	0001_0006	build/phase-3/bitmaps/PART72_f006.png :material-alert-outline: (not found)
9	0001_0005	build/phase-3/bitmaps/PART72_f005.png :material-alert-outline: (not found)
10	0001_0006	build/phase-3/bitmaps/PART72_f006.png :material-alert-outline: (not found)
11	0001_0005	build/phase-3/bitmaps/PART72_f005.png :material-alert-outline: (not found)

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	12
END_FRAME	12
END_ANM	1

Notas de Comportamento:

Peg is a dynamically created (phantom) part. It is not placed by the player but spawned at runtime by another part's behavior (e.g., explosion debris, laser beam, rope end phantom).

Peg is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has custom spawn behavior defined in SOLVE.RES.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-072-peg.md` describes part type `72`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 72`

TC-002: Mass value is positive

- **Given** Peg has `mass = 4000`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Peg has `property_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 0.500`

TC-004: Created Part flag is Yes

- **Given** Identity shows `Created Part: Yes`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == True`

TC-005: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** peg is spawned dynamically and has no player interaction

TC-006: Not used in any built-in level

- **Given** Levels Using This Part shows *This part is not used in any built-in level (may be created dynamically at runtime)*
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** no level references `part_num = 72`

TC-007: Custom spawn behavior

- **Given** Category Rules show `Spawns: custom`
- **When** behavior handler processes
- **Then** custom spawn logic is applied

TC-008: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-009: Special Mechanics category (Category Part Count = 15)

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-010: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** peg is not destroyed

TC-011: Triggers are collision, electrical, proximity

- **Given** Category Rules show `Triggers: collision, electrical, proximity`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-012: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-013: Bitmap dimensions 15×10

- **Given** Physics Properties show `bitmap_w = 15, bitmap_h = 10`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: No SOLVE.RES state transitions (static part)

- **Given** State Machine shows *No SOLVE.RES state data — this part has no programmatic state transitions*
- **When** behavior handler evaluates
- **Then** no state transitions are triggered

TC-015: Spawned at runtime by other parts

- **Given** Behavior Notes: "It is not placed by the player but spawned at runtime by another part's behavior (e.g., explosion debris, laser beam, rope end phantom)"
- **When** parent part triggers spawn
- **Then** peg instance is created dynamically

Peça 73 — Trap Door



Categoria: Special Mechanics
Fases em que aparece: 33 fases
Quantidade de Estados: 3 estado(s)

This trap door drops open when heavy objects land on top of it.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Up	2→3	2	No
Down	3→-1	4	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1000	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	512	Q8.8 = 2.000
unk_6	16	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	48	bitmap frame width
bitmap_h	49	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	53×49	0
3	1	1–7 (7 frames)	61×52	19
4	8	8	22×49	154

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001, 0001_0007	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f001.png :material-check: build/phase-3/bitmaps/PART73_f007.png :material-check:
1	0001_0000, 0001_0001, 0001_0008	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f001.png :material-check: build/phase-3/bitmaps/PART73_f008.png :material-check:
2	0001_0000, 0001_0009, 0001_0002	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f009.png :material-check: build/phase-3/bitmaps/PART73_f002.png :material-check:
3	0001_0000, 0001_0003, 0001_0010	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f003.png :material-check: build/phase-3/bitmaps/PART73_f010.png :material-check:
4	0001_0000, 0001_0004, 0001_0011	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f004.png :material-check: build/phase-3/bitmaps/PART73_f011.png :material-check:
5	0001_0000, 0001_0005, 0001_0012	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f005.png :material-check: build/phase-3/bitmaps/PART73_f012.png :material-check:
6	0001_0000, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f006.png :material-check: build/phase-3/bitmaps/PART73_f013.png :material-check:
7	0001_0000, 0001_0006, 0001_0013	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f006.png :material-check: build/phase-3/bitmaps/PART73_f013.png :material-check:
8	0001_0000, 0001_0006	build/phase-3/bitmaps/PART73_f000.png :material-check: build/phase-3/bitmaps/PART73_f006.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
6	3073	0	build/phase-6/raw-wav/SX_3073_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	26
END_FRAME	9
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Trap Door has 2 semantic states: Up, Down. Transitions are triggered by: collision, electrical, proximity.

Terminal state(s): Down (part is removed from simulation).

This part has custom spawn behavior defined in SOLVE.RES.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-073-trap-door.md` describes part type `73`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 73`

TC-002: Mass value is positive

- **Given** Trap Door has `mass = 1000`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Trap Door has `property_q8 = 512`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 2.000`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** trap door responds to heavy objects landing

TC-006: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-007: SOLVE.RES state transitions (Up → Down)

- **Given** Trap Door has states: Up (`2→3`) and Down (`3→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-008: Terminal state Down (part removed)

- **Given** State Machine shows `Down: 3→-1`
- **When** state reaches Down
- **Then** part is removed from simulation (terminal)

TC-009: Triggers are collision, electrical, proximity

- **Given** Category Rules show `Triggers: collision, electrical, proximity`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-010: Part used in 33 levels

- **Given** Trap Door (type 73)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 33 levels reference `part_num = 73`

TC-011: Custom spawn behavior

- **Given** Category Rules show `Spawns: custom`
- **When** behavior handler processes
- **Then** custom spawn logic is applied

TC-012: Special Mechanics category (Category Part Count = 15)

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Special Mechanics category

TC-013: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** trap door is not destroyed

TC-014: Bitmap dimensions 48×49

- **Given** Physics Properties show `bitmap_w = 48, bitmap_h = 49`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Drops open when heavy objects land

- **Given** Description: "This trap door drops open when heavy objects land on top of it"
- **When** object with sufficient mass lands on trap door
- **Then** trap door transitions to Down state

Peça 74 — Nitroglycerine



Categoria: Explosives / Projectiles

Fases em que aparece: 38 fases

Quantidade de Estados: 2 estado(s)

This tube of nitroglycerine will explode if it's bumped or dropped with enough force. It blows up all kinds of things, including most walls.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Unexploded	1→3	1	No
Exploded	3→-1	—	Yes :material-bomb:

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2200	mass/density (relative scale)
unk_2	1800	category-specific property
property_q8	64	Q8.8 = 0.250
unk_6	32	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	10	bitmap frame width
bitmap_h	29	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–12 (13 frames)	16×29	0
3	13	13	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART74_f000.png :material-check:

Frame	Sprites	PNG References
1	0001_0000	build/phase-3/bitmaps/PART74_f000.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART74_f001.png :material-check:
3	0001_0002	build/phase-3/bitmaps/PART74_f002.png :material-check:
4	0001_0003	build/phase-3/bitmaps/PART74_f003.png :material-check:
5	0001_0004	build/phase-3/bitmaps/PART74_f004.png :material-check:
6	0001_0005	build/phase-3/bitmaps/PART74_f005.png :material-check:
7	0002_0000	build/phase-3/bitmaps/PART74_2_f000.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	74	0	build/phase-6/raw-wav/SX_74_11025.wav
2	74	0	build/phase-6/raw-wav/SX_74_11025.wav
3	74	0	build/phase-6/raw-wav/SX_74_11025.wav
4	74	0	build/phase-6/raw-wav/SX_74_11025.wav
5	74	0	build/phase-6/raw-wav/SX_74_11025.wav
6	74	0	build/phase-6/raw-wav/SX_74_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	8
END_FRAME	8
PLAY_SFX	6
END_ANM	1

Notas de Comportamento:

Nitroglycerine has 2 semantic states: Unexploded, Exploded. Transitions are triggered by: collision, proximity, timer.

State(s) Exploded are explosive — reaching these destroys the part.

Terminal state(s): Exploded (part is removed from simulation).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-074-nitroglycerine.md` describes part type `74`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 74`

TC-002: Mass value is positive

- **Given** Nitroglycerine has `mass = 2200`

- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Nitroglycerine has `property_q8 = 64`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 0.250`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** nitroglycerine responds to collision/proximity triggers

TC-006: Projectile connection supported

- **Given** Connections & Capabilities shows `Projectile: Yes`
- **When** behavior handler evaluates
- **Then** projectile spawn behavior is enabled

TC-007: SOLVE.RES state transitions (Unexploded → Exploded)

- **Given** Nitroglycerine has states: Unexploded (`1→3`) and Exploded (`3→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-008: Exploded state is explosive

- **Given** State Machine shows `Exploded` as explosive
- **When** state reaches Exploded
- **Then** explosion damages nearby parts

TC-009: Terminal state Exploded (part removed)

- **Given** State Machine shows `Exploded: 3→-1`
- **When** state reaches Exploded
- **Then** part is removed from simulation (terminal)

TC-010: Destructible flag is Yes

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** nitroglycerine can be destroyed by other explosives

TC-011: Spawns projectile

- **Given** Category Rules show `Spawns: projectile`
- **When** trigger activates nitroglycerine
- **Then** projectile part instance is spawned

TC-012: Part used in 38 levels

- **Given** Nitroglycerine (type 74)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 38 levels reference `part_num = 74`

TC-013: Explosives / Projectiles category (Category Part Count = 10)

- **Given** Category Rules show `Category Part Count: 10`
- **When** validated
- **Then** 10 parts in Explosives / Projectiles category

TC-014: Animated (Section A) — 14 duration entries

- **Given** ANM shows `Animated (Section A): Yes, Section A Durations: 14 entries`
- **When** animation renderer processes frame timing
- **Then** Section A durations control playback at 60 Hz

TC-015: Explodes on bump or drop with sufficient force

- **Given** Description: "This tube of nitroglycerine will explode if it's bumped or dropped with enough force"
- **When** collision or fall produces sufficient impact energy
- **Then** nitroglycerine transitions to Exploded state

Peça 75 — Match-on-a-Spring



Categoria: Springs / Pneumatics

Fases em que aparece: 52 fases

Quantidade de Estados: 4 estado(s)

Pull the little peg with a rope to make this match pop up lit. Use it to light fuses, pop blimps and balloons, and heat up coffee pots.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Unlit	<code>1→4</code>	1	No
Lit	<code>3→-1</code>	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	2000	mass/density (relative scale)
<code>unk_2</code>	995	category-specific property
<code>property_q8</code>	1024	Q8.8 = 4.000
<code>unk_6</code>	32	category-specific property
<code>dim_w1</code>	0	dimension 1 width
<code>dim_h1</code>	0	dimension 1 height
<code>dim_w2</code>	0	dimension 2 width
<code>dim_h2</code>	0	dimension 2 height
<code>bitmap_w</code>	36	bitmap frame width
<code>bitmap_h</code>	23	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–1 (2 frames)	43×23	0
4	2	2–9 (8 frames)	46×23	50
3	10	10–26 (17 frames)	33×38	192
2	27	27	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0008, 0001_0000, 0001_0009, 0001_0010	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f000.png :material-check: build/phase-3/bitmaps/PART75_f009.png :material-check: build/phase-3/bitmaps/PART75_f010.png :material-check:
1	0001_0008, 0001_0000, 0001_0009, 0001_0010	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f000.png :material-check: build/phase-3/bitmaps/PART75_f009.png :material-check: build/phase-3/bitmaps/PART75_f010.png :material-check:
2	0001_0008, 0001_0000, 0001_0009, 0001_0010	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f000.png :material-check: build/phase-3/bitmaps/PART75_f009.png :material-check: build/phase-3/bitmaps/PART75_f010.png :material-check:
3	0001_0008, 0001_0000, 0001_0009, 0001_0010	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f000.png :material-check: build/phase-3/bitmaps/PART75_f009.png :material-check: build/phase-3/bitmaps/PART75_f010.png :material-check:
4	0001_0008, 0001_0000, 0001_0009, 0001_0010	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f000.png :material-check: build/phase-3/bitmaps/PART75_f009.png :material-check: build/phase-3/bitmaps/PART75_f010.png :material-check:
5	0001_0008, 0001_0010	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f010.png :material-check:
6	0001_0008, 0001_0011	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f011.png :material-check:
7	0001_0008, 0001_0012	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f012.png :material-check:
8	0001_0008, 0001_0012	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f012.png :material-check:
9	0001_0008, 0001_0013	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f013.png :material-check:
10	0001_0008, 0001_0014	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f014.png :material-check:
11	0001_0008, 0001_0015, 0002_0000, 0001_0016	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f015.png :material-check: build/phase-3/bitmaps/PART75_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART75_f016.png :material-check:
12	0001_0008, 0001_0015, 0001_0017, 0002_0001	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f015.png :material-check: build/phase-3/bitmaps/PART75_f017.png :material-check: build/phase-3/bitmaps/PART75_2_f001.png :material-alert-outline: (not found)
13	0001_0008, 0001_0015, 0001_0016, 0002_0002	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f015.png :material-check: build/phase-3/bitmaps/PART75_f016.png :material-check: build/phase-3/bitmaps/PART75_2_f002.png :material-alert-outline: (not found)
14	0001_0008, 0001_0015, 0002_0000, 0001_0017	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f015.png :material-check: build/phase-3/bitmaps/PART75_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART75_f017.png :material-check:
15	0001_0008, 0001_0015, 0001_0017, 0002_0001	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f015.png :material-check: build/phase-3/bitmaps/PART75_f017.png :material-check: build/phase-3/bitmaps/PART75_2_f001.png :material-alert-outline: (not found)
16	0001_0008, 0001_0015, 0001_0017, 0002_0003	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f015.png :material-check: build/phase-3/bitmaps/PART75_f017.png :material-check: build/phase-3/bitmaps/PART75_2_f003.png :material-alert-outline: (not found)
17	0001_0008, 0001_0015, 0001_0017, 0002_0004	build/phase-3/bitmaps/PART75_f008.png :material-check: build/phase-3/bitmaps/PART75_f015.png :material-check: build/phase-3/bitmaps/PART75_f017.png :material-check: build/phase-3/bitmaps/PART75_2_f004.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
7	3075	0	build/phase-6/raw-wav/SX_3075_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	60
END_FRAME	18
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Match-on-a-Spring has 2 semantic states: Unlit, Lit. Transitions are triggered by: collision, electrical.

Terminal state(s): Lit (part is removed from simulation).

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-075-match-on-a-spring.md` describes part type `75`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 75`

TC-002: Mass value is positive

- **Given** Match-on-a-Spring has `mass = 2000`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Match-on-a-Spring has `property_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 4.000`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** match responds to rope pull

TC-006: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-007: SOLVE.RES state transitions (Unlit → Lit)

- **Given** Match-on-a-Spring has states: Unlit (`1→4`) and Lit (`3→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-008: Terminal state Lit (part removed)

- **Given** State Machine shows `Lit: 3→-1`
- **When** state reaches Lit
- **Then** part is removed from simulation (terminal)

TC-009: Triggers are collision and electrical

- **Given** Category Rules show `Triggers: collision, electrical`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Part used in 52 levels

- **Given** Match-on-a-Spring (type 75)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 52 levels reference `part_num = 75`

TC-011: Springs / Pneumatics category (Category Part Count = 7)

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Springs / Pneumatics category

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** match-on-a-spring is not destroyed

TC-013: Animated (Section A) — 35 duration entries

- **Given** ANM shows `Animated (Section A): Yes, Section A Durations: 35 entries`
- **When** animation renderer processes frame timing
- **Then** Section A durations control playback at 60 Hz

TC-014: 18 total animation frames across 4 states

- **Given** ANM shows `Total Frames: 18, Total States: 4`
- **When** animation renderer processes
- **Then** all 18 frames map to correct states

TC-015: Rope pull launches lit match upward

- **Given** Description: "Pull the little peg with a rope to make this match pop up lit"
- **When** rope is pulled with sufficient force
- **Then** match launches upward in Lit state

Peça 76 — Steel Cable



Categoria: Ropes / Belts / Pulleys

Fases em que aparece: 14 fases

Quantidade de Estados: 1 estado(s)

This steel cable works just like rope, except it's much stronger. It can only be cut with tin snips. To use the steel cable: pull it out of the Parts Bin onto the screen. Click on the first object you want tied, then drag the cursor over the second object. When the line turns from red to green the cable is in position. Click again to attach it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1600	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	0	Q8.8 = 0.000
unk_6	0	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	11	bitmap frame width
bitmap_h	10	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	4

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×15	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART76.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1

Opcode	Count
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Steel Cable is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-076-steel-cable.md` describes part type `76`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 76`

TC-002: Mass value is positive

- **Given** Steel Cable has `mass = 1600`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 is zero

- **Given** Steel Cable has `property_q8 = 0`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 0.000`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** steel cable is static constraint

TC-006: Belt connection supported

- **Given** Connections & Capabilities shows `Belt Connection: Yes`
- **When** behavior handler evaluates
- **Then** belt constraint is registered

TC-007: Rope connection supported

- **Given** Connections & Capabilities shows `Rope Connection: Yes`
- **When** behavior handler evaluates
- **Then** rope constraint is registered

TC-008: No triggers (static constraint)

- **Given** Category Rules show `Triggers: *(none)*`
- **When** behavior handler evaluates
- **Then** no triggers are registered

TC-009: Part used in 14 levels

- **Given** Steel Cable (type 76)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 14 levels reference `part_num = 76`

TC-010: Ropes / Belts / Pulleys category (Category Part Count = 4)

- **Given** Category Rules show `Category Part Count: 4`
- **When** validated
- **Then** 4 parts in Ropes / Belts / Pulleys category

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** steel cable is not destroyed

TC-012: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-013: Bitmap dimensions 11×10

- **Given** Physics Properties show `bitmap_w = 11, bitmap_h = 10`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-014: No SOLVE.RES state transitions (static part)

- **Given** State Machine shows *No SOLVE.RES state data — this part has no programmatic state transitions*
- **When** behavior handler evaluates
- **Then** no state transitions are triggered

TC-015: Cut only by tin snips

- **Given** Description: "This steel cable works just like rope, except it's much stronger. It can only be cut with tin snips"
- **When** object other than tin snips contacts cable
- **Then** cable remains intact

Peça 77 — Tin Snips



Categoria: Cutting / Popping

Fases em que aparece: 6 fases

Quantidade de Estados: 4 estado(s)

You can use these tin snips to cut through rope or steel cable, or to pop blimps and balloons. To make them cut, just bump the handles with another object. They can also be used to pop blimps and balloons.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Open	4→5	—	No
Closed	3→-1	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	16	category-specific property
dim_w1	240	dimension 1 width
dim_h1	240	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	56	bitmap frame width
bitmap_h	33	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	3

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
4	0	0	61×33	0
5	1	1	16×8	13
6	2	2	64×21	20
3	3	3	64×21	29

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0002	build/phase-3/bitmaps/PART77_f000.png :material-check: build/phase-3/bitmaps/PART77_f002.png :material-check:
1	0001_0002	build/phase-3/bitmaps/PART77_f002.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART77_f001.png :material-check:

Frame	Sprites	PNG References
3	0001_0001	build/phase-3/bitmaps/PART77_f001.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
2	3077	0	build/phase-6/raw-wav/SX_3077_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	5
END_FRAME	4
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Tin Snips has 2 semantic states: Open, Closed. Transitions are triggered by: collision.

Terminal state(s): Closed (part is removed from simulation).

Compliance Test Cases

TC-001: Part type number matches filename

- Given part-077-tin-snips.md describes part type 77
- When parsed by decoder/d_puzzle.py → nt_part_info_t2_base
- Then part_num == 77

TC-002: Mass value is positive

- Given Tin Snips has mass = 7552
- When validated against build/phase-9/part-properties.json
- Then mass > 0

TC-003: Property Q8.8 within range

- Given Tin Snips has property_q8 = 128
- When converted from Q8.8 fixed-point
- Then property = 0.500

TC-004: Created Part flag is No

- Given Identity shows Created Part: No
- When nt_part_info_t2_base.is_created is accessed
- Then is_created == False

TC-005: Interactive flag is Yes

- Given Identity shows Interactive: Yes
- When simulation runs
- Then tin snips respond to collision

TC-006: SOLVE.RES state transitions (Open → Closed)

- Given Tin Snips has states: Open (4→5) and Closed (3→-1)
- When behavior handler evaluates trigger

- **Then** state transitions follow SOLVE.RES rules

TC-007: Terminal state Closed (part removed)

- **Given** State Machine shows `Closed: 3→-1`
- **When** state reaches Closed
- **Then** part is removed from simulation (terminal)

TC-008: Triggers are collision

- **Given** Category Rules show `Triggers: collision`
- **When** behavior handler evaluates
- **Then** collision trigger is registered

TC-009: Part used in 6 levels

- **Given** Tin Snips (type 77)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 6 levels reference `part_num = 77`

TC-010: Cutting / Popping category (Category Part Count = 3)

- **Given** Category Rules show `Category Part Count: 3`
- **When** validated
- **Then** 3 parts in Cutting / Popping category

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** tin snips are not destroyed

TC-012: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-013: 4 total animation frames across 4 states

- **Given** ANM shows `Total Frames: 4, Total States: 4`
- **When** animation renderer processes
- **Then** all 4 frames map to correct states

TC-014: Cuts rope and steel cable on collision

- **Given** Description: "You can use these tin snips to cut through rope or steel cable"
- **When** collision triggers snips
- **Then** connected rope or steel cable is severed

TC-015: Pops balloons and blimps

- **Given** Description: "They can also be used to pop blimps and balloons"
- **When** snips activate near balloon/blimp
- **Then** balloon or blimp is destroyed



Categoria: Light / Flame Sources
Fases em que aparece: 44 fases
Quantidade de Estados: 4 estado(s)

Bump these flint rocks to start a fire, which can be used to light fuses, pop blimps and balloons, and heat up coffee pots.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Unlit	1→2	1	No
Lit	2→-1	7	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4000	mass/density (relative scale)
unk_2	140	category-specific property
property_q8	96	Q8.8 = 0.375
unk_6	18	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	46	bitmap frame width
bitmap_h	34	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	proximity, electrical
Destructible	No
Spawns	fire
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	Yes
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	48×34	0
2	1	1–11 (11 frames)	48×34	19
5	12	12–26 (15 frames)	48×41	350
7	27	27–31 (5 frames)	48×36	749

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
1	0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
2	0001_0000, 0001_0000, 0003_0004, 0003_0000, 0003_0001	build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f001.png :material-alert-outline: (not found)
3	0001_0000, 0001_0000, 0003_0004, 0003_0000, 0003_0001	build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f001.png :material-alert-outline: (not found)
4	0001_0001, 0001_0001, 0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_f001.png :material-check: build/phase-3/bitmaps/PART78_f001.png :material-check: build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
5	0001_0002, 0001_0002, 0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_f002.png :material-check: build/phase-3/bitmaps/PART78_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
6	0001_0003, 0001_0003, 0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_f003.png :material-check: build/phase-3/bitmaps/PART78_f003.png :material-check: build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
7	0001_0004, 0001_0004, 0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_f004.png :material-check: build/phase-3/bitmaps/PART78_f004.png :material-check: build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
8	0001_0005, 0001_0005, 0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
9	0001_0006, 0001_0006, 0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
10	0001_0000, 0001_0000, 0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
11	0001_0001, 0001_0001, 0003_0004, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_f001.png :material-check: build/phase-3/bitmaps/PART78_f001.png :material-check: build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
12	0003_0004, 0001_0000, 0001_0000, 0002_0005, 0003_0002, 0003_0003	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)
13	0001_0000, 0001_0000, 0003_0004, 0003_0002, 0003_0003, 0002_0006	build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_f000.png :material-check: build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f006.png :material-alert-outline: (not found)
14	0003_0004, 0003_0002, 0003_0003, 0002_0007	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f007.png :material-alert-outline: (not found)
15	0003_0004, 0003_0002, 0003_0003, 0002_0008	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f008.png :material-alert-outline: (not found)
16	0003_0004, 0003_0002, 0003_0003, 0002_0009	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART78_2_f009.png :material-alert-outline: (not found)
17	0003_0004, 0003_0002, 0003_0003, 0002_0005	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f005.png :material-alert-outline: (not found)
18	0003_0004, 0003_0002, 0003_0003, 0002_0006	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f006.png :material-alert-outline: (not found)
19	0003_0004, 0003_0002, 0003_0003, 0002_0007	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f007.png :material-alert-outline: (not found)
20	0003_0004, 0003_0002, 0003_0003, 0002_0008	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f008.png :material-alert-outline: (not found)
21	0003_0004, 0003_0002, 0003_0003, 0002_0009	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f009.png :material-alert-outline: (not found)
22	0003_0004, 0003_0002, 0003_0003, 0002_0005	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f005.png :material-alert-outline: (not found)
23	0003_0004, 0003_0002, 0003_0003, 0002_0006	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART78_2_f006.png :material-alert-outline: (not found)
24	0003_0004, 0003_0002, 0003_0003, 0002_0007	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f007.png :material-alert-outline: (not found)
25	0003_0004, 0003_0002, 0003_0003, 0002_0008	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f008.png :material-alert-outline: (not found)
26	0003_0004, 0003_0002, 0003_0003, 0002_0009	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f009.png :material-alert-outline: (not found)
27	0003_0004, 0003_0002, 0003_0003, 0002_0000	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f000.png :material-alert-outline: (not found)
28	0003_0004, 0003_0002, 0003_0003, 0002_0001	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f001.png :material-alert-outline: (not found)
29	0003_0004, 0003_0002, 0003_0003, 0002_0002	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f002.png :material-alert-outline: (not found)
30	0003_0004, 0003_0002, 0003_0003, 0002_0003	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART78_2_f003.png :material-alert-outline: (not found)
31	0003_0004, 0003_0002, 0003_0003, 0002_0004	build/phase-3/bitmaps/PART78_3_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_3_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART78_2_f004.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
2	3078	0	build/phase-6/raw-wav/SX_3078_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	140
END_FRAME	32
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Flint & Tinder has 2 semantic states: Unlit, Lit. Transitions are triggered by: proximity, electrical.

Terminal state(s): Lit (part is removed from simulation).

This part can ignite nearby combustible parts (balloons, explosives). Uses proximity trigger.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-078-flint-&-tinder.md` describes part type `78`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 78`

TC-002: Mass value is positive

- **Given** Flint & Tinder has `mass = 4000`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Flint & Tinder has `property_q8 = 96`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 0.375`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed

- **Then** `is_created == False`

TC-005: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** flint & tinder responds to proximity/electrical triggers

TC-006: Fire/Flame connection supported

- **Given** Connections & Capabilities shows `Fire/Flame: Yes`
- **When** behavior handler evaluates
- **Then** fire spawn behavior is enabled

TC-007: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-008: SOLVE.RES state transitions (Unlit → Lit)

- **Given** Flint & Tinder has states: Unlit (1→2) and Lit (2→-1)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-009: Terminal state Lit (part removed)

- **Given** State Machine shows `Lit: 2→-1`
- **When** state reaches Lit
- **Then** part is removed from simulation (terminal)

TC-010: Triggers are proximity and electrical

- **Given** Category Rules show `Triggers: proximity, electrical`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-011: Spawns fire

- **Given** Category Rules show `Spawns: fire`
- **When** flint & tinder is bumped
- **Then** fire is spawned at location

TC-012: Part used in 44 levels

- **Given** Flint & Tinder (type 78)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 44 levels reference `part_num = 78`

TC-013: Light / Flame Sources category (Category Part Count = 5)

- **Given** Category Rules show `Category Part Count: 5`
- **When** validated
- **Then** 5 parts in Light / Flame Sources category

TC-014: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** flint & tinder is not destroyed

TC-015: 32 total animation frames across 4 states

- **Given** ANM shows `Total Frames: 32`, `Total States: 4`
- **When** animation renderer processes
- **Then** all 32 frames map to correct states

TC-016: Bump ignition lights fire

- **Given** Description: "Bump these flint rocks to start a fire"
- **When** collision or proximity trigger fires
- **Then** flint & tinder enters Lit state and spawns fire

Peça 79 — Hot Air Balloon



Categoria: Balloons / Airships
Fases em que aparece: 30 fases
Quantidade de Estados: 5 estado(s)

Light the candle to create hot air, which will makes this balloon rise into the air. Tie a rope to the eye hook on the bottom and use it to lift things, pull triggers, and so on.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Lit	1→5	1	No
Lit	5→4	5	No
Popped	4→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	12	mass/density (relative scale)
unk_2	4000	category-specific property
property_q8	64	Q8.8 = 0.250
unk_6	32	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	34	bitmap frame width
bitmap_h	83	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity
Destructible	Yes

Rule	Value
Spawns	(none)
Category Part Count	3

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	40×83	0
5	1	1–5 (5 frames)	40×83	13
4	6	6–7 (2 frames)	40×83	108
2	8	8–9 (2 frames)	80×118	136
3	10	10–11 (2 frames)	17×65	168

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0001_0000	build/phase-3/bitmaps/PART79_f001.png :material-check: build/phase-3/bitmaps/PART79_f000.png :material-check:
1	0001_0001, 0002_0006, 0001_0000	build/phase-3/bitmaps/PART79_f001.png :material-check: build/phase-3/bitmaps/PART79_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART79_f000.png :material-check:
2	0001_0001, 0002_0007, 0001_0000	build/phase-3/bitmaps/PART79_f001.png :material-check: build/phase-3/bitmaps/PART79_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART79_f000.png :material-check:
3	0001_0001, 0002_0008, 0001_0000	build/phase-3/bitmaps/PART79_f001.png :material-check: build/phase-3/bitmaps/PART79_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART79_f000.png :material-check:
4	0001_0001, 0002_0009, 0001_0000	build/phase-3/bitmaps/PART79_f001.png :material-check: build/phase-3/bitmaps/PART79_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART79_f000.png :material-check:
5	0001_0001, 0002_0005, 0001_0000	build/phase-3/bitmaps/PART79_f001.png :material-check: build/phase-3/bitmaps/PART79_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART79_f000.png :material-check:
6	0001_0001, 0001_0000	build/phase-3/bitmaps/PART79_f001.png :material-check: build/phase-3/bitmaps/PART79_f000.png :material-check:

Frame	Sprites	PNG References
7	0001_0001, 0001_0003	build/phase-3/bitmaps/PART79_f001.png :material-check: build/phase-3/bitmaps/PART79_f003.png :material-check:
8	0001_0004, 0001_0002	build/phase-3/bitmaps/PART79_f004.png :material-check: build/phase-3/bitmaps/PART79_f002.png :material-check:
9	0001_0005, 0001_0006, 0001_0002	build/phase-3/bitmaps/PART79_f005.png :material-check: build/phase-3/bitmaps/PART79_f006.png :material-check: build/phase-3/bitmaps/PART79_f002.png :material-check:
10	0001_0007, 0001_0002	build/phase-3/bitmaps/PART79_f007.png :material-check: build/phase-3/bitmaps/PART79_f002.png :material-check:
11	0001_0006, 0001_0002	build/phase-3/bitmaps/PART79_f006.png :material-check: build/phase-3/bitmaps/PART79_f002.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
7	3004	0	build/phase-6/raw-wav/SX_3004_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	30
END_FRAME	12
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Hot Air Balloon has 3 semantic states: Not Lit, Lit, Popped. Transitions are triggered by: collision, proximity.

Terminal state(s): Popped (part is removed from simulation).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-079-hot-air-balloon.md` describes part type `79`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 79`

TC-002: Mass value is positive

- **Given** Hot Air Balloon has `mass = 12`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Hot Air Balloon has `property_q8 = 64`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 0.250`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** balloon responds to candle ignition

TC-006: SOLVE.RES state transitions (Not Lit → Lit → Popped)

- **Given** Hot Air Balloon has states: Not Lit (1→5), Lit (5→4), Popped (4→1)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-007: Terminal state Popped (part removed)

- **Given** State Machine shows `Popped: 4→1`
- **When** state reaches Popped
- **Then** part is removed from simulation (terminal)

TC-008: Triggers are collision and proximity

- **Given** Category Rules show `Triggers: collision, proximity`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-009: Part used in 30 levels

- **Given** Hot Air Balloon (type 79)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 30 levels reference `part_num = 79`

TC-010: Balloons / Airships category (Category Part Count = 3)

- **Given** Category Rules show `Category Part Count: 3`
- **When** validated
- **Then** 3 parts in Balloons / Airships category

TC-011: Destructible flag is Yes

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** balloon can be destroyed

TC-012: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-013: 12 total animation frames across 5 states

- **Given** ANM shows `Total Frames: 12, Total States: 5`
- **When** animation renderer processes
- **Then** all 12 frames map to correct states

TC-014: Candle ignition causes balloon to rise

- **Given** Description: "Light the candle to create hot air, which will makes this balloon rise into the air"
- **When** candle flame contacts balloon candle

- **Then** balloon transitions to Lit state and rises

TC-015: Rope attachment point for lifting

- **Given** Description: "Tie a rope to the eye hook on the bottom and use it to lift things"
- **When** rope is attached to eye hook
- **Then** rising balloon can lift connected objects

Peça 80 — Fireworks



Categoria: Explosives / Projectiles

Fases em que aparece: 17 fases

Quantidade de Estados: 15 estado(s)

This part lets you choose between three different types of fireworks displays. Program the part to get the celebration of your choice. Light the fuse with a flaming part like a candle, rocket, or an Aladdin's lamp.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Set Off	9→19 , 12→20 , 16→21	—	No
Set Off	6→12 , 7→16 , 8→-1	21	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1500	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	32	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	23	bitmap frame width
bitmap_h	86	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity, timer
Destructible	Yes
Spawns	projectile
Category Part Count	10

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	Yes

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
9	0	0	26×86	0
19	1	1–21 (21 frames)	64×106	13
10	22	22–24 (3 frames)	24×87	420
11	25	25–30 (6 frames)	24×86	459
6	31	31–47 (17 frames)	56×57	515
12	48	48	56×105	678
20	49	49–69 (21 frames)	56×106	697
14	70	70–72 (3 frames)	24×86	1078
15	73	73–78 (6 frames)	40×37	1107
7	79	79–94 (16 frames)	40×37	1149
16	95	95	56×100	1353
21	96	96–116 (21 frames)	56×97	1372
17	117	117–119 (3 frames)	96×111	1713
18	120	120–125 (6 frames)	96×111	1734
8	126	126–134 (9 frames)	96×111	1776

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0002_0000	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found)
1	0001_0000, 0002_0000, 0002_0005	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f005.png :material-alert-outline: (not found)
2	0001_0000, 0002_0000, 0002_0005	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f005.png :material-alert-outline: (not found)
3	0001_0000, 0002_0000, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
4	0001_0000, 0002_0001, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
5	0001_0000, 0002_0001, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
6	0001_0000, 0002_0001, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
7	0001_0000, 0002_0002, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
8	0001_0000, 0002_0002, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
9	0001_0000, 0002_0002, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
10	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
11	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
12	0001_0000, 0002_0003, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART80_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
13	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
14	0001_0000, 0002_0004, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
15	0001_0000, 0002_0004, 0001_0000, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
16	0001_0000	build/phase-3/bitmaps/PART80_f000.png :material-check:
17	0001_0001	build/phase-3/bitmaps/PART80_f001.png :material-check:
18	0001_0002, 0002_0008, 0002_0012	build/phase-3/bitmaps/PART80_f002.png :material-check: build/phase-3/bitmaps/PART80_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f012.png :material-alert-outline: (not found)
19	0002_0013, 0002_0014, 0002_0015, 0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_2_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f009.png :material-alert-outline: (not found)
20	0002_0014, 0002_0017, 0002_0015, 0002_0016, 0001_0000, 0002_0011	build/phase-3/bitmaps/PART80_2_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f017.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f016.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART80_2_f011.png :material-alert-outline: (not found)
21	0001_0000, 0002_0010	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f010.png :material-alert-outline: (not found)
22	0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f009.png :material-alert-outline: (not found)
23	0001_0000, 0002_0008	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f008.png :material-alert-outline: (not found)
24	0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f009.png :material-alert-outline: (not found)
25	0001_0000, 0002_0010	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f010.png :material-alert-outline: (not found)
26	0001_0000, 0002_0011	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f011.png :material-alert-outline: (not found)
27	0004_0000	build/phase-3/bitmaps/PART80_4_f000.png :material-alert-outline: (not found)
28	0004_0000	build/phase-3/bitmaps/PART80_4_f000.png :material-alert-outline: (not found)
29	0004_0001	build/phase-3/bitmaps/PART80_4_f001.png :material-alert-outline: (not found)
30	0004_0002	build/phase-3/bitmaps/PART80_4_f002.png :material-alert-outline: (not found)
31	0004_0003	build/phase-3/bitmaps/PART80_4_f003.png :material-alert-outline: (not found)
32	0004_0004	build/phase-3/bitmaps/PART80_4_f004.png :material-alert-outline: (not found)
33	0004_0005	build/phase-3/bitmaps/PART80_4_f005.png :material-alert-outline: (not found)
34	0004_0006	build/phase-3/bitmaps/PART80_4_f006.png :material-alert-outline: (not found)
35	0004_0007	build/phase-3/bitmaps/PART80_4_f007.png :material-alert-outline: (not found)
36	0004_0008	build/phase-3/bitmaps/PART80_4_f008.png :material-alert-outline: (not found)
37	0004_0009	build/phase-3/bitmaps/PART80_4_f009.png :material-alert-outline: (not found)
38	0004_0010	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART80_4_f010.png :material-alert-outline: (not found)
39	0004_0011	build/phase-3/bitmaps/PART80_4_f011.png :material-alert-outline: (not found)
40	0004_0012	build/phase-3/bitmaps/PART80_4_f012.png :material-alert-outline: (not found)
41	0004_0013	build/phase-3/bitmaps/PART80_4_f013.png :material-alert-outline: (not found)
42	0004_0014	build/phase-3/bitmaps/PART80_4_f014.png :material-alert-outline: (not found)
43	0003_0000	build/phase-3/bitmaps/PART80_3_f000.png :material-alert-outline: (not found)
44	0001_0000, 0002_0000	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found)
45	0001_0000, 0002_0000, 0002_0005	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f005.png :material-alert-outline: (not found)
46	0001_0000, 0002_0000, 0002_0005	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f005.png :material-alert-outline: (not found)
47	0001_0000, 0002_0000, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
48	0001_0000, 0002_0001, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
49	0001_0000, 0002_0001, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
50	0001_0000, 0002_0001, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
51	0001_0000, 0002_0002, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
52	0001_0000, 0002_0002, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
53	0001_0000, 0002_0002, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
54	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
55	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
56	0001_0000, 0002_0003, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
57	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
58	0001_0000, 0002_0004, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material-alert-outline: (not found)
59	0001_0000, 0002_0004, 0001_0000, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART80_2_f006.png :material-alert-outline: (not found)
60	0001_0000	build/phase-3/bitmaps/PART80_f000.png :material-check:
61	0001_0001	build/phase-3/bitmaps/PART80_f001.png :material-check:
62	0001_0002, 0002_0008, 0002_0012	build/phase-3/bitmaps/PART80_f002.png :material-check: build/phase-3/bitmaps/PART80_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f012.png :material-alert-outline: (not found)
63	0002_0013, 0002_0014, 0002_0015, 0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_2_f013.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f009.png :material-alert-outline: (not found)
64	0002_0014, 0002_0017, 0002_0015, 0002_0016, 0001_0000, 0002_0011	build/phase-3/bitmaps/PART80_2_f014.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f017.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f015.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f016.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f011.png :material-alert-outline: (not found)
65	0001_0000, 0002_0010	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f010.png :material-alert-outline: (not found)
66	0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f009.png :material-alert-outline: (not found)
67	0001_0000, 0002_0008	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f008.png :material-alert-outline: (not found)
68	0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f009.png :material-alert-outline: (not found)
69	0001_0000, 0002_0010	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f010.png :material-alert-outline: (not found)

Frame	Sprites	PNG References
70	0001_0000, 0002_0011	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f011.png :material-alert-outline: (not found)
71	0005_0000	build/phase-3/bitmaps/PART80_5_f000.png :material-alert-outline: (not found)
72	0005_0000	build/phase-3/bitmaps/PART80_5_f000.png :material-alert-outline: (not found)
73	0005_0001	build/phase-3/bitmaps/PART80_5_f001.png :material-alert-outline: (not found)
74	0005_0002	build/phase-3/bitmaps/PART80_5_f002.png :material-alert-outline: (not found)
75	0005_0003	build/phase-3/bitmaps/PART80_5_f003.png :material-alert-outline: (not found)
76	0005_0004	build/phase-3/bitmaps/PART80_5_f004.png :material-alert-outline: (not found)
77	0005_0005	build/phase-3/bitmaps/PART80_5_f005.png :material-alert-outline: (not found)
78	0005_0006	build/phase-3/bitmaps/PART80_5_f006.png :material-alert-outline: (not found)
79	0005_0007	build/phase-3/bitmaps/PART80_5_f007.png :material-alert-outline: (not found)
80	0005_0008	build/phase-3/bitmaps/PART80_5_f008.png :material-alert-outline: (not found)
81	0005_0009	build/phase-3/bitmaps/PART80_5_f009.png :material-alert-outline: (not found)
82	0005_0010	build/phase-3/bitmaps/PART80_5_f010.png :material-alert-outline: (not found)
83	0005_0011	build/phase-3/bitmaps/PART80_5_f011.png :material-alert-outline: (not found)
84	0005_0012	build/phase-3/bitmaps/PART80_5_f012.png :material-alert-outline: (not found)
85	0005_0013	build/phase-3/bitmaps/PART80_5_f013.png :material-alert-outline: (not found)
86	0003_0000	build/phase-3/bitmaps/PART80_3_f000.png :material-alert-outline: (not found)
87	0001_0000, 0002_0000	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found)
88	0001_0000, 0002_0000, 0002_0005	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f005.png :material-alert-outline: (not found)
89	0001_0000, 0002_0000, 0002_0005	build/phase-3/bitmaps/PART80_f000.png :material-check: build/phase-3/bitmaps/PART80_2_f000.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f005.png :material- alert-outline: (not found)
90	0001_0000, 0002_0000, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f000.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material- alert-outline: (not found)
91	0001_0000, 0002_0001, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f001.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material- alert-outline: (not found)
92	0001_0000, 0002_0001, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f001.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material- alert-outline: (not found)
93	0001_0000, 0002_0001, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f001.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material- alert-outline: (not found)
94	0001_0000, 0002_0002, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f002.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material- alert-outline: (not found)
95	0001_0000, 0002_0002, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f002.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material- alert-outline: (not found)
96	0001_0000, 0002_0002, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f002.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material- alert-outline: (not found)
97	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f003.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material- alert-outline: (not found)
98	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f003.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material- alert-outline: (not found)
99	0001_0000, 0002_0003, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f003.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f006.png :material- alert-outline: (not found)
100	0001_0000, 0002_0003, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f003.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material- alert-outline: (not found)
101	0001_0000, 0002_0004, 0002_0007	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f004.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f007.png :material- alert-outline: (not found)
102	0001_0000, 0002_0004, 0001_0000, 0002_0006	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f004.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f006.png :material- alert-outline: (not found)
103	0001_0000	build/phase-3/bitmaps/PART80_f000.png :material- check:
104	0001_0001	build/phase-3/bitmaps/PART80_f001.png :material- check:
105	0001_0002, 0002_0008, 0002_0012	build/phase-3/bitmaps/PART80_f002.png :material- check: build/phase-3/bitmaps/PART80_2_f008.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f012.png :material- alert-outline: (not found)
106	0002_0013, 0002_0014, 0002_0015, 0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_2_f013.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f014.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f015.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f009.png :material- alert-outline: (not found)
107	0002_0014, 0002_0017, 0002_0015, 0002_0016, 0001_0000, 0002_0011	build/phase-3/bitmaps/PART80_2_f014.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f017.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f015.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART80_2_f016.png :material- alert-outline: (not found) build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f011.png :material- alert-outline: (not found)
108	0001_0000, 0002_0010	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f010.png :material- alert-outline: (not found)
109	0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f009.png :material- alert-outline: (not found)
110	0001_0000, 0002_0008	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f008.png :material- alert-outline: (not found)
111	0001_0000, 0002_0009	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f009.png :material- alert-outline: (not found)
112	0001_0000, 0002_0010	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f010.png :material- alert-outline: (not found)
113	0001_0000, 0002_0011	build/phase-3/bitmaps/PART80_f000.png :material- check: build/phase-3/bitmaps/PART80_2_f011.png :material- alert-outline: (not found)
114	0006_0000	build/phase-3/bitmaps/PART80_6_f000.png :material- alert-outline: (not found)
115	0006_0000	build/phase-3/bitmaps/PART80_6_f000.png :material- alert-outline: (not found)
116	0006_0001	build/phase-3/bitmaps/PART80_6_f001.png :material- alert-outline: (not found)
117	0006_0002	build/phase-3/bitmaps/PART80_6_f002.png :material- alert-outline: (not found)
118	0006_0003	build/phase-3/bitmaps/PART80_6_f003.png :material- alert-outline: (not found)
119	0006_0004	build/phase-3/bitmaps/PART80_6_f004.png :material- alert-outline: (not found)
120	0006_0005	build/phase-3/bitmaps/PART80_6_f005.png :material- alert-outline: (not found)
121	0006_0006	build/phase-3/bitmaps/PART80_6_f006.png :material- alert-outline: (not found)
122	0006_0007	build/phase-3/bitmaps/PART80_6_f007.png :material- alert-outline: (not found)
123	0006_0008	build/phase-3/bitmaps/PART80_6_f008.png :material- alert-outline: (not found)

Frame	Sprites	PNG References
124	0006_0009	build/phase-3/bitmaps/PART80_6_f009.png :material-alert-outline: (not found)
125	0006_0010	build/phase-3/bitmaps/PART80_6_f010.png :material-alert-outline: (not found)
126	0006_0011	build/phase-3/bitmaps/PART80_6_f011.png :material-alert-outline: (not found)
127	0006_0012	build/phase-3/bitmaps/PART80_6_f012.png :material-alert-outline: (not found)
128	0006_0013	build/phase-3/bitmaps/PART80_6_f013.png :material-alert-outline: (not found)
129	0006_0014	build/phase-3/bitmaps/PART80_6_f014.png :material-alert-outline: (not found)
130	0006_0015	build/phase-3/bitmaps/PART80_6_f015.png :material-alert-outline: (not found)
131	0006_0016	build/phase-3/bitmaps/PART80_6_f016.png :material-alert-outline: (not found)
132	0006_0017	build/phase-3/bitmaps/PART80_6_f017.png :material-alert-outline: (not found)
133	0006_0018	build/phase-3/bitmaps/PART80_6_f018.png :material-alert-outline: (not found)
134	0003_0000	build/phase-3/bitmaps/PART80_3_f000.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
27	3080	0	build/phase-6/raw-wav/SX_3080_11025.wav
45	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
71	3080	0	build/phase-6/raw-wav/SX_3080_11025.wav
88	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
114	3080	0	build/phase-6/raw-wav/SX_3080_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	282
END_FRAME	135
PLAY_SFX	6
END_ANM	1

Notas de Comportamento:

Fireworks has 2 semantic states: Not Set Off, Set Off. Transitions are triggered by: collision, proximity, timer.

Terminal state(s): Set Off (part is removed from simulation).

This part can be destroyed by explosives.

This part spawns a **projectile** (cannonball, rocket, missile) when triggered. The projectile is a separate part instance.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 81 — Toaster



Categoria: Electrical
Fases em que aparece: 12 fases
Quantidade de Estados: 9 estado(s)

Plug this toaster into an electrical outlet and push down on the switch to make it work. When the toast is done, it pops into the air. Use it to bump other objects. You can also program it to work as a timer. The darker the toast you choose, the longer the amount of time it takes to pop up.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Toasting	1→8, 6→9, 7→10	—	No
Toasting	2→6, 3→7, 4→-1	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	3776	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	24	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	44	bitmap frame width
bitmap_h	35	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No

Rule	Value
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	48×35	0
8	1	1–4 (4 frames)	48×31	19
2	5	5–15 (11 frames)	40×49	73
6	16	16	48×35	336
9	17	17–20 (4 frames)	48×31	355
3	21	21–31 (11 frames)	40×49	409
7	32	32	48×35	672
10	33	33–36 (4 frames)	48×31	691
4	37	37–47 (11 frames)	40×49	745

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f006.png :material-check:
1	0001_0000, 0001_0001	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
2	0001_0002, 0001_0001	build/phase-3/bitmaps/PART81_f002.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
3	0001_0003, 0001_0001	build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
4	0001_0003, 0001_0001	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
5	0001_0004, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART81_f004.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f006.png :material-check:
6	0001_0007, 0001_0000, 0001_0007, 0001_0001	build/phase-3/bitmaps/PART81_f007.png :material-check: build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f007.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
7	0001_0000, 0001_0001, 0001_0008, 0001_0008	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check:
8	0001_0000, 0001_0001, 0001_0008, 0001_0008	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check:
9	0001_0004, 0001_0001, 0001_0009, 0001_0008	build/phase-3/bitmaps/PART81_f004.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f009.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check:
10	0001_0005, 0001_0001, 0001_0008, 0001_0009	build/phase-3/bitmaps/PART81_f005.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check: build/phase-3/bitmaps/PART81_f009.png :material-check:
11	0001_0003, 0001_0001, 0001_0008, 0001_0008	build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART81_f008.png :material-check:
12	0001_0002, 0001_0001, 0001_0008, 0001_0008	build/phase-3/bitmaps/PART81_f002.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check:
13	0001_0000, 0001_0001, 0001_0008, 0001_0010	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check: build/phase-3/bitmaps/PART81_f010.png :material-check:
14	0001_0000, 0001_0001, 0001_0010, 0001_0008	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f010.png :material-check: build/phase-3/bitmaps/PART81_f008.png :material-check:
15	0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f006.png :material-check:
16	0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f006.png :material-check:
17	0001_0000, 0001_0001	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
18	0001_0002, 0001_0001	build/phase-3/bitmaps/PART81_f002.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
19	0001_0003, 0001_0001	build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
20	0001_0003, 0001_0001	build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:

Frame	Sprites	PNG References
21	0001_0004, 0001_0001, 0001_0015	build/phase-3/bitmaps/PART81_f004.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f015.png :material-check:
22	0001_0011, 0001_0000, 0001_0011, 0001_0001	build/phase-3/bitmaps/PART81_f011.png :material-check: build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f011.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
23	0001_0000, 0001_0001, 0001_0012, 0001_0012	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check:
24	0001_0000, 0001_0001, 0001_0012, 0001_0012	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check:
25	0001_0004, 0001_0001, 0001_0013, 0001_0012	build/phase-3/bitmaps/PART81_f004.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f013.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check:
26	0001_0005, 0001_0001, 0001_0012, 0001_0013	build/phase-3/bitmaps/PART81_f005.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check: build/phase-3/bitmaps/PART81_f013.png :material-check:
27	0001_0003, 0001_0001, 0001_0012, 0001_0012	build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check:
28		

Frame	Sprites	PNG References
	0001_0002, 0001_0001, 0001_0012, 0001_0012	build/phase-3/bitmaps/PART81_f002.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check:
29	0001_0000, 0001_0001, 0001_0012, 0001_0014	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check: build/phase-3/bitmaps/PART81_f014.png :material-check:
30	0001_0000, 0001_0001, 0001_0014, 0001_0012	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f014.png :material-check: build/phase-3/bitmaps/PART81_f012.png :material-check:
31	0001_0000, 0001_0001, 0001_0015	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f015.png :material-check:
32	0001_0000, 0001_0001, 0001_0006	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f006.png :material-check:
33	0001_0000, 0001_0001	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
34	0001_0002, 0001_0001	build/phase-3/bitmaps/PART81_f002.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
35	0001_0003, 0001_0001	build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
36	0001_0003, 0001_0001	build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
37	0001_0004, 0001_0001, 0001_0020	build/phase-3/bitmaps/PART81_f004.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f020.png :material-check:
38	0001_0016, 0001_0000, 0001_0016, 0001_0001	build/phase-3/bitmaps/PART81_f016.png :material-check: build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f016.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:
39	0001_0000, 0001_0001, 0001_0017, 0001_0017	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check:
40	0001_0000, 0001_0001, 0001_0017, 0001_0017	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check:
41	0001_0004, 0001_0001, 0001_0018, 0001_0017	build/phase-3/bitmaps/PART81_f004.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f018.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check:
42	0001_0005, 0001_0001, 0001_0017, 0001_0018	build/phase-3/bitmaps/PART81_f005.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check: build/phase-3/bitmaps/PART81_f018.png :material-check:
43	0001_0003, 0001_0001, 0001_0017, 0001_0017	build/phase-3/bitmaps/PART81_f003.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check:
44	0001_0002, 0001_0001, 0001_0017, 0001_0017	build/phase-3/bitmaps/PART81_f002.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART81_f017.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check:
45	0001_0000, 0001_0001, 0001_0017, 0001_0019	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check: build/phase-3/bitmaps/PART81_f019.png :material-check:
46	0001_0000, 0001_0001, 0001_0019, 0001_0017	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f019.png :material-check: build/phase-3/bitmaps/PART81_f017.png :material-check:
47	0001_0000, 0001_0001, 0001_0020	build/phase-3/bitmaps/PART81_f000.png :material-check: build/phase-3/bitmaps/PART81_f001.png :material-check: build/phase-3/bitmaps/PART81_f020.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
3	3081	0	build/phase-6/raw-wav/SX_3081_11025.wav
19	3081	0	build/phase-6/raw-wav/SX_3081_11025.wav
35	3081	0	build/phase-6/raw-wav/SX_3081_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	159
END_FRAME	48
PLAY_SFX	3
END_ANM	1

Notas de Comportamento:

Toaster has 2 semantic states: Not Toasting, Toasting. Transitions are triggered by: electrical, collision.

Terminal state(s): Toasting (part is removed from simulation).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-081-toaster.md` describes part type `81`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`

- **Then** `part_num == 81`

TC-002: Mass value is positive

- **Given** Toaster has `mass = 3776`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Toaster has `property_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 4.000`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is Yes

- **Given** Identity shows `Interactive: Yes`
- **When** simulation runs
- **Then** toaster responds to electrical activation

TC-006: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-007: SOLVE.RES state transitions (Not Toasting → Toasting)

- **Given** Toaster has states: Not Toasting (`1→8`, `6→9`, `7→10`) and Toasting (`2→6`, `3→7`, `4→-1`)
- **When** behavior handler evaluates trigger
- **Then** state transitions follow SOLVE.RES rules

TC-008: Terminal state Toasting (part removed)

- **Given** State Machine shows `Toasting: 4→-1`
- **When** state reaches Toasting
- **Then** part is removed from simulation (terminal)

TC-009: Triggers are electrical and collision

- **Given** Category Rules show `Triggers: electrical, collision`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-010: Part used in 12 levels

- **Given** Toaster (type 81)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 12 levels reference `part_num = 81`

TC-011: Electrical category (Category Part Count = 9)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Electrical category

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** toaster is not destroyed

TC-013: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-014: 48 total animation frames across 9 states

- **Given** ANM shows `Total Frames: 48, Total States: 9`
- **When** animation renderer processes
- **Then** all 48 frames map to correct states

TC-015: Toast pops up after timer (darkness programmable)

- **Given** Description: "You can also program it to work as a timer. The darker the toast you choose, the longer the amount of time it takes to pop up"
 - **When** toasting timer completes
 - **Then** toast pops upward with velocity
-

Peça 82 — Sand Wall



Categoria: Walls / Floors
Fases em que aparece: 14 fases
Quantidade de Estados: 3 estado(s)

This is a sand wall or floor. Stretch it vertically or horizontally to any length you need. Its surface isn't very slippery. Explosions won't affect it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	48	friction = 0.188
unk_6	128	category-specific property
tile_w	16	base width in tile units
tile_h	16	base height in tile units
collision_w	576	AABB collision width
collision_h	416	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–11 (12 frames)	16×16	0
2	12	12–17 (6 frames)	16×16	84
3	18	18–29 (12 frames)	32×32	126

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART82_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART82_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART82_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART82_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART82_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART82_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART82_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART82_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART82_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART82_f009.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART82_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART82_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART82_f012.png :material-check:
13	0001_0013	build/phase-3/bitmaps/PART82_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART82_f014.png :material-check:
15	0001_0015	build/phase-3/bitmaps/PART82_f015.png :material-check:
16	0001_0016	build/phase-3/bitmaps/PART82_f016.png :material-check:
17	0001_0017	build/phase-3/bitmaps/PART82_f017.png :material-check:

Frame	Sprites	PNG References
18	0001_0018	build/phase-3/bitmaps/PART82_f018.png :material-check:
19	0001_0019	build/phase-3/bitmaps/PART82_f019.png :material-check:
20	0001_0020	build/phase-3/bitmaps/PART82_f020.png :material-check:
21	0001_0021	build/phase-3/bitmaps/PART82_f021.png :material-check:
22	0001_0022	build/phase-3/bitmaps/PART82_f022.png :material-check:
23	0001_0023	build/phase-3/bitmaps/PART82_f023.png :material-check:
24	0001_0024	build/phase-3/bitmaps/PART82_f024.png :material-check:
25	0001_0025	build/phase-3/bitmaps/PART82_f025.png :material-check:
26	0001_0026	build/phase-3/bitmaps/PART82_f026.png :material-check:
27	0001_0027	build/phase-3/bitmaps/PART82_f027.png :material-check:
28	0001_0028	build/phase-3/bitmaps/PART82_f028.png :material-check:
29	0001_0029	build/phase-3/bitmaps/PART82_f029.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	30
END_FRAME	30
END_ANM	1

Notas de Comportamento:

Sand Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-082-sand-wall.md` describes part type `82`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 82`

TC-002: Mass value is positive

- **Given** Sand Wall has `mass = 4153`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Friction value within range

- **Given** Sand Wall has `friction_q8 = 48`
- **When** converted from Q8.8 fixed-point
- **Then** `friction = 0.188` (not very slippery)

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** sand wall is static geometry

TC-006: Walls / Floors category (Category Part Count = 15)

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-007: Destructible flag is Yes (but immune per description)

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** sand wall is NOT destroyed (immune per description: "Explosions won't affect it")

TC-008: No triggers (static wall)

- **Given** Category Rules show `Triggers: *(none)*`
- **When** behavior handler evaluates
- **Then** no triggers are registered

TC-009: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-010: Part used in 14 levels

- **Given** Sand Wall (type 82)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 14 levels reference `part_num = 82`

TC-011: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-012: Tile dimensions 16×16 base

- **Given** Physics Properties show `tile_w = 16, tile_h = 16`
- **When** sprite is loaded
- **Then** base tile dimensions match

TC-013: Collision AABB dimensions 576×416

- **Given** Physics Properties show `collision_w = 576, collision_h = 416`
- **When** physics engine resolves collisions
- **Then** AABB matches sand wall geometry

TC-014: Vertically/horizontally stretchable

- **Given** Description: "Stretch it vertically or horizontally to any length you need"
- **When** editor resizes wall
- **Then** collision geometry updates accordingly

TC-015: 30 total animation frames across 3 states

- **Given** ANM shows `Total Frames: 30, Total States: 3`
- **When** animation renderer processes

- **Then** all 30 frames map to correct states

Peça 83 — Cinder Block Wall



Categoria: Walls / Floors
Fases em que aparece: 32 fases
Quantidade de Estados: 3 estado(s)

This is a cinder block wall or floor. Stretch it vertically or horizontally to any length you need. Its surface isn't very slippery. Explosions will blow holes through it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	1024	friction = 4.000
unk_6	40	category-specific property
tile_w	16	base width in tile units
tile_h	16	base height in tile units
collision_w	576	AABB collision width
collision_h	416	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
4	0	0–3 (4 frames)	16×16	0
1	4	4–7 (4 frames)	16×16	28
3	8	8–19 (12 frames)	16×16	56

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART83_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART83_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART83_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART83_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART83_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART83_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART83_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART83_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART83_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART83_f009.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART83_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART83_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART83_f012.png :material-check:
13	0001_0013	build/phase-3/bitmaps/PART83_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART83_f014.png :material-check:
15	0001_0015	build/phase-3/bitmaps/PART83_f015.png :material-check:
16	0001_0009	build/phase-3/bitmaps/PART83_f009.png :material-check:
17	0001_0016	build/phase-3/bitmaps/PART83_f016.png :material-check:
18	0001_0017	build/phase-3/bitmaps/PART83_f017.png :material-check:
19	0001_0018	build/phase-3/bitmaps/PART83_f018.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	20
END_FRAME	20
END_ANM	1

Notas de Comportamento:

Cinder Block Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-083-cinder-block-wall.md` describes part type `83`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 83`

TC-002: Mass value is positive

- **Given** Cinder Block Wall has `mass = 4153`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Friction value within range

- **Given** Cinder Block Wall has `friction_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `friction = 4.000` (high friction)

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** cinder block wall is static geometry

TC-006: Walls / Floors category (Category Part Count = 15)

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-007: Destructible flag is Yes

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** cinder block wall is destroyed (hole created)

TC-008: No triggers (static wall)

- **Given** Category Rules show `Triggers: *(none)*`
- **When** behavior handler evaluates
- **Then** no triggers are registered

TC-009: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-010: Part used in 32 levels

- **Given** Cinder Block Wall (type 83)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 32 levels reference `part_num = 83`

TC-011: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-012: Tile dimensions 16×16 base

- **Given** Physics Properties show `tile_w = 16, tile_h = 16`
- **When** sprite is loaded
- **Then** base tile dimensions match

TC-013: Collision AABB dimensions 576×416

- **Given** Physics Properties show `collision_w = 576, collision_h = 416`
- **When** physics engine resolves collisions
- **Then** AABB matches cinder block wall geometry

TC-014: Vertically/horizontally stretchable

- **Given** Description: "Stretch it vertically or horizontally to any length you need"
- **When** editor resizes wall
- **Then** collision geometry updates accordingly

TC-015: Explosions blow holes through it

- **Given** Description: "Explosions will blow holes through it"
- **When** explosive part contacts wall
- **Then** hole is created in wall geometry

Peça 84 — Greco-Roman Wall



Categoria: Walls / Floors

Fases em que aparece: 42 fases

Quantidade de Estados: 3 estado(s)

This is a Greco-Roman wall or floor. Stretch it vertically or horizontally to any length you need. It has a pretty slippery surface. Explosions won't affect it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>mass</code>	4153	mass/density (relative scale)
<code>unk_2</code>	0	category-specific property
<code>friction_q8</code>	1024	friction = 4.000
<code>unk_6</code>	16	category-specific property
<code>tile_w</code>	16	base width in tile units
<code>tile_h</code>	16	base height in tile units
<code>collision_w</code>	576	AABB collision width
<code>collision_h</code>	416	AABB collision height

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	16×16	0
3	4	4–12 (9 frames)	16×16	28
2	13	13–24 (12 frames)	24×32	91

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART84_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART84_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART84_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART84_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART84_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART84_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART84_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART84_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART84_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART84_f009.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART84_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART84_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART84_f012.png :material-check:
13		

Frame	Sprites	PNG References
	0001_0013	build/phase-3/bitmaps/PART84_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART84_f014.png :material-check:
15	0001_0015	build/phase-3/bitmaps/PART84_f015.png :material-check:
16	0001_0016	build/phase-3/bitmaps/PART84_f016.png :material-check:
17	0001_0017	build/phase-3/bitmaps/PART84_f017.png :material-check:
18	0001_0018	build/phase-3/bitmaps/PART84_f018.png :material-check:
19	0001_0019	build/phase-3/bitmaps/PART84_f019.png :material-check:
20	0001_0020	build/phase-3/bitmaps/PART84_f020.png :material-check:
21	0001_0021	build/phase-3/bitmaps/PART84_f021.png :material-check:
22	0001_0022	build/phase-3/bitmaps/PART84_f022.png :material-check:
23	0001_0023	build/phase-3/bitmaps/PART84_f023.png :material-check:
24	0001_0024	build/phase-3/bitmaps/PART84_f024.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	25
END_FRAME	25
END_ANM	1

Notas de Comportamento:

Greco-Roman Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-084-greco-roman-wall.md` describes part type `84`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 84`

TC-002: Mass value is positive

- **Given** Greco-Roman Wall has `mass = 4153`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Friction value within range

- **Given** Greco-Roman Wall has `friction_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `friction = 4.000` (high friction — "pretty slippery surface")

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** greco-roman wall is static geometry

TC-006: Walls / Floors category (Category Part Count = 15)

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-007: Destructible flag is Yes (but immune per description)

- **Given** Category Rules show `Destructible: Yes`
- **When** explosive part collides
- **Then** greco-roman wall is NOT destroyed (immune per description: "Explosions won't affect it")

TC-008: No triggers (static wall)

- **Given** Category Rules show `Triggers: *(none)*`
- **When** behavior handler evaluates
- **Then** no triggers are registered

TC-009: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-010: Part used in 42 levels

- **Given** Greco-Roman Wall (type 84)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 42 levels reference `part_num = 84`

TC-011: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-012: Tile dimensions 16×16 base

- **Given** Physics Properties show `tile_w = 16, tile_h = 16`
- **When** sprite is loaded
- **Then** base tile dimensions match

TC-013: Collision AABB dimensions 576×416

- **Given** Physics Properties show `collision_w = 576, collision_h = 416`
- **When** physics engine resolves collisions
- **Then** AABB matches greco-roman wall geometry

TC-014: Vertically/horizontally stretchable

- **Given** Description: "Stretch it vertically or horizontally to any length you need"
- **When** editor resizes wall
- **Then** collision geometry updates accordingly

TC-015: 25 total animation frames across 3 states

- **Given** ANM shows `Total Frames: 25, Total States: 3`
- **When** animation renderer processes

- **Then** all 25 frames map to correct states

Peça 85 — Log Wall



Categoria: Walls / Floors
Fases em que aparece: 18 fases
Quantidade de Estados: 3 estado(s)

This is a log wall or floor. Stretch it vertically or horizontally to any length you need. Its surface isn't very slippery. Explosions will blow holes through it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	96	friction = 0.375
unk_6	96	category-specific property
tile_w	16	base width in tile units
tile_h	16	base height in tile units
collision_w	576	AABB collision width
collision_h	416	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–12 (13 frames)	16×16	0
2	13	13–22 (10 frames)	16×16	91
3	23	23–34 (12 frames)	24×27	161

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART85_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART85_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART85_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART85_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART85_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART85_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART85_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART85_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART85_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART85_f009.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART85_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART85_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART85_f012.png :material-check:
13	0001_0013	build/phase-3/bitmaps/PART85_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART85_f014.png :material-check:
15	0001_0015	build/phase-3/bitmaps/PART85_f015.png :material-check:
16	0001_0016	build/phase-3/bitmaps/PART85_f016.png :material-check:
17	0001_0017	build/phase-3/bitmaps/PART85_f017.png :material-check:
18	0001_0018	build/phase-3/bitmaps/PART85_f018.png :material-check:
19	0001_0019	build/phase-3/bitmaps/PART85_f019.png :material-check:
20	0001_0020	build/phase-3/bitmaps/PART85_f020.png :material-check:
21	0001_0021	build/phase-3/bitmaps/PART85_f021.png :material-check:
22	0001_0022	build/phase-3/bitmaps/PART85_f022.png :material-check:
23	0001_0023	build/phase-3/bitmaps/PART85_f023.png :material-check:
24	0001_0024	build/phase-3/bitmaps/PART85_f024.png :material-check:
25	0001_0025	build/phase-3/bitmaps/PART85_f025.png :material-check:
26	0001_0026	build/phase-3/bitmaps/PART85_f026.png :material-check:
27	0001_0027	build/phase-3/bitmaps/PART85_f027.png :material-check:
28	0001_0028	build/phase-3/bitmaps/PART85_f028.png :material-check:
29	0001_0029	build/phase-3/bitmaps/PART85_f029.png :material-check:
30	0001_0030	build/phase-3/bitmaps/PART85_f030.png :material-check:
31	0001_0031	build/phase-3/bitmaps/PART85_f031.png :material-check:
32	0001_0032	build/phase-3/bitmaps/PART85_f032.png :material-check:

Frame	Sprites	PNG References
33	0001_0033	build/phase-3/bitmaps/PART85_f033.png :material-check:
34	0001_0034	build/phase-3/bitmaps/PART85_f034.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	35
END_FRAME	35
END_ANM	1

Notas de Comportamento:

Log Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-085-log-wall.md` describes part type `85`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 85`

TC-002: Mass value is positive

- **Given** Log Wall has `mass = 4153`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Friction value within range

- **Given** Log Wall has `friction_q8 = 96`
- **When** converted from Q8.8 fixed-point
- **Then** `friction = 0.375` (not very slippery)

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** log wall is static geometry

TC-006: Walls / Floors category (Category Part Count = 15)

- **Given** Category Rules show `Category Part Count: 15`
- **When** validated
- **Then** 15 parts in Walls / Floors category

TC-007: Destructible flag is Yes

- **Given** Category Rules show `Destructible: Yes`

- **When** explosive part collides
- **Then** log wall is destroyed (hole created)

TC-008: No triggers (static wall)

- **Given** Category Rules show `Triggers: *(none)*`
- **When** behavior handler evaluates
- **Then** no triggers are registered

TC-009: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-010: Part used in 18 levels

- **Given** Log Wall (type 85)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 18 levels reference `part_num = 85`

TC-011: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-012: Tile dimensions 16×16 base

- **Given** Physics Properties show `tile_w = 16, tile_h = 16`
- **When** sprite is loaded
- **Then** base tile dimensions match

TC-013: Collision AABB dimensions 576×416

- **Given** Physics Properties show `collision_w = 576, collision_h = 416`
- **When** physics engine resolves collisions
- **Then** AABB matches log wall geometry

TC-014: Vertically/horizontally stretchable

- **Given** Description: "Stretch it vertically or horizontally to any length you need"
- **When** editor resizes wall
- **Then** collision geometry updates accordingly

TC-015: Explosions blow holes through it

- **Given** Description: "Explosions will blow holes through it"
- **When** explosive part contacts wall
- **Then** hole is created in wall geometry

Peça 86 — Tiny Gear



Categoria: Rotating Power Sources

Fases em que aparece: 16 fases

Quantidade de Estados: 2 estado(s)

This tiny gear rotates at twice the speed of a big gear when it's placed next to one. These gears must be placed directly beside or above/below other gears.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	48	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	17	bitmap frame width
bitmap_h	17	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, rope
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	Yes
Rope Connection	Yes
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×17	0
2	1	1–4 (4 frames)	24×17	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART86_f000.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART86_f000.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART86_f001.png :material-check:

Frame	Sprites	PNG References
3	0001_0002	build/phase-3/bitmaps/PART86_f002.png :material-check:
4	0001_0003	build/phase-3/bitmaps/PART86_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	5
END_FRAME	5
END_ANM	1

Notas de Comportamento:

Tiny Gear is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-086-tiny-gear.md` describes part type `86`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 86`

TC-002: Mass value is positive

- **Given** Tiny Gear has `mass = 7552`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: Property Q8.8 within range

- **Given** Tiny Gear has `property_q8 = 1024`
- **When** converted from Q8.8 fixed-point
- **Then** `property = 4.000`

TC-004: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-005: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** tiny gear rotates automatically when connected to motor

TC-006: Electrical connection supported

- **Given** Connections & Capabilities shows `Electrical: Yes`
- **When** behavior handler evaluates
- **Then** electrical triggers are registered

TC-007: Belt connection supported

- **Given** Connections & Capabilities shows `Belt Connection: Yes`
- **When** behavior handler evaluates
- **Then** belt constraint is registered

TC-008: Rope connection supported

- **Given** Connections & Capabilities shows `Rope Connection: Yes`
- **When** behavior handler evaluates
- **Then** rope constraint is registered

TC-009: Triggers are collision, electrical, rope

- **Given** Category Rules show `Triggers: collision, electrical, rope`
- **When** behavior handler evaluates
- **Then** all three trigger types are registered

TC-010: Part used in 16 levels

- **Given** Tiny Gear (type 86)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 16 levels reference `part_num = 86`

TC-011: Rotating Power Sources category (Category Part Count = 7)

- **Given** Category Rules show `Category Part Count: 7`
- **When** validated
- **Then** 7 parts in Rotating Power Sources category

TC-012: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** tiny gear is not destroyed

TC-013: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`
- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-014: Bitmap dimensions 17×17

- **Given** Physics Properties show `bitmap_w = 17, bitmap_h = 17`
- **When** sprite is loaded
- **Then** frame dimensions match

TC-015: Rotates at 2× speed of big gear

- **Given** Description: "This tiny gear rotates at twice the speed of a big gear when it's placed next to one"
 - **When** gear train is connected to motor
 - **Then** tiny gear angular velocity = 2× big gear angular velocity
-

Peça 87 — Programmable Ball



Categoria: Balls

Fases em que aparece: 11 fases

Quantidade de Estados: 7 estado(s)

This ball can be programmed to vary in appearance, mass, elasticity, density, and friction.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2832	mass/density (relative scale)
damping	200	drag/air-resistance or density
cor_q8	128	COR = 0.500
radius	16	collision radius in internal units
sprite_w	0	sprite width
sprite_h	0	sprite height
data_offset	27	offset to per-state data or count

Regras de Categoria:

Rule	Value
Triggers	collision, trap
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	32×27	0
3	1	1	32×27	7
7	2	2	32×27	14
8	3	3	32×27	21
9	4	4	32×27	28
5	5	5–12 (8 frames)	32×27	35
4	13	13–20 (8 frames)	32×27	91

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0020	build/phase-3/bitmaps/PART87_f020.png :material-check:
1	0001_0018	build/phase-3/bitmaps/PART87_f018.png :material-check:
2	0001_0016	build/phase-3/bitmaps/PART87_f016.png :material-check:
3	0001_0017	build/phase-3/bitmaps/PART87_f017.png :material-check:
4		

Frame	Sprites	PNG References
	0001_0019	build/phase-3/bitmaps/PART87_f019.png :material-check:
5	0001_0008	build/phase-3/bitmaps/PART87_f008.png :material-check:
6	0001_0009	build/phase-3/bitmaps/PART87_f009.png :material-check:
7	0001_0010	build/phase-3/bitmaps/PART87_f010.png :material-check:
8	0001_0011	build/phase-3/bitmaps/PART87_f011.png :material-check:
9	0001_0012	build/phase-3/bitmaps/PART87_f012.png :material-check:
10	0001_0013	build/phase-3/bitmaps/PART87_f013.png :material-check:
11	0001_0014	build/phase-3/bitmaps/PART87_f014.png :material-check:
12	0001_0015	build/phase-3/bitmaps/PART87_f015.png :material-check:
13	0001_0000	build/phase-3/bitmaps/PART87_f000.png :material-check:
14	0001_0001	build/phase-3/bitmaps/PART87_f001.png :material-check:
15	0001_0002	build/phase-3/bitmaps/PART87_f002.png :material-check:
16	0001_0003	build/phase-3/bitmaps/PART87_f003.png :material-check:
17	0001_0004	build/phase-3/bitmaps/PART87_f004.png :material-check:
18	0001_0005	build/phase-3/bitmaps/PART87_f005.png :material-check:
19	0001_0006	build/phase-3/bitmaps/PART87_f006.png :material-check:
20	0001_0007	build/phase-3/bitmaps/PART87_f007.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	21
END_FRAME	21
END_ANM	1

Notas de Comportamento:

Programmable Ball is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Compliance Test Cases

TC-001: Part type number matches filename

- **Given** `part-087-programmable-ball.md` describes part type `87`
- **When** parsed by `decoder/d_puzzle.py` → `nt_part_info_t2_base`
- **Then** `part_num == 87`

TC-002: Mass value is positive

- **Given** Programmable Ball has `mass = 2832`
- **When** validated against `build/phase-9/part-properties.json`
- **Then** `mass > 0`

TC-003: COR within valid range [0.0, 1.0]

- **Given** Programmable Ball has `cor_q8 = 128`
- **When** converted from Q8.8 fixed-point
- **Then** `0.0 <= (128 >> 8) <= 1.0` i.e. `COR = 0.500`

TC-004: Radius is non-negative

- **Given** Programmable Ball has `radius = 16`
- **When** physics engine initializes collision
- **Then** `radius >= 0`

TC-005: Damping value is non-negative

- **Given** Programmable Ball has `damping = 200`
- **When** physics engine applies drag
- **Then** `damping >= 0`

TC-006: Part used in 11 levels

- **Given** Programmable Ball (type 87)
- **When** checked against `build/phase-5/yaml/*.yaml`
- **Then** 11 levels reference `part_num = 87`

TC-007: Triggers are collision and trap

- **Given** Category Rules show `Triggers: collision, trap`
- **When** behavior handler evaluates
- **Then** both trigger types are registered

TC-008: Created Part flag is No

- **Given** Identity shows `Created Part: No`
- **When** `nt_part_info_t2_base.is_created` is accessed
- **Then** `is_created == False`

TC-009: Interactive flag is No

- **Given** Identity shows `Interactive: No`
- **When** simulation runs
- **Then** programmable ball physics are automatic

TC-010: Balls category (Category Part Count = 9)

- **Given** Category Rules show `Category Part Count: 9`
- **When** validated
- **Then** 9 parts in Balls category

TC-011: Destructible flag is No

- **Given** Category Rules show `Destructible: No`
- **When** explosive part collides
- **Then** programmable ball is not destroyed

TC-012: No electrical/belt/rope/fire/laser/projectile connections

- **Given** Connections & Capabilities shows all connection flags as No
- **When** checked against `category_rules`
- **Then** only default connections

TC-013: Single default state (no SOLVE.RES transitions)

- **Given** State Machine shows *Single default state — no named transitions*
- **When** behavior handler evaluates
- **Then** no state transitions are triggered

TC-014: Static animation (no Section A durations)

- **Given** ANM shows `Animated (Section A): No`

- **When** animation renderer processes frame timing
- **Then** Section A durations is empty

TC-015: 21 total animation frames across 7 states (programmed variants)

- **Given** ANM shows `Total Frames: 21`, `Total States: 7`
- **When** animation renderer processes
- **Then** all 21 frames map to correct programmed states

TC-016: Programmable appearance, mass, elasticity, density, friction

- **Given** Description: "This ball can be programmed to vary in appearance, mass, elasticity, density, and friction"
 - **When** player programs ball parameters
 - **Then** ball physics and appearance reflect programmed values
-

Peça 88 — Accelerator Tube



Categoria: Pipe Systems
Fases em que aparece: 38 fases
Quantidade de Estados: 4 estado(s)

This accelerator tube can be connected to any type of large pipe. It will speed up or change the direction of any object passing through it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	38	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	<code>collision</code> , <code>electrical</code>
Destructible	No
Spawns	(none)
Category Part Count	4

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
20	0	0	40×48	0
16	1	1–4 (4 frames)	40×48	19
21	5	5	48×38	95
22	6	6–9 (4 frames)	48×38	114

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0018, 0001_0019, 0001_0020	build/phase-3/bitmaps/PART88_f018.png :material-check: build/phase-3/bitmaps/PART88_f019.png :material-check: build/phase-3/bitmaps/PART88_f020.png :material-check:
1	0001_0018, 0001_0019, 0001_0020	build/phase-3/bitmaps/PART88_f018.png :material-check: build/phase-3/bitmaps/PART88_f019.png :material-check: build/phase-3/bitmaps/PART88_f020.png :material-check:
2	0001_0018, 0001_0019, 0001_0021	build/phase-3/bitmaps/PART88_f018.png :material-check: build/phase-3/bitmaps/PART88_f019.png :material-check: build/phase-3/bitmaps/PART88_f021.png :material-check:
3	0001_0018, 0001_0019, 0001_0022	build/phase-3/bitmaps/PART88_f018.png :material-check: build/phase-3/bitmaps/PART88_f019.png :material-check: build/phase-3/bitmaps/PART88_f022.png :material-check:
4	0001_0018, 0001_0019, 0001_0023	build/phase-3/bitmaps/PART88_f018.png :material-check: build/phase-3/bitmaps/PART88_f019.png :material-check: build/phase-3/bitmaps/PART88_f023.png :material-check:
5	0001_0000, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART88_f000.png :material-check: build/phase-3/bitmaps/PART88_f002.png :material-check: build/phase-3/bitmaps/PART88_f001.png :material-check:
6	0001_0000, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART88_f000.png :material-check: build/phase-3/bitmaps/PART88_f002.png :material-check: build/phase-3/bitmaps/PART88_f001.png :material-check:
7	0001_0000, 0001_0001, 0001_0003	build/phase-3/bitmaps/PART88_f000.png :material-check: build/phase-3/bitmaps/PART88_f001.png :material-check: build/phase-3/bitmaps/PART88_f003.png :material-check:
8	0001_0000, 0001_0001, 0001_0004	build/phase-3/bitmaps/PART88_f000.png :material-check: build/phase-3/bitmaps/PART88_f001.png :material-check: build/phase-3/bitmaps/PART88_f004.png :material-check:
9	0001_0000, 0001_0001, 0001_0005	build/phase-3/bitmaps/PART88_f000.png :material-check: build/phase-3/bitmaps/PART88_f001.png :material-check: build/phase-3/bitmaps/PART88_f005.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	30
END_FRAME	10
END_ANM	1

Notas de Comportamento:

Accelerator Tube is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 89 — Trans-Roto-Matic



Categoria: Special Mechanics
Fases em que aparece: 14 fases
Quantidade de Estados: 2 estado(s)

This handy gadget turns translational motion (back and forth) into rotational energy (around in circles). Tie a rope to the eye hook, then tug it with another object to make the gear turn. Hitch a belt to the gear and use it to rotate pinwheels, conveyors, and other rotating parts.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1000	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	24	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	83	bitmap frame width
bitmap_h	49	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	88×49	0
2	1	1–8 (8 frames)	88×49	31

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0001_0000, 0001_0002, 0001_0007, 0001_0008	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f002.png :material-check: build/phase-3/bitmaps/ PART89_f007.png :material-check: build/phase-3/bitmaps/ PART89_f008.png :material-check:
1	0001_0001, 0001_0000, 0001_0002, 0001_0007, 0001_0008	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f002.png :material-check: build/phase-3/bitmaps/ PART89_f007.png :material-check: build/phase-3/bitmaps/ PART89_f008.png :material-check:
2	0001_0001, 0001_0002, 0001_0000, 0001_0007, 0001_0009	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f002.png :material-check: build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f007.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART89_f009.png :material-check:
3	0001_0001, 0001_0003, 0001_0000, 0001_0007, 0001_0010	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f003.png :material-check: build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f007.png :material-check: build/phase-3/bitmaps/ PART89_f010.png :material-check:
4	0001_0001, 0001_0004, 0001_0000, 0001_0008, 0001_0005	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f004.png :material-check: build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f008.png :material-check: build/phase-3/bitmaps/ PART89_f005.png :material-check:
5	0001_0001, 0001_0004, 0001_0000, 0001_0009, 0001_0005	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f004.png :material-check: build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f009.png :material-check: build/phase-3/bitmaps/ PART89_f005.png :material-check:
6	0001_0001, 0001_0004, 0001_0000, 0001_0010, 0001_0006	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f004.png :material-check: build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f010.png :material-check: build/phase-3/bitmaps/ PART89_f006.png :material-check:
7	0001_0001, 0001_0003, 0001_0000, 0001_0008, 0001_0005	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f003.png :material-check: build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f008.png :material-check: build/phase-3/bitmaps/ PART89_f005.png :material-check:
8	0001_0001, 0001_0002, 0001_0000, 0001_0005, 0001_0008	build/phase-3/bitmaps/ PART89_f001.png :material-check: build/phase-3/bitmaps/ PART89_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART89_f000.png :material-check: build/phase-3/bitmaps/ PART89_f005.png :material-check: build/phase-3/bitmaps/ PART89_f008.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
6	3089	0	build/phase-6/raw-wav/SX_3089_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	45
END_FRAME	9
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Trans-Roto-Matic is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has custom spawn behavior defined in SOLVE.RES.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 90 — Roto-Trans Converter



Categoria: Special Mechanics
Fases em que aparece: 15 fases
Quantidade de Estados: 2 estado(s)

This nifty contraption turns rotational energy (circular) into translational motion (back and forth movement). Hitch the little wheel to a rotational motor with a belt, then tie a rope to the eye hook and hitch it to something you want to lift or pull (like a teeter-totter, phazer trigger, or laundry basket).

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1000	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	24	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	22	bitmap frame width
bitmap_h	69	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×69	0
2	1	1–8 (8 frames)	24×69	25

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0003, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f003.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:
1		

Frame	Sprites	PNG References
	0001_0000, 0001_0003, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f003.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:
2	0001_0000, 0001_0004, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f004.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:
3	0001_0000, 0001_0005, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f005.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:
4	0001_0000, 0001_0006, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f006.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:
5	0001_0000, 0001_0007, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f007.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:
6	0001_0000, 0001_0008, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f008.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:
7	0001_0000, 0001_0009, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f009.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:

Frame	Sprites	PNG References
8	0001_0000, 0001_0010, 0001_0002, 0001_0001	build/phase-3/bitmaps/PART90_f000.png :material-check: build/phase-3/bitmaps/PART90_f010.png :material-check: build/phase-3/bitmaps/PART90_f002.png :material-check: build/phase-3/bitmaps/PART90_f001.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3090	0	build/phase-6/raw-wav/SX_3090_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	36
END_FRAME	9
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Roto-Trans Converter is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has custom spawn behavior defined in SOLVE.RES.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 91 — Red Laser



Categoria: Lasers / Optics
Fases em que aparece: 30 fases
Quantidade de Estados: 8 estado(s)

When this red laser gun is plugged into an outlet (and the switch is on) it will fire a red beam. When the beam strikes a red Laser-Activated Plug, it will generate energy that can be used to power anything hooked up to that outlet. Laser beams can be bounced and directed by using angled mirrors. They may also be fired into Laser Mixers, which will blend the colors of the beams. If an object passes through a laser beam, it will temporarily cut off the energy flow. Use laser beams to light fuses and candles, or to pop balloons.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Operating	1→6 , 6→7 , 7→8 , 8→9	8	No
Operating	9→4 , 4→3 , 3→2 , 2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	24	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	51	bitmap frame width
bitmap_h	51	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical , proximity , collision
Destructible	No
Spawns	laser_beam
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	Yes
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	48×18	0
6	1	1	24×42	13
7	2	2	48×18	26
8	3	3	24×42	39
9	4	4–15 (12 frames)	48×18	52

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
4	16	16–27 (12 frames)	24×42	228
3	28	28–39 (12 frames)	0×0	0
2	40	40	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0002_0000, 0001_0001	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f001.png :material-check:
1	0002_0001, 0001_0004	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f004.png :material-check:
2	0002_0000, 0001_0001	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f001.png :material-check:
3	0002_0001, 0001_0004	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f004.png :material-check:
4	0002_0000, 0001_0001	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f001.png :material-check:
5	0002_0000, 0001_0000	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f000.png :material-check:
6	0002_0000, 0001_0001	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f001.png :material-check:
7	0002_0000, 0001_0002	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f002.png :material-check:
8	0002_0001, 0001_0004	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f004.png :material-check:
9	0002_0001, 0001_0005	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f005.png :material-check:
10	0002_0001, 0001_0004	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f004.png :material-check:
11	0002_0001, 0001_0003	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f003.png :material-check:
12	0002_0000, 0001_0001	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f001.png :material-check:
13	0002_0000, 0001_0000	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f000.png :material-check:
14		

Frame	Sprites	PNG References
	0002_0000, 0001_0001	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f001.png :material-check:
15	0002_0000, 0001_0002	build/phase-3/bitmaps/PART91_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f002.png :material-check:
16	0002_0001, 0001_0004	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f004.png :material-check:
17	0002_0001, 0001_0005	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f005.png :material-check:
18	0002_0001, 0001_0004	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f004.png :material-check:
19	0002_0001, 0001_0003	build/phase-3/bitmaps/PART91_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART91_f003.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
4	91	0	build/phase-6/raw-wav/SX_91_11025.wav
5	91	0	build/phase-6/raw-wav/SX_91_11025.wav
6	91	0	build/phase-6/raw-wav/SX_91_11025.wav
7	91	0	build/phase-6/raw-wav/SX_91_11025.wav
9	91	0	build/phase-6/raw-wav/SX_91_11025.wav
10	91	0	build/phase-6/raw-wav/SX_91_11025.wav
11	91	0	build/phase-6/raw-wav/SX_91_11025.wav
13	91	0	build/phase-6/raw-wav/SX_91_11025.wav
14	91	0	build/phase-6/raw-wav/SX_91_11025.wav
15	91	0	build/phase-6/raw-wav/SX_91_11025.wav
17	91	0	build/phase-6/raw-wav/SX_91_11025.wav
18	91	0	build/phase-6/raw-wav/SX_91_11025.wav
19	91	0	build/phase-6/raw-wav/SX_91_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	40
END_FRAME	20
PLAY_SFX	13
END_ANM	1

Notas de Comportamento:

Red Laser has 2 semantic states: Not Operating, Operating. Transitions are triggered by: electrical, proximity, collision.

Terminal state(s): Operating (part is removed from simulation).

This part emits a **laser beam**. The beam interacts with mirrors, mixers, and laser-activated plugs.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 92 — Green Laser



Categoria: Lasers / Optics
Fases em que aparece: 23 fases
Quantidade de Estados: 8 estado(s)

When this green laser gun is plugged into an outlet (and the switch is on) it will fire a green laser beam. When the beam strikes a green Laser-Activated Plug, it will generate energy that can be used to power anything hooked up to that outlet. Laser beams can be bounced and directed by using angled mirrors. They may also be fired into Laser Mixers, which will blend the colors of the beams. If an object passes through a laser beam, it will temporarily cut off the energy flow. Use laser beams to light fuses and candles, or to pop balloons.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Operating	1→6 , 6→7 , 7→8 , 8→9	8	No
Operating	9→4 , 4→3 , 3→2 , 2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	24	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	51	bitmap frame width
bitmap_h	51	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, proximity, collision
Destructible	No
Spawns	laser_beam
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	Yes
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	48×18	0
6	1	1	24×42	13
7	2	2	48×18	26
8	3	3	24×42	39
9	4	4–15 (12 frames)	48×18	52
4	16	16–27 (12 frames)	24×42	228
3	28	28–39 (12 frames)	0×0	0
2	40	40	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0002_0000, 0001_0001	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f001.png :material-check:
1	0002_0001, 0001_0004	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f004.png :material-check:
2	0002_0000, 0001_0001	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f001.png :material-check:
3	0002_0001, 0001_0004	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f004.png :material-check:
4	0002_0000, 0001_0001	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f001.png :material-check:

Frame	Sprites	PNG References
5	0002_0000, 0001_0000	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f000.png :material-check:
6	0002_0000, 0001_0001	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f001.png :material-check:
7	0002_0000, 0001_0002	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f002.png :material-check:
8	0002_0001, 0001_0004	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f004.png :material-check:
9	0002_0001, 0001_0005	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f005.png :material-check:
10	0002_0001, 0001_0004	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f004.png :material-check:
11	0002_0001, 0001_0003	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f003.png :material-check:
12	0002_0000, 0001_0001	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f001.png :material-check:
13	0002_0000, 0001_0000	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f000.png :material-check:
14	0002_0000, 0001_0001	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f001.png :material-check:
15	0002_0000, 0001_0002	build/phase-3/bitmaps/PART92_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f002.png :material-check:
16	0002_0001	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found)
17	0002_0001, 0001_0005, 0001_0004	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f005.png :material-check: build/phase-3/bitmaps/PART92_f004.png :material-check:
18	0002_0001, 0001_0004, 0001_0004	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f004.png :material-check: build/phase-3/bitmaps/PART92_f004.png :material-check:
19	0002_0001, 0001_0003, 0001_0004	build/phase-3/bitmaps/PART92_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART92_f003.png :material-check: build/phase-3/bitmaps/PART92_f004.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
4	91	0	build/phase-6/raw-wav/SX_91_11025.wav
5	91	0	build/phase-6/raw-wav/SX_91_11025.wav
6	91	0	build/phase-6/raw-wav/SX_91_11025.wav
7	91	0	build/phase-6/raw-wav/SX_91_11025.wav
9	91	0	build/phase-6/raw-wav/SX_91_11025.wav
10	91	0	build/phase-6/raw-wav/SX_91_11025.wav
11	91	0	build/phase-6/raw-wav/SX_91_11025.wav
13	91	0	build/phase-6/raw-wav/SX_91_11025.wav
14	91	0	build/phase-6/raw-wav/SX_91_11025.wav
15	91	0	build/phase-6/raw-wav/SX_91_11025.wav
16	91	0	build/phase-6/raw-wav/SX_91_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	42
END_FRAME	20
PLAY_SFX	11
END_ANM	1

Notas de Comportamento:

Green Laser has 2 semantic states: Not Operating, Operating. Transitions are triggered by: electrical, proximity, collision.

Terminal state(s): Operating (part is removed from simulation).

This part emits a **laser beam**. The beam interacts with mirrors, mixers, and laser-activated plugs.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 93 — Blue Laser



Categoria: Lasers / Optics
Fases em que aparece: 32 fases
Quantidade de Estados: 8 estado(s)

When this blue laser gun is plugged into an outlet (and the switch is on) it will fire a blue laser beam. When the beam strikes a blue Laser-Activated Plug, it will generate energy that can be used to power anything hooked up to that outlet. Laser beams can be bounced and directed by using angled mirrors. They may also be fired into Laser Mixers, which will blend the colors of the beams. If an object passes through a laser beam, it will temporarily cut off the energy flow. Use laser beams to light fuses and candles, or to pop balloons.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Operating	1→5 , 5→6 , 6→7 , 7→9	7	No
Operating	9→4 , 4→3 , 3→2 , 2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	24	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	51	bitmap frame width
bitmap_h	51	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical , proximity , collision
Destructible	No
Spawns	laser_beam
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	Yes
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	48×18	0
5	1	1	24×42	13
6	2	2	48×18	26
7	3	3	24×42	39
9	4	4–15 (12 frames)	48×18	52
4	16	16–27 (12 frames)	24×42	228
3	28	28–39 (12 frames)	0×0	0
2	40	40	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0002_0000, 0001_0001	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f001.png :material-check:
1	0002_0001, 0001_0004	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f004.png :material-check:
2	0002_0000, 0001_0001	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f001.png :material-check:
3	0002_0001, 0001_0004	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f004.png :material-check:
4	0002_0000, 0001_0001	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f001.png :material-check:
5	0002_0000, 0001_0000	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f000.png :material-check:
6	0002_0000, 0001_0001	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f001.png :material-check:
7	0002_0000, 0001_0002	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f002.png :material-check:
8	0002_0001, 0001_0004	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f004.png :material-check:
9	0002_0001, 0001_0005	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f005.png :material-check:
10	0002_0001, 0001_0004	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f004.png :material-check:
11	0002_0001, 0001_0003	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f003.png :material-check:

Frame	Sprites	PNG References
12	0002_0000, 0001_0001	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f001.png :material-check:
13	0002_0000, 0001_0000	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f000.png :material-check:
14	0002_0000, 0001_0001	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f001.png :material-check:
15	0002_0000, 0001_0002	build/phase-3/bitmaps/PART93_2_f000.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f002.png :material-check:
16	0002_0001, 0001_0004	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f004.png :material-check:
17	0002_0001, 0001_0005	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f005.png :material-check:
18	0002_0001, 0001_0004	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f004.png :material-check:
19	0002_0001, 0001_0003	build/phase-3/bitmaps/PART93_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART93_f003.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
4	91	0	build/phase-6/raw-wav/SX_91_11025.wav
5	91	0	build/phase-6/raw-wav/SX_91_11025.wav
6	91	0	build/phase-6/raw-wav/SX_91_11025.wav
7	91	0	build/phase-6/raw-wav/SX_91_11025.wav
9	91	0	build/phase-6/raw-wav/SX_91_11025.wav
10	91	0	build/phase-6/raw-wav/SX_91_11025.wav
11	91	0	build/phase-6/raw-wav/SX_91_11025.wav
13	91	0	build/phase-6/raw-wav/SX_91_11025.wav
14	91	0	build/phase-6/raw-wav/SX_91_11025.wav
15	91	0	build/phase-6/raw-wav/SX_91_11025.wav
17	91	0	build/phase-6/raw-wav/SX_91_11025.wav
18	91	0	build/phase-6/raw-wav/SX_91_11025.wav
19	91	0	build/phase-6/raw-wav/SX_91_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	40
END_FRAME	20

Opcode	Count
PLAY_SFX	13
END_ANM	1

Notas de Comportamento:

Blue Laser has 2 semantic states: Not Operating, Operating. Transitions are triggered by: electrical, proximity, collision.
Terminal state(s): Operating (part is removed from simulation).

This part emits a **laser beam**. The beam interacts with mirrors, mixers, and laser-activated plugs.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 94 — Angled Mirror



Categoria: Lasers / Optics
Fases em que aparece: 26 fases
Quantidade de Estados: 1 estado(s)

This angled mirror can be used to deflect and change the direction of laser beams. It can be positioned in four different angles.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	28	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	23	bitmap frame width
bitmap_h	23	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, proximity, collision
Destructible	No
Spawns	laser_beam
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	Yes
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	24×23	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART94_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART94_f001.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART94_f001.png :material-check:
3	0001_0000	build/phase-3/bitmaps/PART94_f000.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Angled Mirror is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part emits a **laser beam**. The beam interacts with mirrors, mixers, and laser-activated plugs.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
 - **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
-

Peça 95 — Laser Mixer



Categoria: Lasers / Optics
Fases em que aparece: 13 fases
Quantidade de Estados: 32 estado(s)

This Laser Mixer will blend together the colors of any laser beams passing through it. For instance, a red beam and a blue beam will become violet. The violet beam could then be used to provide energy to a violet Laser-Activated Plug.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1000	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	24	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	40	bitmap frame width
bitmap_h	40	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, proximity, collision
Destructible	No
Spawns	laser_beam
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	Yes
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	40×40	0
2	1	1–12 (12 frames)	40×40	7
3	13	13–24 (12 frames)	40×40	235
4	25	25–36 (12 frames)	40×40	463
5	37	37–48 (12 frames)	40×40	679
6	49	49–60 (12 frames)	40×40	907
7	61	61–72 (12 frames)	40×40	1123
8	73	73–84 (12 frames)	40×40	1351
16	85	85–87 (3 frames)	40×40	1579
17	88	88–99 (12 frames)	40×40	1618
19	100	100–108 (9 frames)	40×40	1846
20	109	109–120 (12 frames)	40×40	2017
21	121	121–132 (12 frames)	0×0	0
22	133	133–144 (12 frames)	0×0	0
23	145	145–156 (12 frames)	0×0	0
24	157	157–168 (12 frames)	0×0	0
25	169	169–171 (3 frames)	0×0	0
26	172	172–183 (12 frames)	0×0	0
27	184	184–195 (12 frames)	0×0	0
28	196	196–207 (12 frames)	0×0	0
29	208	208–219 (12 frames)	0×0	0
30	220	220–231 (12 frames)	0×0	0
31	232	232–243 (12 frames)	0×0	0
32	244	244–255 (12 frames)	0×0	0
33	256	256–258 (3 frames)	0×0	0
9	259	259–270 (12 frames)	0×0	0
10	271	271–282 (12 frames)	0×0	0
11	283	283–294 (12 frames)	0×0	0
12	295	295–306 (12 frames)	0×0	0
13	307	307–318 (12 frames)	0×0	0
14	319	319–330 (12 frames)	0×0	0
15	331	331	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART95_f000.png :material-check:
1	0001_0000, 0001_0001, 0001_0022	build/phase-3/bitmaps/PART95_f000.png :material-check: build/phase-3/bitmaps/PART95_f001.png :material-check: build/phase-3/bitmaps/PART95_f022.png :material-check:
2	0001_0000, 0001_0002, 0001_0022	

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Frame	Sprites	PNG References
104	0001_0029, 0001_0055, 0001_0043	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f055.png :material-check: build/phase-3/bitmaps/PART95_f043.png :material-check:
105	0001_0029, 0001_0055, 0001_0042	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f055.png :material-check: build/phase-3/bitmaps/PART95_f042.png :material-check:
106	0001_0029, 0001_0055, 0001_0043	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f055.png :material-check: build/phase-3/bitmaps/PART95_f043.png :material-check:
107	0001_0029, 0001_0056, 0001_0047	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f056.png :material-check: build/phase-3/bitmaps/PART95_f047.png :material-check:
108	0001_0029, 0001_0056, 0001_0046	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f056.png :material-check: build/phase-3/bitmaps/PART95_f046.png :material-check:
109	0001_0029, 0001_0056, 0001_0045	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f056.png :material-check: build/phase-3/bitmaps/PART95_f045.png :material-check:
110	0001_0029, 0001_0056, 0001_0046	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f056.png :material-check: build/phase-3/bitmaps/PART95_f046.png :material-check:
111	0001_0029, 0001_0057, 0001_0050	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f057.png :material-check: build/phase-3/bitmaps/PART95_f050.png :material-check:
112	0001_0029, 0001_0057, 0001_0049	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f057.png :material-check: build/phase-3/bitmaps/PART95_f049.png :material-check:
113	0001_0029, 0001_0057, 0001_0048	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f057.png :material-check: build/phase-3/bitmaps/PART95_f048.png :material-check:
114	0001_0029, 0001_0057, 0001_0049	build/phase-3/bitmaps/PART95_f029.png :material-check: build/phase-3/bitmaps/PART95_f057.png :material-check: build/phase-3/bitmaps/PART95_f049.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	336
END_FRAME	115
END_ANM	1

Notas de Comportamento:

Laser Mixer is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part emits a **laser beam**. The beam interacts with mirrors, mixers, and laser-activated plugs.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 96 — Laser-Activated Plug



Categoria: Lasers / Optics
Fases em que aparece: 25 fases
Quantidade de Estados: 10 estado(s)

When a laser beam of the right color strikes this laser-activated plug, it will provide electrical power to any part plugged into the outlet. It can be programmed to accept laser beams of any color. But if the plug is blue, for instance, it will only accept a blue laser beam.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	32	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, proximity, collision
Destructible	No
Spawns	laser_beam
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	Yes
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	32×32	0
3	1	1	32×32	7
4	2	2	32×32	14
5	3	3	32×32	21
6	4	4	32×32	28
7	5	5	32×32	35
8	6	6	32×32	42
10	7	7–13 (7 frames)	32×32	49
11	14	14–20 (7 frames)	32×32	140
12	21	21–27 (7 frames)	32×32	231

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART96_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART96_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART96_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART96_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART96_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART96_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART96_f006.png :material-check:
7	0001_0000, 0001_0007	build/phase-3/bitmaps/PART96_f000.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
8	0001_0001, 0001_0007	build/phase-3/bitmaps/PART96_f001.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
9	0001_0002, 0001_0007	build/phase-3/bitmaps/PART96_f002.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
10	0001_0003, 0001_0007	build/phase-3/bitmaps/PART96_f003.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
11	0001_0004, 0001_0007	build/phase-3/bitmaps/PART96_f004.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
12	0001_0005, 0001_0007	build/phase-3/bitmaps/PART96_f005.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
13	0001_0006, 0001_0007	build/phase-3/bitmaps/PART96_f006.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
14	0001_0000, 0001_0007	build/phase-3/bitmaps/PART96_f000.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
15	0001_0001, 0001_0007	build/phase-3/bitmaps/PART96_f001.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
16	0001_0002, 0001_0007	build/phase-3/bitmaps/PART96_f002.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
17	0001_0003, 0001_0007	build/phase-3/bitmaps/PART96_f003.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
18	0001_0004, 0001_0007	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART96_f004.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
19	0001_0005, 0001_0007	build/phase-3/bitmaps/PART96_f005.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
20	0001_0006, 0001_0007	build/phase-3/bitmaps/PART96_f006.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
21	0001_0000, 0001_0007, 0001_0007	build/phase-3/bitmaps/PART96_f000.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
22	0001_0001, 0001_0007, 0001_0007	build/phase-3/bitmaps/PART96_f001.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
23	0001_0002, 0001_0007, 0001_0007	build/phase-3/bitmaps/PART96_f002.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
24	0001_0003, 0001_0007, 0001_0007	build/phase-3/bitmaps/PART96_f003.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
25	0001_0004, 0001_0007, 0001_0007	build/phase-3/bitmaps/PART96_f004.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
26	0001_0005, 0001_0007, 0001_0007	build/phase-3/bitmaps/PART96_f005.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:
27	0001_0006, 0001_0007, 0001_0007	build/phase-3/bitmaps/PART96_f006.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check: build/phase-3/bitmaps/PART96_f007.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	56
END_FRAME	28
END_ANM	1

Notas de Comportamento:

Laser-Activated Plug is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part emits a **laser beam**. The beam interacts with mirrors, mixers, and laser-activated plugs.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 97 — Large Pipes



Categoria: Pipe Systems
Fases em que aparece: 16 fases
Quantidade de Estados: 2 estado(s)

Connect sections of this large pipe together and drop balls or other things inside. Attach sections of curved pipe and t-connectors to control the direction that objects go. You can also attach an accelerator tube to speed up or reverse the direction of objects moving through the pipes.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	48	dimension 1 width
dim_h1	48	dimension 1 height
dim_w2	576	dimension 2 width
dim_h2	416	dimension 2 height
bitmap_w	16	bitmap frame width
bitmap_h	38	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical
Destructible	No
Spawns	(none)
Category Part Count	4

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–2 (3 frames)	16×38	0
2	3	3–5 (3 frames)	40×16	21

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART97_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART97_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART97_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART97_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART97_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART97_f005.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	6
END_FRAME	6
END_ANM	1

Notas de Comportamento:

Large Pipes is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 98 — T-Connector



Categoria: Pipe Systems
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Use this T-Connector to hitch sections of large or curved pipe together. Drop balls and other objects into the openings. Attach an accelerator tube to speed up or reverse the direction of objects moving through the pipes.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000

Property	Raw Value	Interpretation
unk_6	16	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	80	bitmap frame width
bitmap_h	80	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical
Destructible	No
Spawns	(none)
Category Part Count	4

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	80×80	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART98.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

T-Connector is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 99 — Grass Incline



Categoria: Inclines
Fases em que aparece: 8 fases
Quantidade de Estados: 1 estado(s)

Roll balls and other parts up or down this grass incline. Use it to control the direction of balloons and other objects. It can be stretched or shrunk, which changes the angle. It's not affected by explosives.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1000	mass/density (relative scale)
unk_2	1000	category-specific property
friction_q8	128	friction = 0.500
unk_6	128	category-specific property
tile_w	16	base width in tile units
tile_h	16	base height in tile units
collision_w	64	AABB collision width
collision_h	0	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	33	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	6

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No

Capability	Supported
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	16×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART99_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART99_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART99_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART99_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Grass Incline is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).

Peça 100 — Log Incline



Categoria: Inclines
Fases em que aparece: 15 fases
Quantidade de Estados: 1 estado(s)

Roll balls and other parts up or down this log incline. Use it to control the direction of balloons and other objects. It can be stretched or shrunk, which changes the angle. It's not affected by explosives.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4193	mass/density (relative scale)
unk_2	0	category-specific property

Property	Raw Value	Interpretation
friction_q8	96	friction = 0.375
unk_6	96	category-specific property
tile_w	16	base width in tile units
tile_h	0	base height in tile units
collision_w	64	AABB collision width
collision_h	0	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	33	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	6

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	16×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART100_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART100_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART100_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART100_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Log Incline is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).

Peça 101 — Granite Incline



Categoria: Inclines
Fases em que aparece: 47 fases
Quantidade de Estados: 1 estado(s)

Roll balls and other parts up or down this granite incline. Use it to control the direction of balloons and other objects. It can be stretched or shrunk, which changes the angle. It's not affected by explosives.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4053	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	1024	friction = 4.000
unk_6	16	category-specific property
tile_w	16	base width in tile units
tile_h	0	base height in tile units
collision_w	64	AABB collision width
collision_h	0	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	6

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No

Capability	Supported
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	16×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART101_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART101_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART101_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART101_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Granite Incline is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).

Peça 102 — Brick Incline



Categoria: Inclines
Fases em que aparece: 25 fases
Quantidade de Estados: 1 estado(s)

Roll balls and other parts up or down this brick incline. It can be stretched or shrunk, which changes the angle. Use it to control the direction of balloons and other objects. It's not affected by explosives.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4053	mass/density (relative scale)
unk_2	1000	category-specific property
friction_q8	1024	friction = 4.000
unk_6	24	category-specific property
tile_w	16	base width in tile units
tile_h	0	base height in tile units
collision_w	64	AABB collision width
collision_h	0	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	6

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	16×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART102_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART102_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART102_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART102_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4

Opcode	Count
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Brick Incline is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).

Peça 103 — Archway



Categoria: Walls / Floors
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

This big granite archway can be used to fill spaces so objects can't pass through. You can set things on top of it, or bounce things off it. But you can't blow it up.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	1024	friction = 4.000
unk_6	22	category-specific property
tile_w	0	base width in tile units
tile_h	0	base height in tile units
collision_w	0	AABB collision width
collision_h	0	AABB collision height
bitmap_w	64	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	64x64	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART103.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Archway is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 104 — Wooden Barrier



Categoria: Walls / Floors
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

This large wooden barrier can be used to fill spaces so objects can't pass through. You can set things on top of it, or bounce things off it. But you can't blow it up.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	1024	friction = 4.000
unk_6	16	category-specific property
tile_w	0	base width in tile units
tile_h	0	base height in tile units
collision_w	0	AABB collision width
collision_h	0	AABB collision height
bitmap_w	64	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	64×64	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART104.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Wooden Barrier is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 105 — Scaffold Barrier



Categoria: Walls / Floors
Fases em que aparece: 10 fases
Quantidade de Estados: 1 estado(s)

This chunk of metal scaffolding can be used to fill spaces so objects can't pass through. You can set things on top of it, or bounce things off it. But you can't blow it up.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	1024	friction = 4.000
unk_6	24	category-specific property
tile_w	0	base width in tile units
tile_h	0	base height in tile units
collision_w	0	AABB collision width
collision_h	0	AABB collision height
bitmap_w	64	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	64x64	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART105.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Scaffold Barrier is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 106 — Lattice Archway



Categoria: Walls / Floors
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

This large lattice archway can be used to fill spaces so objects can't pass through. You can set things on top of it, or bounce things off it. But you can't blow it up.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
friction_q8	1024	friction = 4.000
unk_6	20	category-specific property
tile_w	0	base width in tile units
tile_h	0	base height in tile units
collision_w	0	AABB collision width
collision_h	0	AABB collision height
bitmap_w	64	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0	64x64	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART106.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Lattice Archway is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 107 — Electric Mixer



Categoria: Electrical
Fases em que aparece: 10 fases
Quantidade de Estados: 2 estado(s)

Plug this electric mixer into an outlet to make it run.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Mixing	2→1	2	No
Mixing	1→-1	1	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	0	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	16	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	49	bitmap frame width
bitmap_h	55	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	56×55	0
1	1	1–4 (4 frames)	56×59	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART107_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART107_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART107_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART107_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART107_f004.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	107	0	build/phase-6/raw-wav/SX_107_11025.wav
2	107	0	build/phase-6/raw-wav/SX_107_11025.wav
3	107	0	build/phase-6/raw-wav/SX_107_11025.wav
4	107	0	build/phase-6/raw-wav/SX_107_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	5
END_FRAME	5
PLAY_SFX	4
END_ANM	1

Notas de Comportamento:

Electric Mixer has 2 semantic states: Not Mixing, Mixing. Transitions are triggered by: electrical, collision.

Terminal state(s): Mixing (part is removed from simulation).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 108 — Leaky Bucket



Categoria: Containers
Fases em que aparece: 11 fases
Quantidade de Estados: 8 estado(s)

There's a hole in the bottom of this bucket. You can program how fast the water leaks out. The faster it drips, the heavier the bucket is when you start the puzzle. As the contents drip out, the bucket gets lighter. Tie a rope to the top and use pulleys to connect it to another object.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Dripping	1→4	1	No
Empty	4→10	4	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	7552	mass/density (relative scale)
unk_2	100	category-specific property
property_q8	32	Q8.8 = 0.125
unk_6	48	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	35	bitmap frame width
bitmap_h	50	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	40×50	0
9	1	1–7 (7 frames)	40×50	7
8	8	8–17 (10 frames)	40×50	88
7	18	18–29 (12 frames)	40×50	202
6	30	30–50 (21 frames)	40×60	350
5	51	51–79 (29 frames)	0×0	0
4	80	80–89 (10 frames)	0×0	0
10	90	90	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART108_f000.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART108_f000.png :material-check:
2	0001_0000, 0001_0001	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f001.png :material-check:
3	0001_0000, 0001_0002	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f002.png :material-check:
4	0001_0000, 0001_0003	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f003.png :material-check:
5	0001_0000, 0001_0004	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f004.png :material-check:
6	0001_0000, 0001_0005	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f005.png :material-check:
7	0001_0000	build/phase-3/bitmaps/PART108_f000.png :material-check:
8	0001_0000	build/phase-3/bitmaps/PART108_f000.png :material-check:
9	0001_0000, 0001_0001	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f001.png :material-check:
10	0001_0000, 0001_0001	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f001.png :material-check:
11	0001_0000, 0001_0002	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f002.png :material-check:
12	0001_0000, 0001_0003	build/phase-3/bitmaps/PART108_f000.png :material-check: build/phase-3/bitmaps/PART108_f003.png :material-check:
13	0001_0000, 0001_0004	

[illegible]

Frame	Sprites	PNG References
39	0001_0000	build/phase-3/bitmaps/PART108_f000.png :material-check:
40	0001_0000	build/phase-3/bitmaps/PART108_f000.png :material-check:
41	0001_0006	build/phase-3/bitmaps/PART108_f006.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
2	3108	0	build/phase-6/raw-wav/SX_3108_11025.wav
9	3108	0	build/phase-6/raw-wav/SX_3108_11025.wav
19	3108	0	build/phase-6/raw-wav/SX_3108_11025.wav
28	3108	0	build/phase-6/raw-wav/SX_3108_11025.wav
36	3108	0	build/phase-6/raw-wav/SX_3108_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	71
END_FRAME	42
PLAY_SFX	5
END_ANM	1

Notas de Comportamento:

Leaky Bucket has 2 semantic states: Dripping, Empty. Transitions are triggered by: collision.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 109 — Blimp



Categoria: Balloons / Airships
Fases em que aparece: 47 fases
Quantidade de Estados: 7 estado(s)

This blimp will fly in a straight line until it bumps into something and reverses direction. It will pop if it bumps into a moving gear, a tack, or certain other sharp objects, and it will blow up if it touches a flame or explosion.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Normal	1→2	1	No
Popped	2→5	7	No
Blown Up	5→-1	6	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	9	mass/density (relative scale)
unk_2	28000	category-specific property
property_q8	64	Q8.8 = 0.250
unk_6	32	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	53	bitmap frame width
bitmap_h	36	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, proximity
Destructible	Yes
Spawns	(none)
Category Part Count	3

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–4 (5 frames)	56×36	0
3	5	5–7 (3 frames)	48×41	73
4	8	8–10 (3 frames)	48×41	100
2	11	11–15 (5 frames)	56×35	127

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
7	16	16	32×68	170
5	17	17–21 (5 frames)	56×35	177
6	22	22–23 (2 frames)	32×68	316

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001	build/phase-3/bitmaps/PART109_f000.png :material-check: build/phase-3/bitmaps/PART109_f001.png :material-check:
1	0001_0000, 0001_0002	build/phase-3/bitmaps/PART109_f000.png :material-check: build/phase-3/bitmaps/PART109_f002.png :material-check:
2	0001_0000, 0001_0003	build/phase-3/bitmaps/PART109_f000.png :material-check: build/phase-3/bitmaps/PART109_f003.png :material-check:
3	0001_0000, 0001_0004	build/phase-3/bitmaps/PART109_f000.png :material-check: build/phase-3/bitmaps/PART109_f004.png :material-check:
4	0001_0000, 0001_0005	build/phase-3/bitmaps/PART109_f000.png :material-check: build/phase-3/bitmaps/PART109_f005.png :material-check:
5	0001_0006	build/phase-3/bitmaps/PART109_f006.png :material-check:
6	0001_0007	build/phase-3/bitmaps/PART109_f007.png :material-check:
7	0001_0006	build/phase-3/bitmaps/PART109_f006.png :material-check:
8	0001_0006	build/phase-3/bitmaps/PART109_f006.png :material-check:
9	0001_0007	build/phase-3/bitmaps/PART109_f007.png :material-check:
10	0001_0006	build/phase-3/bitmaps/PART109_f006.png :material-check:
11	0001_0000, 0001_0005	build/phase-3/bitmaps/PART109_f000.png :material-check: build/phase-3/bitmaps/PART109_f005.png :material-check:
12	0001_0008	build/phase-3/bitmaps/PART109_f008.png :material-check:
13	0001_0008	build/phase-3/bitmaps/PART109_f008.png :material-check:
14	0001_0009	build/phase-3/bitmaps/PART109_f009.png :material-check:

Frame	Sprites	PNG References
15	0001_0010	build/phase-3/bitmaps/PART109_f010.png :material-check:
16	0001_0011	build/phase-3/bitmaps/PART109_f011.png :material-check:
17	0001_0000, 0001_0005	build/phase-3/bitmaps/PART109_f000.png :material-check: build/phase-3/bitmaps/PART109_f005.png :material-check:
18	0001_0008, 0002_0006, 0002_0007, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART109_f008.png :material-check: build/phase-3/bitmaps/PART109_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f000.png :material-alert-outline: (not found)
19	0001_0008, 0002_0006, 0002_0007, 0002_0009, 0002_0000	build/phase-3/bitmaps/PART109_f008.png :material-check: build/phase-3/bitmaps/PART109_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f000.png :material-alert-outline: (not found)
20	0001_0009, 0002_0007, 0002_0006, 0002_0001, 0002_0008	build/phase-3/bitmaps/PART109_f009.png :material-check: build/phase-3/bitmaps/PART109_2_f007.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f001.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f008.png :material-alert-outline: (not found)
21	0001_0010, 0002_0002, 0002_0009, 0002_0008, 0002_0007	build/phase-3/bitmaps/PART109_f010.png :material-check: build/phase-3/bitmaps/PART109_2_f002.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f008.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f007.png :material-alert-outline: (not found)
22	0001_0011, 0002_0003, 0002_0005, 0002_0008, 0002_0009, 0002_0006	build/phase-3/bitmaps/PART109_f011.png :material-check: build/phase-3/bitmaps/PART109_2_f003.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f005.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f008.png :material-

Frame	Sprites	PNG References
		alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f006.png :material-alert-outline: (not found)
23	0001_0011, 0002_0004, 0002_0009, 0002_0006, 0002_0005	build/phase-3/bitmaps/PART109_f011.png :material-check: build/phase-3/bitmaps/PART109_2_f004.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f009.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f006.png :material-alert-outline: (not found) build/phase-3/bitmaps/PART109_2_f005.png :material-alert-outline: (not found)

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	109	0	build/phase-6/raw-wav/SX_109_11025.wav
2	109	0	build/phase-6/raw-wav/SX_109_11025.wav
3	109	0	build/phase-6/raw-wav/SX_109_11025.wav
4	109	0	build/phase-6/raw-wav/SX_109_11025.wav
5	109	0	build/phase-6/raw-wav/SX_109_11025.wav
6	109	0	build/phase-6/raw-wav/SX_109_11025.wav
7	109	0	build/phase-6/raw-wav/SX_109_11025.wav
8	109	0	build/phase-6/raw-wav/SX_109_11025.wav
9	109	0	build/phase-6/raw-wav/SX_109_11025.wav
10	109	0	build/phase-6/raw-wav/SX_109_11025.wav
12	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav
18	3019	0	build/phase-6/raw-wav/SX_3019_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	56
END_FRAME	24
PLAY_SFX	12
END_ANM	1

Notas de Comportamento:

Blimp has 3 semantic states: Normal, Popped, Blown Up. Transitions are triggered by: collision, proximity.

Terminal state(s): Blown Up (part is removed from simulation).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 110 — Elm Tree



Categoria: Scenery
Fases em que aparece: 7 fases
Quantidade de Estados: 0 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	900	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	0	bitmap frame width
bitmap_h	0	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART110.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Elm Tree is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 111 — Spruce Tree



Categoria: Scenery
Fases em que aparece: 10 fases
Quantidade de Estados: 0 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	900	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	59	bitmap frame width
bitmap_h	88	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No

Capability	Supported
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART111.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Spruce Tree is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 112 — Cactus



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 0 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	900	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	47	bitmap frame width
bitmap_h	98	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART112.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Cactus is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 113 — Sun



Categoria: Scenery
Fases em que aparece: 27 fases
Quantidade de Estados: 0 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	1200	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	56	bitmap frame width
bitmap_h	53	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART113.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Sun is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 114 — Big Cloud



Categoria: Scenery
Fases em que aparece: 24 fases
Quantidade de Estados: 0 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	1100	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	90	bitmap frame width
bitmap_h	41	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART114.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1

Opcode	Count
END_ANM	1

Notas de Comportamento:

Big Cloud is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 115 — Little Cloud



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	1100	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	65	bitmap frame width
bitmap_h	40	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No

Capability	Supported
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	72×40	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART115.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Little Cloud is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 116 — Pinball Flipper



Categoria: Springs / Pneumatics
Fases em que aparece: 29 fases
Quantidade de Estados: 2 estado(s)

This pinball flipper will flick any object that drops on top of it. It can be flipped left or right.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	32	category-specific property

Property	Raw Value	Interpretation
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	45	bitmap frame width
bitmap_h	41	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	48×41	0
3	1	1	64×17	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART116_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART116_f001.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3116	0	build/phase-6/raw-wav/SX_3116_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	2
END_FRAME	2

Opcode	Count
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Pinball Flipper is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 117 — Pool Cue



Categoria: Springs / Pneumatics
Fases em que aparece: 29 fases
Quantidade de Estados: 24 estado(s)

This pool cue is spring- loaded and ready to shoot anytime something bumps the button on the back end. It can be rotated to shoot from eight different angles. Use it to hit pool balls and other things.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Hit	3→4, 12→13, 15→17, 19→20, 22→23, 25→26, 28→29, 31→32	31	No
Hit	4→12, 13→15, 17→19, 20→22, 23→25, 26→28, 29→31, 32→-1	33	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1500	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	28	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	78	bitmap frame width
bitmap_h	78	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical
Destructible	No
Spawns	(none)
Category Part Count	7

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0	48×9	0
4	1	1–5 (5 frames)	32×9	7
9	6	6	48×9	44
12	7	7	40×33	51
13	8	8–12 (5 frames)	32×26	58
14	13	13	40×33	95
15	14	14	16×43	102
17	15	15–19 (5 frames)	16×32	109
18	20	20	16×43	146
19	21	21	40×33	153
20	22	22–26 (5 frames)	32×26	160
21	27	27	40×33	197
22	28	28	48×9	204
23	29	29–33 (5 frames)	32×9	211
24	34	34	48×9	248
25	35	35	40×33	255
26	36	36–40 (5 frames)	32×26	262
27	41	41	40×33	299
28	42	42	16×43	306
29	43	43–47 (5 frames)	16×32	313
30	48	48	16×43	350
31	49	49	40×33	357
32	50	50–54 (5 frames)	32×26	364
33	55	55	40×33	401

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART117_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART117_f001.png :material-check:
2	0001_0001	build/phase-3/bitmaps/PART117_f001.png :material-check:
3	0001_0002	build/phase-3/bitmaps/PART117_f002.png :material-check:
4	0001_0003	build/phase-3/bitmaps/PART117_f003.png :material-check:
5	0001_0004	build/phase-3/bitmaps/PART117_f004.png :material-check:
6	0001_0000	build/phase-3/bitmaps/PART117_f000.png :material-check:
7	0001_0010	build/phase-3/bitmaps/PART117_f010.png :material-check:
8	0001_0011	build/phase-3/bitmaps/PART117_f011.png :material-check:
9	0001_0011	build/phase-3/bitmaps/PART117_f011.png :material-check:
10	0001_0012	build/phase-3/bitmaps/PART117_f012.png :material-check:
11	0001_0013	build/phase-3/bitmaps/PART117_f013.png :material-check:
12	0001_0014	build/phase-3/bitmaps/PART117_f014.png :material-check:
13	0001_0010	build/phase-3/bitmaps/PART117_f010.png :material-check:
14	0001_0005	build/phase-3/bitmaps/PART117_f005.png :material-check:
15	0001_0006	build/phase-3/bitmaps/PART117_f006.png :material-check:
16	0001_0006	build/phase-3/bitmaps/PART117_f006.png :material-check:
17	0001_0007	build/phase-3/bitmaps/PART117_f007.png :material-check:
18	0001_0008	build/phase-3/bitmaps/PART117_f008.png :material-check:
19	0001_0009	build/phase-3/bitmaps/PART117_f009.png :material-check:
20	0001_0005	build/phase-3/bitmaps/PART117_f005.png :material-check:
21	0001_0010	build/phase-3/bitmaps/PART117_f010.png :material-check:
22	0001_0011	build/phase-3/bitmaps/PART117_f011.png :material-check:
23	0001_0011	build/phase-3/bitmaps/PART117_f011.png :material-check:
24	0001_0012	build/phase-3/bitmaps/PART117_f012.png :material-check:
25	0001_0013	build/phase-3/bitmaps/PART117_f013.png :material-check:
26	0001_0014	build/phase-3/bitmaps/PART117_f014.png :material-check:
27	0001_0010	build/phase-3/bitmaps/PART117_f010.png :material-check:
28	0001_0000	build/phase-3/bitmaps/PART117_f000.png :material-check:
29	0001_0001	build/phase-3/bitmaps/PART117_f001.png :material-check:
30	0001_0001	build/phase-3/bitmaps/PART117_f001.png :material-check:
31	0001_0002	build/phase-3/bitmaps/PART117_f002.png :material-check:
32	0001_0003	build/phase-3/bitmaps/PART117_f003.png :material-check:
33	0001_0004	build/phase-3/bitmaps/PART117_f004.png :material-check:
34	0001_0000	build/phase-3/bitmaps/PART117_f000.png :material-check:
35	0001_0010	build/phase-3/bitmaps/PART117_f010.png :material-check:
36	0001_0011	build/phase-3/bitmaps/PART117_f011.png :material-check:
37		

Frame	Sprites	PNG References
	0001_0011	build/phase-3/bitmaps/PART117_f011.png :material-check:
38	0001_0012	build/phase-3/bitmaps/PART117_f012.png :material-check:
39	0001_0013	build/phase-3/bitmaps/PART117_f013.png :material-check:
40	0001_0014	build/phase-3/bitmaps/PART117_f014.png :material-check:
41	0001_0010	build/phase-3/bitmaps/PART117_f010.png :material-check:
42	0001_0005	build/phase-3/bitmaps/PART117_f005.png :material-check:
43	0001_0006	build/phase-3/bitmaps/PART117_f006.png :material-check:
44	0001_0006	build/phase-3/bitmaps/PART117_f006.png :material-check:
45	0001_0007	build/phase-3/bitmaps/PART117_f007.png :material-check:
46	0001_0008	build/phase-3/bitmaps/PART117_f008.png :material-check:
47	0001_0009	build/phase-3/bitmaps/PART117_f009.png :material-check:
48	0001_0005	build/phase-3/bitmaps/PART117_f005.png :material-check:
49	0001_0010	build/phase-3/bitmaps/PART117_f010.png :material-check:
50	0001_0011	build/phase-3/bitmaps/PART117_f011.png :material-check:
51	0001_0011	build/phase-3/bitmaps/PART117_f011.png :material-check:
52	0001_0012	build/phase-3/bitmaps/PART117_f012.png :material-check:
53	0001_0013	build/phase-3/bitmaps/PART117_f013.png :material-check:
54	0001_0014	build/phase-3/bitmaps/PART117_f014.png :material-check:
55	0001_0010	build/phase-3/bitmaps/PART117_f010.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
1	3117	0	build/phase-6/raw-wav/SX_3117_11025.wav
8	3117	0	build/phase-6/raw-wav/SX_3117_11025.wav
15	3117	0	build/phase-6/raw-wav/SX_3117_11025.wav
22	3117	0	build/phase-6/raw-wav/SX_3117_11025.wav
29	3117	0	build/phase-6/raw-wav/SX_3117_11025.wav
36	3117	0	build/phase-6/raw-wav/SX_3117_11025.wav
43	3117	0	build/phase-6/raw-wav/SX_3117_11025.wav
50	3117	0	build/phase-6/raw-wav/SX_3117_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	56
END_FRAME	56
PLAY_SFX	8
END_ANM	1

Notas de Comportamento:

Pool Cue has 2 semantic states: Not Hit, Hit. Transitions are triggered by: collision, electrical.

Terminal state(s): Hit (part is removed from simulation).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 118 — Pool Table Wall



Categoria: Pool Table
Fases em que aparece: 7 fases
Quantidade de Estados: 3 estado(s)

You can build your own virtual pool table with these felt-covered walls. Balls will bounce off them. Add pockets wherever you want them. Flip these walls if necessary.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	12	category-specific property
dim_w1	32	dimension 1 width
dim_h1	32	dimension 1 height
dim_w2	576	dimension 2 width
dim_h2	416	dimension 2 height
bitmap_w	32	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	2

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–10 (11 frames)	16×32	0
2	11	11–21 (11 frames)	32×16	77
3	22	22	32×32	154

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART118_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART118_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART118_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART118_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART118_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART118_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART118_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART118_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART118_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART118_f009.png :material-check:
10	0001_0010	build/phase-3/bitmaps/PART118_f010.png :material-check:
11	0001_0011	build/phase-3/bitmaps/PART118_f011.png :material-check:
12	0001_0012	build/phase-3/bitmaps/PART118_f012.png :material-check:
13	0001_0013	build/phase-3/bitmaps/PART118_f013.png :material-check:
14	0001_0014	build/phase-3/bitmaps/PART118_f014.png :material-check:
15	0001_0015	build/phase-3/bitmaps/PART118_f015.png :material-check:
16	0001_0016	build/phase-3/bitmaps/PART118_f016.png :material-check:
17	0001_0017	build/phase-3/bitmaps/PART118_f017.png :material-check:
18	0001_0018	build/phase-3/bitmaps/PART118_f018.png :material-check:
19	0001_0019	build/phase-3/bitmaps/PART118_f019.png :material-check:
20	0001_0020	build/phase-3/bitmaps/PART118_f020.png :material-check:
21	0001_0021	build/phase-3/bitmaps/PART118_f021.png :material-check:
22	0001_0022	build/phase-3/bitmaps/PART118_f022.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	23
END_FRAME	23
END_ANM	1

Notas de Comportamento:

Pool Table Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).

Peça 119 — Pool Table Pocket



Categoria: Pool Table
Fases em que aparece: 7 fases
Quantidade de Estados: 16 estado(s)

These pool table pockets can be rotated to eight different angles. Use them with sections of pool table wall to build your own billiards game. Anything that drops into one of these pockets will disappear.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Hole Empty	1→3, 3→4, 4→5, 5→6, 6→7, 7→8, 8→9, 9→11	—	No
Hole Full	11→12, 12→13, 13→14, 14→15, 15→16, 16→17, 17→18, 18→-1	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	1000	mass/density (relative scale)
unk_2	200	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	28	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	64	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision
Destructible	No
Spawns	(none)
Category Part Count	2

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	64×64	0
3	1	1	32×48	7
4	2	2	64×64	14
5	3	3	48×32	21
6	4	4	64×64	28
7	5	5	32×48	35
8	6	6	64×64	42
9	7	7	48×32	49
11	8	8	64×64	56
12	9	9	32×48	63
13	10	10	64×64	70
14	11	11	48×32	77
15	12	12	64×64	84
16	13	13	32×48	91
17	14	14	64×64	98
18	15	15	48×32	105

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART119_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART119_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART119_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART119_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART119_f004.png :material-check:

Frame	Sprites	PNG References
5	0001_0005	build/phase-3/bitmaps/PART119_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART119_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART119_f007.png :material-check:
8	0001_0000	build/phase-3/bitmaps/PART119_f000.png :material-check:
9	0001_0001	build/phase-3/bitmaps/PART119_f001.png :material-check:
10	0001_0002	build/phase-3/bitmaps/PART119_f002.png :material-check:
11	0001_0003	build/phase-3/bitmaps/PART119_f003.png :material-check:
12	0001_0004	build/phase-3/bitmaps/PART119_f004.png :material-check:
13	0001_0005	build/phase-3/bitmaps/PART119_f005.png :material-check:
14	0001_0006	build/phase-3/bitmaps/PART119_f006.png :material-check:
15	0001_0007	build/phase-3/bitmaps/PART119_f007.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	16
END_FRAME	16
END_ANM	1

Notas de Comportamento:

Pool Table Pocket has 2 semantic states: Hole Empty, Hole Full. Transitions are triggered by: collision.

Terminal state(s): Hole Full (part is removed from simulation).

Casos de Teste Sugeridos (Gerados Automáticos):

- T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 120 — Electrical Outlet



Categoria: Electrical
Fases em que aparece: 32 fases
Quantidade de Estados: 4 estado(s)

This electrical outlet has juice running to it at all times. Plug in any electrical part and it will work.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	25	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, collision
Destructible	No
Spawns	(none)
Category Part Count	9

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	32×32	0
2	1	1	32×32	7
3	2	2	32×32	20
4	3	3	32×32	33

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART120_f000.png :material-check:
1	0001_0000, 0001_0001	build/phase-3/bitmaps/PART120_f000.png :material-check: build/phase-3/bitmaps/PART120_f001.png :material-check:
2	0001_0000, 0001_0001	build/phase-3/bitmaps/PART120_f000.png :material-check: build/phase-3/bitmaps/PART120_f001.png :material-check:
3	0001_0000, 0001_0001, 0001_0001	build/phase-3/bitmaps/PART120_f000.png :material-check: build/phase-3/bitmaps/PART120_f001.png :material-check: build/phase-3/bitmaps/PART120_f001.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	8
END_FRAME	4

Opcode	Count
END_ANM	1

Notas de Comportamento:

Electrical Outlet is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 121 — Vine Tile



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	64	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	64×64	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART121.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Vine Tile is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 122 — tile



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	96	bitmap frame width
bitmap_h	80	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	96×80	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART122.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

tile is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 123 — tile



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	64	bitmap frame width
bitmap_h	64	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	64x64	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART123.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

tile is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the appearance field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (queue_free).
- **T-06 (Cenário / Background):** Validar se o z_index é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 124 — tile



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	80	bitmap frame width
bitmap_h	48	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	80×48	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART124.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

tile is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 125 — Yellow Brick Wall



Categoria: Walls / Floors
Fases em que aparece: 32 fases
Quantidade de Estados: 2 estado(s)

This is a yellow brick wall or floor. Stretch it vertically or horizontally to any length you need. Its surface is pretty slippery. Explosions will blow holes through it.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
friction_q8	1024	friction = 4.000
unk_6	20	category-specific property
tile_w	16	base width in tile units
tile_h	16	base height in tile units
collision_w	576	AABB collision width

Property	Raw Value	Interpretation
collision_h	416	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	16	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	4	4–7 (4 frames)	16×16	28
1	8	8–19 (12 frames)	40×16	56

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART125_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART125_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART125_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART125_f003.png :material-check:
4	0001_0004	build/phase-3/bitmaps/PART125_f004.png :material-check:
5	0001_0005	build/phase-3/bitmaps/PART125_f005.png :material-check:
6	0001_0006	build/phase-3/bitmaps/PART125_f006.png :material-check:
7	0001_0007	build/phase-3/bitmaps/PART125_f007.png :material-check:
8	0001_0008	build/phase-3/bitmaps/PART125_f008.png :material-check:
9	0001_0009	build/phase-3/bitmaps/PART125_f009.png :material-check:
10	0001_0008	build/phase-3/bitmaps/PART125_f008.png :material-check:
11	0001_0009	build/phase-3/bitmaps/PART125_f009.png :material-check:
12	0001_0008	build/phase-3/bitmaps/PART125_f008.png :material-check:
13		

Frame	Sprites	PNG References
	0001_0009	build/phase-3/bitmaps/PART125_f009.png :material-check:
14	0001_0010	build/phase-3/bitmaps/PART125_f010.png :material-check:
15	0001_0010	build/phase-3/bitmaps/PART125_f010.png :material-check:
16	0001_0010	build/phase-3/bitmaps/PART125_f010.png :material-check:
17	0001_0011	build/phase-3/bitmaps/PART125_f011.png :material-check:
18	0001_0011	build/phase-3/bitmaps/PART125_f011.png :material-check:
19	0001_0011	build/phase-3/bitmaps/PART125_f011.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	20
END_FRAME	20
END_ANM	1

Notas de Comportamento:

Yellow Brick Wall is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 126 — Yellow Brick Incline



Categoria: Inclines
Fases em que aparece: 38 fases
Quantidade de Estados: 1 estado(s)

Roll balls and other parts up or down this yellow brick incline. Use it to control the direction of balloons and other objects. It can be stretched or shrunk, which changes the angle. It's not affected by explosives.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	4153	mass/density (relative scale)
unk_2	1000	category-specific property
friction_q8	1024	friction = 4.000

Property	Raw Value	Interpretation
unk_6	20	category-specific property
tile_w	16	base width in tile units
tile_h	0	base height in tile units
collision_w	64	AABB collision width
collision_h	0	AABB collision height
bitmap_w	16	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	No
Spawns	(none)
Category Part Count	6

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0–3 (4 frames)	16×32	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART126_f000.png :material-check:
1	0001_0001	build/phase-3/bitmaps/PART126_f001.png :material-check:
2	0001_0002	build/phase-3/bitmaps/PART126_f002.png :material-check:
3	0001_0003	build/phase-3/bitmaps/PART126_f003.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	4
END_FRAME	4
END_ANM	1

Notas de Comportamento:

Yellow Brick Incline is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).

Peça 128 — purple mountain



Categoria: Scenery
Fases em que aparece: 11 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	189	bitmap frame width
bitmap_h	71	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	192×71	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART128.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

purple mountain is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 129 — desert mesa



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	87	bitmap frame width
bitmap_h	72	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
6	0	0	88x72	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART129.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

desert mesa is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 130 — Glass Building



Categoria: Scenery
Fases em que aparece: 7 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	37	bitmap frame width
bitmap_h	111	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	40×111	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART130.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Glass Building is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 131 — apartment building



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	77	bitmap frame width
bitmap_h	81	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	80×81	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART131.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

apartment building is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 132 — office building



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	77	bitmap frame width
bitmap_h	44	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No

Capability	Supported
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	80×44	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART132.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

office building is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 133 — car



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	110	bitmap frame width
bitmap_h	35	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	112×35	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART133.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

car is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 134 — lamp post



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	20	bitmap frame width
bitmap_h	109	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×109	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART134.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

lamp post is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 135 — fire hydrant



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	29	bitmap frame width
bitmap_h	38	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No

Capability	Supported
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	32x38	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART135.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

fire hydrant is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 136 — Message Computer



Categoria: Special Mechanics
Fases em que aparece: 4 fases
Quantidade de Estados: 3 estado(s)

This little computer is a handy way to relay messages one letter at a time. A letter will appear on the monitor if something bumps the keyboard. The computer may be programmed to display any letter in the alphabet, numbers 0 through 9, and several symbols. Line them up side by side to spell out a whole message.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Off	4→5	—	No
On	5→-1	5	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2400	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	128	Q8.8 = 0.500
unk_6	12	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	46	bitmap frame width
bitmap_h	43	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
4	0	0	48×43	0
5	1	1	48×43	13
6	2	2–8 (7 frames)	48×43	20

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000, 0001_0001	build/phase-3/bitmaps/PART136_f000.png :material-check: build/phase-3/bitmaps/PART136_f001.png :material-check:
1	0001_0000	build/phase-3/bitmaps/PART136_f000.png :material-check:
2	0001_0000, 0002_0000	build/phase-3/bitmaps/PART136_f000.png :material-check: build/phase-3/bitmaps/PART136_2_f000.png :material-alert-outline: (not found)
3	0001_0000, 0002_0001	build/phase-3/bitmaps/PART136_f000.png :material-check: build/phase-3/bitmaps/PART136_2_f001.png :material-alert-outline: (not found)
4	0001_0000, 0002_0002	build/phase-3/bitmaps/PART136_f000.png :material-check: build/phase-3/bitmaps/PART136_2_f002.png :material-alert-outline: (not found)
5	0001_0000, 0002_0003	build/phase-3/bitmaps/PART136_f000.png :material-check: build/phase-3/bitmaps/PART136_2_f003.png :material-alert-outline: (not found)
6	0001_0000, 0002_0004	build/phase-3/bitmaps/PART136_f000.png :material-check: build/phase-3/bitmaps/PART136_2_f004.png :material-alert-outline: (not found)
7	0001_0000, 0002_0005	build/phase-3/bitmaps/PART136_f000.png :material-check: build/phase-3/bitmaps/PART136_2_f005.png :material-alert-outline: (not found)
8	0001_0000, 0002_0006	build/phase-3/bitmaps/PART136_f000.png :material-check: build/phase-3/bitmaps/PART136_2_f006.png :material-alert-outline: (not found)

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	17
END_FRAME	9
END_ANM	1

Notas de Comportamento:

Message Computer has 2 semantic states: Off, On. Transitions are triggered by: collision, electrical, proximity.

Terminal state(s): On (part is removed from simulation).

This part has custom spawn behavior defined in SOLVE.RES.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 137 — Egg Timer



Categoria: Special Mechanics
Fases em que aparece: 46 fases
Quantidade de Estados: 26 estado(s)

Program this egg timer to a desired amount of time. Then drop something on the top knob to make it start.
When the time is up, an arm pops out, bumping anything in its way.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Unactivated	22→37, 23→38, 26→39, 28→40, 30→41, 36→42, 33→43, 34→44	—	No
Time Out	19→-1, 19→-1, 19→-1, 19→-1, 19→-1, 19→-1, 19→-1, 19→-1	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
unk_2	120	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	26	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	49	bitmap frame width
bitmap_h	40	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No

Capability	Supported
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
22	0	0	50×40	0
37	1	1	50×36	37
21	2	2	50×33	74
23	3	3	50×40	105
38	4	4	50×36	142
24	5	5	50×33	179
26	6	6	50×40	210
39	7	7	50×36	247
25	8	8	50×33	284
28	9	9	50×40	315
40	10	10	50×36	352
27	11	11	50×33	389
30	12	12	50×40	420
41	13	13	50×36	457
29	14	14	50×33	494
36	15	15	50×40	525
42	16	16	50×36	562
31	17	17	50×33	599
33	18	18	50×40	630
43	19	19	50×36	667
32	20	20	50×33	704
34	21	21	50×40	735
44	22	22	50×36	772
35	23	23	50×33	809
19	24	24–31 (8 frames)	50×33	840
20	32	32	50×33	1186

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0003, 0001_0004, 0001_0001, 0001_0000, 0001_0002, 0001_0006	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART137_f006.png :material-check:
1	0001_0004, 0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0006	build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f006.png :material-check:
2	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0006	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f006.png :material-check:
3	0001_0003, 0001_0004, 0001_0001, 0001_0000, 0001_0002, 0001_0007	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f007.png :material-check:
4	0001_0004, 0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0007	build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f007.png :material-check:
5	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0007	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART137_f007.png :material-check:
6	0001_0003, 0001_0004, 0001_0001, 0001_0000, 0001_0002, 0001_0008	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f008.png :material-check:
7	0001_0004, 0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0008	build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f008.png :material-check:
8	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0008	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f008.png :material-check:
9	0001_0003, 0001_0004, 0001_0001, 0001_0000, 0001_0002, 0001_0009	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f009.png :material-check:
10	0001_0004, 0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0009	build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/

Frame	Sprites	PNG References
		PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f009.png :material-check:
11	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0009	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f009.png :material-check:
12	0001_0003, 0001_0004, 0001_0001, 0001_0000, 0001_0002, 0001_0010	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f010.png :material-check:
13	0001_0004, 0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0010	build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f010.png :material-check:
14	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0010	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f010.png :material-check:
15	0001_0003, 0001_0004, 0001_0001, 0001_0000, 0001_0002, 0001_0011	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART137_f011.png :material-check:
16	0001_0004, 0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0011	build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f011.png :material-check:
17	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0011	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f011.png :material-check:
18	0001_0003, 0001_0004, 0001_0001, 0001_0000, 0001_0002, 0001_0012	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f012.png :material-check:
19	0001_0004, 0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0012	build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f012.png :material-check:
20	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0012	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART137_f012.png :material-check:
21	0001_0003, 0001_0004, 0001_0001, 0001_0000, 0001_0002, 0001_0005	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check:
22	0001_0004, 0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0005	build/phase-3/bitmaps/ PART137_f004.png :material-check: build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check:
23	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0005	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check:
24	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0005	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check:
25	0001_0003, 0001_0000, 0001_0005, 0001_0001, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check:

Frame	Sprites	PNG References
		build/phase-3/bitmaps/ PART137_f002.png :material-check:
26	0001_0003, 0001_0000, 0001_0005, 0001_0001, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:
27	0001_0003, 0001_0000, 0001_0005, 0001_0001, 0001_0001, 0001_0001, 0001_0001, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:
28	0001_0003, 0001_0000, 0001_0005, 0001_0001, 0001_0001, 0001_0001, 0001_0001, 0001_0001, 0001_0001, 0001_0001, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:
29		

Frame	Sprites	PNG References
	0001_0003, 0001_0000, 0001_0005, 0001_0001, 0001_0001, 0001_0001, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:
30	0001_0003, 0001_0000, 0001_0005, 0001_0001, 0001_0002	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check:
31	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0005	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check:
32	0001_0003, 0001_0001, 0001_0000, 0001_0002, 0001_0005	build/phase-3/bitmaps/ PART137_f003.png :material-check: build/phase-3/bitmaps/ PART137_f001.png :material-check: build/phase-3/bitmaps/ PART137_f000.png :material-check: build/phase-3/bitmaps/ PART137_f002.png :material-check: build/phase-3/bitmaps/ PART137_f005.png :material-check:

Efeitos Sonoros (SFX):

Frame	Sound ID	Bytecode Offset	WAV Reference
25	3337	0	build/phase-6/raw-wav/SX_3337_11025.wav

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	197

Opcode	Count
END_FRAME	33
PLAY_SFX	1
END_ANM	1

Notas de Comportamento:

Egg Timer has 2 semantic states: Unactivated, Time Out. Transitions are triggered by: collision, electrical, proximity.

Terminal state(s): Time Out, Time Out, Time Out, Time Out, Time Out, Time Out, Time Out, Time Out (part is removed from simulation).

This part has custom spawn behavior defined in SOLVE.RES.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 138 — Candle



Categoria: Light / Flame Sources
Fases em que aparece: 26 fases
Quantidade de Estados: 2 estado(s)

This candle can be lit with a laser beam, a phazer, another flaming part (like a rocket or an oil lamp), or with a magnifying glass and light source. Use it to light fuses, make the coffee pot percolate, or pop blimps and balloons.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Unlit	1→2	1	No
Lit	2→-1	2	No

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
unk_2	45	category-specific property
property_q8	54	Q8.8 = 0.211
unk_6	28	category-specific property
dim_w1	0	dimension 1 width

Property	Raw Value	Interpretation
dim_h1	0	dimension 1 height
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	24	bitmap frame width
bitmap_h	33	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	proximity, electrical
Destructible	No
Spawns	fire
Category Part Count	5

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	Yes
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
1	0	0	24×33	0
2	1	1–5 (5 frames)	24×39	7

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART138.png :material-check:
1	0001_0000, 0002_0005	build/phase-3/bitmaps/PART138.png :material-check: build/phase-3/bitmaps/PART138_2_f005.png :material-alert-outline: (not found)
2	0001_0000, 0002_0006	build/phase-3/bitmaps/PART138.png :material-check: build/phase-3/bitmaps/PART138_2_f006.png :material-alert-outline: (not found)
3	0001_0000, 0002_0007	build/phase-3/bitmaps/PART138.png :material-check: build/phase-3/bitmaps/PART138_2_f007.png :material-alert-outline: (not found)
4	0001_0000, 0002_0008	build/phase-3/bitmaps/PART138.png :material-check: build/phase-3/bitmaps/PART138_2_f008.png :material-alert-outline: (not found)
5	0001_0000, 0002_0009	

Frame	Sprites	PNG References
		build/phase-3/bitmaps/PART138.png :material-check: build/phase-3/bitmaps/PART138_2_f009.png :material-alert-outline: (not found)

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	11
END_FRAME	6
END_ANM	1

Notas de Comportamento:

Candle has 2 semantic states: Unlit, Lit. Transitions are triggered by: proximity, electrical.

Terminal state(s): Lit (part is removed from simulation).

This part can ignite nearby combustible parts (balloons, explosives). Uses proximity trigger.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 139 — Teeter-Totter



Categoria: Special Mechanics
Fases em que aparece: 53 fases
Quantidade de Estados: 1 estado(s)

Drop something heavy on the high end of this teeter-totter to catapult an object off the low end. Tie a rope to either end and use it to hoist and lower objects, pull the phazer trigger, or cause other reactions.

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
mass	2000	mass/density (relative scale)
unk_2	1000	category-specific property
property_q8	1024	Q8.8 = 4.000
unk_6	14	category-specific property
dim_w1	0	dimension 1 width
dim_h1	0	dimension 1 height

Property	Raw Value	Interpretation
dim_w2	0	dimension 2 width
dim_h2	0	dimension 2 height
bitmap_w	81	bitmap frame width
bitmap_h	40	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	collision, electrical, proximity
Destructible	No
Spawns	custom
Category Part Count	15

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
5	0	0–2 (3 frames)	88×40	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0001, 0001_0000	build/phase-3/bitmaps/PART139_f001.png :material-check: build/phase-3/bitmaps/PART139_f000.png :material-check:
1	0001_0002, 0001_0000	build/phase-3/bitmaps/PART139_f002.png :material-check: build/phase-3/bitmaps/PART139_f000.png :material-check:
2	0001_0003, 0001_0000	build/phase-3/bitmaps/PART139_f003.png :material-check: build/phase-3/bitmaps/PART139_f000.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	6
END_FRAME	3
END_ANM	1

Notas de Comportamento:

Teeter-Totter is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part has custom spawn behavior defined in SOLVE.RES.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-02 (Física e Gravidade):** Instanciar no ar e verificar se a peça sofre aceleração no eixo Y e se a Hitbox colide perfeitamente com o chão (ID 1).
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.

Peça 140 — Small Cactus



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	900	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	29	bitmap frame width
bitmap_h	65	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	32×65	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART140.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Small Cactus is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 141 — Small Maple Tree



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	900	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	43	bitmap frame width
bitmap_h	50	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	48×50	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART141.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Small Maple Tree is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 142 — Small Pine Tree



Categoria: Scenery
Fases em que aparece: 9 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	900	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	30	bitmap frame width
bitmap_h	62	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	32×62	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART142.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Small Pine Tree is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 143 — Small Cloud



Categoria: Scenery
Fases em que aparece: 24 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	55	bitmap frame width
bitmap_h	27	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	56×27	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART143.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

Small Cloud is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 144 — small mountain



Categoria: Scenery
Fases em que aparece: 10 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	127	bitmap frame width
bitmap_h	63	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	128×63	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART144.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

small mountain is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 145 — medium mountain



Categoria: Scenery
Fases em que aparece: 11 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	159	bitmap frame width
bitmap_h	59	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	160×59	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART145.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

medium mountain is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 146 — small mesa



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	69	bitmap frame width
bitmap_h	24	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
3	0	0	72×24	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART146.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

small mesa is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 147 — medium mesa



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

scenery part

Detalhes dos Estados (SOLVE.RES):

No SOLVE.RES state data — this part has no programmatic state transitions.

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	76	bitmap frame width
bitmap_h	67	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	(none)
Destructible	Yes
Spawns	(none)
Category Part Count	26

Conexões e Capacidades:

Capability	Supported
Electrical	No

Capability	Supported
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	No
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	80×67	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART147.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_FRAME	1
END_ANM	1

Notas de Comportamento:

medium mesa is a static part with no programmatic state transitions. Its visual appearance is determined by the ANM state selected at placement time (via the `appearance` field in the level file).

This part can be destroyed by explosives.

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).
- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 148 — Laser Detector



Categoria: Lasers / Optics
Fases em que aparece: 6 fases
Quantidade de Estados: 4 estado(s)

This laser detector can receive laser beams of any color. When a beam strikes the black eye, the green light turns on. If the beam is broken, the red light flashes. If the beam returns, both lights will flash.

Detalhes dos Estados (SOLVE.RES):

SOLVE.RES State Transitions

State Name	Triggers (self→other)	ANM State	Explosive
Not Activated	2→3	—	No
Activated	3→4	—	No
Alarm On	4→-1	—	No

Propriedades Físicas:

Property	Raw Value	Interpretation
unk_2	0	scenery property (900 for trees, 1100-1200 for clouds/sun)
bitmap_w	37	bitmap frame width
bitmap_h	32	bitmap frame height

Regras de Categoria:

Rule	Value
Triggers	electrical, proximity, collision
Destructible	No
Spawns	laser_beam
Category Part Count	8

Conexões e Capacidades:

Capability	Supported
Electrical	Yes
Belt Connection	No
Rope Connection	No
Fire/Flame	No
Laser	Yes
Projectile	No

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame	Frame Range	Dimensions	Bytecode Offset
2	0	0	40×32	0
3	1	1–19 (19 frames)	40×32	7
4	20	20–38 (19 frames)	0×0	0
5	39	39	0×0	0

Detalhes dos Sprites dos Frames:

Frame	Sprites	PNG References
0	0001_0000	build/phase-3/bitmaps/PART148_f000.png :material-check:
1	0001_0000, 0001_0001	build/phase-3/bitmaps/PART148_f000.png :material-check: build/phase-3/bitmaps/PART148_f001.png :material-check:
2	0001_0000	build/phase-3/bitmaps/PART148_f000.png :material-check:
3	0001_0000	build/phase-3/bitmaps/PART148_f000.png :material-check:

Frame	Sprites	PNG References
4	0001_0000, 0001_0002	build/phase-3/bitmaps/PART148_f000.png :material-check: build/phase-3/bitmaps/PART148_f002.png :material-check:
5	0001_0000, 0001_0001, 0001_0002	build/phase-3/bitmaps/PART148_f000.png :material-check: build/phase-3/bitmaps/PART148_f001.png :material-check: build/phase-3/bitmaps/PART148_f002.png :material-check:
6	0001_0000	build/phase-3/bitmaps/PART148_f000.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	11
END_FRAME	7
END_ANM	1

Notas de Comportamento:

Laser Detector has 3 semantic states: Not Activated, Activated, Alarm On. Transitions are triggered by: electrical, proximity, collision.

Terminal state(s): Alarm On (part is removed from simulation).

This part emits a **laser beam**. The beam interacts with mirrors, mixers, and laser-activated plugs.

This part has timed frame-by-frame animation (Section A durations control playback speed at 60 Hz).

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-01 (Máquina de Estados):** Forçar interação física via script correspondente aos pares de gatilho (SOLVE.RES) e validar se o `state_counter` e o frame ANM atualizam de acordo.
- **T-03 (Elétrica):** Conectar um cabo de energia na âncora, aplicar voltagem e atestar propagação do sinal para o estado ligado.
- **T-05 (Destruição):** Disparar uma explosão adjacente e confirmar que a peça entra em estado terminal ou é varrida da memória (`queue_free`).

Peça 150 — Pine Forest



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Ice wall panel (104x45)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	104	bitmap frame width
bitmap_h	45	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	104x45	 build/phase-3/bitmaps/PART150_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 151 — Cave Opening



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Ice cave wall rough (88x47)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	47	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x47	 build/phase-3/bitmaps/PART151_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
	1

Opcode	Count
END_ANM	
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 152 — Forest Trees



Categoria: Scenery
Fases em que aparece: 8 fases
Quantidade de Estados: 1 estado(s)

Ice bank/slope section (72x36)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	36	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x36	build/phase-3/bitmaps/PART152_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 153 — Rocky Terrain



Categoria: Scenery
Fases em que aparece: 7 fases
Quantidade de Estados: 1 estado(s)

Dark scenery section (72x33)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	33	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x33	 build/phase-3/bitmaps/PART153_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 154 — Rocky Mountain



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Dark floor tile (80x34)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	34	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x34	 build/phase-3/bitmaps/PART154_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 155 — Trees on Rocks



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Dark platform ledge (104x26)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	104	bitmap frame width
bitmap_h	26	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	104x26	 build/phase-3/bitmaps/PART155_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
	1

Opcode	Count
END_ANM	
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 156 — Trees on Terrain



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Dark ledge shorter (88x19)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	19	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x19	 <code>build/phase-3/bitmaps/PART156_f000.png</code> :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 157 — Snow Mountains



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Snow-topped ice formation (120x32)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	120	bitmap frame width
bitmap_h	32	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	120x32	 build/phase-3/bitmaps/PART157_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 158 — Iceberg



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Iceberg with white snowcap (72x42)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	42	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x42	 build/phase-3/bitmaps/PART158_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 159 — Snow Peaks



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)
Ice spire/pillar (80x31)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	31	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x31	 build/phase-3/bitmaps/PART159_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
	1

Opcode	Count
END_ANM	
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 160 — Sand Dune



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Wood plank horizontal (112x21)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	21	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x21	build/phase-3/bitmaps/PART160_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 161 — Desert Dune



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Shorter wood plank (88x19)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	19	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x19	 build/phase-3/bitmaps/PART161_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 162 — Sand Hill



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Thin wood beam (72x13)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	13	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x13	 build/phase-3/bitmaps/PART162_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 163 — Tropical Island



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 0 estado(s)

Wood platform/top (112x30)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	30	bitmap frame height

No physics properties — this is a static scenery part.

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x30	 build/phase-3/bitmaps/PART163_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 164 — Red Canyon



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Red-painted wood ledge (112x22)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	22	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x22	build/phase-3/bitmaps/PART164_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 165 — Red Rock



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Red-painted wood beam (96x16)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	96	bitmap frame width
bitmap_h	16	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	96x16	 build/phase-3/bitmaps/PART165_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 166 — Red Formation



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Red wood strip narrow (56x26)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	56	bitmap frame width
bitmap_h	26	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	56x26	 build/phase-3/bitmaps/PART166_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 167 — Castle Fortress



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Building/wall section tall (96x124)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	96	bitmap frame width
bitmap_h	124	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	96x124	 build/phase-3/bitmaps/PART167_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 168 — Planet Rings



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)
Large background decor (160x92)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	160	bitmap frame width
bitmap_h	92	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	160x92	build/phase-3/bitmaps/PART168_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 169 — Moon



Categoria: Scenery
Fases em que aparece: 11 fases
Quantidade de Estados: 1 estado(s)
Ice background panel (64x61)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	64	bitmap frame width
bitmap_h	61	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	64x61	 build/phase-3/bitmaps/PART169_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 170 — Moon Crater



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)
Wood background panel (72x66)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	66	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x66	 build/phase-3/bitmaps/PART170_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 171 — Owl



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Background detail (48x45)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	48	bitmap frame width
bitmap_h	45	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	48x45	 build/phase-3/bitmaps/PART171_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 172 — Spiral Galaxy



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Ice ruin/arch structure (80x84)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	84	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x84	 build/phase-3/bitmaps/PART172_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 173 — Gold Nugget



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	83	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x83	 build/phase-3/bitmaps/PART173_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 174 — Gold Rock



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	48	bitmap frame width
bitmap_h	51	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	48x51	 build/phase-3/bitmaps/PART174_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 175 — Globe



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	48	bitmap frame width
bitmap_h	41	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	48x41	 build/phase-3/bitmaps/PART175_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 176 — Comet



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	48	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x48	 build/phase-3/bitmaps/PART176_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 177 — Star Glow



Categoria: Scenery
Fases em que aparece: 8 fases
Quantidade de Estados: 1 estado(s)

Snow block detail (56x57)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	56	bitmap frame width
bitmap_h	57	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	56x57	 build/phase-3/bitmaps/PART177_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 178 — Star Burst



Categoria: Scenery
Fases em que aparece: 10 fases
Quantidade de Estados: 1 estado(s)

Small ice connector joint (24x23)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	23	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x23	 build/phase-3/bitmaps/PART178_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 179 — Star Sparkle



Categoria: Scenery
Fases em que aparece: 9 fases
Quantidade de Estados: 1 estado(s)

Tiny ice connector joint (24x17)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	17	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x17	 build/phase-3/bitmaps/PART179_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 180 — Stalactites



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Wood roof/canopy (96x54)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	96	bitmap frame width
bitmap_h	54	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	96x54	 build/phase-3/bitmaps/PART180_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 181 — Greenhouse Dome



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Ice cap/ledge section (88x30)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	30	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x30	 build/phase-3/bitmaps/PART181_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 182 — Sphinx Pyramid



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Log cabin wall section (88x44)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	44	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x44	 build/phase-3/bitmaps/PART182_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 183 — Futuristic Tower



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Background block (88x57)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	57	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x57	 build/phase-3/bitmaps/PART183_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 184 — Sci-Fi City



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Ice block tile (80x47)

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>bitmap_w</code>	80	bitmap frame width
<code>bitmap_h</code>	47	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x47	 <code>build/phase-3/bitmaps/PART184_f000.png</code> :material-close:

Opcodes do Bytecode:

Opcode	Count
<code>DRAW_BMP</code>	1
<code>END_ANM</code>	1
<code>END_FRAME</code>	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 185 — Cityscape



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Wood tile (56x40)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	56	bitmap frame width
bitmap_h	40	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	56x40	 build/phase-3/bitmaps/PART185_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 186 — Metropolis



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Ice floor tile (72x44)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	44	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x44	 build/phase-3/bitmaps/PART186_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 187 — City Buildings



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)
Background cap piece (88x33)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	33	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x33	 build/phase-3/bitmaps/PART187_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 188 — Ice Castle



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Background insert (64x33)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	64	bitmap frame width
bitmap_h	33	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	64x33	 build/phase-3/bitmaps/PART188_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 189 — Crystal Tower



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Background detail panel (56x50)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	56	bitmap frame width
bitmap_h	50	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	56x50	 build/phase-3/bitmaps/PART189_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 190 — City Skyline



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Background panel (88x53)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	53	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x53	 build/phase-3/bitmaps/PART190_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 191 — Totem Pole



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Vertical wood post (24x83)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	83	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x83	 build/phase-3/bitmaps/PART191_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 192 — Satellite Tower



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Tall ice spire/formation (72x103)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	103	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x103	 build/phase-3/bitmaps/PART192_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 193 — Radio Tower



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Short ice column section (32x56)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	56	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x56	 build/phase-3/bitmaps/PART193_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 194 — Dish Antenna



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Background post (32x45)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	45	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x45	 build/phase-3/bitmaps/PART194_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 195 — Glass Dome



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Snow bank top (64x30)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	64	bitmap frame width
bitmap_h	30	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	64x30	 build/phase-3/bitmaps/PART195_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 196 — Crystal Formation



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Wood structure block (72x54)

Propriedades Físicas:

Property	Raw Value	Interpretation
<code>bitmap_w</code>	72	bitmap frame width
<code>bitmap_h</code>	54	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x54	 <code>build/phase-3/bitmaps/PART196_f000.png</code> :material-close:

Opcodes do Bytecode:

Opcodes	Count
<code>DRAW_BMP</code>	1
<code>END_ANM</code>	1
<code>END_FRAME</code>	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 197 — Mountain Range



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Wide wood beam (136x16)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	136	bitmap frame width
bitmap_h	16	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	136x16	 build/phase-3/bitmaps/PART197_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 198 — Volcano



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Wood frame/arch (80x57)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	57	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x57	 build/phase-3/bitmaps/PART198_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 199 — Erupting Volcano



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Large wood panel (128x73)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	128	bitmap frame width
bitmap_h	73	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	128x73	 build/phase-3/bitmaps/PART199_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 200 — Red Coral Branch



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Red decorative trim (112x11)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	11	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x11	 build/phase-3/bitmaps/PART200_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 201 — Red Seaweed



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Wide red decorative trim (128x24)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	128	bitmap frame width
bitmap_h	24	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	128x24	 build/phase-3/bitmaps/PART201_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 202 — Red Coral



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Red decorative beam (168x18)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	168	bitmap frame width
bitmap_h	18	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	168x18	 build/phase-3/bitmaps/PART202_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 203 — Dark Space



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Solid dark filler (72x72)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	72	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x72	 build/phase-3/bitmaps/PART203_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 204 — Satellite



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	64	bitmap frame width
bitmap_h	79	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	64x79	 build/phase-3/bitmaps/PART204_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 205 — Blue Space



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Blue ice/water surface (72x72)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	72	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x72	 build/phase-3/bitmaps/PART205_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 206 — Ocean Wave



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Ice strip horizontal (112x32)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	32	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
2	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x32	 build/phase-3/bitmaps/PART206_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 207 — Breaking Wave



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Ice ledge section (112x18)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	18	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x18	 build/phase-3/bitmaps/PART207_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 208 — Surf Wave



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Ice edge trim (72x14)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	14	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x14	 build/phase-3/bitmaps/PART208_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 209 — Rising Bubbles



Categoria: Scenery
Fases em que aparece: 8 fases
Quantidade de Estados: 1 estado(s)

Tall ice column (16x81)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	81	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	16x81	 build/phase-3/bitmaps/PART209_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 210 — Bubble Stream



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Ice pillar section (16x74)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	74	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	16x74	 build/phase-3/bitmaps/PART210_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 211 — Floating Bubbles



Categoria: Scenery
Fases em que aparece: 7 fases
Quantidade de Estados: 1 estado(s)

Short ice pillar (16x40)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	40	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	16x40	 build/phase-3/bitmaps/PART211_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 212 — Coral Arch



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Wood fence/railing (80x49)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	49	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x49	 build/phase-3/bitmaps/PART212_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 213 — Coral Branch



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Wood railing section (56x44)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	56	bitmap frame width
bitmap_h	44	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	56x44	 build/phase-3/bitmaps/PART213_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 214 — Coral Formation



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Short wood railing (56x28)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	56	bitmap frame width
bitmap_h	28	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	56x28	 build/phase-3/bitmaps/PART214_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 215 — Green Mountain



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

Wood cube block (72x54)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	54	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x54	 build/phase-3/bitmaps/PART215_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 216 — Mossy Rock



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Wood board/plank (80x36)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	36	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x36	 build/phase-3/bitmaps/PART216_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 217 — Green Cliff



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Wood slab wide (96x45)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	96	bitmap frame width
bitmap_h	45	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	96x45	 build/phase-3/bitmaps/PART217_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 218 — Shipwreck



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Large wood block (112x91)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	91	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x91	 build/phase-3/bitmaps/PART218_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 219 — Submarine



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Wide wood panel (144x64)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	144	bitmap frame width
bitmap_h	64	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	144x64	 build/phase-3/bitmaps/PART219_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 220 — Ancient Ruins



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Wood arch/doorway (88x71)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	71	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x71	 build/phase-3/bitmaps/PART220_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 221 — Stone Temple



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Tall wood block (88x58)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	88	bitmap frame width
bitmap_h	58	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	88x58	 build/phase-3/bitmaps/PART221_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 222 — Ornate Gate



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	48	bitmap frame width
bitmap_h	16	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	48x16	 build/phase-3/bitmaps/PART222_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 223 — Bamboo Fence



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Small wood tile (24x31)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	31	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x31	 build/phase-3/bitmaps/PART223_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 224 — Stone Wall



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	27	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x27	 build/phase-3/bitmaps/PART224_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 225 — Stone Block



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Wood detail piece (40x35)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	40	bitmap frame width
bitmap_h	35	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	40x35	 build/phase-3/bitmaps/PART225_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 226 — Hot Rod Car



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Wood top decoration (136x61)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	136	bitmap frame width
bitmap_h	61	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	136x61	 build/phase-3/bitmaps/PART226_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 227 — Clown Fish



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Wood patch/tile (64x25)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	64	bitmap frame width
bitmap_h	25	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	64x25	 build/phase-3/bitmaps/PART227_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 228 — Anchor



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Medium wood block (48x42)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	48	bitmap frame width
bitmap_h	42	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	48x42	 build/phase-3/bitmaps/PART228_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 229 — Treasure Chest



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Wood texture tile (56x42)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	56	bitmap frame width
bitmap_h	42	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	56x42	 build/phase-3/bitmaps/PART229_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 230 — Ship Logo



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Background strip (40x16)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	40	bitmap frame width
bitmap_h	16	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	40x16	 build/phase-3/bitmaps/PART230_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 231 — Comet Streak



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Tiny wood strip (24x12)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	12	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x12	 build/phase-3/bitmaps/PART231_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 232 — Golden Shell



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	40	bitmap frame width
bitmap_h	16	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	40x16	 build/phase-3/bitmaps/PART232_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 233 — Shell



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Small wood tile (32x21)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	21	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x21	 build/phase-3/bitmaps/PART233_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 234 — Shell Spiral



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Small wood strip (32x18)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	18	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32×18	 build/phase-3/bitmaps/PART234_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 235 — Red Fan Coral



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	13	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32×13	 build/phase-3/bitmaps/PART235_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 236 — Starfish



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Wood filler joint (24x19)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	19	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x19	 build/phase-3/bitmaps/PART236_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 237 — Metal Gears



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

Tiny wood joint (16x18)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	18	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	16x18	 build/phase-3/bitmaps/PART237_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 238 — Orange Coral



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Small vertical wood (24x35)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	35	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x35	 build/phase-3/bitmaps/PART238_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 239 — Coral Branch



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Square wood tile (32x25)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	25	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x25	 build/phase-3/bitmaps/PART239_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 240 — Green Coral



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Vertical wood section (24x35)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	35	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x35	 build/phase-3/bitmaps/PART240_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 241 — Coral Tree



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Small square wood (32x25)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	25	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x25	 build/phase-3/bitmaps/PART241_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 242 — Seaweed Tall



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Tall thin wood (40x52)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	40	bitmap frame width
bitmap_h	52	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	40x52	 build/phase-3/bitmaps/PART242_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 243 — Kelp Forest



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)
Wood cluster/clump (32x31)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	31	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x31	 build/phase-3/bitmaps/PART243_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 244 — Kelp Plant



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Tall wood strip (32x68)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	68	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x68	 build/phase-3/bitmaps/PART244_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 245 — Seaweed Vine



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

Tall thin wood divider (16x81)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	81	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	16x81	 build/phase-3/bitmaps/PART245_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 246 — Seaweed Long



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

Wood divider strip (16x66)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	66	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	16x66	 build/phase-3/bitmaps/PART246_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 247 — Plant Stem



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Narrow wood strip (16x46)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
bitmap_h	46	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	16x46	 build/phase-3/bitmaps/PART247_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 248 — Sea Urchin



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Dark wood tile (32x21)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	21	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x21	 build/phase-3/bitmaps/PART248_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 249 — Red Crab



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

Dark wood detail (32x22)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	22	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x22	 build/phase-3/bitmaps/PART249_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 250 — Jellyfish



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Crystal ice detail (32x27)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	27	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x27	 build/phase-3/bitmaps/PART250_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 251 — Octopus



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Wood texture panel (48x33)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	48	bitmap frame width
bitmap_h	33	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	48x33	 build/phase-3/bitmaps/PART251_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 252 — Striped Fish



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Mini wood tile (24x19)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	19	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x19	 build/phase-3/bitmaps/PART252_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 253 — Angelfish



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 0 estado(s)

Micro wood strip (24x14)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	14	bitmap frame height

No physics properties — this is a static scenery part.

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x14	 build/phase-3/bitmaps/PART253_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 254 — Goldfish



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Small wood strip (32x14)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	14	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x14	 build/phase-3/bitmaps/PART254_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 255 — Seahorse



Categoria: Scenery
Fases em que aparece: 5 fases
Quantidade de Estados: 1 estado(s)

Mini tall wood (24x35)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	24	bitmap frame width
bitmap_h	35	bitmap frame height

No physics properties — this is a static scenery part.

Maapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	24x35	 <code>build/phase-3/bitmaps/PART255_f000.png</code> :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 256 — Shark



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Background ledge (48x15)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	48	bitmap frame width
bitmap_h	15	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	48x15	 build/phase-3/bitmaps/PART256_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 257 — Fish School



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Nano wood tile (16x9)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	16	bitmap frame width
	9	bitmap frame height

Property	Raw Value	Interpretation
bitmap_h		

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	16x9	  build/phase-3/bitmaps/PART257_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 258 — Fish Scatter



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Micro flat wood (32x11)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	11	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x11	

Frame	Dimensions	PNG References
		 build/phase-3/bitmaps/PART258_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 259 — Fish Group



Categoria: Scenery
Fases em que aparece: 8 fases
Quantidade de Estados: 1 estado(s)

Small flat wood (32x21)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	21	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x21	 build/phase-3/bitmaps/PART259_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 260 — Great White Shark



Categoria: Scenery
Fases em que aparece: 6 fases
Quantidade de Estados: 1 estado(s)

Wide flat ice surface (120x35)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	120	bitmap frame width
bitmap_h	35	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	120x35	 build/phase-3/bitmaps/PART260_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 261 — Dolphin



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Ice surface tile (104x35)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	104	bitmap frame width
bitmap_h	35	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	104x35	 build/phase-3/bitmaps/PART261_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 262 — Orca Whale



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Wide background decor (176x91)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	176	bitmap frame width
bitmap_h	91	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	176x91	 build/phase-3/bitmaps/PART262_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 263 — Gray Whale



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Wide wood decoration (216x88)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	216	bitmap frame width
bitmap_h	88	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	216x88	 build/phase-3/bitmaps/PART263_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 264 — Lionfish



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	48	bitmap frame width
bitmap_h	53	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	48x53	 build/phase-3/bitmaps/PART264_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 265 — Stingray



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	20	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x20	 build/phase-3/bitmaps/PART265_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 266 — White Coral



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	40	bitmap frame width
bitmap_h	25	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	40x25	 build/phase-3/bitmaps/PART266_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 267 — Tree Trunk



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 1 estado(s)

Small wood section (32x26)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	32	bitmap frame width
bitmap_h	26	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	32x26	 build/phase-3/bitmaps/PART267_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 268 — Stained Glass



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Tall pole/flagpole (56x111)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	56	bitmap frame width
bitmap_h	111	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	56x111	 build/phase-3/bitmaps/PART268_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 269 — Abstract Art



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Large rock/wood wall (80x101)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	101	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x101	 build/phase-3/bitmaps/PART269_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 270 — Landscape Painting



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Large ice block (104x76)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	104	bitmap frame width
bitmap_h	76	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	104×76	 build/phase-3/bitmaps/PART270_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 271 — Blue Tiles



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	64	bitmap frame width
bitmap_h	64	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	64×64	 build/phase-3/bitmaps/PART271_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 272 — Computer Monitor



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Dark background tile (80x62)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	80	bitmap frame width
bitmap_h	62	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	80x62	 build/phase-3/bitmaps/PART272_f000.png :material-close:

Opcodes do Bytecode:

Opcodes	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 273 — Metal Vent



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Background vertical strip (40x60)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	40	bitmap frame width
bitmap_h	60	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	40x60	 build/phase-3/bitmaps/PART273_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 274 — Einstein Bust



Categoria: Scenery
Fases em que aparece: 2 fases
Quantidade de Estados: 1 estado(s)

Background decoration (40x46)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	40	bitmap frame width
bitmap_h	46	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	40x46	 build/phase-3/bitmaps/PART274_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 275 — Workshop Board



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 0 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	120	bitmap frame width
bitmap_h	125	bitmap frame height

No physics properties — this is a static scenery part.

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	120x125	 build/phase-3/bitmaps/PART275_f000.png :material-check:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	2
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 276 — Cliff Edge



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 0 estado(s)

Large ice end piece (120x100)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	120	bitmap frame width
bitmap_h	100	bitmap frame height

No physics properties — this is a static scenery part.

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	120x100	 build/phase-3/bitmaps/PART276_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 277 — Treasure Map



Categoria: Scenery
Fases em que aparece: 0 fases
Quantidade de Estados: 1 estado(s)

Scenery decoration tile

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	120	bitmap frame width
bitmap_h	113	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	120x113	 build/phase-3/bitmaps/PART277_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

Peça 278 — Periodic Table



Categoria: Scenery
Fases em que aparece: 1 fases
Quantidade de Estados: 1 estado(s)

Dark background block (112x66)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	112	bitmap frame width
bitmap_h	66	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	112x66	 build/phase-3/bitmaps/PART278_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
	1

Opcode	Count
END_ANM	
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 279 — Wooden Door



Categoria: Scenery
Fases em que aparece: 3 fases
Quantidade de Estados: 1 estado(s)

Tree canopy/foilage (64x80)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	64	bitmap frame width
bitmap_h	80	bitmap frame height

No physics properties — this is a static scenery part.

Mapeamento de Sprite por Estado (ANM):

State ID	First Frame
1	0

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	64x80	 build/phase-3/bitmaps/PART279_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.
-

Peça 280 — Stone Texture



Categoria: Scenery
Fases em que aparece: 4 fases
Quantidade de Estados: 0 estado(s)

Wood panel backdrop (72x64)

Propriedades Físicas:

Property	Raw Value	Interpretation
bitmap_w	72	bitmap frame width
bitmap_h	64	bitmap frame height

No physics properties — this is a static scenery part.

Detalhes dos Sprites dos Frames:

Frame	Dimensions	PNG References
0	72x64	 build/phase-3/bitmaps/PART280_f000.png :material-close:

Opcodes do Bytecode:

Opcode	Count
DRAW_BMP	1
END_ANM	1
END_FRAME	1

Casos de Teste Sugeridos (Gerados Automáticos):

- **T-06 (Cenário / Background):** Validar se o `z_index` é negativo e se a ferramenta permite colocar peças ativas sobrepostas a esta.

9. Catálogo Detalhado de Fases e Dependências

Este documento cataloga exaustivamente todas as fases do jogo. Para cada fase, detalhamos o objetivo, dicas, **setup exato das peças** (inventário + cenário), e um **Mapa de Dependências** que estipula quais sistemas (`Solvers`), áudios e volumes de frames visuais (`.ANM`) o `LevelLoader` da Godot precisará puxar para a memória.

ELEV1.yaml - Under the Sea

Objetivo: Get Mel home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

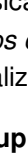
Dicas Ativas (2): - Mel could walk through this pipe... if there were one here. - This looks like a good place for an anti-gravity pad.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - **Efeitos Sonoros (SFX):** *Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 9 arquivos `.ANM` carregados. - Totaliza aproximadamente **257 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pipe Wall:** 6 peça(s)

-  **Curved Pipe Wall:** 6 peça(s)
-  **Mel Schlemming:** 1 peça(s)
-  **Large Curved Pipe:** 1 peça(s)
-  **Mel's House:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Greco-Roman Wall:** 2 peça(s)
-  **Large Pipes:** 1 peça(s)
-  **Granite Incline:** 1 peça(s)
-  **Color Block:** 4 peça(s)
-  **Pine Forest:** 4 peça(s)
-  **Cave Opening:** 1 peça(s)
-  **Forest Trees:** 2 peça(s)
-  **Rocky Terrain:** 2 peça(s)
-  **Moon:** 1 peça(s)
-  **Star Burst:** 1 peça(s)
-  **Star Sparkle:** 2 peça(s)
-  **City Skyline:** 1 peça(s)
-  **Dark Space:** 1 peça(s)
-  **Rising Bubbles:** 2 peça(s)
-  **Bubble Stream:** 2 peça(s)
-  **Floating Bubbles:** 2 peça(s)
-  **Green Mountain:** 2 peça(s)
-  **Mossy Rock:** 2 peça(s)
-  **Green Cliff:** 1 peça(s)
-  **Shipwreck:** 1 peça(s)
-  **Ancient Ruins:** 2 peça(s)
-  **Stone Temple:** 2 peça(s)
-  **Bamboo Fence:** 1 peça(s)
-  **Stone Block:** 1 peça(s)
-  **Anchor:** 1 peça(s)
-  **Shell:** 1 peça(s)
-  **Shell Spiral:** 1 peça(s)
-  **Starfish:** 1 peça(s)
-  **Metal Gears:** 1 peça(s)
-  **Orange Coral:** 4 peça(s)
-  **Coral Branch:** 3 peça(s)
-  **Green Coral:** 1 peça(s)
-  **Seaweed Tall:** 1 peça(s)
-  **Kelp Forest:** 2 peça(s)
- **Kelp Plant:** 1 peça(s)

-  **Seaweed Vine:** 3 peça(s)
 -  **Seaweed Long:** 1 peça(s)
 -  **Plant Stem:** 1 peça(s)
 -  **Red Crab:** 1 peça(s)
 -  **Jellyfish:** 12 peça(s)
 -  **Octopus:** 1 peça(s)
 -  **Striped Fish:** 2 peça(s)
 -  **Angelfish:** 4 peça(s)
 -  **Goldfish:** 3 peça(s)
 -  **Seahorse:** 1 peça(s)
 -  **Great White Shark:** 1 peça(s)
 -  **Dolphin:** 1 peça(s)
-

ELEV2.yaml - Yellow Brick Path

Objetivo: Get Newton Mouse into his hole.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Holes in the floor are a blast. - Newton is just crazy about cheese.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1010_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 13 arquivos .ANM carregados. - Totaliza aproximadamente **420 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 1 peça(s)
 -  **Curie Cat:** 2 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Wood Wall:** 3 peça(s)
 - **Cheese:** 4 peça(s)
 - **Remote Control Explosives:** 1 peça(s)
 - **Mouse Hole:** 1 peça(s)
 - **Hot Air Balloon:** 1 peça(s)
 - **Cinder Block Wall:** 4 peça(s)
 - **Greco-Roman Wall:** 10 peça(s)
 - **Yellow Brick Wall:** 5 peça(s)
-

ELEV3.yaml - Space Hoops

Objetivo: Put the basketball into the basket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Jack has been known to be a good shot.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1023_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **128 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 3 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 2 peça(s)
-  **Basketball:** 1 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Gear:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Caution Wall:** 1 peça(s)
-  **Pinball Bumper:** 1 peça(s)
-  **Anti-Gravity Pad:** 2 peça(s)
-  **Boxes:** 1 peça(s)
-  **Tiny Gear:** 1 peça(s)
-  **Pinball Flipper:** 1 peça(s)
-  **Yellow Brick Incline:** 5 peça(s)
-  **Snow Mountains:** 3 peça(s)
-  **Iceberg:** 4 peça(s)
-  **Snow Peaks:** 2 peça(s)
-  **Planet Rings:** 1 peça(s)
-  **Moon:** 1 peça(s)
-  **Stalactites:** 1 peça(s)
-  **Futuristic Tower:** 1 peça(s)
-  **City Buildings:** 1 peça(s)
-  **Ice Castle:** 1 peça(s)
-  **Crystal Tower:** 1 peça(s)
-  **Totem Pole:** 1 peça(s)
-  **Radio Tower:** 1 peça(s)
-  **Dish Antenna:** 1 peça(s)

ELEV4.yaml - Bump and Roll

Objetivo: Get the tennis ball into the aquarium.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - This ball needs a little punch to get going. - Bowling ball in the side pocket! Get it? - A little time management maybe...

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1009_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos .ANM carregados. - Totaliza aproximadamente **309 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
 -  **Brick Wall:** 1 peça(s)
 -  **Wood incline:** 3 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Wood Wall:** 2 peça(s)
 -  **Caution Wall:** 2 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Greco-Roman Wall:** 2 peça(s)
 -  **Granite Incline:** 2 peça(s)
 -  **Brick Incline:** 3 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Yellow Brick Wall:** 2 peça(s)
 -  **Yellow Brick Incline:** 2 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

ELEV5.yaml - Midnight Blast

Objetivo: Fire the cannon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - This ball needs a little extra help to get going. - Is it conveyor or conveyer? Something to be pondered I suppose.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1002_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos .ANM carregados. - Totaliza aproximadamente **254 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 3 peça(s)

-  **Conveyor Belt:** 1 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Cannon:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Pinball:** 3 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **desert mesa:** 2 peça(s)
 -  **small mesa:** 3 peça(s)
 -  **medium mesa:** 1 peça(s)
 -  **Moon:** 1 peça(s)
 -  **Star Glow:** 1 peça(s)
 -  **Star Burst:** 3 peça(s)
 -  **Star Sparkle:** 4 peça(s)
 -  **Sphinx Pyramid:** 1 peça(s)
-

ELEV6.yaml - Bill's Aquarium

Objetivo: Break Bill's aquarium.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Use a corner pipe to turn the ball in the proper direction. - In some situations, one just isn't enough.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de**

Áudio: - Música de Fundo: `SX_1011_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via*

`AudioDispatcher` pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **343 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 1 peça(s)
-  **Belt:** 1 peça(s)
-  **Fish Tank:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Thumb Tack:** 1 peça(s)
-  **Large Curved Pipe:** 2 peça(s)
-  **Anti-Gravity Pad:** 2 peça(s)

-  **Trap Door:** 1 peça(s)
 -  **Cinder Block Wall:** 8 peça(s)
 -  **Programmable Ball:** 1 peça(s)
 -  **Accelerator Tube:** 2 peça(s)
 -  **Large Pipes:** 1 peça(s)
 -  **Granite Incline:** 17 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

ELEV7.yaml - Caged Balloons

Objetivo: Pop both of the balloons.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Sometimes you have to really atTACK a puzzle. - There is a way to pull this balloon down with the parts available.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1007_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 15 arquivos .ANM carregados. - Totaliza aproximadamente **286 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 2 peça(s)
 -  **Thumb Tack:** 2 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Anti-Gravity Pad:** 2 peça(s)
 -  **Cinder Block Wall:** 8 peça(s)
 -  **Greco-Roman Wall:** 6 peça(s)
 -  **Archway:** 4 peça(s)
 -  **Spruce Tree:** 1 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Glass Building:** 4 peça(s)
 -  **Small Maple Tree:** 1 peça(s)
 -  **Small Pine Tree:** 1 peça(s)
 -  **Small Cloud:** 1 peça(s)
 -  **small mountain:** 2 peça(s)
-

ELEV8.yaml - Air Time

Objetivo: Put all of the blimps in the hangar on the right. They all need to be there at the same time.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - There must be a way to turn these three blimps around without disturbing the others.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1003_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos .ANM carregados. - Totaliza aproximadamente **65 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pinball Bumper:** 3 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Cinder Block Wall:** 8 peça(s)
 -  **Scaffold Barrier:** 6 peça(s)
 -  **Blimp:** 6 peça(s)
 -  **Elm Tree:** 7 peça(s)
 -  **Spruce Tree:** 9 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 7 peça(s)
 -  **Small Cloud:** 3 peça(s)
-

ELEV9.yaml - Pulley Popper

Objetivo: Pop the silver balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - This ball needs to make its way into the beach bucket. - There must be a way to get this bucket through the wall below.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1006_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 16 arquivos .ANM carregados. - Totaliza aproximadamente **311 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 3 peça(s)
-  **Balloon:** 1 peça(s)
-  **Pulley:** 5 peça(s)
-  **Rope:** 2 peça(s)
-  **Fish Tank:** 3 peça(s)
-  **Bucket:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Boat Cleat:** 3 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Wood Wall:** 9 peça(s)
-  **Thumb Tack:** 1 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)

-  **Steel Cable:** 1 peça(s)
 -  **Programmable Ball:** 1 peça(s)
 -  **Leaky Bucket:** 1 peça(s)
-

ELEV10.yaml - Separation

Objetivo: Put the baseball in the aquarium and the pinball in the cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (4): - A light ball will roll over a trap door, but a heavy ball will fall through. - Bombs away! - Room for a Kaboom! - What to do with this gap? Something should SPRING to mind.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `GravitySolver (Standard)` - **Recursos de Áudio:** - Música de Fundo: `SX_1000_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 11 arquivos `.ANM` carregados. - Totaliza aproximadamente **109 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 4 peça(s)
 -  **Remote Control:** 2 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Caution Wall:** 4 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Boxes:** 2 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Brick Incline:** 22 peça(s)
-

ELEV11.yaml - Beach Blast

Objetivo: Shoot off all the fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - There is a way to light these five rockets at the same time. But it can only be done at the speed of light. - Leave the lighting of this last one up to Mel.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver (Standard)` - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1019_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos `.ANM` carregados. - Totaliza aproximadamente **566 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 3 peça(s)
-  **Balloon:** 2 peça(s)

-  **Flashlight:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Wood Wall:** 8 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Fireworks:** 6 peça(s)
 -  **Sand Wall:** 2 peça(s)
 -  **Log Wall:** 6 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **tile:** 4 peça(s)
 -  **Color Block:** 2 peça(s)
 -  **Glass Building:** 1 peça(s)
 -  **apartment building:** 1 peça(s)
 -  **Candle:** 1 peça(s)
 -  **Sand Hill:** 1 peça(s)
 -  **Tropical Island:** 1 peça(s)
 -  **Star Glow:** 1 peça(s)
 -  **Star Burst:** 1 peça(s)
 -  **Star Sparkle:** 2 peça(s)
 -  **Dark Space:** 1 peça(s)
 -  **Ocean Wave:** 1 peça(s)
 -  **Breaking Wave:** 1 peça(s)
 -  **Surf Wave:** 1 peça(s)
 -  **Comet Streak:** 1 peça(s)
 -  **Metal Gears:** 4 peça(s)
 -  **Angelfish:** 1 peça(s)
 -  **Fish School:** 1 peça(s)
 -  **Fish Scatter:** 1 peça(s)
 -  **Fish Group:** 1 peça(s)
 -  **Tree Trunk:** 1 peça(s)
-

ELEV12.yaml - The Dream

Objetivo: Put the balls into the boxes. The green ball goes in the aquarium, the blue one in the crate, the yellow one in the basket, and the red one in the cardboard box.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - An incline might be helpful here.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1002_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 12 arquivos .ANM carregados. - Totaliza aproximadamente **94 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Anti-Gravity Pad:** 4 peça(s)
 -  **Boxes:** 4 peça(s)
 -  **Greco-Roman Wall:** 8 peça(s)
 -  **Programmable Ball:** 4 peça(s)
 -  **Granite Incline:** 4 peça(s)
 -  **Scaffold Barrier:** 5 peça(s)
 -  **Blimp:** 2 peça(s)
 -  **Sun:** 3 peça(s)
 -  **Big Cloud:** 15 peça(s)
 -  **Small Cloud:** 9 peça(s)
 -  **small mountain:** 4 peça(s)
 -  **medium mountain:** 2 peça(s)
-

ELEV13.yaml - Late for the World Cup

Objetivo: Put the soccer ball into the basket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - This all looks a little off-balance. - A large curved pipe will help get the ball around the corner. - Anti-gravity will move the basketball into the air, but it will still need help to get over the wall.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1018_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 15 arquivos .ANM carregados. - Totaliza aproximadamente **114 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Brick Wall:** 9 peça(s)
-  **Basketball:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Large Curved Pipe:** 2 peça(s)
-  **Soccer Ball:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Boxes:** 1 peça(s)
-  **Accelerator Tube:** 1 peça(s)
-  **Brick Incline:** 6 peça(s)

-  **Big Cloud:** 5 peça(s)
 -  **tile:** 8 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
 -  **Small Cloud:** 9 peça(s)
 -  **Snow Mountains:** 1 peça(s)
 -  **Iceberg:** 1 peça(s)
 -  **Snow Peaks:** 2 peça(s)
 -  **Greenhouse Dome:** 1 peça(s)
 -  **Ocean Wave:** 1 peça(s)
 -  **Breaking Wave:** 1 peça(s)
 -  **Surf Wave:** 6 peça(s)
-

ELEV14.yaml - Trap Newton

Objetivo: Trap Newton Mouse with the laundry basket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Get Newton to fall on the plunger of the remote bomb. - Use the remote bomb to blow up this wall and allow the pinball through. - Hmm....What would cut this rope?

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1005_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **133 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 7 peça(s)
-  **Rope:** 1 peça(s)
-  **Laundry Basket:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Cheese:** 2 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Brick Incline:** 1 peça(s)
-  **Glass Building:** 3 peça(s)
-  **apartment building:** 2 peça(s)
-  **office building:** 1 peça(s)
-  **car:** 1 peça(s)
-  **lamp post:** 4 peça(s)

ELEV15.yaml - Balloon Traveler

Objetivo: Use the tack on the left side of the screen to pop the balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Anti-gravity pads have an interesting effect on balloons. Place one here and see what happens!

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1002_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 5 arquivos .ANM carregados. - Totaliza aproximadamente **73 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Anti-Gravity Pad:** 4 peça(s)
 -  **Greco-Roman Wall:** 9 peça(s)
 -  **Granite Incline:** 12 peça(s)
-

ELEV16.yaml - Kitty in a Cage

Objetivo: Lure Curie Cat into the cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - There must be something that will work with this plug. - Here is some rotational energy waiting to be tapped.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1011_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos .ANM carregados. - Totaliza aproximadamente **402 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 5 peça(s)
-  **Pulley:** 5 peça(s)
-  **Belt:** 3 peça(s)
-  **Rope:** 1 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Gear:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Generator:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Cheese:** 1 peça(s)
-  **Can Opener:** 1 peça(s)

-  **Boxes:** 1 peça(s)
 -  **Tiny Gear:** 1 peça(s)
 -  **Trans-Roto-Matic:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Scaffold Barrier:** 8 peça(s)
-

ELEV17.yaml - The Food Machine.

Objetivo: Open the can, start the mixer, and heat up some coffee.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1007_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 16 arquivos `.ANM` carregados. - Totaliza aproximadamente **212 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Mouse Motor:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Generator:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Can Opener:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Electric Mixer:** 1 peça(s)
 -  **Yellow Brick Wall:** 8 peça(s)
 -  **Yellow Brick Incline:** 8 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

ELEV18.yaml - Night in the Mountains

Objetivo: Help Mel get back to his cabin.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - On occasion, parts will be used in unusual ways. This accelerator tube for instance...

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 22 arquivos .ANM carregados. - Totaliza aproximadamente **375 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 2 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Log Wall:** 8 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Log Incline:** 6 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Elm Tree:** 1 peça(s)
 -  **Spruce Tree:** 7 peça(s)
 -  **Little Cloud:** 2 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **purple mountain:** 2 peça(s)
 -  **Candle:** 1 peça(s)
 -  **Small Maple Tree:** 1 peça(s)
 -  **Small Pine Tree:** 12 peça(s)
 -  **small mountain:** 1 peça(s)
 -  **medium mountain:** 3 peça(s)
 -  **Pine Forest:** 8 peça(s)
 -  **Cave Opening:** 1 peça(s)
 -  **Forest Trees:** 2 peça(s)
 -  **Rocky Terrain:** 2 peça(s)
 -  **Moon:** 1 peça(s)
 -  **Star Glow:** 1 peça(s)
 -  **Star Burst:** 2 peça(s)
 -  **Star Sparkle:** 3 peça(s)
 -  **Dark Space:** 1 peça(s)
-

ELEV19.yaml - Double Whammy

Objetivo: Blow up both missiles.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Here's a nice spot for something tipsy.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1004_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos `.ANM` carregados. - Totaliza aproximadamente **187 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Topsy Trailer:** 1 peça(s)
 -  **Pulley:** 3 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Missile:** 2 peça(s)
 -  **Match-on-a-Spring:** 2 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Yellow Brick Incline:** 5 peça(s)
 -  **Candle:** 1 peça(s)
-

ELEV20.yaml - Deep Sea Explosion

Objetivo: Make the soccer ball float off the top of the screen.

Ambiente: - Gravidade: 238 - Pressão do Ar: 128

Dicas Ativas (2): - This basketball may be the key to the whole thing. - Use a rope to pull down on the remote bomb's plunger. A teeter-totter might help.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 16 arquivos `.ANM` carregados. - Totaliza aproximadamente **191 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Topsy Trailer:** 1 peça(s)
-  **Pulley:** 2 peça(s)
-  **Basketball:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Large Curved Pipe:** 1 peça(s)

-  **Soccer Ball:** 1 peça(s)
 -  **Sand Wall:** 7 peça(s)
 -  **Log Wall:** 1 peça(s)
 -  **Grass Incline:** 7 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Treasure Chest:** 1 peça(s)
 -  **Shell:** 1 peça(s)
 -  **Shell Spiral:** 1 peça(s)
 -  **Metal Gears:** 7 peça(s)
 -  **Orange Coral:** 1 peça(s)
 -  **Seaweed Tall:** 1 peça(s)
 -  **Kelp Plant:** 1 peça(s)
 -  **Seaweed Vine:** 1 peça(s)
 -  **Seaweed Long:** 1 peça(s)
 -  **Plant Stem:** 1 peça(s)
 -  **Red Crab:** 1 peça(s)
 -  **Jellyfish:** 1 peça(s)
 -  **Octopus:** 1 peça(s)
 -  **Striped Fish:** 1 peça(s)
 -  **Goldfish:** 1 peça(s)
 -  **Seahorse:** 1 peça(s)
 -  **Shark:** 1 peça(s)
 -  **Fish School:** 2 peça(s)
 -  **Fish Group:** 1 peça(s)
 -  **Great White Shark:** 2 peça(s)
-

ELEV21.yaml - Newton Drops a Message

Objetivo: Reveal Newton's message and get Curie Cat off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Who needs gravity in a spot like this? - What if this ball were to get blown to the left?

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1021_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 24 arquivos .ANM carregados. - Totaliza aproximadamente **537 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Balloon:** 1 peça(s)
-  **Curie Cat:** 1 peça(s)

-  **Remote Control:** 2 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Cheese:** 3 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Caution Wall:** 13 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Alligator:** 1 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 2 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Yellow Brick Wall:** 8 peça(s)
 -  **Yellow Brick Incline:** 4 peça(s)
 -  **Message Computer:** 7 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

ELEV22.yaml - Home Sweet Home

Objetivo: Help Newton Mouse get home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Some explosives might help remodel this wall. - Newton looks small enough to go through these pipes.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1007_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos .ANM carregados. - Totaliza aproximadamente **163 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Remote Control:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Pipe Wall:** 1 peça(s)
-  **Wood Wall:** 7 peça(s)
-  **Cheese:** 3 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Large Curved Pipe:** 8 peça(s)

-  **Mouse Hole:** 1 peça(s)
 -  **Accelerator Tube:** 4 peça(s)
 -  **Large Pipes:** 1 peça(s)
-

ELEV23.yaml - Feeding Curie

Objetivo: Open a can of cat food for Curie Cat.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Getting this bucket to fall may take some explosive thinking. - Some extra GEAR may be needed to get the power from this motor to the generator. - This generator needs to turn to get power flowing through the outlet.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1017_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 15 arquivos `.ANM` carregados. - Totaliza aproximadamente **419 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Pulley:** 4 peça(s)
 -  **Belt:** 3 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Curie Cat:** 1 peça(s)
 -  **Gear:** 4 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Generator:** 1 peça(s)
 -  **Mandrill Motor:** 1 peça(s)
 -  **Wood Wall:** 2 peça(s)
 -  **Can Opener:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Lattice Archway:** 9 peça(s)
 -  **Forest Trees:** 3 peça(s)
 -  **Rocky Terrain:** 2 peça(s)
-

ELEV24.yaml - Fishtank Target

Objetivo: Blow up the fish tank with the missile.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Find a source of rotational energy to drive this machine. - Use the inclines to adjust the missile's trajectory.

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1007_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 22 arquivos .ANM carregados. - Totaliza aproximadamente **328 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 2 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Belt:** 3 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Gear:** 6 peça(s)
 -  **Fish Tank:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Tiny Gear:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Brick Incline:** 6 peça(s)
 -  **Spruce Tree:** 8 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 1 peça(s)
 -  **purple mountain:** 1 peça(s)
 -  **Small Pine Tree:** 6 peça(s)
 -  **Small Cloud:** 2 peça(s)
 -  **medium mountain:** 2 peça(s)
 -  **Pine Forest:** 2 peça(s)
 -  **Forest Trees:** 1 peça(s)
 -  **Rocky Terrain:** 1 peça(s)
-

ELEV25.yaml - Ghost Dynamite

Objetivo: Light the dynamite with the candle.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Try connecting a rope from the match to a balloon. - Some anti-gravity will keep the candle floating.
- Try putting a balloon in here.

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1000_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher*

pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **198 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 10 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Wood Wall:** 9 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Anti-Gravity Pad:** 7 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

ELEV26.yaml - The Merry Mixer

Objetivo: Turn on the electric mixer.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Use hedge trimmers to cut the balloon loose. - A springboard can bounce the ball up, over, and into the pipe. - Anti-gravity pads have a funny effect on balloons.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de**

Áudio: - Música de Fundo: `SX_1020_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via*

`AudioDispatcher` pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **153 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 10 peça(s)
-  **Balloon:** 1 peça(s)
-  **Pulley:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Pinball:** 1 peça(s)

-  **Thumb Tack:** 1 peça(s)
 -  **Large Curved Pipe:** 2 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Brick Incline:** 15 peça(s)
 -  **Electric Mixer:** 1 peça(s)
-

HH1.yaml - Shootin' Hoops

Objetivo: Put the basketball into the box on the right.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **175 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 1 peça(s)
 -  **Conveyor Belt:** 2 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Basketball:** 1 peça(s)
 -  **Gear:** 2 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Springboard:** 3 peça(s)
 -  **Cheese:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Large Curved Pipe:** 2 peça(s)
 -  **Pool Ball:** 5 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Large Pipes:** 1 peça(s)
 -  **Grass Incline:** 1 peça(s)
 -  **Log Incline:** 1 peça(s)
 -  **Granite Incline:** 1 peça(s)
 -  **Brick Incline:** 1 peça(s)
 -  **Yellow Brick Incline:** 4 peça(s)
-

HH2.yaml - Pool Shark

Objetivo: Put the eight ball in the corner pocket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1007_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos .ANM carregados. - Totaliza aproximadamente **239 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 1 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Large Curved Pipe:** 6 peça(s)
 -  **Super Ball:** 1 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Pinball Bumper:** 4 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Accelerator Tube:** 2 peça(s)
 -  **Large Pipes:** 2 peça(s)
 -  **Grass Incline:** 3 peça(s)
 -  **Log Incline:** 1 peça(s)
 -  **Granite Incline:** 1 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Pool Table Wall:** 3 peça(s)
 -  **Pool Table Pocket:** 1 peça(s)
 -  **Yellow Brick Wall:** 1 peça(s)
 -  **Yellow Brick Incline:** 3 peça(s)
 -  **Teeter-Totter:** 2 peça(s)
-

HH3.yaml - High Impact Fishing

Objetivo: Break all of the fish tanks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1001_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 11 arquivos .ANM carregados. - Totaliza aproximadamente **242 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Topsy Trailer:** 4 peça(s)

-  **Pulley:** 3 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Fish Tank:** 4 peça(s)
 -  **Electric Fan:** 3 peça(s)
 -  **Large Curved Pipe:** 4 peça(s)
 -  **Pool Ball:** 3 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Large Pipes:** 3 peça(s)
 -  **Electrical Outlet:** 3 peça(s)
-

HH4.yaml - Plunk it in the Pocket

Objetivo: Put each pool ball into the nearest pocket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de**

Áudio: - Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via*

AudioDispatcher pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 28 arquivos `.ANM` carregados. - Totaliza aproximadamente **484 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 2 peça(s)
-  **Conveyor Belt:** 2 peça(s)
-  **Mouse Motor:** 3 peça(s)
-  **Belt:** 5 peça(s)
-  **Jack-in-the-box:** 3 peça(s)
-  **Gear:** 2 peça(s)
-  **Electric Fan:** 2 peça(s)
-  **Boxing Glove:** 4 peça(s)
-  **Tennis Ball:** 2 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Vacuum:** 2 peça(s)
-  **Cheese:** 3 peça(s)
-  **Large Curved Pipe:** 3 peça(s)
-  **Super Ball:** 1 peça(s)
-  **Pool Ball:** 4 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Nitroglycerine:** 6 peça(s)
-  **Toaster:** 1 peça(s)
-  **Accelerator Tube:** 1 peça(s)
-  **Grass Incline:** 1 peça(s)
-  **Log Incline:** 1 peça(s)

-  **Granite Incline:** 1 peça(s)
 -  **Brick Incline:** 1 peça(s)
 -  **Pool Cue:** 5 peça(s)
 -  **Pool Table Wall:** 6 peça(s)
 -  **Pool Table Pocket:** 6 peça(s)
 -  **Electrical Outlet:** 3 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
-

HH5.yaml - Lawn Bowling

Objetivo: Put the bowling ball into the small glass aquarium.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1013_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 12 arquivos `.ANM` carregados. - Totaliza aproximadamente **166 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Conveyor Belt:** 3 peça(s)
 -  **Mouse Motor:** 2 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Springboard:** 2 peça(s)
 -  **Grass Floor:** 3 peça(s)
 -  **Alligator:** 2 peça(s)
 -  **Pinball Bumper:** 2 peça(s)
 -  **Anti-Gravity Pad:** 2 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Grass Incline:** 8 peça(s)
-

HH6.yaml - Crazy Cannon

Objetivo: Fire the cannon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1016_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **300 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 2 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Gear:** 3 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Cannon:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Nitroglycerine:** 4 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Steel Cable:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Programmable Ball:** 1 peça(s)
 -  **Blimp:** 2 peça(s)
-

HH7.yaml - The Balloon Buster's Ball

Objetivo: Pop all of the balloons.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1020_11025.wav` - *Efeitos Sonoros*

(SFX): *Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais**

(ANM & PNG): - Requer 12 arquivos `.ANM` carregados. - Totaliza aproximadamente **111 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 2 peça(s)
-  **Balloon:** 11 peça(s)
-  **Belt:** 5 peça(s)
-  **Rope:** 3 peça(s)
-  **Gear:** 10 peça(s)
-  **Dynamite:** 5 peça(s)
-  **Remote Control:** 4 peça(s)
-  **Boat Cleat:** 6 peça(s)
-  **Hedge Trimmers:** 5 peça(s)
-  **Thumb Tack:** 11 peça(s)

-  **Remote Control Explosives:** 4 peça(s)
 -  **Nitroglycerine:** 6 peça(s)
-

HH8.yaml - Setting Sail

Objetivo: Make some toast.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1018_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 24 arquivos `.ANM` carregados. - Totaliza aproximadamente **294 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 1 peça(s)
 -  **Topsy Trailer:** 1 peça(s)
 -  **Conveyor Belt:** 4 peça(s)
 -  **Pulley:** 4 peça(s)
 -  **Belt:** 4 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Gear:** 8 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Mandrill Motor:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Aladdin's Lamp:** 1 peça(s)
 -  **Caution Wall:** 2 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Toaster:** 1 peça(s)
 -  **Log Incline:** 1 peça(s)
 -  **Granite Incline:** 1 peça(s)
 -  **Teeter-Totter:** 2 peça(s)
-

HH9.yaml - Score!

Objetivo: Put the bowling ball into the basket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de**

Áudio: - Música de Fundo: `SX_1017_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 16 arquivos `.ANM` carregados. - Totaliza aproximadamente **202 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Wood incline:** 6 peça(s)
 -  **Mouse Motor:** 2 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Jack-in-the-box:** 2 peça(s)
 -  **Gear:** 2 peça(s)
 -  **Boxing Glove:** 2 peça(s)
 -  **Springboard:** 3 peça(s)
 -  **Vacuum:** 1 peça(s)
 -  **Caution Wall:** 2 peça(s)
 -  **Large Curved Pipe:** 2 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Large Pipes:** 1 peça(s)
-

HH10.yaml - Fire One

Objetivo: Launch the missile.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos `.ANM` carregados. - Totaliza aproximadamente **348 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
-  **Balloon:** 1 peça(s)
-  **Conveyor Belt:** 3 peça(s)
-  **Pulley:** 1 peça(s)
-  **Rope:** 2 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boat Cleat:** 3 peça(s)

-  **Electric Fan:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Aladdin's Lamp:** 1 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Greco-Roman Wall:** 1 peça(s)
 -  **Granite Incline:** 4 peça(s)
 -  **Electrical Outlet:** 2 peça(s)
-

HH11.yaml - The Rocketeer's Revenge

Objetivo: Launch the rocket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. -

Totaliza aproximadamente **451 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 4 peça(s)
-  **Topsy Trailer:** 1 peça(s)
-  **Balloon:** 2 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 2 peça(s)
-  **Rope:** 1 peça(s)
-  **Gear:** 2 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Generator:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Boxing Glove:** 2 peça(s)
-  **Rocket:** 1 peça(s)
-  **Aladdin's Lamp:** 1 peça(s)
-  **Pipe Wall:** 3 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Alligator:** 1 peça(s)

-  **Flint & Tinder:** 1 peça(s)
 -  **Sand Wall:** 1 peça(s)
 -  **Brick Incline:** 2 peça(s)
 -  **Blimp:** 2 peça(s)
 -  **Electrical Outlet:** 2 peça(s)
-

HH12.yaml - Pool Trick

Objetivo: Shoot each of the numbered balls into the nearest pocket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `GravitySolver (Standard)` - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos `.ANM` carregados. - Totaliza aproximadamente **164 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 3 peça(s)
 -  **Pinball:** 4 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Pool Ball:** 8 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Anti-Gravity Pad:** 4 peça(s)
 -  **Pool Cue:** 9 peça(s)
 -  **Pool Table Wall:** 6 peça(s)
 -  **Pool Table Pocket:** 6 peça(s)
-

HH13.yaml - Big Red Head to Head

Objetivo: Launch the rocket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` - `ElectricalPropagationSystem` - `GravitySolver (Standard)` - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **406 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 8 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Belt:** 2 peça(s)
-  **Gear:** 2 peça(s)
-  **Bike Pump:** 1 peça(s)

-  **Electric Switch & Outlet:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Tennis Ball:** 3 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Brick Incline:** 6 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Egg Timer:** 2 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH14.yaml - Blimp Blaster

Objetivo: Shoot down the blimp with a missile.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1004_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 13 arquivos `.ANM` carregados. - Totaliza aproximadamente **208 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 2 peça(s)
-  **Pulley:** 4 peça(s)
-  **Rope:** 2 peça(s)
-  **Boat Cleat:** 3 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Aladdin's Lamp:** 1 peça(s)
-  **Missile:** 2 peça(s)
-  **Match-on-a-Spring:** 2 peça(s)
-  **Cinder Block Wall:** 2 peça(s)
-  **Blimp:** 1 peça(s)
-  **Pinball Flipper:** 1 peça(s)

-  **Candle:** 1 peça(s)
-

HH15.yaml - Mouse Race

Objetivo: Help Newton Mouse get home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 13 arquivos `.ANM` carregados. - Totaliza aproximadamente **210 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 1 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Laundry Basket:** 1 peça(s)
 -  **Gear:** 2 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Cheese:** 12 peça(s)
 -  **Mouse Hole:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Greco-Roman Wall:** 19 peça(s)
 -  **Granite Incline:** 12 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

HH16.yaml - Coffee Cage

Objetivo: Put the pot in the cage and brew some coffee.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1001_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **225 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Rope:** 1 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Thumb Tack:** 1 peça(s)

-  **Coffee Pot:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Cinder Block Wall:** 2 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH17.yaml - Rocket Rivalry

Objetivo: See who can launch the rocket!

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1005_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 35 arquivos `.ANM` carregados. - Totaliza aproximadamente **681 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Pulley:** 2 peça(s)
-  **Belt:** 2 peça(s)
-  **Rope:** 1 peça(s)
-  **Gear:** 2 peça(s)
-  **Bucket:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Springboard:** 2 peça(s)
-  **Pinball:** 1 peça(s)
-  **Tennis Ball:** 1 peça(s)
-  **Aladdin's Lamp:** 1 peça(s)
-  **Mel Schlemming:** 1 peça(s)
-  **Large Curved Pipe:** 1 peça(s)
-  **Super Ball:** 1 peça(s)
- **Alligator:** 1 peça(s)

-  **Coffee Pot:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Greco-Roman Wall:** 2 peça(s)
 -  **Trans-Roto-Matic:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Granite Incline:** 6 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

HH18.yaml - Too Many Mice

Objetivo: Help Newton and his pals each find a mouse hole.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1006_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **224 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Pinwheel:** 1 peça(s)
-  **Newton Mouse:** 6 peça(s)
-  **Cheese:** 5 peça(s)
-  **Thumb Tack:** 1 peça(s)
-  **Coffee Pot:** 1 peça(s)
-  **Mouse Hole:** 2 peça(s)
-  **Anti-Gravity Pad:** 3 peça(s)
-  **Flint & Tinder:** 1 peça(s)
-  **Greco-Roman Wall:** 16 peça(s)

-  **Granite Incline:** 27 peça(s)
-

HH19.yaml - See-Saw Asylum

Objetivo: Pop the balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1004_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **226 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Trap Door:** 2 peça(s)
 -  **Greco-Roman Wall:** 8 peça(s)
 -  **Granite Incline:** 9 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Teeter-Totter:** 2 peça(s)
-

HH20.yaml - Bill's Bad Day

Objetivo: Break Bill's aquarium.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **342 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 2 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Fish Tank:** 1 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Wood Wall:** 8 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
-

HH21.yaml - Zap Trap.

Objetivo: Hit the sensor with the red laser.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 16 arquivos `.ANM` carregados. - Totaliza aproximadamente **317 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Wood Wall:** 1 peça(s)
-  **Pool Ball:** 1 peça(s)
-  **Boxes:** 1 peça(s)
-  **Flint & Tinder:** 1 peça(s)
-  **Hot Air Balloon:** 1 peça(s)
-  **Red Laser:** 1 peça(s)

-  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Laser-Activated Plug:** 2 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Laser Detector:** 1 peça(s)
-

HH22.yaml - Laser Lighter

Objetivo: Light the hot air balloon at the bottom left and make it float off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 15 arquivos `.ANM` carregados. - Totaliza aproximadamente **145 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Greco-Roman Wall:** 5 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Granite Incline:** 10 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Candle:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH23.yaml - Balloons or Bust

Objetivo: Pop both balloons.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 12 arquivos `.ANM` carregados. - Totaliza aproximadamente **135 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 2 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH24.yaml - Punch Mel

Objetivo: Knock Mel off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1015_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos .ANM carregados. - Totaliza aproximadamente **282 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 2 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Brick Incline:** 4 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH25.yaml - Pass to the Glass

Objetivo: Get the soccer ball into the aquarium.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1015_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 13 arquivos .ANM carregados. - Totaliza aproximadamente **134 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Large Curved Pipe:** 3 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Greco-Roman Wall:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Granite Incline:** 6 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH26.yaml - Slam Dunk

Objetivo: Get the basketball into the cardboard box at the bottom of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1015_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos .ANM carregados. - Totaliza aproximadamente **232 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Pulley:** 4 peça(s)
-  **Basketball:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Tennis Ball:** 3 peça(s)
-  **Pipe Wall:** 1 peça(s)
-  **Thumb Tack:** 1 peça(s)

-  **Remote Control Explosives:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Trap Door:** 2 peça(s)
 -  **Nitroglycerine:** 2 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH27.yaml - Sharpshooter

Objetivo: Light the candle with the laser.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **156 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Log Wall:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

HH28.yaml - Ragtime Rollers

Objetivo: Put the bowling ball into the aquarium and the silver ball into the cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1021_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos .ANM carregados. - Totaliza aproximadamente **170 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Boxes:** 2 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Cinder Block Wall:** 5 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **T-Connector:** 1 peça(s)
 -  **Granite Incline:** 13 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH29.yaml - Loony Lamp

Objetivo: Light the oil lamp.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1020_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos .ANM carregados. - Totaliza aproximadamente **201 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Rope:** 3 peça(s)
-  **Laundry Basket:** 1 peça(s)
-  **Bucket:** 1 peça(s)

-  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Aladdin's Lamp:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Yellow Brick Wall:** 4 peça(s)
 -  **Yellow Brick Incline:** 9 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Candle:** 2 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH30.yaml - Little Hindenburg

Objetivo: Destroy the blimp with the remote bomb.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1018_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. -

Totaliza aproximadamente **297 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Rope:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Wood Wall:** 2 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Trap Door:** 1 peça(s)
-  **Green Laser:** 1 peça(s)
-  **Blue Laser:** 1 peça(s)
- **Angled Mirror:** 4 peça(s)

-  **Blimp:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Electrical Outlet:** 2 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH31.yaml - Puppy Popper

Objetivo: Pop the doggy balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` - `GravitySolver (Standard)` - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1017_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **120 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Bucket:** 2 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Springboard:** 2 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Granite Incline:** 2 peça(s)
 -  **Yellow Brick Incline:** 2 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

HH32.yaml - Mixer Moxie

Objetivo: Turn on the electric mixer.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` - `ElectricalPropagationSystem` - `GravitySolver (Standard)` - **Recursos de Áudio:** - Música de Fundo: `SX_1017_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 21 arquivos `.ANM` carregados. - Totaliza aproximadamente **497 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)

-  **Balloon:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Cinder Block Wall:** 6 peça(s)
 -  **Log Incline:** 1 peça(s)
 -  **Electric Mixer:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

HH33.yaml - Shootin' Newton

Objetivo: Get Newton Mouse into the aquarium.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1007_11025.wav` - *Efeitos Sonoros*

(SFX): *Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 21 arquivos `.ANM` carregados. - Totaliza aproximadamente **612 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 2 peça(s)
-  **Pulley:** 2 peça(s)
-  **Belt:** 3 peça(s)
-  **Rope:** 1 peça(s)
-  **Gear:** 3 peça(s)
-  **Bucket:** 1 peça(s)
-  **Cannon:** 1 peça(s)
-  **Dynamite:** 5 peça(s)
-  **Remote Control:** 2 peça(s)
-  **Rocket:** 1 peça(s)

-  **Newton Mouse:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Nitroglycerine:** 6 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Trans-Roto-Matic:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Yellow Brick Wall:** 7 peça(s)
-

HH34.yaml - Blimp Buster

Objetivo: Destroy the blimp!

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1009_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **210 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Greco-Roman Wall:** 2 peça(s)
 -  **Granite Incline:** 1 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

HH35.yaml - The Pop Shop

Objetivo: Pop the silver balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1014_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **226 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 2 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Gear:** 1 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Pinwheel:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Cinder Block Wall:** 1 peça(s)
 -  **Tiny Gear:** 1 peça(s)
 -  **Trans-Roto-Matic:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

HH36.yaml - Pop the Purple Puppy

Objetivo: Pop the balloon before it escapes!

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **128 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)

-  **Pulley:** 3 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Laundry Basket:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Cinder Block Wall:** 2 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Granite Incline:** 3 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH37.yaml - 'Gator on Guard

Objetivo: Put Curie Cat in the cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **645 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Rope:** 2 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Grass Floor:** 1 peça(s)
-  **Alligator:** 1 peça(s)
-  **Boxes:** 1 peça(s)
-  **Trap Door:** 1 peça(s)

-  **Match-on-a-Spring**: 1 peça(s)
 -  **Log Wall**: 7 peça(s)
 -  **Log Incline**: 8 peça(s)
 -  **Blimp**: 1 peça(s)
 -  **Egg Timer**: 1 peça(s)
 -  **Teeter-Totter**: 2 peça(s)
-

HH38.yaml - The Age of Aquarium

Objetivo: Break both fish tanks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - LaserRaycastSystem - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1008_11025.wav - Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos .ANM carregados. - Totaliza aproximadamente **507 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball**: 1 peça(s)
 -  **Brick Wall**: 2 peça(s)
 -  **Fish Tank**: 2 peça(s)
 -  **Cannon**: 1 peça(s)
 -  **Flashlight**: 1 peça(s)
 -  **Magnifying Glass**: 1 peça(s)
 -  **Boxing Glove**: 1 peça(s)
 -  **Springboard**: 1 peça(s)
 -  **Pinball**: 1 peça(s)
 -  **Large Curved Pipe**: 4 peça(s)
 -  **Missile**: 1 peça(s)
 -  **Flint & Tinder**: 1 peça(s)
 -  **Brick Incline**: 7 peça(s)
 -  **Blimp**: 2 peça(s)
 -  **Pinball Flipper**: 1 peça(s)
 -  **Egg Timer**: 1 peça(s)
 -  **Teeter-Totter**: 1 peça(s)
-

HH39.yaml - Rootin' Tootin' Newton

Objetivo: Get Newton Mouse into the caution box at the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo:

SX_1013_11025.wav - Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos `.ANM` carregados. - Totaliza aproximadamente **207 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Pulley:** 5 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Bucket:** 2 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Cheese:** 5 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Caution Wall:** 10 peça(s)
 -  **Large Curved Pipe:** 3 peça(s)
 -  **Super Ball:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Large Pipes:** 3 peça(s)
 -  **T-Connector:** 1 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH40.yaml - The Perky Percolator

Objetivo: Make coffee by placing a flame under it.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1012_11025.wav - Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 22 arquivos `.ANM` carregados. - Totaliza aproximadamente **275 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Rope:** 2 peça(s)
-  **Bucket:** 1 peça(s)
-  **Dynamite:** 1 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Boxing Glove:** 1 peça(s)

-  **Hedge Trimmers:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Greco-Roman Wall:** 4 peça(s)
 -  **Granite Incline:** 3 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH41.yaml - Blimp Exodus

Objetivo: Get the blimp off the screen any way you can.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1011_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **410 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 1 peça(s)
-  **Rope:** 2 peça(s)
-  **Laundry Basket:** 1 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Dynamite:** 7 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Nitroglycerine:** 2 peça(s)
-  **Flint & Tinder:** 1 peça(s)

-  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 2 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Yellow Brick Wall:** 8 peça(s)
 -  **Yellow Brick Incline:** 6 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH42.yaml - Omega Launch

Objetivo: Launch the rocket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher`*

pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 29 arquivos `.ANM` carregados. -

Totaliza aproximadamente **397 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boat Cleat:** 3 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Boxes:** 1 peça(s)
-  **Trap Door:** 2 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Steel Cable:** 1 peça(s)
-  **Tin Snips:** 1 peça(s)

-  **Flint & Tinder:** 1 peça(s)
 -  **Cinder Block Wall:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Granite Incline:** 4 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH43.yaml - 'Gator in the Garden

Objetivo: Help Mel Schlemming get back to his cozy home in the Everglades.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 24 arquivos `.ANM` carregados. - Totaliza aproximadamente **473 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 2 peça(s)
-  **Balloon:** 1 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Pulley:** 2 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Mel Schlemming:** 1 peça(s)
-  **Mel's House:** 1 peça(s)
-  **Grass Floor:** 1 peça(s)
-  **Alligator:** 1 peça(s)
-  **Trap Door:** 1 peça(s)
-  **Nitroglycerine:** 1 peça(s)
-  **Cinder Block Wall:** 1 peça(s)
-  **Large Pipes:** 1 peça(s)
-  **Blimp:** 1 peça(s)
-  **Pinball Flipper:** 1 peça(s)
-  **Egg Timer:** 1 peça(s)

-  **Teeter-Totter:** 1 peça(s)
-

HH44.yaml - Skyrockets in Flight

Objetivo: Launch the fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1001_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **308 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Wood Wall:** 6 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Log Incline:** 3 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH45.yaml - Log Hangar

Objetivo: Get the balloon into the log box at the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos `.ANM` carregados. - Totaliza aproximadamente **285 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Pulley:** 4 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Log Wall:** 7 peça(s)
 -  **Log Incline:** 4 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH46.yaml - Bye-Bye Bucket

Objetivo: Knock the bucket off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 25 arquivos `.ANM` carregados. - Totaliza aproximadamente **400 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Pulley:** 2 peça(s)
-  **Rope:** 2 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)

-  **Magnifying Glass:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Greco-Roman Wall:** 7 peça(s)
 -  **Granite Incline:** 4 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH47.yaml - Bucket Battle

Objetivo: Put the bucket on top of the caution wall.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 13 arquivos `.ANM` carregados. - Totaliza aproximadamente **122 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Rope:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Caution Wall:** 8 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 3 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Leaky Bucket:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH48.yaml - Hot Air Heroics

Objetivo: Light the balloon so that it floats off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1004_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **485 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Mouse Motor:** 1 peça(s)
 -  **Pulley:** 5 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Generator:** 1 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Mandrill Motor:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HH49.yaml - Bigshot

Objetivo: Fire the cannon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1004_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher`*

pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 24 arquivos .ANM carregados. - Totaliza aproximadamente **580 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Pulley:** 5 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Cannon:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Greco-Roman Wall:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 2 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

HH50.yaml - Last Launch

Objetivo: Launch all three fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1005_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos .ANM carregados. - Totaliza aproximadamente **417 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 5 peça(s)
-  **Rope:** 2 peça(s)

-  **Bucket:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Fireworks:** 3 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Yellow Brick Wall:** 7 peça(s)
 -  **Yellow Brick Incline:** 3 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HLEV1.yaml - Things That Go Boom!!!

Objetivo: Set off the fireworks on the left side of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 25 arquivos `.ANM` carregados. - Totaliza aproximadamente **628 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 21 peça(s)
-  **Topsy Trailer:** 1 peça(s)
-  **Pulley:** 1 peça(s)
-  **Rope:** 2 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Dynamite:** 4 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Flashlight:** 1 peça(s)

-  **Captain Z Super Phazer:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Pipe Wall:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Nitroglycerine:** 2 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Brick Incline:** 3 peça(s)
 -  **Blimp:** 2 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HLEV2.yaml - Curie and the Fish Squish

Objetivo: Break all six fish tanks and help Curie Cat exit the right side of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - LaserRaycastSystem -

Recursos de Áudio: - Música de Fundo: SX_1010_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 11 arquivos .ANM carregados. - Totaliza aproximadamente **508 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Curie Cat:** 1 peça(s)
 -  **Fish Tank:** 6 peça(s)
 -  **Cannon:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Caution Wall:** 12 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Yellow Brick Incline:** 4 peça(s)
 -  **Candle:** 1 peça(s)
-

HLEV3.yaml - Newton's Cafe

Objetivo: Help Newton Mouse treat all his pals to a cheese dinner.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1017_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos .ANM carregados. - Totaliza aproximadamente **174 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 2 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Newton Mouse:** 6 peça(s)
 -  **Wood Wall:** 11 peça(s)
 -  **Cheese:** 2 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
-

HLEV4.yaml - Curious Curie

Objetivo: Help Curie Cat get out of the basement. Make her exit the bottom of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1020_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 21 arquivos .ANM carregados. - Totaliza aproximadamente **658 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
-  **Topsy Trailer:** 2 peça(s)
-  **Pulley:** 2 peça(s)
-  **Rope:** 2 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Tennis Ball:** 2 peça(s)
-  **Remote Control Explosives:** 1 peça(s)

-  **Coffee Pot:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Greco-Roman Wall:** 27 peça(s)
 -  **Granite Incline:** 5 peça(s)
 -  **Pool Cue:** 1 peça(s)
-

HLEV5.yaml - Cage the Kitty

Objetivo: Put Curie Cat in the cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Can you think of anything that might "phaze" this dynamite? - Pipes. Newton Mouse. Does that GENERATE any ideas?

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1016_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **563 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 4 peça(s)
-  **Balloon:** 1 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Pulley:** 1 peça(s)
-  **Belt:** 1 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Dynamite:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Cheese:** 1 peça(s)
-  **Large Curved Pipe:** 4 peça(s)
-  **Alligator:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Boxes:** 1 peça(s)
-  **Steel Cable:** 1 peça(s)
-  **Cinder Block Wall:** 1 peça(s)
-  **Greco-Roman Wall:** 2 peça(s)
-  **Log Wall:** 3 peça(s)
-  **Accelerator Tube:** 2 peça(s)

-  **Granite Incline:** 6 peça(s)
 -  **Yellow Brick Wall:** 1 peça(s)
-

HLEV6.yaml - Target Practice

Objetivo: Use the boxing glove to pop the balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - The cannon ball would roll a lot better if there wasn't a wall in the way. - Place the cannon here to give you the push you need.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **310 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Cannon:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Large Curved Pipe:** 6 peça(s)
 -  **Grass Floor:** 1 peça(s)
 -  **Sand Wall:** 3 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Large Pipes:** 1 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Sand Dune:** 2 peça(s)
 -  **Desert Dune:** 2 peça(s)
 -  **Sand Hill:** 3 peça(s)
 -  **Tropical Island:** 2 peça(s)
-

HLEV7.yaml - Plural Pop

Objetivo: Pop both of the balloons.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Anti-gravity will move the ball in the right direction. - Once you get the ball up here, you'll need to "accelerate" it over to the right.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1007_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **199 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Brick Wall:** 11 peça(s)
 -  **Balloon:** 2 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Large Curved Pipe:** 2 peça(s)
 -  **Anti-Gravity Pad:** 2 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Accelerator Tube:** 2 peça(s)
 -  **Brick Incline:** 8 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Small Cloud:** 3 peça(s)
-

HLEV8.yaml - Laser Coffee.

Objetivo: Get the coffee pot off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Did you know three colored laser beams mix to make a single white one?

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 22 arquivos `.ANM` carregados. - Totaliza aproximadamente **357 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 1 peça(s)
-  **Belt:** 1 peça(s)
-  **Electric Switch & Outlet:** 2 peça(s)
-  **Pinball:** 1 peça(s)

-  **Wood Wall:** 2 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 13 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Spruce Tree:** 2 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 2 peça(s)
 -  **purple mountain:** 2 peça(s)
 -  **Small Pine Tree:** 4 peça(s)
 -  **Small Cloud:** 2 peça(s)
 -  **small mountain:** 1 peça(s)
 -  **medium mountain:** 3 peça(s)
-

HLEV9.yaml - Newton Goes Bowling

Objetivo: Put the bowling ball into the cardboard box.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - You'll need help to turn on the generator as you push down the plunger. (A bucket-load of help!)

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1003_11025.wav` - **Efeitos Sonoros (SFX):** *Despachados dinamicamente via `AudioDispatcher`*

pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 27 arquivos `.ANM` carregados. -

Totaliza aproximadamente **273 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Pulley:** 2 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Bucket:** 2 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Electric Motor:** 1 peça(s)

-  **Cheese:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Log Wall:** 9 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Angled Mirror:** 2 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Log Incline:** 12 peça(s)
 -  **Elm Tree:** 1 peça(s)
 -  **Spruce Tree:** 12 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 2 peça(s)
 -  **purple mountain:** 3 peça(s)
 -  **Small Pine Tree:** 10 peça(s)
 -  **Small Cloud:** 3 peça(s)
 -  **small mountain:** 3 peça(s)
 -  **medium mountain:** 3 peça(s)
-

HLEV10.yaml - Toast Roast

Objetivo: It's breakfast time. Let's roast some toast.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Even a mouse could pop open the Jack-in-the-box. - Use mouse power to get the ball over to the toaster.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de**

Áudio: - Música de Fundo: `SX_0_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via*

`AudioDispatcher` pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **461 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 2 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 2 peça(s)
-  **Belt:** 2 peça(s)
-  **Basketball:** 1 peça(s)
-  **Rope:** 2 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Electric Switch & Outlet:** 2 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Electric Fan:** 1 peça(s)
-  **Baseball:** 1 peça(s)

-  **Hedge Trimmers:** 1 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Toaster:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Yellow Brick Wall:** 1 peça(s)
 -  **Color Block:** 2 peça(s)
 -  **Ocean Wave:** 1 peça(s)
 -  **Surf Wave:** 3 peça(s)
 -  **Rising Bubbles:** 2 peça(s)
 -  **Bubble Stream:** 2 peça(s)
 -  **Floating Bubbles:** 3 peça(s)
 -  **Green Mountain:** 1 peça(s)
 -  **Mossy Rock:** 1 peça(s)
 -  **Green Cliff:** 1 peça(s)
 -  **Shipwreck:** 1 peça(s)
 -  **Submarine:** 1 peça(s)
 -  **Ancient Ruins:** 1 peça(s)
 -  **Stone Temple:** 1 peça(s)
-

HLEV11.yaml - Clash of the Critters

Objetivo: Help Curie Cat and Newton Mouse blow each other off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Carefully placed explosives could do a double duty.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_0_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **488 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 7 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Cannon:** 2 peça(s)
-  **Remote Control:** 2 peça(s)
-  **Flashlight:** 2 peça(s)
-  **Lava Lamp:** 1 peça(s)
-  **Magnifying Glass:** 2 peça(s)
-  **Newton Mouse:** 1 peça(s)

-  **Cheese:** 1 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Can Opener:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Brick Incline:** 14 peça(s)
 -  **Electrical Outlet:** 2 peça(s)
-

HLEV12.yaml - Cheese Please!

Objetivo: Put Newton Mouse into the cage at the bottom. Put the center piece of cheese in with him.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Whatever is placed here should help the ball get to the timer, and clear the way for Newton. - What happens when cheese chunks collide?

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **406 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 4 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Remote Control:** 2 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Cheese:** 5 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Cinder Block Wall:** 9 peça(s)
 -  **Granite Incline:** 3 peça(s)
 -  **Brick Incline:** 1 peça(s)
 -  **Scaffold Barrier:** 7 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

HLEV13.yaml - Danger Blimp

Objetivo: Get the blimp into the hangar on the left.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1023_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **262 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 6 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 3 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Yellow Brick Wall:** 1 peça(s)
 -  **purple mountain:** 3 peça(s)
 -  **Small Cloud:** 2 peça(s)
 -  **small mountain:** 2 peça(s)
 -  **medium mountain:** 5 peça(s)
-

HLEV14.yaml - Pompeian Rhapsody

Objetivo: Light the candle and turn on the lava lamp.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - There must be some trick to a match-on-a-stick!

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1006_11025.wav` - *Efeitos Sonoros*

(SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG)**: - Requer 12 arquivos `.ANM` carregados. - Totaliza aproximadamente **120 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon**: 1 peça(s)
 -  **Mouse Motor**: 1 peça(s)
 -  **Pulley**: 2 peça(s)
 -  **Belt**: 1 peça(s)
 -  **Rope**: 2 peça(s)
 -  **Lava Lamp**: 1 peça(s)
 -  **Anti-Gravity Pad**: 1 peça(s)
 -  **Match-on-a-Spring**: 1 peça(s)
 -  **Greco-Roman Wall**: 13 peça(s)
 -  **Roto-Trans Converter**: 1 peça(s)
 -  **Granite Incline**: 11 peça(s)
 -  **Color Block**: 7 peça(s)
 -  **Candle**: 1 peça(s)
-

HLEV15.yaml - Blimp-Be-Gone

Objetivo: Blow up the blimp before it can get off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Something as small as a mouse can start off this whole puzzle. - Something you See (or Saw) in the parts bin might be a tip-off. - The blimp's starting point is also the missile's launching point.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo:

`SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG)**: - Requer 21 arquivos `.ANM` carregados. - Totaliza aproximadamente **299 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt**: 1 peça(s)
-  **Mouse Motor**: 1 peça(s)
-  **Pulley**: 4 peça(s)
-  **Belt**: 1 peça(s)
-  **Rope**: 2 peça(s)
-  **Bucket**: 2 peça(s)
-  **Dynamite**: 2 peça(s)
-  **Electric Switch & Outlet**: 1 peça(s)
-  **Remote Control**: 1 peça(s)
-  **Boat Cleat**: 1 peça(s)
-  **Pinball**: 1 peça(s)

-  **Tennis Ball:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 8 peça(s)
 -  **Anti-Gravity Pad:** 2 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Yellow Brick Wall:** 3 peça(s)
 -  **Yellow Brick Incline:** 9 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

HLEV16.yaml - Newton's Special Brew

Objetivo: Get the coffee to start brewing.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - A perfect place to catch some wind. - A grand place for Newton's favorite dairy product.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` - `ElectricalPropagationSystem` - `GravitySolver (Standard)` - **Recursos de Áudio:** - Música de Fundo: `SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 26 arquivos `.ANM` carregados. - Totaliza aproximadamente **300 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Electric Fan:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Pinwheel:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Cheese:** 1 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Coffee Pot:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Boxes:** 1 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Steel Cable:** 1 peça(s)

-  **Cinder Block Wall:** 9 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Granite Incline:** 19 peça(s)
 -  **Glass Building:** 9 peça(s)
 -  **apartment building:** 1 peça(s)
 -  **office building:** 1 peça(s)
 -  **lamp post:** 1 peça(s)
 -  **fire hydrant:** 1 peça(s)
-

HLEV17.yaml - Mel on the Move

Objetivo: Help Mel get safely home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Something just isn't connecting. What Mel needs is a "phazer-totter." - Think Jericho, 800 B.C. Think "boom!"

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos `.ANM` carregados. - Totaliza aproximadamente **383 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 17 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Pulley:** 4 peça(s)
 -  **Rope:** 4 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Captain Z Super Phazer:** 3 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Wood Wall:** 11 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Caution Wall:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Anti-Gravity Pad:** 2 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Wooden Barrier:** 3 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 3 peça(s)
-

HLEV18.yaml - Edison's Breakfast

Objetivo: Turn on the can opener.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Reach deep down into your evil side. What would you really like to do to this happy face balloon? - Some toast would go nicely with that delicious can of kitty food.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1001_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 28 arquivos `.ANM` carregados. - Totaliza aproximadamente **560 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 6 peça(s)
 -  **Topsy Trailer:** 1 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Curie Cat:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Wood Wall:** 4 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Can Opener:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Toaster:** 1 peça(s)
 -  **Wooden Barrier:** 13 peça(s)
 -  **Blimp:** 1 peça(s)
-

HLEV19.yaml - Mel's Mine Field

Objetivo: Get Mel or any of his friends to the house.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1001_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos .ANM carregados. - Totaliza aproximadamente **363 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 7 peça(s)
 -  **Dynamite:** 2 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Pipe Wall:** 3 peça(s)
 -  **Wood Wall:** 5 peça(s)
 -  **Mel Schlemming:** 10 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Nitroglycerine:** 14 peça(s)
 -  **Cinder Block Wall:** 3 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Granite Incline:** 1 peça(s)
 -  **Elm Tree:** 5 peça(s)
 -  **Spruce Tree:** 6 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 1 peça(s)
-

HLEV20.yaml - Mel Takes a Hike

Objetivo: Set off the fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (4): - If Mel could only get over to those flint rocks, I'll bet he could spark some interesting events. - This wall needs to go. - Gravity, in this case, is working against us in two ways. - Color is important when it comes to switches.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1004_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos .ANM carregados. - Totaliza aproximadamente **639 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)

-  **Conveyor Belt:** 1 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Gear:** 1 peça(s)
 -  **Electric Switch & Outlet:** 2 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Tennis Ball:** 2 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Cinder Block Wall:** 4 peça(s)
 -  **Greco-Roman Wall:** 9 peça(s)
 -  **Tiny Gear:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 2 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Granite Incline:** 9 peça(s)
 -  **Moon:** 1 peça(s)
 -  **Spiral Galaxy:** 1 peça(s)
 -  **Star Glow:** 1 peça(s)
 -  **Star Burst:** 2 peça(s)
 -  **Star Sparkle:** 2 peça(s)
 -  **Dark Space:** 1 peça(s)
-

HLEV21.yaml - The Big Balloon Bummer

Objetivo: Pop all the balloons.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` -

`ElectricalPropagationSystem` - `GravitySolver (Standard)` - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1007_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher`*

pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 32 arquivos `.ANM` carregados. -

Totaliza aproximadamente **286 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Balloon:** 4 peça(s)

-  **Conveyor Belt:** 2 peça(s)
-  **Mouse Motor:** 2 peça(s)
-  **Pulley:** 2 peça(s)
-  **Belt:** 2 peça(s)
-  **Rope:** 1 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Pinball:** 2 peça(s)
-  **Tennis Ball:** 1 peça(s)
-  **Thumb Tack:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Hot Air Balloon:** 1 peça(s)
-  **Greco-Roman Wall:** 10 peça(s)
-  **Log Wall:** 3 peça(s)
-  **Red Laser:** 1 peça(s)
-  **Granite Incline:** 9 peça(s)
-  **Sun:** 1 peça(s)
-  **Big Cloud:** 1 peça(s)
-  **Pinball Flipper:** 1 peça(s)
-  **Color Block:** 1 peça(s)
-  **purple mountain:** 1 peça(s)
-  **desert mesa:** 1 peça(s)
-  **Small Cloud:** 3 peça(s)
-  **medium mountain:** 4 peça(s)
-  **small mesa:** 2 peça(s)
-  **medium mesa:** 1 peça(s)
-  **Erupting Volcano:** 1 peça(s)
-  **Red Coral Branch:** 1 peça(s)
-  **Red Coral:** 2 peça(s)

HLEV22.yaml - Jolly Jolly Jumpin' Jack

Objetivo: Make Jack pop out of his box.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1006_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 32 arquivos .ANM carregados. - Totaliza aproximadamente **468 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 2 peça(s)
-  **Pulley:** 5 peça(s)
-  **Belt:** 3 peça(s)
-  **Basketball:** 1 peça(s)
-  **Rope:** 2 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Gear:** 2 peça(s)
-  **Bucket:** 2 peça(s)
-  **Cannon:** 1 peça(s)
-  **Dynamite:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Hedge Trimmers:** 2 peça(s)
-  **Tennis Ball:** 2 peça(s)
-  **Wood Wall:** 2 peça(s)
-  **Caution Wall:** 20 peça(s)
-  **Large Curved Pipe:** 2 peça(s)
-  **Nitroglycerine:** 1 peça(s)
-  **Blue Laser:** 1 peça(s)
-  **Angled Mirror:** 2 peça(s)
-  **Large Pipes:** 1 peça(s)
-  **Sun:** 1 peça(s)
-  **Big Cloud:** 1 peça(s)
-  **Color Block:** 1 peça(s)
-  **Egg Timer:** 1 peça(s)
-  **Small Cloud:** 4 peça(s)
-  **small mesa:** 2 peça(s)
-  **medium mesa:** 2 peça(s)
-  **Red Canyon:** 3 peça(s)
-  **Red Rock:** 3 peça(s)

-  **Red Formation:** 2 peça(s)
 -  **Volcano:** 2 peça(s)
 -  **Erupting Volcano:** 1 peça(s)
 -  **Red Coral Branch:** 3 peça(s)
 -  **Red Seaweed:** 2 peça(s)
 -  **Red Coral:** 3 peça(s)
-

HLEV23.yaml - Rocket Racket

Objetivo: Make the rocket fly off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 24 arquivos `.ANM` carregados. - Totaliza aproximadamente **453 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
-  **Balloon:** 2 peça(s)
-  **Conveyor Belt:** 2 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 3 peça(s)
-  **Gear:** 5 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Dynamite:** 4 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Tennis Ball:** 1 peça(s)
-  **Wood Wall:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Missile:** 1 peça(s)
-  **Greco-Roman Wall:** 18 peça(s)
-  **Tiny Gear:** 3 peça(s)
-  **Blue Laser:** 3 peça(s)
-  **Angled Mirror:** 3 peça(s)
-  **Laser-Activated Plug:** 1 peça(s)
-  **Granite Incline:** 8 peça(s)
-  **Electrical Outlet:** 1 peça(s)
-  **Candle:** 2 peça(s)
-  **Teeter-Totter:** 1 peça(s)

HLEV24.yaml - Trick Shot

Objetivo: Put the 2 ball into the middle pocket at the bottom of the screen, and put the 3 ball into the top left corner pocket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1012_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos .ANM carregados. - Totaliza aproximadamente **123 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Pool Ball:** 9 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Pool Table Wall:** 6 peça(s)
 -  **Pool Table Pocket:** 6 peça(s)
 -  **Color Block:** 1 peça(s)
-

HLEV25.yaml - Fire Show Shenanigans

Objetivo: Set off all the fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1023_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos .ANM carregados. - Totaliza aproximadamente **406 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 1 peça(s)
-  **Wood incline:** 7 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Tennis Ball:** 2 peça(s)
-  **Pipe Wall:** 9 peça(s)
-  **Curved Pipe Wall:** 2 peça(s)

-  **Remote Control Explosives:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Fireworks:** 11 peça(s)
 -  **tile:** 3 peça(s)
 -  **Color Block:** 2 peça(s)
 -  **Candle:** 1 peça(s)
 -  **Dark Space:** 1 peça(s)
-

HLEV26.yaml - Late for Breakfast

Objetivo: Make some toast and launch the balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1000_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 35 arquivos `.ANM` carregados. - Totaliza aproximadamente **625 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 3 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Generator:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Tennis Ball:** 1 peça(s)
-  **Wood Wall:** 1 peça(s)
-  **Caution Wall:** 1 peça(s)
-  **Large Curved Pipe:** 1 peça(s)
-  **Soccer Ball:** 1 peça(s)
-  **Hot Air Balloon:** 1 peça(s)
-  **Toaster:** 1 peça(s)
-  **Sand Wall:** 2 peça(s)
-  **Cinder Block Wall:** 1 peça(s)
-  **Greco-Roman Wall:** 4 peça(s)
-  **Accelerator Tube:** 1 peça(s)

-  **Trans-Roto-Matic:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Archway:** 1 peça(s)
 -  **Wooden Barrier:** 3 peça(s)
 -  **Scaffold Barrier:** 1 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 2 peça(s)
 -  **Small Cloud:** 5 peça(s)
 -  **Sand Dune:** 3 peça(s)
 -  **Desert Dune:** 2 peça(s)
 -  **Sand Hill:** 1 peça(s)
 -  **Tropical Island:** 2 peça(s)
-

HLEV27.yaml - Curie and the Cannon

Objetivo: Fire the cannon and put the laundry basket over Curie Cat.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1007_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 32 arquivos `.ANM` carregados. - Totaliza aproximadamente **683 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 1 peça(s)
-  **Belt:** 2 peça(s)
-  **Rope:** 1 peça(s)
-  **Laundry Basket:** 1 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Gear:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Cannon:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Electric Fan:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)

-  **Pinwheel:** 1 peça(s)
 -  **Pipe Wall:** 4 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Greco-Roman Wall:** 1 peça(s)
 -  **Tiny Gear:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Granite Incline:** 1 peça(s)
 -  **Archway:** 8 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 2 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Candle:** 1 peça(s)
 -  **Small Cloud:** 3 peça(s)
 -  **Ship Logo:** 1 peça(s)
-

HLEV28.yaml - Patriotic Pop-A-Thon

Objetivo: Pop the balloon in the top right corner of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` - `ElectricalPropagationSystem` - `GravitySolver (Standard)` - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1000_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 27 arquivos `.ANM` carregados. - Totaliza aproximadamente **671 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 5 peça(s)
-  **Balloon:** 2 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Pulley:** 4 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 4 peça(s)
-  **Cannon:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)

-  **Boat Cleat:** 2 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Hedge Trimmers:** 2 peça(s)
 -  **Wood Wall:** 8 peça(s)
 -  **Vacuum:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Laser-Activated Plug:** 2 peça(s)
 -  **Color Block:** 9 peça(s)
 -  **Message Computer:** 3 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
 -  **Star Burst:** 50 peça(s)
-

HLEV29.yaml - Immobilized Mel

Objetivo: Help Mel get to the floor at the bottom of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1017_11025.wav - **Efeitos Sonoros (SFX):** Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 16 arquivos .ANM carregados. - Totaliza aproximadamente **331 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 1 peça(s)
-  **Remote Control:** 2 peça(s)
-  **Springboard:** 2 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Tennis Ball:** 6 peça(s)
-  **Pipe Wall:** 12 peça(s)
-  **Wood Wall:** 5 peça(s)
-  **Cheese:** 1 peça(s)
-  **Mel Schlemming:** 1 peça(s)

-  **Remote Control Explosives:** 2 peça(s)
 -  **Caution Wall:** 17 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Nitroglycerine:** 5 peça(s)
 -  **Cinder Block Wall:** 2 peça(s)
 -  **Blue Space:** 3 peça(s)
 -  **Bubble Stream:** 1 peça(s)
 -  **Floating Bubbles:** 3 peça(s)
 -  **Coral Arch:** 1 peça(s)
 -  **Coral Branch:** 1 peça(s)
 -  **Coral Formation:** 1 peça(s)
 -  **Green Mountain:** 1 peça(s)
 -  **Mossy Rock:** 1 peça(s)
 -  **Seaweed Tall:** 1 peça(s)
 -  **Kelp Forest:** 1 peça(s)
 -  **Seaweed Long:** 2 peça(s)
 -  **Sea Urchin:** 1 peça(s)
 -  **Red Crab:** 1 peça(s)
 -  **Striped Fish:** 2 peça(s)
 -  **Angelfish:** 1 peça(s)
 -  **Goldfish:** 4 peça(s)
 -  **Seahorse:** 1 peça(s)
 -  **Shark:** 2 peça(s)
 -  **Fish Scatter:** 2 peça(s)
 -  **Fish Group:** 1 peça(s)
 -  **Great White Shark:** 2 peça(s)
 -  **Tree Trunk:** 1 peça(s)
 -  **Stone Texture:** 8 peça(s)
-

HLEV30.yaml - Free Smiley

Objetivo: Make the smiley face balloon fly off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo:

`SX_1013_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 22 arquivos `.ANM` carregados. - Totaliza aproximadamente **364 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 2 peça(s)

-  **Balloon:** 2 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Cinder Block Wall:** 2 peça(s)
 -  **Greco-Roman Wall:** 8 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Spruce Tree:** 2 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 2 peça(s)
 -  **Little Cloud:** 2 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **Small Cloud:** 2 peça(s)
 -  **small mountain:** 1 peça(s)
 -  **Trees on Rocks:** 1 peça(s)
 -  **Einstein Bust:** 1 peça(s)
 -  **Wooden Door:** 1 peça(s)
 -  **Stone Texture:** 1 peça(s)
-

HLEV31.yaml - Rocketropolis

Objetivo: Launch the rocket off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1014_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 26 arquivos `.ANM` carregados. - Totaliza aproximadamente **454 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 2 peça(s)
-  **Gear:** 3 peça(s)

-  **Bike Pump:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Tennis Ball:** 2 peça(s)
 -  **Pipe Wall:** 11 peça(s)
 -  **Curved Pipe Wall:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Toaster:** 1 peça(s)
 -  **Tiny Gear:** 3 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Scaffold Barrier:** 18 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 2 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Glass Building:** 3 peça(s)
 -  **apartment building:** 1 peça(s)
 -  **office building:** 1 peça(s)
 -  **Small Cloud:** 1 peça(s)
 -  **Sphinx Pyramid:** 1 peça(s)
 -  **Futuristic Tower:** 1 peça(s)
 -  **Cityscape:** 1 peça(s)
 -  **Metropolis:** 1 peça(s)
 -  **City Buildings:** 3 peça(s)
 -  **City Skyline:** 2 peça(s)
-

MLEV1.yaml - Mountain Man Mel

Objetivo: Get Mel to his cabin.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - What a lovely spot for a balloon!

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros*

(SFX): *Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais**

(ANM & PNG): - Requer 27 arquivos `.ANM` carregados. - Totaliza aproximadamente **689 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 3 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Curie Cat:** 1 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Bucket:** 2 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Mandrill Motor:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Alligator:** 1 peça(s)
 -  **Sand Wall:** 2 peça(s)
 -  **Log Wall:** 5 peça(s)
 -  **Log Incline:** 3 peça(s)
 -  **Elm Tree:** 2 peça(s)
 -  **Spruce Tree:** 3 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 1 peça(s)
 -  **Little Cloud:** 8 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **purple mountain:** 2 peça(s)
 -  **Small Maple Tree:** 3 peça(s)
 -  **Small Pine Tree:** 4 peça(s)
 -  **Small Cloud:** 1 peça(s)
-

MLEV2.yaml - Mondrian's Toy Box

Objetivo: Make all five Jack-in-the-boxes pop open.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Find a way to turn these gears. They all appear to be connected.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo:

`SX_1019_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos `.ANM` carregados. - Totaliza aproximadamente **144 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Belt:** 14 peça(s)
 -  **Jack-in-the-box:** 5 peça(s)
 -  **Gear:** 7 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Pinwheel:** 1 peça(s)
 -  **Pipe Wall:** 18 peça(s)
 -  **Tiny Gear:** 15 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Color Block:** 7 peça(s)
-

MLEV3.yaml - Gimpy Blimpy

Objetivo: Get the balloon off the top of the screen without popping any blimps.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - I wonder if a little tube of something explosive might fit between the blimps?

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1001_11025.wav - Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 12 arquivos .ANM carregados. - Totaliza aproximadamente **282 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Pipe Wall:** 15 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Log Wall:** 1 peça(s)
 -  **Blimp:** 6 peça(s)
 -  **Yellow Brick Wall:** 1 peça(s)
 -  **Yellow Brick Incline:** 5 peça(s)
-

MLEV4.yaml - Pipe Gripe

Objetivo: Get the Pinball off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - A little acceleration might be in order right about here. - This looks like the only place on the screen the ball can get out. - The shortest route is a straight line... but sometimes that just isn't an option.

Mapa de Dependências: - Sistemas Físicos Necessários: - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1016_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos .ANM carregados. - Totaliza aproximadamente **45 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pinball:** 1 peça(s)
 -  **Pipe Wall:** 6 peça(s)
 -  **Large Curved Pipe:** 11 peça(s)
 -  **Accelerator Tube:** 10 peça(s)
 -  **Large Pipes:** 6 peça(s)
 -  **T-Connector:** 2 peça(s)
-

MLEV5.yaml - Up and Away!

Objetivo: Launch the missile.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Try some negative gravity on this pinball.

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - LaserRaycastSystem - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1009_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 12 arquivos .ANM carregados. - Totaliza aproximadamente **175 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Pipe Wall:** 2 peça(s)
-  **Large Curved Pipe:** 8 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Missile:** 1 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Hot Air Balloon:** 1 peça(s)
-  **Accelerator Tube:** 5 peça(s)
-  **Rocky Mountain:** 2 peça(s)
-  **Trees on Rocks:** 2 peça(s)
-  **Trees on Terrain:** 3 peça(s)
-  **Moon Crater:** 1 peça(s)
-  **Owl:** 1 peça(s)
-  **Red Coral Branch:** 4 peça(s)

-  **Red Seaweed:** 3 peça(s)
 -  **Red Coral:** 3 peça(s)
-

MLEV6.yaml - Mouse House Munchies

Objetivo: Feed Newton Mouse the cheese.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - The balloon tied to this rope is dying to "cut loose."

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1013_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **323 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 5 peça(s)
 -  **Balloon:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Cannon:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Baseball:** 2 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Pipe Wall:** 3 peça(s)
 -  **Cheese:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Large Curved Pipe:** 4 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Brick Incline:** 3 peça(s)
 -  **tile:** 1 peça(s)
 -  **Computer Monitor:** 1 peça(s)
 -  **Metal Vent:** 2 peça(s)
 -  **Cliff Edge:** 1 peça(s)
 -  **Periodic Table:** 1 peça(s)
-

MLEV7.yaml - Laser Launch

Objetivo: Launch the rocket off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Use a conveyor belt to move the rocket into the pit on the right. - Use some pipes to get the ball going in the right direction.

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1015_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 24 arquivos .ANM carregados. - Totaliza aproximadamente **350 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Topsy Trailer:** 1 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Gear:** 3 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Large Curved Pipe:** 2 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 3 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Large Pipes:** 1 peça(s)
 -  **Yellow Brick Wall:** 15 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
-

MLEV8.yaml - Fire Ball

Objetivo: Launch the fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - The right combination of parts will force the trap door to open. - A plumber's nightmare--a soccer ball in the pipes.

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1001_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **331 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
 -  **Pulley:** 3 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 3 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 3 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Greco-Roman Wall:** 10 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Granite Incline:** 16 peça(s)
 -  **Leaky Bucket:** 1 peça(s)
-

MLEV9.yaml - Fourth of July

Objetivo: Launch all the fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - If there was just a little more time, you could light the candle.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1023_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **369 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 1 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)

-  **Electric Motor:** 1 peça(s)
 -  **Caution Wall:** 6 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Fireworks:** 4 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 2 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Yellow Brick Incline:** 7 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

MLEV10.yaml - Beach Ball Bonanza

Objetivo: Put the striped ball into the bucket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - The mouse motor can power the generator. - It would be FANTastic to turn this pinwheel, which could then turn the conveyor belt.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1013_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **363 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 2 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 4 peça(s)
-  **Gear:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Dynamite:** 2 peça(s)
-  **Electric Fan:** 1 peça(s)
-  **Generator:** 1 peça(s)
-  **Pinwheel:** 1 peça(s)
-  **Tennis Ball:** 1 peça(s)
-  **Wood Wall:** 3 peça(s)
-  **Anti-Gravity Pad:** 3 peça(s)
-  **Nitroglycerine:** 1 peça(s)

-  **Sand Wall:** 9 peça(s)
 -  **Tiny Gear:** 2 peça(s)
 -  **Programmable Ball:** 1 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 2 peça(s)
 -  **Yellow Brick Incline:** 12 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Small Cloud:** 2 peça(s)
 -  **small mesa:** 1 peça(s)
-

MLEV11.yaml - Cat Bounce

Objetivo: Pop the balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Use some mouse power to generate electricity. - Curie Cat comes running whenever she hears the can opener.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo:

`SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **462 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Mouse Motor:** 2 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Curie Cat:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Generator:** 2 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Newton Mouse:** 2 peça(s)
 -  **Cheese:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Can Opener:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Cinder Block Wall:** 10 peça(s)
 -  **Granite Incline:** 1 peça(s)
 -  **Archway:** 3 peça(s)
-

MLEV12.yaml - Laser Target

Objetivo: Shoot the sensor at the bottom right with the laser.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Pull down on the chain. - This is a dual-purpose pinball. - It's done with mirrors.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1019_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **317 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
 -  **Wood incline:** 8 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Remote Control:** 2 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Wood Wall:** 4 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Cinder Block Wall:** 3 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Scaffold Barrier:** 9 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Candle:** 1 peça(s)
 -  **Laser Detector:** 1 peça(s)
-

MLEV13.yaml - Breakfast Buffet

Objetivo: Make coffee and toast.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Place the mouse motor here so the tennis ball can roll over it. - With a little luck, a timer will push the tennis ball to the toaster.

Mapa de Dependências: - Sistemas Físicos Necessários: - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1016_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 21 arquivos .ANM carregados. - Totaliza aproximadamente **236 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 10 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Gear:** 2 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Toaster:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Brick Incline:** 27 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 2 peça(s)
 -  **Color Block:** 2 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Pine Forest:** 5 peça(s)
 -  **Cave Opening:** 1 peça(s)
 -  **Forest Trees:** 4 peça(s)
 -  **Rocky Terrain:** 1 peça(s)
 -  **Moon:** 1 peça(s)
 -  **Star Glow:** 2 peça(s)
 -  **Star Burst:** 5 peça(s)
 -  **Star Sparkle:** 8 peça(s)
-

MLEV14.yaml - Pipe Mixer

Objetivo: Turn on the electric mixer.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - There's too much gravity here!

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_0_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 16 arquivos `.ANM` carregados. - Totaliza aproximadamente **165 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 7 peça(s)
 -  **Topsy Trailer:** 1 peça(s)
 -  **Pulley:** 3 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Aladdin's Lamp:** 1 peça(s)
 -  **Pipe Wall:** 7 peça(s)
 -  **Curved Pipe Wall:** 1 peça(s)
 -  **Large Curved Pipe:** 2 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Accelerator Tube:** 2 peça(s)
 -  **Electric Mixer:** 1 peça(s)
 -  **Leaky Bucket:** 2 peça(s)
 -  **Blimp:** 1 peça(s)
-

MLEV15.yaml - Fishing in Rome

Objetivo: Break the fish tank.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - What goes up doesn't always come down.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1019_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 16 arquivos `.ANM` carregados. - Totaliza aproximadamente **357 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Jack-in-the-box:** 1 peça(s)
-  **Fish Tank:** 1 peça(s)
-  **Cannon:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Magnifying Glass:** 2 peça(s)

-  **Boxing Glove:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Greco-Roman Wall:** 12 peça(s)
 -  **Granite Incline:** 4 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Big Cloud:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
 -  **Small Cloud:** 2 peça(s)
-

MLEV16.yaml - Bomb Baffle

Objetivo: Set off both bombs.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - See what happens if you pull down and push up at the same time. - This appears to be the only object with enough potential energy to do anything.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1005_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **247 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 2 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Remote Control:** 2 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Caution Wall:** 9 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Leaky Bucket:** 2 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Yellow Brick Incline:** 6 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

MLEV17.yaml - Bowl Me Over

Objetivo: Put the bowling ball into the cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - You could get some real mouse power going here. - Keep it moving...

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de**

Áudio: - Música de Fundo: `SX_1008_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via*

AudioDispatcher pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **202 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Brick Wall:** 7 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Boxing Glove:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Trap Door:** 2 peça(s)
 -  **Trans-Roto-Matic:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Brick Incline:** 11 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
-

MLEV18.yaml - Mel in a Muddle

Objetivo: Get Mel home in one piece!

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - A timer here would not only help Mel across, but also get the pinball moving. - Mel is a little off-balance in this one.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo:

`SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via* `AudioDispatcher` *pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 22 arquivos .ANM carregados. - Totaliza aproximadamente **380 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 14 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Remote Control:** 2 peça(s)
 -  **Boat Cleat:** 3 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Trans-Roto-Matic:** 1 peça(s)
 -  **Brick Incline:** 10 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

MLEV19.yaml - Mel's Amazing Maze

Objetivo: Get Mel home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - A remote bomb plunger goes here. Set the bomb itself to the left so it can blow a hole in the floor.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `GravitySolver` (Standard) - **Recursos de Áudio:** -

Música de Fundo: `SX_0_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos .ANM carregados. - Totaliza aproximadamente **394 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Wood incline:** 8 peça(s)
-  **Remote Control:** 2 peça(s)
-  **Flashlight:** 1 peça(s)

-  **Springboard:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Wood Wall:** 12 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Caution Wall:** 9 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

MLEV20.yaml - Illuminate the Lab

Objetivo: Turn on the lava lamp on top of Mel's underwater laboratory.

Ambiente: - Gravidade: 272 - Pressão do Ar: 128

Dicas Ativas (2): - This pinball needs to float. - Push down on this teeter-totter.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **364 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Rope:** 3 peça(s)
-  **Fish Tank:** 1 peça(s)
-  **Bucket:** 2 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Lava Lamp:** 1 peça(s)
-  **Pinball:** 2 peça(s)
-  **Pipe Wall:** 1 peça(s)
-  **Curved Pipe Wall:** 2 peça(s)
-  **Mel Schlemming:** 1 peça(s)
-  **Caution Wall:** 3 peça(s)
-  **Large Curved Pipe:** 2 peça(s)
-  **Anti-Gravity Pad:** 2 peça(s)
-  **Sand Wall:** 6 peça(s)
-  **Grass Incline:** 8 peça(s)
-  **Color Block:** 2 peça(s)
-  **Egg Timer:** 1 peça(s)

-  **Rising Bubbles:** 1 peça(s)
 -  **Floating Bubbles:** 1 peça(s)
 -  **Ship Logo:** 1 peça(s)
 -  **Starfish:** 1 peça(s)
 -  **Green Coral:** 1 peça(s)
 -  **Coral Tree:** 1 peça(s)
 -  **Seaweed Long:** 4 peça(s)
 -  **Plant Stem:** 1 peça(s)
 -  **Sea Urchin:** 1 peça(s)
 -  **Red Crab:** 1 peça(s)
 -  **Jellyfish:** 1 peça(s)
 -  **Striped Fish:** 1 peça(s)
 -  **Angelfish:** 2 peça(s)
 -  **Fish School:** 2 peça(s)
 -  **Fish Group:** 6 peça(s)
 -  **Dolphin:** 3 peça(s)
 -  **Orca Whale:** 1 peça(s)
-

MLEV21.yaml - Blow Up

Objetivo: Make the balloon fly off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Don't forget to cut the rope, or the balloon will never take off! - Give the baseball some help over to the remote plunger.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1006_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **211 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Pulley:** 2 peça(s)
-  **Rope:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Dynamite:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Remote Control Explosives:** 1 peça(s)

-  **Caution Wall:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Yellow Brick Wall:** 8 peça(s)
 -  **Yellow Brick Incline:** 9 peça(s)
-

MLEV22.yaml - Gravity Cavity

Objetivo: Put the soccer ball in the caution box with the 1-ball.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - How can one CONVEY the marvels of soccer?

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **177 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Caution Wall:** 4 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 12 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
 -  **Candle:** 1 peça(s)
-

MLEV23.yaml - The Sky is Falling!

Objetivo: Mel is in trouble. Help him get to his house in the upper right.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Find a way to stop these buckets from falling. - Unless he can find a way to float, Mel will fall through this hole.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1001_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 24 arquivos `.ANM` carregados. - Totaliza aproximadamente **360 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Rope:** 4 peça(s)
 -  **Bucket:** 4 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 4 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Wood Wall:** 3 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Nitroglycerine:** 4 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Elm Tree:** 4 peça(s)
 -  **tile:** 1 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
 -  **Glass Building:** 1 peça(s)
 -  **office building:** 1 peça(s)
 -  **car:** 1 peça(s)
 -  **lamp post:** 4 peça(s)
 -  **fire hydrant:** 1 peça(s)
 -  **Small Maple Tree:** 3 peça(s)
-

MLEV24.yaml - Stop the Invasion

Objetivo: Blow up all of the blimps.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - This blimp will help in the destruction of the others.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1023_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos `.ANM` carregados. - Totaliza aproximadamente **207 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Pulley:** 3 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Missile:** 2 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Cinder Block Wall:** 5 peça(s)
 -  **Blimp:** 8 peça(s)
 -  **Moon:** 1 peça(s)
 -  **Star Glow:** 3 peça(s)
 -  **Star Burst:** 17 peça(s)
 -  **Star Sparkle:** 36 peça(s)
 -  **Stalactites:** 1 peça(s)
 -  **Greenhouse Dome:** 1 peça(s)
 -  **City Skyline:** 1 peça(s)
 -  **Totem Pole:** 1 peça(s)
 -  **Satellite Tower:** 1 peça(s)
 -  **Radio Tower:** 1 peça(s)
 -  **Dish Antenna:** 1 peça(s)
 -  **Glass Dome:** 2 peça(s)
 -  **Crystal Formation:** 1 peça(s)
-

MLEV25.yaml - Mel-O-Drama

Objetivo: Help Mel make it home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Mel could use a little help from his mouse friends.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1004_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 23 arquivos `.ANM` carregados. - Totaliza aproximadamente **556 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 1 peça(s)
-  **Topsy Trailer:** 1 peça(s)

-  **Basketball:** 3 peça(s)
 -  **Curie Cat:** 1 peça(s)
 -  **Baseball:** 2 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Newton Mouse:** 2 peça(s)
 -  **Pipe Wall:** 2 peça(s)
 -  **Curved Pipe Wall:** 2 peça(s)
 -  **Wood Wall:** 9 peça(s)
 -  **Cheese:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Big Cloud:** 1 peça(s)
 -  **Little Cloud:** 2 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **desert mesa:** 1 peça(s)
 -  **Small Pine Tree:** 1 peça(s)
 -  **Small Cloud:** 2 peça(s)
 -  **small mesa:** 1 peça(s)
 -  **medium mesa:** 1 peça(s)
 -  **Red Canyon:** 1 peça(s)
 -  **Red Rock:** 2 peça(s)
 -  **Red Formation:** 1 peça(s)
 -  **Castle Fortress:** 1 peça(s)
-

MLEV26.yaml - Lunar Base Take-off

Objetivo: Launch the rocket off the top of the screen.

Ambiente: - Gravidade: 0 - Pressão do Ar: 0

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1023_11025.wav` - **Efeitos Sonoros (SFX):** *Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 25 arquivos `.ANM` carregados. - Totaliza aproximadamente **517 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 3 peça(s)

-  **Flashlight:** 1 peça(s)
 -  **Captain Z Super Phazer:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Greco-Roman Wall:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Scaffold Barrier:** 19 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
 -  **Moon:** 1 peça(s)
 -  **Star Glow:** 1 peça(s)
 -  **Star Burst:** 1 peça(s)
 -  **Star Sparkle:** 6 peça(s)
 -  **Greenhouse Dome:** 1 peça(s)
 -  **Futuristic Tower:** 1 peça(s)
 -  **Sci-Fi City:** 3 peça(s)
 -  **Cityscape:** 3 peça(s)
 -  **Metropolis:** 3 peça(s)
 -  **City Buildings:** 3 peça(s)
 -  **Ice Castle:** 2 peça(s)
 -  **Crystal Tower:** 1 peça(s)
 -  **Totem Pole:** 1 peça(s)
 -  **Radio Tower:** 1 peça(s)
 -  **Glass Dome:** 1 peça(s)
 -  **Dark Space:** 1 peça(s)
-

MLEV27.yaml - Flight through the Alps

Objetivo: Launch the balloon off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1017_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 29 arquivos `.ANM` carregados. - Totaliza aproximadamente **338 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Belt:** 1 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Cannon:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Lava Lamp:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Solar Panel:** 1 peça(s)
-  **Pinwheel:** 1 peça(s)
-  **Pinball Bumper:** 1 peça(s)
-  **Soccer Ball:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Steel Cable:** 1 peça(s)
-  **Tin Snips:** 1 peça(s)
-  **Hot Air Balloon:** 1 peça(s)
-  **Cinder Block Wall:** 4 peça(s)
-  **Red Laser:** 1 peça(s)
-  **Blue Laser:** 1 peça(s)
-  **Granite Incline:** 10 peça(s)
-  **Scaffold Barrier:** 10 peça(s)
-  **purple mountain:** 4 peça(s)
-  **Teeter-Totter:** 1 peça(s)
-  **small mountain:** 3 peça(s)
-  **medium mountain:** 5 peça(s)
-  **Pine Forest:** 5 peça(s)
-  **Cave Opening:** 1 peça(s)

-  **Forest Trees:** 6 peça(s)
 -  **Rocky Terrain:** 1 peça(s)
 -  **Moon:** 1 peça(s)
-

MLEV28.yaml - Soccer Shocker

Objetivo: Put the soccer ball into the wooden crate.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 31 arquivos `.ANM` carregados. - Totaliza aproximadamente **381 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 2 peça(s)
-  **Balloon:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Boat Cleat:** 2 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Tennis Ball:** 1 peça(s)
-  **Thumb Tack:** 1 peça(s)
-  **Large Curved Pipe:** 3 peça(s)
-  **Alligator:** 1 peça(s)
-  **Pool Ball:** 1 peça(s)
-  **Soccer Ball:** 1 peça(s)
-  **Boxes:** 1 peça(s)
-  **Steel Cable:** 1 peça(s)
-  **Tin Snips:** 1 peça(s)
-  **Flint & Tinder:** 1 peça(s)
-  **Hot Air Balloon:** 1 peça(s)
-  **Accelerator Tube:** 2 peça(s)
-  **Blimp:** 1 peça(s)
-  **Sun:** 1 peça(s)
-  **Big Cloud:** 1 peça(s)
-  **Pool Cue:** 1 peça(s)

-  **Yellow Brick Wall:** 14 peça(s)
 -  **Yellow Brick Incline:** 11 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Small Cloud:** 2 peça(s)
 -  **Mountain Range:** 1 peça(s)
 -  **Volcano:** 1 peça(s)
 -  **Blue Space:** 1 peça(s)
 -  **Ocean Wave:** 1 peça(s)
 -  **Breaking Wave:** 1 peça(s)
 -  **Surf Wave:** 2 peça(s)
 -  **Tree Trunk:** 1 peça(s)
-

MLEV29.yaml - Pinball Puzzler

Objetivo: Make the pinball in the upper left corner fall off the bottom of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1006_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 11 arquivos .ANM carregados. - Totaliza aproximadamente **94 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Springboard:** 3 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Caution Wall:** 18 peça(s)
 -  **Large Curved Pipe:** 11 peça(s)
 -  **Accelerator Tube:** 3 peça(s)
 -  **T-Connector:** 1 peça(s)
 -  **Pinball Flipper:** 1 peça(s)
 -  **tile:** 2 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

MLEV30.yaml - All Terrain Pinball

Objetivo: Put the pinball into the wicker basket and activate the computer terminal.

Ambiente: - Gravidade: 143 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - PipeNetwork (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: SX_1021_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos .ANM carregados. - Totaliza aproximadamente **232 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 13 peça(s)
-  **Pinball:** 1 peça(s)
-  **Wood Wall:** 6 peça(s)
-  **Grass Floor:** 1 peça(s)
-  **Alligator:** 1 peça(s)
-  **Pinball Bumper:** 3 peça(s)
-  **Boxes:** 3 peça(s)
-  **Sand Wall:** 2 peça(s)
-  **Log Wall:** 7 peça(s)
-  **Accelerator Tube:** 2 peça(s)
-  **Log Incline:** 8 peça(s)
-  **Spruce Tree:** 1 peça(s)
-  **Sun:** 1 peça(s)
-  **Big Cloud:** 1 peça(s)
-  **Pinball Flipper:** 2 peça(s)
-  **tile:** 2 peça(s)
-  **Color Block:** 3 peça(s)
-  **Message Computer:** 1 peça(s)
-  **Small Maple Tree:** 1 peça(s)
-  **Small Pine Tree:** 2 peça(s)
-  **Small Cloud:** 2 peça(s)
-  **Cave Opening:** 1 peça(s)
-  **Rising Bubbles:** 2 peça(s)
-  **Bubble Stream:** 3 peça(s)
-  **Coral Arch:** 1 peça(s)
-  **Coral Branch:** 1 peça(s)
-  **Green Mountain:** 1 peça(s)
-  **Hot Rod Car:** 1 peça(s)
-  **Clown Fish:** 1 peça(s)
-  **Shell:** 1 peça(s)
-  **Metal Gears:** 5 peça(s)
-  **Orange Coral:** 2 peça(s)
-  **Seaweed Tall:** 6 peça(s)
-  **Kelp Forest:** 1 peça(s)
-  **Kelp Plant:** 2 peça(s)
-  **Seaweed Vine:** 1 peça(s)
-  **Seaweed Long:** 1 peça(s)
-  **Plant Stem:** 1 peça(s)
-  **Sea Urchin:** 1 peça(s)

-  **Red Crab:** 1 peça(s)
 -  **Jellyfish:** 1 peça(s)
 -  **Octopus:** 1 peça(s)
 -  **Striped Fish:** 3 peça(s)
 -  **Seahorse:** 1 peça(s)
 -  **Shark:** 1 peça(s)
 -  **Fish Scatter:** 1 peça(s)
 -  **Fish Group:** 2 peça(s)
 -  **Great White Shark:** 1 peça(s)
 -  **Dolphin:** 1 peça(s)
 -  **Tree Trunk:** 1 peça(s)
-

RLEV1.yaml - Exit Laughing

Objetivo: Make the happy face balloon exit the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1018_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **354 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
-  **Balloon:** 4 peça(s)
-  **Pulley:** 2 peça(s)
-  **Rope:** 3 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Dynamite:** 1 peça(s)
-  **Captain Z Super Phazer:** 2 peça(s)
-  **Coffee Pot:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Fireworks:** 1 peça(s)
-  **Toaster:** 1 peça(s)
-  **Greco-Roman Wall:** 18 peça(s)
-  **Trans-Roto-Matic:** 1 peça(s)
-  **Granite Incline:** 1 peça(s)
-  **Yellow Brick Incline:** 1 peça(s)
-  **Egg Timer:** 1 peça(s)
-  **Teeter-Totter:** 1 peça(s)
-  **Rising Bubbles:** 2 peça(s)

-  **Bubble Stream:** 1 peça(s)
 -  **Floating Bubbles:** 1 peça(s)
 -  **Treasure Chest:** 1 peça(s)
 -  **Ship Logo:** 1 peça(s)
 -  **Shell:** 1 peça(s)
 -  **Shell Spiral:** 2 peça(s)
 -  **Metal Gears:** 4 peça(s)
 -  **Orange Coral:** 2 peça(s)
 -  **Coral Branch:** 2 peça(s)
 -  **Green Coral:** 2 peça(s)
 -  **Seaweed Tall:** 2 peça(s)
 -  **Kelp Forest:** 4 peça(s)
 -  **Kelp Plant:** 1 peça(s)
 -  **Seaweed Vine:** 3 peça(s)
 -  **Striped Fish:** 2 peça(s)
 -  **Goldfish:** 1 peça(s)
 -  **Seahorse:** 1 peça(s)
 -  **Fish Group:** 4 peça(s)
 -  **Great White Shark:** 1 peça(s)
 -  **Orca Whale:** 1 peça(s)
 -  **Gray Whale:** 1 peça(s)
-

RLEV2.yaml - The Critter-Gitter Cat Trap

Objetivo: Trap Curie Cat in the laundry basket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **448 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Topsy Trailer:** 3 peça(s)
-  **Balloon:** 1 peça(s)
-  **Rope:** 2 peça(s)
-  **Laundry Basket:** 1 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Pipe Wall:** 14 peça(s)

-  **Coffee Pot:** 1 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Fireworks:** 1 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
 -  **Stained Glass:** 1 peça(s)
 -  **Abstract Art:** 1 peça(s)
 -  **Landscape Painting:** 1 peça(s)
 -  **Einstein Bust:** 1 peça(s)
-

RLEV3.yaml - Lil' Mel's Jail Break

Objetivo: Assist Mel in his daring escape and journey home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1018_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 28 arquivos `.ANM` carregados. - Totaliza aproximadamente **602 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
-  **Conveyor Belt:** 2 peça(s)
-  **Belt:** 2 peça(s)
-  **Rope:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Pinball:** 2 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Mel Schlemming:** 1 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Caution Wall:** 17 peça(s)
-  **Large Curved Pipe:** 5 peça(s)
-  **Mel's House:** 1 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Boxes:** 1 peça(s)
-  **Greco-Roman Wall:** 1 peça(s)
-  **Accelerator Tube:** 1 peça(s)
-  **Trans-Roto-Matic:** 1 peça(s)
-  **Red Laser:** 1 peça(s)
-  **Green Laser:** 1 peça(s)
-  **Blue Laser:** 1 peça(s)

-  **Angled Mirror:** 5 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Large Pipes:** 2 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Yellow Brick Wall:** 1 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **apartment building:** 2 peça(s)
 -  **office building:** 3 peça(s)
-

RLEV4.yaml - Laser Balloon Blitz

Objetivo: Pop the balloon.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1012_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 26 arquivos `.ANM` carregados. - Totaliza aproximadamente **530 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 1 peça(s)
-  **Dynamite:** 1 peça(s)
-  **Electric Fan:** 1 peça(s)
-  **Generator:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Caution Wall:** 9 peça(s)
-  **Missile:** 1 peça(s)
-  **Nitroglycerine:** 1 peça(s)
-  **Steel Cable:** 1 peça(s)
-  **Trans-Roto-Matic:** 1 peça(s)
-  **Roto-Trans Converter:** 1 peça(s)
-  **Red Laser:** 1 peça(s)
-  **Green Laser:** 1 peça(s)
-  **Angled Mirror:** 9 peça(s)
-  **Laser Mixer:** 1 peça(s)
-  **Laser-Activated Plug:** 1 peça(s)
-  **Blimp:** 4 peça(s)
- **Sun:** 1 peça(s)

-  **Big Cloud:** 1 peça(s)
 -  **purple mountain:** 1 peça(s)
 -  **Small Cloud:** 3 peça(s)
 -  **small mountain:** 2 peça(s)
 -  **medium mountain:** 1 peça(s)
 -  **Pine Forest:** 3 peça(s)
 -  **Cave Opening:** 1 peça(s)
 -  **Forest Trees:** 1 peça(s)
 -  **Rocky Terrain:** 2 peça(s)
 -  **Trees on Terrain:** 1 peça(s)
 -  **Snow Mountains:** 1 peça(s)
 -  **Iceberg:** 1 peça(s)
 -  **Castle Fortress:** 1 peça(s)
-

RLEV5.yaml - Balloon Over Manhattan

Objetivo: Make the balloon fly off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` - `ElectricalPropagationSystem` - `GravitySolver (Standard)` - **Recursos de Áudio:** - Música de Fundo: `SX_1006_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **184 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 6 peça(s)
-  **Pulley:** 6 peça(s)
-  **Belt:** 2 peça(s)
-  **Rope:** 3 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Solar Panel:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Wood Wall:** 4 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Hot Air Balloon:** 1 peça(s)
-  **Cinder Block Wall:** 6 peça(s)
-  **Trans-Roto-Matic:** 1 peça(s)
-  **Roto-Trans Converter:** 2 peça(s)
-  **Glass Building:** 3 peça(s)

-  **office building**: 1 peça(s)
 -  **Stalactites**: 1 peça(s)
 -  **Greenhouse Dome**: 1 peça(s)
 -  **Sphinx Pyramid**: 1 peça(s)
 -  **Futuristic Tower**: 1 peça(s)
 -  **Sci-Fi City**: 1 peça(s)
 -  **Cityscape**: 1 peça(s)
 -  **Crystal Tower**: 1 peça(s)
 -  **City Skyline**: 4 peça(s)
 -  **Totem Pole**: 1 peça(s)
 -  **Satellite Tower**: 1 peça(s)
 -  **Radio Tower**: 1 peça(s)
 -  **Glass Dome**: 1 peça(s)
-

RLEV6.yaml - Pool Ball Sinker

Objetivo: Sink the eight ball.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` -

`ElectricalPropagationSystem` - `GravitySolver (Standard)` - **Recursos de Áudio:** - Música de Fundo:

`SX_1010_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 20 arquivos `.ANM` carregados. - Totaliza aproximadamente **310 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball**: 1 peça(s)
-  **Belt**: 1 peça(s)
-  **Jack-in-the-box**: 1 peça(s)
-  **Electric Switch & Outlet**: 1 peça(s)
-  **Remote Control**: 1 peça(s)
-  **Boxing Glove**: 1 peça(s)
-  **Springboard**: 1 peça(s)
-  **Pinball**: 2 peça(s)
-  **Electric Motor**: 1 peça(s)
-  **Remote Control Explosives**: 1 peça(s)
-  **Pool Ball**: 2 peça(s)
-  **Anti-Gravity Pad**: 1 peça(s)
-  **Trap Door**: 2 peça(s)
-  **Cinder Block Wall**: 2 peça(s)
-  **Greco-Roman Wall**: 1 peça(s)
-  **Granite Incline**: 4 peça(s)
-  **Pinball Flipper**: 1 peça(s)

-  **Pool Cue:** 1 peça(s)
 -  **Pool Table Wall:** 2 peça(s)
 -  **Pool Table Pocket:** 1 peça(s)
-

RLEV7.yaml - Mega-Launch

Objetivo: Launch the missile, the rocket, and the fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 27 arquivos `.ANM` carregados. - Totaliza aproximadamente **585 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 1 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Rocket:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Thumb Tack:** 1 peça(s)
-  **Caution Wall:** 9 peça(s)
-  **Large Curved Pipe:** 2 peça(s)
-  **Anti-Gravity Pad:** 1 peça(s)
-  **Missile:** 1 peça(s)
-  **Trap Door:** 1 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Tin Snips:** 1 peça(s)
-  **Flint & Tinder:** 1 peça(s)
-  **Fireworks:** 1 peça(s)
-  **Accelerator Tube:** 1 peça(s)
-  **Large Pipes:** 1 peça(s)
-  **Yellow Brick Wall:** 1 peça(s)

-  **Yellow Brick Incline:** 8 peça(s)
-

RLEV8.yaml - Newton's Dilemma

Objetivo: Get Newton safely home to his mouse hole.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1014_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **430 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 9 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Curie Cat:** 1 peça(s)
 -  **Gear:** 2 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Wood Wall:** 15 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Cheese:** 6 peça(s)
 -  **Pinball Bumper:** 1 peça(s)
 -  **Mouse Hole:** 1 peça(s)
 -  **Anti-Gravity Pad:** 3 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Tiny Gear:** 2 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

RLEV9.yaml - Brain Teaser Breakout

Objetivo: Lift the basket off of Newton so he can eat the cheese.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1001_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos `.ANM` carregados. - Totaliza aproximadamente **349 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 3 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Laundry Basket:** 1 peça(s)
 -  **Dynamite:** 4 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Cheese:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Log Wall:** 15 peça(s)
 -  **Log Incline:** 10 peça(s)
 -  **Leaky Bucket:** 1 peça(s)
 -  **Teeter-Totter:** 2 peça(s)
-

RLEV10.yaml - Jack's Big Fan

Objetivo: Make the Jack-in-the-box pop out.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **441 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 14 peça(s)
-  **Belt:** 1 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Cannon:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Electric Fan:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Springboard:** 1 peça(s)

-  **Pinwheel:** 1 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Brick Incline:** 13 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

RLEV11.yaml - Illuminated Schlemming

Objetivo: Light both candles and get Mel home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos `.ANM` carregados. - Totaliza aproximadamente **629 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Cannon:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Rocket:** 4 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Wood Wall:** 2 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Greco-Roman Wall:** 14 peça(s)
 -  **Granite Incline:** 3 peça(s)
 -  **Candle:** 2 peça(s)
 -  **Laser Detector:** 1 peça(s)
-

RLEV12.yaml - Schlemming in Dreamland

Objetivo: Mel is sleep walking. Help him get home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 19 arquivos `.ANM` carregados. - Totaliza aproximadamente **393 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 5 peça(s)
 -  **Rope:** 5 peça(s)
 -  **Boat Cleat:** 5 peça(s)
 -  **Thumb Tack:** 2 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Alligator:** 2 peça(s)
 -  **Anti-Gravity Pad:** 2 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Nitroglycerine:** 2 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Sand Wall:** 10 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **purple mountain:** 2 peça(s)
 -  **Teeter-Totter:** 2 peça(s)
 -  **Small Cloud:** 3 peça(s)
 -  **small mountain:** 2 peça(s)
 -  **medium mountain:** 1 peça(s)
 -  **Moon:** 1 peça(s)
 -  **Star Glow:** 3 peça(s)
 -  **Star Burst:** 16 peça(s)
 -  **Star Sparkle:** 9 peça(s)
-

RLEV13.yaml - Lasermania

Objetivo: Use the lasers to power the electric can-opener.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1007_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 21 arquivos `.ANM` carregados. - Totaliza aproximadamente **448 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 1 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Belt:** 3 peça(s)
 -  **Gear:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Generator:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Wood Wall:** 5 peça(s)
 -  **Can Opener:** 1 peça(s)
 -  **Nitroglycerine:** 4 peça(s)
 -  **Greco-Roman Wall:** 2 peça(s)
 -  **Tiny Gear:** 2 peça(s)
 -  **Red Laser:** 4 peça(s)
 -  **Green Laser:** 5 peça(s)
 -  **Blue Laser:** 5 peça(s)
 -  **Angled Mirror:** 16 peça(s)
 -  **Laser Mixer:** 5 peça(s)
 -  **Laser-Activated Plug:** 6 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
-

RLEV14.yaml - Ballistic Balloon

Objetivo: Put the balloon inside the laundry basket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1005_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 32 arquivos `.ANM` carregados. -

Totaliza aproximadamente **613 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 6 peça(s)
-  **Balloon:** 1 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Belt:** 2 peça(s)
-  **Rope:** 1 peça(s)
-  **Laundry Basket:** 1 peça(s)
-  **Cannon:** 1 peça(s)

-  **Electric Switch & Outlet:** 2 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 2 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Flashlight:** 2 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Magnifying Glass:** 2 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Pinwheel:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 10 peça(s)
 -  **Anti-Gravity Pad:** 4 peça(s)
 -  **Steel Cable:** 2 peça(s)
 -  **Tin Snips:** 1 peça(s)
 -  **Hot Air Balloon:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Wooden Barrier:** 1 peça(s)
 -  **Yellow Brick Wall:** 1 peça(s)
 -  **Yellow Brick Incline:** 3 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

RLEV15.yaml - Missile vs. Blimp

Objetivo: Remove the green blimp from the skies.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS

`Teleport`) - **Recursos de Áudio:** - Música de Fundo: `SX_1013_11025.wav` - *Efeitos Sonoros (SFX): Despachados*

dinamicamente via `AudioDispatcher` pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** -

Requer 31 arquivos `.ANM` carregados. - Totaliza aproximadamente **651 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 7 peça(s)
-  **Wood incline:** 3 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 1 peça(s)

-  **Pulley:** 1 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Rope:** 4 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Cannon:** 1 peça(s)
 -  **Dynamite:** 1 peça(s)
 -  **Electric Switch & Outlet:** 2 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Boat Cleat:** 3 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Baseball:** 3 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Magnifying Glass:** 2 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Cheese:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 12 peça(s)
 -  **Large Curved Pipe:** 2 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Missile:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Steel Cable:** 2 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Teeter-Totter:** 3 peça(s)
-

RLEV16.yaml - Rodent Restaurant

Objetivo: Newton Mouse is ready for breakfast. Brew up some coffee and start the electric mixer.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 41 arquivos `.ANM` carregados. - Totaliza aproximadamente **814 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 5 peça(s)
-  **Topsy Trailer:** 2 peça(s)

-  **Conveyor Belt:** 3 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Belt:** 5 peça(s)
 -  **Basketball:** 1 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Gear:** 3 peça(s)
 -  **Bike Pump:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Mandrill Motor:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Pinwheel:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Tennis Ball:** 2 peça(s)
 -  **Wood Wall:** 5 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Super Ball:** 1 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 3 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Trans-Roto-Matic:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Granite Incline:** 4 peça(s)
 -  **Electric Mixer:** 1 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

RLEV17.yaml - Fireworks Berserk

Objetivo: Set off the fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 45 arquivos `.ANM` carregados. - Totaliza aproximadamente **1422 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 5 peça(s)
-  **Conveyor Belt:** 3 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Pulley:** 2 peça(s)
-  **Belt:** 3 peça(s)
-  **Basketball:** 1 peça(s)
-  **Rope:** 3 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Fish Tank:** 1 peça(s)
-  **Cannon:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Electric Fan:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Mandrill Motor:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Pinwheel:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)
-  **Pinball:** 2 peça(s)
-  **Wood Wall:** 2 peça(s)
-  **Cheese:** 1 peça(s)
-  **Mel Schlemming:** 1 peça(s)
-  **Remote Control Explosives:** 1 peça(s)
-  **Caution Wall:** 7 peça(s)
-  **Large Curved Pipe:** 1 peça(s)
-  **Grass Floor:** 1 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Fireworks:** 1 peça(s)

-  **Greco-Roman Wall:** 1 peça(s)
 -  **Programmable Ball:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 4 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Yellow Brick Wall:** 1 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **lamp post:** 1 peça(s)
 -  **Teeter-Totter:** 3 peça(s)
 -  **medium mesa:** 1 peça(s)
-

RLEV18.yaml - Big Job, Little Mouse

Objetivo: Newton Mouse has some major painting to do. Help him lift the leaky paint bucket on the right up above his scaffold.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de**

Áudio: - Música de Fundo: `SX_1007_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via*

AudioDispatcher pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 28 arquivos `.ANM` carregados. - Totaliza aproximadamente **676 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Brick Wall:** 7 peça(s)
-  **Topsy Trailer:** 2 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Pulley:** 3 peça(s)
-  **Belt:** 4 peça(s)
-  **Basketball:** 1 peça(s)
-  **Rope:** 3 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Gear:** 6 peça(s)
-  **Bucket:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Boat Cleat:** 5 peça(s)
-  **Mandrill Motor:** 1 peça(s)

-  **Hedge Trimmers:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Vacuum:** 1 peça(s)
 -  **Caution Wall:** 1 peça(s)
 -  **Can Opener:** 1 peça(s)
 -  **Steel Cable:** 2 peça(s)
 -  **Toaster:** 1 peça(s)
 -  **Cinder Block Wall:** 4 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Brick Incline:** 8 peça(s)
 -  **Archway:** 1 peça(s)
 -  **Leaky Bucket:** 2 peça(s)
 -  **Egg Timer:** 1 peça(s)
 -  **Rising Bubbles:** 1 peça(s)
 -  **Bamboo Fence:** 1 peça(s)
 -  **Wooden Door:** 1 peça(s)
 -  **Stone Texture:** 13 peça(s)
-

RLEV19.yaml - Fandango

Objetivo: Help Newton Mouse cool off by turning on the fan.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1009_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 40 arquivos `.ANM` carregados. - Totaliza aproximadamente **886 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 9 peça(s)
-  **Balloon:** 1 peça(s)
-  **Conveyor Belt:** 3 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Pulley:** 1 peça(s)
-  **Belt:** 5 peça(s)
-  **Basketball:** 2 peça(s)
-  **Rope:** 4 peça(s)
-  **Bike Pump:** 1 peça(s)
-  **Electric Switch & Outlet:** 2 peça(s)
-  **Remote Control:** 1 peça(s)

-  **Electric Fan:** 1 peça(s)
 -  **Generator:** 1 peça(s)
 -  **Lava Lamp:** 1 peça(s)
 -  **Mandrill Motor:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Newton Mouse:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Electric Motor:** 2 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 2 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Soccer Ball:** 2 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Toaster:** 1 peça(s)
 -  **Programmable Ball:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Trans-Roto-Matic:** 1 peça(s)
 -  **Roto-Trans Converter:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 5 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Scaffold Barrier:** 2 peça(s)
 -  **Sun:** 1 peça(s)
 -  **Teeter-Totter:** 3 peça(s)
-

RLEV20.yaml - It's Hip to Drip

Objetivo: Put the bowling ball into the steel cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo:

`SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. - Totaliza aproximadamente **251 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)

-  **Topsy Trailer:** 1 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Tennis Ball:** 2 peça(s)
 -  **Pipe Wall:** 2 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Leaky Bucket:** 2 peça(s)
 -  **Yellow Brick Wall:** 8 peça(s)
 -  **Yellow Brick Incline:** 13 peça(s)
 -  **Egg Timer:** 1 peça(s)
-

RLEV21.yaml - Mind-Mixer

Objetivo: Turn on the electric mixer.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1016_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 30 arquivos `.ANM` carregados. - Totaliza aproximadamente **524 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Wood incline:** 4 peça(s)
-  **Conveyor Belt:** 1 peça(s)
-  **Mouse Motor:** 2 peça(s)
-  **Belt:** 1 peça(s)
-  **Basketball:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Flashlight:** 1 peça(s)
-  **Generator:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Solar Panel:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Pinball:** 1 peça(s)

-  **Wood Wall:** 12 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Large Curved Pipe:** 1 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 2 peça(s)
 -  **Tiny Gear:** 2 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 4 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Electric Mixer:** 1 peça(s)
 -  **Pool Cue:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Color Block:** 1 peça(s)
 -  **Blue Space:** 10 peça(s)
 -  **Rising Bubbles:** 1 peça(s)
 -  **Bubble Stream:** 3 peça(s)
 -  **Floating Bubbles:** 1 peça(s)
 -  **Coral Arch:** 1 peça(s)
 -  **Coral Branch:** 1 peça(s)
 -  **Green Mountain:** 1 peça(s)
 -  **Mossy Rock:** 1 peça(s)
 -  **Green Cliff:** 1 peça(s)
 -  **Seaweed Tall:** 1 peça(s)
 -  **Kelp Plant:** 1 peça(s)
 -  **Seaweed Vine:** 1 peça(s)
 -  **Jellyfish:** 1 peça(s)
 -  **Fish Scatter:** 1 peça(s)
 -  **Fish Group:** 2 peça(s)
-

RLEV22.yaml - The Ultimate Schlemming

Objetivo: Get Mel home. Don't allow the red light to blink on the laser-detector.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1017_11025.wav` - *Efeitos Sonoros (SFX): Despachados*

dinamicamente via `AudioDispatcher` pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** -
Requer 29 arquivos `.ANM` carregados. - Totaliza aproximadamente **650 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 2 peça(s)
 -  **Conveyor Belt:** 1 peça(s)
 -  **Mouse Motor:** 1 peça(s)
 -  **Pulley:** 7 peça(s)
 -  **Belt:** 1 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Bucket:** 1 peça(s)
 -  **Remote Control:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Solar Panel:** 1 peça(s)
 -  **Springboard:** 1 peça(s)
 -  **Electric Motor:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Caution Wall:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Log Wall:** 12 peça(s)
 -  **Programmable Ball:** 1 peça(s)
 -  **Accelerator Tube:** 1 peça(s)
 -  **Blue Laser:** 2 peça(s)
 -  **Angled Mirror:** 1 peça(s)
 -  **Log Incline:** 6 peça(s)
 -  **Leaky Bucket:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **Candle:** 1 peça(s)
 -  **Laser Detector:** 1 peça(s)
-

RLEV23.yaml - Blast-A-Blimp

Objetivo: Blow up the blimp with the missile.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - Sistemas Físicos Necessários: - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1000_11025.wav` - **Efeitos Sonoros (SFX):** *Despachados dinamicamente via `AudioDispatcher`*

pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 30 arquivos .ANM carregados. - Totaliza aproximadamente **478 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 1 peça(s)
-  **Pulley:** 2 peça(s)
-  **Belt:** 1 peça(s)
-  **Rope:** 2 peça(s)
-  **Bike Pump:** 2 peça(s)
-  **Bucket:** 1 peça(s)
-  **Boat Cleat:** 1 peça(s)
-  **Generator:** 1 peça(s)
-  **Captain Z Super Phazer:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Pinwheel:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Thumb Tack:** 1 peça(s)
-  **Caution Wall:** 3 peça(s)
-  **Missile:** 1 peça(s)
-  **Trap Door:** 1 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Blue Laser:** 1 peça(s)
-  **Laser-Activated Plug:** 1 peça(s)
-  **Blimp:** 1 peça(s)
-  **Sun:** 1 peça(s)
-  **Big Cloud:** 2 peça(s)
-  **Pool Cue:** 1 peça(s)
-  **Yellow Brick Wall:** 4 peça(s)
-  **Yellow Brick Incline:** 9 peça(s)
-  **Color Block:** 1 peça(s)
-  **Egg Timer:** 1 peça(s)
-  **Candle:** 1 peça(s)
-  **Small Cloud:** 2 peça(s)
-  **Rising Bubbles:** 2 peça(s)
-  **Floating Bubbles:** 1 peça(s)
-  **Coral Branch:** 1 peça(s)
-  **Submarine:** 1 peça(s)
-  **Kelp Forest:** 1 peça(s)
- **Kelp Plant:** 2 peça(s)

-  **Jellyfish:** 1 peça(s)
 -  **Angelfish:** 1 peça(s)
 -  **Shark:** 1 peça(s)
 -  **Fish Group:** 3 peça(s)
 -  **Great White Shark:** 1 peça(s)
-

RLEV24.yaml - Enchanted Kingdom

Objetivo: Help Mel exit the right side of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver (Ropes & Belts)` - `ElectricalPropagationSystem` - `GravitySolver (Standard)` - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 21 arquivos `.ANM` carregados. - Totaliza aproximadamente **400 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 3 peça(s)
-  **Balloon:** 3 peça(s)
-  **Pulley:** 5 peça(s)
-  **Rope:** 4 peça(s)
-  **Laundry Basket:** 1 peça(s)
-  **Cannon:** 1 peça(s)
-  **Electric Switch & Outlet:** 1 peça(s)
-  **Lava Lamp:** 1 peça(s)
-  **Magnifying Glass:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Mel Schlemming:** 1 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Hot Air Balloon:** 1 peça(s)
-  **Greco-Roman Wall:** 4 peça(s)
-  **Red Laser:** 1 peça(s)
-  **Granite Incline:** 4 peça(s)
-  **Blimp:** 3 peça(s)
-  **Elm Tree:** 2 peça(s)
-  **Little Cloud:** 5 peça(s)
-  **Color Block:** 1 peça(s)
-  **Small Maple Tree:** 1 peça(s)
-  **Small Pine Tree:** 1 peça(s)
-  **Forest Trees:** 6 peça(s)
-  **Rocky Mountain:** 7 peça(s)
-  **Trees on Terrain:** 1 peça(s)

-  **Red Canyon:** 1 peça(s)
 -  **Castle Fortress:** 1 peça(s)
 -  **Moon:** 1 peça(s)
 -  **Wooden Door:** 1 peça(s)
 -  **Stone Texture:** 4 peça(s)
-

RLEV25.yaml - Poolin' Around

Objetivo: Sink one ball into each hole.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1021_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos .ANM carregados. - Totaliza aproximadamente **123 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Wood incline:** 2 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Pool Ball:** 16 peça(s)
 -  **Pool Cue:** 5 peça(s)
 -  **Pool Table Wall:** 6 peça(s)
 -  **Pool Table Pocket:** 6 peça(s)
-

RLEV26.yaml - Beam Scream

Objetivo: Break the laser beam.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1020_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 26 arquivos .ANM carregados. - Totaliza aproximadamente **354 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 2 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 2 peça(s)
-  **Basketball:** 1 peça(s)
-  **Dynamite:** 1 peça(s)
-  **Electric Switch & Outlet:** 2 peça(s)
-  **Electric Fan:** 1 peça(s)
-  **Baseball:** 1 peça(s)
-  **Springboard:** 1 peça(s)

-  **Pinwheel:** 1 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Pipe Wall:** 2 peça(s)
 -  **Alligator:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Anti-Gravity Pad:** 1 peça(s)
 -  **Boxes:** 1 peça(s)
 -  **Trap Door:** 1 peça(s)
 -  **Nitroglycerine:** 3 peça(s)
 -  **Cinder Block Wall:** 18 peça(s)
 -  **Green Laser:** 2 peça(s)
 -  **Archway:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
 -  **purple mountain:** 1 peça(s)
 -  **medium mountain:** 1 peça(s)
 -  **Laser Detector:** 1 peça(s)
 -  **Snow Mountains:** 2 peça(s)
 -  **Iceberg:** 2 peça(s)
 -  **Snow Peaks:** 1 peça(s)
-

RLEV27.yaml - Basketball Brain-Buster

Objetivo: Send the basketball off the top of the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo:

`SX_1001_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 21 arquivos `.ANM` carregados. - Totaliza aproximadamente **239 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 2 peça(s)
-  **Mouse Motor:** 1 peça(s)
-  **Belt:** 2 peça(s)
-  **Basketball:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Bucket:** 1 peça(s)
-  **Remote Control:** 1 peça(s)
-  **Boxing Glove:** 1 peça(s)
-  **Springboard:** 1 peça(s)
-  **Pinwheel:** 1 peça(s)

-  **Pinball:** 1 peça(s)
 -  **Pipe Wall:** 4 peça(s)
 -  **Remote Control Explosives:** 1 peça(s)
 -  **Large Curved Pipe:** 3 peça(s)
 -  **Coffee Pot:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Accelerator Tube:** 2 peça(s)
 -  **Yellow Brick Wall:** 17 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
-

RLEV28.yaml - Kitty Catcher

Objetivo: Put Curie Cat in the hole on the left. Then trap her with the laundry basket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1001_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 21 arquivos `.ANM` carregados. - Totaliza aproximadamente **470 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
-  **Conveyor Belt:** 4 peça(s)
-  **Mouse Motor:** 5 peça(s)
-  **Pulley:** 1 peça(s)
-  **Belt:** 8 peça(s)
-  **Basketball:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Laundry Basket:** 1 peça(s)
-  **Curie Cat:** 1 peça(s)
-  **Jack-in-the-box:** 1 peça(s)
-  **Gear:** 6 peça(s)
-  **Bucket:** 1 peça(s)
-  **Hedge Trimmers:** 1 peça(s)
-  **Newton Mouse:** 3 peça(s)
-  **Grass Floor:** 3 peça(s)
-  **Sand Wall:** 7 peça(s)
-  **Greco-Roman Wall:** 1 peça(s)
-  **Tiny Gear:** 2 peça(s)
-  **Granite Incline:** 2 peça(s)

-  **Yellow Brick Wall:** 10 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
-

RLEV29.yaml - TIM2 Contest Champion!

Objetivo: Congratulations to Richard and William Smith for creating this First Prize Puzzle in the TIM2 Puzzle Contest! Check out their work, and see if you can help Mel get home.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1009_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 24 arquivos `.ANM` carregados. - Totaliza aproximadamente **657 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 8 peça(s)
 -  **Wood incline:** 7 peça(s)
 -  **Conveyor Belt:** 7 peça(s)
 -  **Mouse Motor:** 2 peça(s)
 -  **Belt:** 14 peça(s)
 -  **Jack-in-the-box:** 1 peça(s)
 -  **Gear:** 11 peça(s)
 -  **Electric Switch & Outlet:** 1 peça(s)
 -  **Electric Fan:** 1 peça(s)
 -  **Generator:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Springboard:** 5 peça(s)
 -  **Pinwheel:** 1 peça(s)
 -  **Wood Wall:** 6 peça(s)
 -  **Electric Motor:** 2 peça(s)
 -  **Vacuum:** 1 peça(s)
 -  **Cheese:** 1 peça(s)
 -  **Mel Schlemming:** 1 peça(s)
 -  **Mel's House:** 1 peça(s)
 -  **Grass Floor:** 1 peça(s)
 -  **Alligator:** 2 peça(s)
 -  **Tiny Gear:** 7 peça(s)
 -  **Brick Incline:** 1 peça(s)
 -  **Teeter-Totter:** 2 peça(s)
-

TLEV1.yaml - Balls

Objetivo: Knock the eight ball off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Note how the super ball on the left bounces.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1000_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos .ANM carregados. - Totaliza aproximadamente **113 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
 -  **Brick Wall:** 3 peça(s)
 -  **Basketball:** 2 peça(s)
 -  **Baseball:** 2 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Tennis Ball:** 2 peça(s)
 -  **Super Ball:** 2 peça(s)
 -  **Pool Ball:** 2 peça(s)
 -  **Soccer Ball:** 2 peça(s)
 -  **Programmable Ball:** 2 peça(s)
-

TLEV2.yaml - Walls and Inclines

Objetivo: Put both bowling balls into the large column box in the center.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - A wall can be stretched to fit correctly. - Inclines could help get things rolling in the right direction.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1001_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 18 arquivos .ANM carregados. - Totaliza aproximadamente **292 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
-  **Brick Wall:** 2 peça(s)
-  **Wood incline:** 2 peça(s)
-  **Pipe Wall:** 3 peça(s)
-  **Curved Pipe Wall:** 2 peça(s)
-  **Wood Wall:** 3 peça(s)
-  **Caution Wall:** 3 peça(s)
-  **Grass Floor:** 3 peça(s)
-  **Sand Wall:** 3 peça(s)
-  **Cinder Block Wall:** 3 peça(s)

-  **Greco-Roman Wall:** 3 peça(s)
 -  **Log Wall:** 3 peça(s)
 -  **Grass Incline:** 2 peça(s)
 -  **Log Incline:** 2 peça(s)
 -  **Granite Incline:** 2 peça(s)
 -  **Brick Incline:** 6 peça(s)
 -  **Yellow Brick Wall:** 2 peça(s)
 -  **Yellow Brick Incline:** 6 peça(s)
-

TLEV3.yaml - Gears and Belts

Objetivo: Get all of the basketballs into the piped area.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (4): - Notice that this conveyor belt moves faster than the one above. This is because it is attached to the small gear. - This conveyor belt moves slower because it is connected to a large gear. - The small gear will spin two times faster than the large gear. - Look at the machine on the left side and try to duplicate what you see.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos `.ANM` carregados. - Totaliza aproximadamente **97 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 3 peça(s)
 -  **Mouse Motor:** 2 peça(s)
 -  **Belt:** 5 peça(s)
 -  **Basketball:** 3 peça(s)
 -  **Gear:** 8 peça(s)
 -  **Tennis Ball:** 2 peça(s)
 -  **Pipe Wall:** 3 peça(s)
 -  **Tiny Gear:** 3 peça(s)
-

TLEV4.yaml - Teeter-Totters & Ropes

Objetivo: Put the three bowling balls into the wicker baskets.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Notice how this steel cable goes through the pulleys. This allows the teeter-totter to pull up on an object placed at a higher level. - Look at the right side of the screen and try to duplicate what you see.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos `.ANM` carregados. - Totaliza aproximadamente **75 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 3 peça(s)
 -  **Wood incline:** 3 peça(s)
 -  **Topsy Trailer:** 3 peça(s)
 -  **Pulley:** 3 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Wood Wall:** 3 peça(s)
 -  **Boxes:** 4 peça(s)
 -  **Steel Cable:** 1 peça(s)
 -  **Teeter-Totter:** 2 peça(s)
-

TLEV5.yaml - Switches and Plugs

Objetivo: Start both of the electric mixers.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (6): - Notice that these electric plugs do not have a switch. They are always on. - A ball bumping this switch turns on the power. - This electric switch has been flipped upside-down. In this position, it must be bumped from below to flip it on. - A switch that has been turned on may also be turned off by flipping it back to its original position. - Notice that both plugs and switches have two outlets. - This ball will turn on a switch placed beneath it. A switch or plug must be set in position before the electric part you wish to plug into it.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** -

Música de Fundo: `SX_1002_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 17 arquivos `.ANM` carregados. -

Totaliza aproximadamente **359 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 1 peça(s)
-  **Electric Switch & Outlet:** 5 peça(s)
-  **Electric Fan:** 2 peça(s)
-  **Baseball:** 1 peça(s)
-  **Pinball:** 1 peça(s)
-  **Tennis Ball:** 2 peça(s)
-  **Electric Motor:** 1 peça(s)
-  **Vacuum:** 3 peça(s)
-  **Super Ball:** 1 peça(s)
-  **Can Opener:** 1 peça(s)
-  **Toaster:** 1 peça(s)
-  **Red Laser:** 1 peça(s)
-  **Green Laser:** 1 peça(s)
-  **Blue Laser:** 1 peça(s)

-  **Brick Incline:** 1 peça(s)
 -  **Electric Mixer:** 2 peça(s)
 -  **Electrical Outlet:** 2 peça(s)
-

TLEV6.yaml - Motors & Generators

Objetivo: Put both bowling balls between the logs.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Connect a belt to this wheel from any rotating power source. The generator will make electricity. - Plug this motor into any powered outlet and it will run. - If this conveyor belt is rotating the wrong direction, flip the motor.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1003_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 9 arquivos `.ANM` carregados. - Totaliza aproximadamente **175 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
 -  **Conveyor Belt:** 2 peça(s)
 -  **Mouse Motor:** 2 peça(s)
 -  **Belt:** 4 peça(s)
 -  **Generator:** 2 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Electric Motor:** 2 peça(s)
 -  **Log Wall:** 3 peça(s)
-

TLEV7.yaml - Lights and Solar Panels

Objetivo: Turn on all of the lights.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Pull down on this chain to turn on a lava lamp. Rope or steel cable can be attached to the chain. - Bump the button on top of a flashlight to turn it on. - The solar panel works like a generator. Get any light to shine on the panel to power the plug.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1004_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 15 arquivos `.ANM` carregados. - Totaliza aproximadamente **156 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 1 peça(s)
-  **Topsy Trailer:** 1 peça(s)

-  **Conveyor Belt:** 2 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Basketball:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Flashlight:** 2 peça(s)
 -  **Lava Lamp:** 2 peça(s)
 -  **Solar Panel:** 2 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Electric Motor:** 2 peça(s)
 -  **Steel Cable:** 1 peça(s)
 -  **Programmable Ball:** 1 peça(s)
 -  **Granite Incline:** 2 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

TLEV8.yaml - Fire

Objetivo: Light both candles, both lamps, both flint-rocks, and the match-on-a-spring.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (7): - Any light shone through a magnifying glass will start a flame or light a fuse. - You may need to flip a magnifying glass to make it work. - To light the match, pull out this pin. A rope or steel cable can be attached to the eye hook. - Hit the flint rocks with a falling object to light a fire. - Check out how the other lamp is lit when you run the puzzle. - Notice how the other candle is lit when you run the puzzle. - Check out how the other flint rock is lit when you run the puzzle.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1005_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **127 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Pulley:** 1 peça(s)
-  **Rope:** 1 peça(s)
-  **Flashlight:** 4 peça(s)
-  **Baseball:** 2 peça(s)
-  **Magnifying Glass:** 4 peça(s)
-  **Tennis Ball:** 2 peça(s)
-  **Aladdin's Lamp:** 2 peça(s)
-  **Caution Wall:** 6 peça(s)
-  **Soccer Ball:** 1 peça(s)
-  **Match-on-a-Spring:** 1 peça(s)
-  **Flint & Tinder:** 2 peça(s)
-  **Candle:** 2 peça(s)

-  **Teeter-Totter:** 1 peça(s)
-

TLEV9.yaml - Explosives

Objetivo: Blow up both remote bombs.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (6): - The nitroglycerine will explode if dropped or hit with enough force. - A light source and magnifying glass will light fuses. - Push or pull down on the plunger to set off the bomb. A rope or steel cable can be tied to the plunger. - Notice that not all walls can be blown up. - Notice that some walls can be destroyed. - This side is a mirror image of the other.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1006_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 13 arquivos `.ANM` carregados. - Totaliza aproximadamente **144 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Dynamite:** 2 peça(s)
 -  **Remote Control:** 2 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Flashlight:** 2 peça(s)
 -  **Magnifying Glass:** 2 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Caution Wall:** 5 peça(s)
 -  **Nitroglycerine:** 2 peça(s)
 -  **Log Wall:** 8 peça(s)
 -  **Programmable Ball:** 1 peça(s)
-

TLEV10.yaml - Things That Launch

Objetivo: Launch all the rockets, missiles, and fireworks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (4): - A fuse can be lit with fire or by shining light through a magnifying glass. - When a missile collides with something, it will explode. - When lit, this part will explode into fireworks before leaving the screen. It's a programmable part. - Missiles and rockets can be launched horizontally, fireworks can't.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `GravitySolver` (Standard) - `LaserRaycastSystem` -

Recursos de Áudio: - Música de Fundo: `SX_1020_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 14 arquivos `.ANM` carregados. - Totaliza aproximadamente **520 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Flashlight:** 3 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Magnifying Glass:** 3 peça(s)
 -  **Rocket:** 2 peça(s)
 -  **Tennis Ball:** 4 peça(s)
 -  **Pipe Wall:** 2 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Missile:** 2 peça(s)
 -  **Flint & Tinder:** 2 peça(s)
 -  **Fireworks:** 2 peça(s)
 -  **Sand Wall:** 2 peça(s)
 -  **Cinder Block Wall:** 2 peça(s)
 -  **Greco-Roman Wall:** 2 peça(s)
 -  **Yellow Brick Wall:** 3 peça(s)
-

TLEV11.yaml - Cannons

Objetivo: Fire all of the cannons.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Direct flame may be used to light fuses. - Magnified light may be used to light fuses. - Cannons can be flipped in six directions.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo:

`SX_1009_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos*

comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos `.ANM` carregados. - Totaliza aproximadamente **174 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
 -  **Pulley:** 1 peça(s)
 -  **Rope:** 1 peça(s)
 -  **Cannon:** 6 peça(s)
 -  **Flashlight:** 4 peça(s)
 -  **Baseball:** 5 peça(s)
 -  **Magnifying Glass:** 4 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
-

TLEV12.yaml - Fan, Pinwheel, Bike Pump

Objetivo: Put a bowling ball into each steel cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Plug a fan into any electric power source to make it run. - Use wind from the fan or bike pump to turn the pinwheel. The pinwheel may be flipped if you want to change the direction it rotates. - Push down on the bike pump and air will come out the front.

Mapa de Dependências: - Sistemas Físicos Necessários: - `ConstraintSolver` (Ropes & Belts) - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1001_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos `.ANM` carregados. - Totaliza aproximadamente **153 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 4 peça(s)
 -  **Conveyor Belt:** 4 peça(s)
 -  **Belt:** 4 peça(s)
 -  **Bike Pump:** 2 peça(s)
 -  **Electric Fan:** 2 peça(s)
 -  **Pinwheel:** 4 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Boxes:** 4 peça(s)
 -  **Cinder Block Wall:** 1 peça(s)
 -  **Electrical Outlet:** 2 peça(s)
-

TLEV13.yaml - Balloons

Objetivo: Pop the happy face balloon, but none of the others.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Hedge trimmers, tin snips, and tacks will pop balloons. - Fire and explosions will pop balloons. - Balloons can pull some objects upward.

Mapa de Dependências: - Sistemas Físicos Necessários: - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1011_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos `.ANM` carregados. - Totaliza aproximadamente **85 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Brick Wall:** 1 peça(s)
 -  **Topsy Trailer:** 1 peça(s)
 -  **Balloon:** 6 peça(s)
 -  **Pulley:** 4 peça(s)
 -  **Rope:** 3 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Thumb Tack:** 1 peça(s)
 -  **Nitroglycerine:** 1 peça(s)
-

TLEV14.yaml - Blimps

Objetivo: Pop all of the blimps.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - A burning blimp can light fuses, candles, and oil lamps. - A blimp will explode if it comes in contact with any fire. - If a blimp bumps into hedge trimmers, tin snips, or a tack it will pop.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1002_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 5 arquivos .ANM carregados. - Totaliza aproximadamente **103 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Thumb Tack:** 2 peça(s)
 -  **Flint & Tinder:** 2 peça(s)
 -  **Greco-Roman Wall:** 1 peça(s)
 -  **Blimp:** 4 peça(s)
 -  **Candle:** 1 peça(s)
-

TLEV15.yaml - Hot Air Balloons

Objetivo: Launch all of the hot air balloons.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Light the candle in the balloon to make it rise. - Ropes and cables can be tied to balloons.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1004_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 12 arquivos .ANM carregados. - Totaliza aproximadamente **116 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 1 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Boat Cleat:** 1 peça(s)
 -  **Flashlight:** 1 peça(s)
 -  **Magnifying Glass:** 1 peça(s)
 -  **Hedge Trimmers:** 1 peça(s)
 -  **Pinball:** 3 peça(s)
 -  **Match-on-a-Spring:** 1 peça(s)
 -  **Flint & Tinder:** 1 peça(s)
 -  **Hot Air Balloon:** 3 peça(s)
 -  **Blimp:** 1 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

TLEV16.yaml - Anti-Gravity Pad

Objetivo: Put one balloon into each yellow brick box.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - These anti-gravity pads will reverse the gravitational pull on an object. - Notice how the bowling ball rises instead of dropping. - The anti-gravity pad makes this balloon sink instead of rising.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1012_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 7 arquivos .ANM carregados. - Totaliza aproximadamente **198 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Balloon:** 3 peça(s)
 -  **Basketball:** 1 peça(s)
 -  **Rocket:** 1 peça(s)
 -  **Caution Wall:** 3 peça(s)
 -  **Anti-Gravity Pad:** 6 peça(s)
 -  **Yellow Brick Wall:** 9 peça(s)
-

TLEV17.yaml - Things That Bounce

Objetivo: Put all of the pinballs into a cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (4): - Attach a belt from a motor to the Jack-in-the-box to pop it open. Anything on top will be thrown off when it pops open. - An object bouncing on the springboard will get higher with each bounce. - The pinball bumper will bounce an object off at any angle desired. - Drop something on top of this and it will be flipped into the air.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ConstraintSolver (Ropes & Belts) - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1013_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 11 arquivos .ANM carregados. - Totaliza aproximadamente **111 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Mouse Motor:** 2 peça(s)
-  **Belt:** 2 peça(s)
-  **Jack-in-the-box:** 2 peça(s)
-  **Baseball:** 2 peça(s)
-  **Springboard:** 2 peça(s)
-  **Pinball:** 6 peça(s)
-  **Pinball Bumper:** 2 peça(s)
-  **Boxes:** 2 peça(s)
-  **Greco-Roman Wall:** 3 peça(s)
-  **Granite Incline:** 5 peça(s)

-  **Pinball Flipper:** 2 peça(s)
-

TLEV18.yaml - Boxing Glove

Objetivo: Get both balloons off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Hit the button on the back of this glove to make it punch. Flip it to punch in the other direction.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1019_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos .ANM carregados. - Totaliza aproximadamente **77 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Balloon:** 2 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Boxing Glove:** 2 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Yellow Brick Wall:** 3 peça(s)
 -  **Yellow Brick Incline:** 2 peça(s)
-

TLEV19.yaml - Electric Mixer, Can Opener

Objetivo: Turn on both electric mixers and open both cans.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Plug the mixer into an electric outlet to turn it on. - Plug the can opener into an electric outlet to make it open the can.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ElectricalPropagationSystem - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1014_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 4 arquivos .ANM carregados. - Totaliza aproximadamente **41 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Electric Switch & Outlet:** 4 peça(s)
 -  **Baseball:** 4 peça(s)
 -  **Can Opener:** 2 peça(s)
 -  **Electric Mixer:** 2 peça(s)
-

TLEV20.yaml - Toasters

Objetivo: Make toast in each of the toasters.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - A toaster must be plugged into an outlet to work. - After a toaster is plugged into an outlet, the handle must be pushed down. - When the toast pops up, it can push up on objects.

Mapa de Dependências: - Sistemas Físicos Necessários: - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1021_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos `.ANM` carregados. - Totaliza aproximadamente **87 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Topsy Trailer:** 1 peça(s)
 -  **Baseball:** 2 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Pool Ball:** 1 peça(s)
 -  **Toaster:** 3 peça(s)
 -  **Electrical Outlet:** 3 peça(s)
-

TLEV21.yaml - Large Pipes

Objetivo: Put one bowling ball into each steel cage.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - This accelerator can be flipped to force objects up, down, left, or right. The arrows indicate the direction of acceleration. - Drop something into this curved pipe and it will come out the other side.

Mapa de Dependências: - Sistemas Físicos Necessários: - `GravitySolver` (Standard) - `PipeNetwork` (BFS Teleport) - **Recursos de Áudio:** - Música de Fundo: `SX_1013_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 7 arquivos `.ANM` carregados. - Totaliza aproximadamente **61 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
 -  **Pipe Wall:** 2 peça(s)
 -  **Large Curved Pipe:** 4 peça(s)
 -  **Boxes:** 2 peça(s)
 -  **Accelerator Tube:** 2 peça(s)
 -  **Large Pipes:** 2 peça(s)
 -  **Yellow Brick Incline:** 2 peça(s)
-

TLEV22.yaml - Egg Timer

Objetivo: Get all of the pinballs off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - This timer can be programmed so that the little arm will pop out after a set amount of time. Push the button on top to turn it on.

Mapa de Dependências: - Sistemas Físicos Necessários: - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos `.ANM` carregados. - Totaliza aproximadamente **106 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Remote Control:** 2 peça(s)
 -  **Baseball:** 4 peça(s)
 -  **Pinball:** 4 peça(s)
 -  **Wood Wall:** 6 peça(s)
 -  **Remote Control Explosives:** 2 peça(s)
 -  **Egg Timer:** 4 peça(s)
-

TLEV23.yaml - Message Machine

Objetivo: Turn on all of the computers to read the message.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Drop something on the keyboard to turn on the computer.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1016_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos .ANM carregados. - Totaliza aproximadamente **66 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
 -  **Basketball:** 1 peça(s)
 -  **Baseball:** 1 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Soccer Ball:** 1 peça(s)
 -  **Yellow Brick Incline:** 1 peça(s)
 -  **Message Computer:** 6 peça(s)
-

TLEV24.yaml - Pool Balls, Walls, Cues

Objetivo: Put the 6-ball into the bottom right corner pocket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (6): - Bump this button on the back of the pool cue to make it spring forward. - Change the angle at which the pool ball is hit to create a different trajectory. - These pockets can be rotated to fit any angle on a pool table. If you get a ball close enough to the hole, it will go in. - This wall can be stretched just like any other. It can also be flipped to a vertical position. - The pool cue can be rotated to 8 different angles. - The number on the pool ball can be programmed from 1 to 15. All of the pool balls react the same way.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1021_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos .ANM carregados. - Totaliza aproximadamente **151 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Baseball:** 2 peça(s)
 -  **Tennis Ball:** 2 peça(s)
 -  **Pool Ball:** 2 peça(s)
 -  **Pool Cue:** 4 peça(s)
 -  **Pool Table Wall:** 8 peça(s)
 -  **Pool Table Pocket:** 8 peça(s)
 -  **Yellow Brick Wall:** 3 peça(s)
 -  **Yellow Brick Incline:** 5 peça(s)
-

TLEV25.yaml - Coffee Pot

Objetivo: Blow both hot air balloons off the screen.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Make a fire under the coffee pot and it will blow steam out of its spout.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1017_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 5 arquivos .ANM carregados. - Totaliza aproximadamente **106 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Baseball:** 2 peça(s)
 -  **Coffee Pot:** 2 peça(s)
 -  **Flint & Tinder:** 2 peça(s)
 -  **Hot Air Balloon:** 2 peça(s)
 -  **Sand Wall:** 5 peça(s)
-

TLEV26.yaml - Lasers

Objetivo: Turn on all of the lasers.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Plug lasers into a power plug to make them work. - Lasers can be flipped up, down, left, or right.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - ElectricalPropagationSystem - GravitySolver (Standard) - LaserRaycastSystem - **Recursos de Áudio:** - Música de Fundo: SX_1001_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos .ANM carregados. - Totaliza aproximadamente **80 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Electric Switch & Outlet:** 2 peça(s)
-  **Baseball:** 2 peça(s)
-  **Red Laser:** 2 peça(s)

-  **Green Laser:** 2 peça(s)
 -  **Blue Laser:** 2 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
-

TLEV27.yaml - Laser-Activated Plugs

Objetivo: Start the electric mixer.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - When a laser of the correct color hits this sensor the power will be turned on. - Use a blue laser to turn on this switch. - These plugs are programmable by color. Program the switch to match the laser color.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1015_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos `.ANM` carregados. - Totaliza aproximadamente **97 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Laser-Activated Plug:** 3 peça(s)
 -  **Electric Mixer:** 1 peça(s)
 -  **Electrical Outlet:** 1 peça(s)
-

TLEV28.yaml - Laser Mixers

Objetivo: Open all of the cans.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Combine two or more laser beams through this mixer to create new laser colors. - Mix the colors to make a new color that will match each switch.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1005_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 7 arquivos `.ANM` carregados. - Totaliza aproximadamente **227 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Can Opener:** 3 peça(s)
-  **Red Laser:** 2 peça(s)
-  **Green Laser:** 2 peça(s)
-  **Blue Laser:** 2 peça(s)
-  **Laser Mixer:** 3 peça(s)
-  **Laser-Activated Plug:** 3 peça(s)

-  **Electrical Outlet:** 3 peça(s)
-

TLEV29.yaml - Laser Mirrors

Objetivo: Start the electric mixer.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Flip the mirrors to angle the lasers in the correct direction. - Try placing the mirror here. Small adjustments may be needed to get the correct angle on the laser.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1016_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos `.ANM` carregados. - Totaliza aproximadamente **216 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Angled Mirror:** 5 peça(s)
 -  **Laser Mixer:** 1 peça(s)
 -  **Laser-Activated Plug:** 1 peça(s)
 -  **Electric Mixer:** 1 peça(s)
 -  **Electrical Outlet:** 2 peça(s)
-

TLEV30.yaml - Laser Detector

Objetivo: Turn on the green light in each of the laser detectors.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - When a laser hits the eye of this detector, the green light will blink. If the beam is shut off, the red light will blink. If the beam then returns, both lights will blink.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - `LaserRaycastSystem` - **Recursos de Áudio:** - Música de Fundo: `SX_1017_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 5 arquivos `.ANM` carregados. - Totaliza aproximadamente **71 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Red Laser:** 1 peça(s)
 -  **Green Laser:** 1 peça(s)
 -  **Blue Laser:** 1 peça(s)
 -  **Electrical Outlet:** 3 peça(s)
 -  **Laser Detector:** 3 peça(s)
-

TLEV31.yaml - Translation & Rotation

Objetivo: Put the pinballs into the aquariums.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (5): - Pull on this arm with a rope to rotate the center gear. - The rotation of this gear can be used to power another part. - Connect a belt from a motor to this wheel to move the arm up and down. - A rope connected to this arm will pull on the object attached to the other end. - Push on this end to turn the center gear.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1019_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 10 arquivos `.ANM` carregados. - Totaliza aproximadamente **86 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Conveyor Belt:** 2 peça(s)
 -  **Mouse Motor:** 2 peça(s)
 -  **Pulley:** 2 peça(s)
 -  **Belt:** 4 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Pinball:** 2 peça(s)
 -  **Tennis Ball:** 2 peça(s)
 -  **Boxes:** 2 peça(s)
 -  **Trans-Roto-Matic:** 2 peça(s)
 -  **Roto-Trans Converter:** 2 peça(s)
-

TLEV32.yaml - Buckets

Objetivo: Put the two beach buckets up inside the yellow bricks.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - The amount of liquid that drips out of the bucket can be programmed. The faster the bucket drips, the more quickly it becomes lighter. - Ropes and cables can be tied to buckets. - Many objects can be dropped into buckets.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1020_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos `.ANM` carregados. - Totaliza aproximadamente **94 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Pulley:** 4 peça(s)
-  **Rope:** 2 peça(s)
-  **Bucket:** 2 peça(s)
-  **Greco-Roman Wall:** 3 peça(s)
-  **Leaky Bucket:** 2 peça(s)
-  **Yellow Brick Wall:** 6 peça(s)

TLEV33.yaml - Snips & Trimmers

Objetivo: Put one bucket into each basket.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - The hedge trimmers can only cut rope. They won't work on steel cable. - Tin snips can cut rope or steel cable.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ConstraintSolver` (Ropes & Belts) -

`GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1021_11025.wav` - *Efeitos Sonoros*

(SFX): *Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais**

(ANM & PNG): - Requer 11 arquivos `.ANM` carregados. - Totaliza aproximadamente **93 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Rope:** 3 peça(s)
 -  **Bucket:** 3 peça(s)
 -  **Boat Cleat:** 5 peça(s)
 -  **Baseball:** 2 peça(s)
 -  **Hedge Trimmers:** 2 peça(s)
 -  **Pinball:** 1 peça(s)
 -  **Tennis Ball:** 1 peça(s)
 -  **Boxes:** 5 peça(s)
 -  **Steel Cable:** 2 peça(s)
 -  **Tin Snips:** 3 peça(s)
 -  **Leaky Bucket:** 2 peça(s)
-

TLEV34.yaml - Cats and Aquariums

Objetivo: Put all five cats between the pipes.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Fish aquariums will attract cats. Once they are broken they will attract cats from a greater distance. - Bump the cat to make her take a few steps forward.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `GravitySolver` (Standard) - **Recursos de Áudio:** -

Música de Fundo: `SX_1018_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher`*

pelos comportamentos das peças. - **Recursos Visuais (ANM & PNG):** - Requer 8 arquivos `.ANM` carregados. -

Totaliza aproximadamente **362 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Curie Cat:** 5 peça(s)
-  **Fish Tank:** 2 peça(s)
-  **Baseball:** 2 peça(s)
-  **Tennis Ball:** 1 peça(s)
-  **Pipe Wall:** 3 peça(s)

-  **Cinder Block Wall:** 6 peça(s)
 -  **Granite Incline:** 9 peça(s)
 -  **Teeter-Totter:** 1 peça(s)
-

TLEV35.yaml - Mice and Cheese

Objetivo: Get all of the mice to a piece of cheese.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Cheese will attract mice if they can see it. They can't see it from a long distance. - Curie Cat will chase any mouse she sees. A caught mouse is instant cat food.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1003_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 4 arquivos .ANM carregados. - Totaliza aproximadamente **272 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Curie Cat:** 2 peça(s)
 -  **Newton Mouse:** 5 peça(s)
 -  **Cheese:** 3 peça(s)
 -  **Log Wall:** 5 peça(s)
-

TLEV36.yaml - Mouse Hole

Objetivo: Get a mouse into each mouse hole.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - If a mouse encounters a mouse hole as it is running along it will go inside.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - GravitySolver (Standard) - **Recursos de Áudio:** - Música de Fundo: SX_1002_11025.wav - *Efeitos Sonoros (SFX): Despachados dinamicamente via AudioDispatcher pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 4 arquivos .ANM carregados. - Totaliza aproximadamente **82 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Newton Mouse:** 2 peça(s)
 -  **Cheese:** 2 peça(s)
 -  **Mouse Hole:** 2 peça(s)
 -  **Yellow Brick Wall:** 2 peça(s)
-

TLEV37.yaml - Mandrill Motor

Objetivo: Put a bowling ball inside each crate.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (3): - Release this shade and Pavlov Mandrill will walk on the treadmill, trying to get the bananas. - Connect a belt to this wheel to turn another part. - Pull down on this lever to flip up the shade.

Mapa de Dependências: - Sistemas Físicos Necessários: - `ConstraintSolver` (Ropes & Belts) - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1007_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 7 arquivos `.ANM` carregados. - Totaliza aproximadamente **113 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 2 peça(s)
 -  **Conveyor Belt:** 2 peça(s)
 -  **Belt:** 2 peça(s)
 -  **Rope:** 2 peça(s)
 -  **Laundry Basket:** 2 peça(s)
 -  **Mandrill Motor:** 2 peça(s)
 -  **Boxes:** 2 peça(s)
-

TLEV38.yaml - Mel Schlemming

Objetivo: Get one Schlemming into each house.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Mel can be programmed to walk, run, or stand still. - If Mel bumps into something solid he will turn around.

Mapa de Dependências: - Sistemas Físicos Necessários: - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1004_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 3 arquivos `.ANM` carregados. - Totaliza aproximadamente **207 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Mel Schlemming:** 5 peça(s)
 -  **Caution Wall:** 5 peça(s)
 -  **Mel's House:** 4 peça(s)
-

TLEV39.yaml - Edison Alligator

Objetivo: Put the pinball into the aquarium.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (2): - Edison Alligator will happily eat mice or Mel Schlemming for dinner. - Edison will toss any object that lands on his nose.

Mapa de Dependências: - Sistemas Físicos Necessários: - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1005_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 9 arquivos `.ANM` carregados. - Totaliza aproximadamente **260 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Bowling Ball:** 1 peça(s)
-  **Newton Mouse:** 1 peça(s)

-  **Pinball:** 1 peça(s)
-  **Cheese:** 1 peça(s)
-  **Mel Schlemming:** 1 peça(s)
-  **Grass Floor:** 4 peça(s)
-  **Alligator:** 3 peça(s)
-  **Boxes:** 1 peça(s)
-  **Grass Incline:** 6 peça(s)

TLEV40.yaml - Vacuum

Objetivo: Vacuum up both mice.

Ambiente: - Gravidade: 272 - Pressão do Ar: 67

Dicas Ativas (1): - Plug the vacuum into an electric outlet and it will vacuum up anything within range.

Mapa de Dependências: - **Sistemas Físicos Necessários:** - `ElectricalPropagationSystem` - `GravitySolver` (Standard) - **Recursos de Áudio:** - Música de Fundo: `SX_1006_11025.wav` - *Efeitos Sonoros (SFX): Despachados dinamicamente via `AudioDispatcher` pelos comportamentos das peças.* - **Recursos Visuais (ANM & PNG):** - Requer 6 arquivos `.ANM` carregados. - Totaliza aproximadamente **165 frames** de sprite PNG na memória de vídeo.

Setup de Peças (Agrupadas por Tipo):

-  **Electric Switch & Outlet:** 1 peça(s)
-  **Newton Mouse:** 2 peça(s)
-  **Wood Wall:** 2 peça(s)
-  **Vacuum:** 2 peça(s)
-  **Cheese:** 2 peça(s)
-  **Electrical Outlet:** 1 peça(s)

10. Catálogo de User Journeys (Fluxos de UX)

Este documento especifica os 14 diagramas de interação e fluxo de usuário ponta-a-ponta, extraídos do motor original. Ele estabelece as regras exatas das máquinas de estado da Interface Gráfica (UI) que a versão da Godot deve implementar.

Complete specification for all 14 user journeys in The Incredible Machine 2. Cross-referenced with the original game manual (SierraChest walkthrough), the Python reference editor (`tim2_editor/`), and reverse-engineered ANM/BMP assets.

Journey Overview

ID	Journey	Original Game	Python Editor	Standalone Script
J01	Sign-In Window	Yes	*	<code>j01_signin.py</code>
J02	Main Menu	Yes		<code>j02_mainmenu.py</code>
J03	Puzzle Select	Yes		<code>j03_puzzle_select.py</code>
J04	Parts Bin — Browse & Place	Yes		<code>j04_parts_bin.py</code>
J05	Part Handles — Flip/Stretch/Program/Delete	Yes		<code>j05_part_handles.py</code>

ID	Journey	Original Game	Python Editor	Standalone Script
J06	Connections — Rope/Belt/Electrical	Yes	•	j06_connections.py
J07	Programmable Parts — 14 dialogs	Yes	•	j07_programmable.py
J08	Simulation — Start/Observe/Pause/Reset	Yes	•	j08_simulation.py
J09	Simulation — Solve Conditions	Yes	•	j09_solve.py
J10	Save & Load	Yes	•	j10_save_load.py
J11	Player Preferences	Yes		j11_preferences.py
J12	Music & Sound	Yes		j12_music_sound.py
J13	Hints & Info Dialogs	Yes		j13_hints_info.py
J14	Head-to-Head	Yes		j14_head_to_head.py

J01 — Sign-In Window



Screen Spec

- ANM: `SIGN_IN.ANM` (279×125 canvas)
- BMP: `SIGN_IN.BMP`
- Original features per manual:
 - Dark blue player name input field
 - Light blue player registry (all signed-in names)
 - Green help question mark button
 - Done button

Interaction Flow

```
Window opens → type name → Enter/Add → name in registry
→ click name to select → Done → proceed to Main Menu
```

States

State	ANM State	Visual
Idle	1	Window visible, empty input
Editing	2	Cursor in name field
Selected	3	Name highlighted in registry
Done	4	Window closing

Test Script

- Asset checks: `SIGN_IN.ANM` exists, has valid Section C/B/D
- Behavioral: input parsing, registry management, selection
- Manual: `python3 scripts/phase-20/journeys/j01_signin.py --interactive`

J02 — Main Menu



Screen Spec

- ANM: MAINMENU.ANM (672×377 canvas, 11 states, 116 frames)
- BMP: MAINMENU.BMP , MAINBUTN.BMP

Button Layout (per manual)

Row	Left	Center	Right
Upper	Puzzle Play	Professor Tim's Workshop	Head-to-Head
Lower	Guided Tour	Sign-In	Tutorial
Right icons	Computer (preferences)	Stop (quit)	Question mark (help)

Interaction Flow

Main Menu → hover button → click → navigate to target screen
Computer icon → J11 Preferences
Stop icon → Quit dialog → Exit

States

State	Description
idle	Menu visible, all buttons ready
hover_X	Button X highlighted
click_X	Navigate to target screen
quit	Exit game

J03 — Puzzle Select



Screen Spec

- BMP: PUZDISP.BMP , PUZTYPE1.BMP through PUZTYPE8.BMP
- 206 levels across 7 categories

Categories (per manual)

Category	Level Prefix	Count
Tutorial	TLEV*	~40
Easy	ELEV*	26
Medium	MLEV*	30
Hard	HLEV*	31
Really Hard	RLEV*	29
Head-to-Head	HH*	50
Workshop (custom)	user-saved	varies

Interaction Flow

```
Select category → scroll puzzle list → click puzzle → LOAD
Light bulb icon: solved (on) / unsolved (off)
Double-click = quick load
```

J04 — Parts Bin → Browse & Place



Screen Spec

- ANM: `BINSOLVE.ANM` (8 states, 9 frames)
- BMP: `ICONS1-10.BMP`, `PARTSBIN.BMP`
- Data: `build/phase-9/part-name-catalog.json` (149 parts × 19 categories)

Hotkeys (per manual)

Key	Category
B	Balls
W	Walls
P	Pipes
I	Inclines
A	Arches
O	Pool Stuff
G	Gears & Belts
R	Ropes
F	Flames
E	Electrical
L	Lasers
M	Mouse Cage
C	Critters

Interaction Flow

```
Parts Bin visible → scroll categories → hover part → part name tooltip
→ click part → part attached to cursor → click canvas → part placed at grid snap
→ right-click existing part → duplicate copy
```

In-Editor Status

- Catalog panel: Implemented (`tim2_editor/main.py:draw_catalog()`)
- Grid snap: (`16px grid, place_part()`)
- Hotkeys: Not implemented
- Duplicate: Not implemented
- Tooltips: Not implemented

J05 — Part Handles



Handle Types (per manual)

Handle	Icon	Action
Recycle	Trash can	Remove part back to Parts Bin
Flip	Curved arrows	Rotate/flip part orientation
Size/Stretch	Double arrow	Click-drag to resize
Program	Computer	Open programming dialog (J07)
Info	Magnifying glass	Open info dialog (J13)
Solve	Checkered flag	Program solution condition (Workshop only)
Lock	Padlock	Lock part in position (Workshop only)

ANM Reference

- HANDLES.ANM (16 states, 108 frames, animated)
- BUTTONS.ANM (21 states, 49 frames)

Interaction Flow

```
Part placed on canvas → cursor over part → handles appear as overlay
→ click handle → execute action
→ flip: rotate clockwise 90°/flip horizontal/flip vertical
→ stretch: click-drag to resize, click to set
→ recycle: remove part
→ program: open dialog window
→ info: open description window
```

J06 — Connections

Connection Types (per manual)

Rope (part type 10, steel cable type 76): - Attaches to 16 part types: Teeter-Totter, Leaky Bucket, Lava Lamp, Pulley, Boat Cleat, Phazer, Mandrill Motor, Trans-Roto-Matic, Laundry Basket, Balloon, Topsy Trailer, Roto-Trans-Converter, Bucket, Hot Air Balloon, Match-on-a-spring, Remote Control Bomb

Belt (part type 8): - Attaches to 11 rotating part types: Large Gears, Small Gears, Pinwheel, Conveyor Belt, Mouse Motor, Generator, Electric Motor, Trans-Roto-Matic, Roto-Trans-Converter, Jack-in-the-Box, Mandrill Motor

Electrical (outlet connections): - 5 power sources: Generator, Electric Switch & Outlet, Electrical Outlet, Solar Panel, Laser-Activated Plug

Interaction Flow

```
Select Rope/Belt from Parts Bin → cursor shows line
→ click first part (anchor turns red) → drag to second part
→ line turns green when valid connection → click to attach
→ connection line drawn between parts (center-to-center)
```

Part Struct Fields

- `connected_1`, `connected_2`: Part indices for rope/belt connections
- `outlet_plugged_1`, `outlet_plugged_2`: Part indices for electrical
- `belt_anchor_x/y`, `rope_1_anchor_x/y`, `rope_2_anchor_x/y`: Anchor positions

In-Editor Status

- Connection line rendering: `(main.py:draw_part())` lines 330-337
- Click-and-drag connection tool: Not implemented
- Red/green line color feedback: Not implemented

J08 — Simulation → Run/Pause/Reset



Game Loop (per godot-specification.md §1)

```
60 Hz tick rate
  Pass 0: advance state machines
  Pass 1: apply gravity
  Pass 2-3: AABB collision detection + response
  Pass 4-5: proximity interactions
  Pass 6-7: constraint enforcement (3 passes)
```

Controls (per manual)

Action	Trigger
Start	Click green flag (Parts Bin)
Stop/Pause	Click checkered flag
Reset	Click broom icon (remove unlocked parts)
Timeout	Auto-stop after ~10,000 ticks (if unsolved)

Visual Feedback

- Parts animate via ANM state machine (Section C → Section D bytecode)
- Moving parts update position each tick
- Collisions trigger state transitions + sound effects
- Goal Bar shows progress

In-Editor Status

- Simulation loop: `(world.py:step())` 5 passes
- Green flag / play toggle: `(space bar, toggle_preview())`
- Reset on stop: `(world.reset())`
- ANM rendering during sim: (gray rectangles placeholder)
- Sound on collision: Not implemented

J09 — Simulation → Solve Conditions



Condition Types

Type	Rect	Check
Position	(x, y, w, h)	Part bbox overlaps target rect
Off-screen	(-1, -1, -1, -1)	Part outside viewport
Off-top	(-500, -2000, 1640, 2000)	Part above screen
Off-bottom	(-500, 400, 1640, 3000)	Part below screen
State-only	(0, 0, 0, 0)	part.state_counter == target_state

Solution Struct (per level file)

```
conditions_count: N
delay: ticks_before_end
conditions:
  - part_index: INT16LE
    part_state_1: UINT16LE
    part_state_2: UINT16LE
    part_count: UINT16LE
    rect: { x, y, width, height }
```

Interaction Flow

```
Simulation runs → each tick: check all conditions
→ all met → delay_ticks → puzzle solved overlay
→ SOLVE.ANM victory animation
→ menu: Replay / Next Puzzle / Back to Select
```

In-Editor Status

- Solution checker: (solver.py)
- Green target rects: (main.py:draw() lines 242-252)
- Victory overlay: SOLVE.ANM not integrated
- Victory menu: Not implemented

J10 — Save & Load



Actions (per manual)

Action	Shortcut	Description
Save	Ctrl+S	Serialize level to YAML file
Load	Ctrl+L	Load YAML file, populate editor
New	N key	Clear editor, start empty level

Serialization Format

```
level_info: magic, title, goal, bg_color
environment: pressure, gravity, music_track
hints: count, entries
```



```
parts: fixed_count, moving_count, part_entries
solution: conditions_count, delay, conditions
```

Fidelity Requirements

- All 48-byte part struct fields preserved
- Hex flags (`0xNNNN`) preserved
- Connections (connected_1/2, outlet_plugged_1/2) preserved
- Solution conditions preserved
- Float/int round-trip tolerance

In-Editor Status

- Save: (`save_yaml()`) → 269 lines of YAML output)
 - Load: (`load_yaml()`) → regex parser, all 206 levels)
 - New: (`new_level()`)
 - Shortcuts: (`Ctrl+S`, `Ctrl+L`, `N`)
-

J11 — Player Preferences

Toggles (per manual)

Preference	Default	Effect
Ambient Animation	ON	Parts wiggle/twinkle when idle
Scenery Displayed	ON	Background scenery visible
Background	ON	Wallpaper/tiles/color blocks
Info Handles	ON	Magnifying glass on part hover
Sound Effects	ON	Play sounds on collisions/actions

Storage

- JSON file in user data directory
 - Keyed by player name (from J01 sign-in)
 - Loaded on game start, saved on preference change
-

J12 — Music & Sound

Music System

- Tracks: IDs 1000–1023 (24 tracks)
- Format: FM synthesis via ADL.DRV (OPL2)
- Access: Click guitar icon on control panel

Sound Effects

- FM synthesis: 14 patches in TIM2.SX (IDs < 300)
- Digital audio: 113 RAW files at 11025 Hz (IDs 3001–3754)
- Access via ANM Section D bytecode: `PLAY_SFX sound_id`

Controls

Control	Action
Music list	Scroll, click to select
Note button	Restart current track
Switch	Turn music on/off
POW button	Preview selected sound effect
Volume arrows	Increase/decrease volume
Done	Close panel

J13 — Hints & Info Dialogs

Info Dialogs

- ANM: `INFO00.ANM` through `INFO148.ANM` (one per part type)
- Each displays: part name, description, usage tips
- Access: click magnifying glass handle on part OR click ? then click part
- Text source: `build/phase-9/part-name-catalog.json` description field

Hints System

- ANM: `HINTLOCK.ANM` (4 states, 43 frames)
- Each level can have 0–5 hints
- Hint data: position (x, y), icon flip, text
- Access: click Hand icon on control panel → pointing hands appear → click to read

Professor Help

- ANM: `PROFESOR.ANM` (329×138 canvas)
 - Professor Tim provides contextual tips
 - Triggered by tutorial and guided tour modes
-

J14 — Head-to-Head Mode



Screen Spec

- ANM: `HEDTOHED.ANM` (639×399, 5 states, 21 frames)
- 50 head-to-head puzzles

Gameplay

- Two players alternate turns placing parts
- Turn indicator shows current player
- Each player builds toward completing the puzzle
- First player to trigger solution wins
- Score tracking per puzzle

In-Editor Status

- Not implemented in Python editor

- Standalone test script only

Validation

- **Master runner:** `scripts/phase-20/01-validate-user-journeys.py`
- **Per-journey scripts:** `scripts/phase-20/journeys/j{NN}_{name}.py`
- **Logs:** `build/phase-20/logs/J{NN}_{timestamp}.log`
- **Report:** `build/phase-20/user-journey-report.json`

Running Tests

```
<a id="list-all-journeys"></a>
# List all journeys
python3 scripts/phase-20/01-validate-user-journeys.py --list

<a id="run-all-headless"></a>
# Run all headless
python3 scripts/phase-20/01-validate-user-journeys.py

<a id="run-one-interactively-with-pygame-window"></a>
# Run one interactively (with pygame window)
python3 scripts/phase-20/01-validate-user-journeys.py --interactive J04

<a id="run-specific-journey-headless"></a>
# Run specific journey headless
python3 scripts/phase-20/01-validate-user-journeys.py --journey J08
```

Sound & Animation Integration

Every user journey references specific sound effects and animation states. The following table shows which journeys trigger audio and ANM playback:

Sound References by Journey

Journey	Sound Type	Sound IDs	Source
J01 Sign-In	—	—	No sounds in original
J02 Main Menu	UI click	3000 (menu select)	MAINMENU ANM PLAY_SFX
J03 Puzzle Select	UI click, solved jingle	3000, custom	PUZLTYPE BMP
J04 Parts Bin	Part pickup/place	5, 3005 (generic pickup)	BINSOLVE ANM
J05 Part Handles	Flip, stretch click	per-part sounds	BUTTONS/HANDLES ANM
J06 Connections	Rope snap, belt attach	3006 (click)	Connection validation
J07 Programmable	Open/close dialog	3000 (UI)	INFO ANMs
J08 Simulation	Collision sounds	per-part sounds (61 unique IDs)	PLAY_SFX in ANM bytecode
J09 Solve	Victory fanfare	3000, SOLVE sequence	SOLVE ANM
J10 Save/Load	Disk write	—	OS-level
J11 Preferences	Toggle click	3000	UI feedback
J12 Music & Sound	Track playback, SFX preview	1000–1023 (music), 3001–3754 (SFX)	ADL.DRV + RAW files
J13 Hints	Hint reveal	3000	HINTLOCK ANM
J14 Head-to-Head	Turn change, victory	3000, custom	HEDTOHED ANM

Animation States by Journey

Journey	ANM	States Used	Frame Range
J01 Sign-In	SIGN_IN	2 (editing), 3 (selected), 5 (idle)	0–7
J02 Main Menu	MAINMENU	1–11 (11 states)	0–460 (animated)
J03 Puzzle Select	—	(static BMPs)	PUZDISP, PUZTYPE1-8
J04 Parts Bin	BINSOLVE	1–8	0–8
J05 Part Handles	HANDLES, BUTTONS	1–16 (HANDLES), 1–21 (BUTTONS)	0–107, 0–48
J06 Connections	—	Dynamic (line drawing)	Draw via DRAW_LINE opcode
J07 Programmable	INFO0–INFO148	1 (info display)	1–72 (varies per part)
J08 Simulation	PART0–PART261	All part states (89 unique IDs)	0–189 (varies)
J09 Solve	SOLVE	1 (solved overlay)	0–N
J10 Save/Load	LOADSAVE	1,3,5	0–10
J11 Preferences	—	(button highlights)	—
J12 Music & Sound	—	Track list, volume bar	Draw via DRAW_RECT
J13 Hints	HINTLOCK	1–4	0–216 (animated)
J14 Head-to-Head	HEDTOHED	1–5	0–20

Validation Requirements for Sounds

Every journey test script that references sounds must: 1. Check that the sound ID exists in `behaviors.json` per-part `sounds[]` 2. Verify WAV file exists: `build/phase-6/raw-wav/SX_{ID}_11025.wav` 3. In interactive mode: play the sound when the associated action occurs 4. Log every sound triggered with `sound_id` and WAV path

Validation Requirements for Animations

Every journey test script that references ANM states must: 1. Check that the ANM JSON exists in `build/phase-4/anm/` 2. Verify that the state ID exists in ANM Section C 3. Verify frame metadata exists in Section B 4. Verify PNG files exist for each `DRAW_BMP` sprite reference 5. Log `state→frame→sprite` resolutions

Animation Frame Timing (Standard 60 Hz)

Timing Type	Description
Static ANM	Section A empty — frame changes only on state transition
Animated ANM	Section A has durations — frames advance per tick
Idle animation	Parts wiggle/twinkle when ambient animation is ON (preference)
State transition	Instant frame change when <code>state_counter</code> changes
Sub-state countdown	Temporary state with timed reversion (offsets 0xAE–0xB0)

See Also

- `build/phase-15/handoff/godot-specification.md` §1 — Game loop (moved from `docs/godot-specification.md`)
- `tim2_specs/state-machine-specification.md` — Full state machine reference
- `tim2_specs/validation-directives.md` — Validation architecture and patterns
- `tim2_specs/user-journey-specification.md` — Full user journey reference
- `tim2_editor/main.py` — Python editor reference implementation
- `tim2_editor/game.py` — Game mode
- `tim2_editor/sandbox.py` — Interactive part viewer
- `scripts/phase-20/journeys/` — Per-journey test scripts

11. Editor de Níveis Standalone (Godot 4.6+)

O Editor de Fases é o pilar da versão "Clean Room". O objetivo principal desta arquitetura é portar a flexibilidade do protótipo em Python (`tim2_editor`) para a engine da Godot 4.6+, permitindo aos jogadores construir, salvar, carregar e testar as engenhocas *sem depender do jogo base compilado*.

Visão Geral do Layout

A UI do editor não deve sobrepor o viewport principal para manter a legibilidade. Ela é dividida nos seguintes containers de controle:



1. **Toolbar Superior** (`MarginContainer > HBoxContainer`):
2. Botões globais de ação (Salvar, Carregar, Limpar, Testar).
3. **Viewport Central** (`SubViewportContainer > SubViewport > Node2D`):
4. A área "canvas" de jogo. Suporta resolução base de 560x377 escalada.
5. Possui lógica isolada para zoom (`Mouse Wheel`) e translação da câmera (`Middle Mouse Drag`).
6. **Painel de Catálogo Direito** (`PanelContainer > ScrollContainer > VBoxContainer`):
7. Apresenta as 149 peças + cenário, agrupadas por Categoria.
8. Ícones extraídos dos primeiros frames (`PART{N}_f000.png`).
9. **Overlay Dinâmico** (`Handles Overlay`):
10. Renderiza alças de manipulação contextuais quando o jogador clica numa peça na viewport.

Estrutura da Árvore de Nós (Scene Tree)

O `LevelEditor.tscn` deve ser instanciado como uma hierarquia desacoplada:

```
LevelEditor (Control)
├── UI_CanvasLayer (CanvasLayer)
│   ├── TopMenu (HBoxContainer)
│   │   ├── BtnLoad (Button)
│   │   ├── BtnSave (Button)
│   │   ├── BtnPlay (Button)
│   │   └── EnvSettingsDialog (PopupPanel)
│   ├── SideCatalog (PanelContainer)
│   │   ├── Scroll (ScrollContainer)
│   │   │   └── PartGrid (GridContainer)
│   │   │       ├── PartBtn0 (TextureButton)
│   │   │       └── PartBtnN (TextureButton)
│   └── ModeIndicator (Label)
└── GameSpace (SubViewportContainer)
    ├── LevelViewport (SubViewport)
    │   ├── Camera2D (Handles Zoom/Pan)
    │   ├── Background (ColorRect/TextureRect)
    │   ├── GridOverlay (Node2D - custom _draw)
    │   └── PartContainer (Node2D)
    │       ├── PlacedPartA (Area2D + Sprite2D)
    │       └── PlacedPartB (Area2D + Sprite2D)
```

Máquina de Estados da Ferramenta (Tool Modes)

A classe `EditorController.gd` intercepta a entrada de mouse e teclado em `_unhandled_input(event)`. Apenas um modo fica ativo por vez.

1. `MODE_PLACE` (Atalho: P)

- **Ação:** Instancia uma nova peça na cena.
- **Fluxo:** O jogador clica no `PartBtn0` do catálogo. O editor entra em `MODE_PLACE`. O cursor exibe uma versão transparente/fantasma do ícone (`modulate = Color(1, 1, 1, 0.5)`). Ao clicar no *GameSpace*, o script instancia a cena respectiva atrelada à grade (*Snap-to-Grid* em 16px).

2. `MODE_SELECT` (Atalho: S)

- **Ação:** Manipular uma peça existente.
- **Fluxo:** Clicar num nó `Area2D` dentro do `PartContainer` ativa seu destaque.
- **Handles (Alças):**
 -  **Girar/Inverter (Flip):** Rotaciona a peça ou espelha o eixo X/Y (alterando os `flags`).
 -  **Deletar:** Remove o nó da memória com `queue_free()`.
- **Programar:** Abre o pop-up modal para definir comportamentos avançados.

3. `MODE_CONNECT` (Atalho: C)

- **Ação:** Cria lógicas físicas entre duas peças.
- **Fluxo:** Essencial para Cordas, Correias e Eletricidade.
- Clique inicial no "Nó de Ancoragem" da Peça A.
- A Engine traça uma `Line2D` elástica seguindo a posição global do cursor do Mouse.
- O cursor brilha "Verde" se passar por cima de uma âncora válida (ex: engrenagem se for correia), ou "Vermelho" caso seja inválido.
- Clique secundário na Peça B finaliza o vínculo e registra os IDs matemáticos (`connected_1`, `connected_2`).

4. `MODE_PREVIEW` (Atalho: Espaço)

- **Ação:** Testa o código de física real da fase.
- **Fluxo:** Ao clicar em Play, o editor serializa toda a fase atual e desativa a UI. O sistema instancia o `PhysicsSolver` de 60Hz. Se o usuário clicar em "Stop", a fase é destruída e reconstruída com os dados de antes do Play para garantir estado imutável.

Fluxos e Jornadas Restritas de Edição (User Journeys)

J07 - Peças Programáveis (Program Dialogs)

Algumas peças não são estáticas. O editor precisa injetar caixas de diálogo modais exclusivas quando o botão "Programar" é acionado na UI. - **Laser:** Definir o canal RGB de emissão. - **Balão de Fala:** Configurar a String que aparecerá quando disparado. - **Interruptores:** Definir qual o valor elétrico que propaga na ativação.

J10 - Save & Load Serializado (YAML)

O motor nativo da Godot possui `ResourceSaver`, porém, a especificação dita que os dados do puzzle devem estar totalmente dissociados da Engine (para possibilitar modding aberto e validação limpa via Python). - O editor processa um parser YAML nativo (`LevelSerializer.gd`). - Ao acionar "Save", o construtor formata os dicionários, as condições de vitória (`SolutionCondition`), as matrizes de `fixed_parts` e joga um arquivo `nome.yaml` usando o `FileAccess`.

Funcionalidades QoL (Quality of Life)

O editor moderno ultrapassa o limite clássico de 1994, oferecendo: 1. **Histórico de Ações (Undo/Redo)**: Usa a classe embutida `UndoRedo` da Godot, mapeando o instanciamento/destruição de peças na memória. 2. **Seleção em Massa (Box Select)**: O botão esquerdo arrastado em área neutra cria um Box. Múltiplas peças podem ser arrastadas e apagadas de uma só vez. 3. **Menu "Goal & Hints"**: Modal (`PopupPanel`) focado para o Level Designer escrever a dica de vitória ("Goal") e preencher os nós esféricos vermelhos que o jogador poderá clicar no futuro para obter a tradução livre da dica.

Mapeamento Avançado de Casos de Uso e Resolução de Ambiguidades

Durante a arquitetura do editor, diversas ambiguidades (*edge cases*) emergem na interação do usuário. Abaixo, detalhamos cada Caso de Uso com suas respectivas resoluções de ambiguidade (User Journeys expandidas).

UC01: Iniciar Novo Nível (New Level)

- **Ambiguidade**: O que acontece se o jogador tiver trabalho não salvo ao clicar em "Novo"?
- **Resolução (User Journey)**:
 - O usuário clica em "Novo" (ou pressiona `N`).
 - A engine verifica o sinalizador interno `is_dirty`. Se verdadeiro, um modal de aviso surge no meio da tela: *"Você tem alterações não salvas. Deseja descartá-las?"*
 - Ao confirmar, a `SceneTree` do `PartContainer` é purgada (`queue_free` em todos os nós) e as configurações globais (gravidade, pressão) retornam aos valores *default*. O sinalizador `is_dirty` volta a `false`.

UC02: Adicionar e Posicionar Peça (Place Part)

- **Ambiguidade**: Como o usuário sabe se o local de drop é válido antes de clicar? O que acontece se o mouse sair do `GameSpace`?
- **Resolução (User Journey)**:
 - O usuário clica na peça desejada no Catálogo Lateral. O cursor do mouse passa a exibir um "fantasma" transparente da peça.
 - Ao mover o mouse pela viewport, a peça fantasma segue a ancoragem do *snap-to-grid* (16x16 pixels).
 - Um shader/código de colisão roda em tempo real: se a *Bounding Box* da peça fantasma sobrepor qualquer outra peça existente, o fantasma fica destacado na cor **Vermelha** (inválido). Se estiver livre, fica em sua cor normal.
 - Tentar clicar com o fantasma vermelho gera um feedback sonoro de erro.
 - Clicar fora dos limites do `SubViewport` (ex: em cima do menu) ou clicar com o Botão Direito cancela a ação `MODE_PLACE` e devolve o cursor ao estado normal.

UC03: Estabelecer Vínculo Físico (Connect Parts)

- **Ambiguidade**: O que acontece se o usuário tentar ligar uma Corda a um pino Elétrico? E como cancelar um traçado pela metade?
- **Resolução (User Journey)**:
 - O usuário ativa `MODE_CONNECT` e clica numa âncora mecânica (Peça A). Uma `Line2D` elástica começa a seguir a posição do mouse.
 - O sistema filtra imediatamente os alvos válidos em tela: como a Peça A é mecânica (Motor), apenas as âncoras mecânicas (Engrenagens/Polias) ganham um leve brilho.
 - Se o cursor passar por cima de um pino elétrico, a linha fica **Vermelha** e um "X" aparece, bloqueando o clique.
 - Se o usuário pressionar `ESC` ou clicar com o **Botão Direito**, o vínculo em andamento é destruído sumariamente e a ferramenta reseta.
 - Um clique válido numa âncora verde finaliza a linha geométrica e consolida o ID matemático.

UC04: Programar Comportamento Específico (Program Part)

- **Ambiguidade:** Como o usuário descobre visualmente que uma peça é programável sem ter que sair testando todas? E se ele cancelar a edição?
- **Resolução (User Journey):**
 - No modo `MODE_SELECT`, ao passar o mouse (*hover*) sobre uma peça programável (ex: *Laser* ou *Interruptor*), um pequeno ícone de "Computador" (Handle de Programação) aparece flutuando anexado a ela.
 - Ao clicar nesse ícone (ou com o Botão Direito > Programar), a simulação é pausada e um Modal pop-up de configuração se abre sobrepondo o editor.
 - O usuário altera os parâmetros (ex: troca o laser para Vermelho).
 - Se clicar em **Cancelar** ou no **[X]**, nenhuma modificação é feita nos *Flags* da peça. Apenas clicar em **Confirmar/ Aplicar** salva as máscaras de bits (`flags_1`, `flags_2`) no banco de dados da peça na `SceneTree`.

UC05: Definir Condições de Vitória (Set Win Condition)

- **Ambiguidade:** É possível ter múltiplas condições de vitória? Como visualizá-las e deletá-las no canvas?
- **Resolução (User Journey):**
 - O usuário clica no menu superior "Goal/Win Setup".
 - O catálogo lateral é temporariamente substituído pela aba de *Goals*.
 - O usuário pode adicionar múltiplas condições clicando em "Add Target Rect".
 - No *GameSpace*, polígonos semi-transparentes verdes aparecem. Eles possuem *Handles* de arrastar as bordas (stretch) para definir a zona exata de alvo.
 - Cada polígono possui um botão flutuante **[X]** no seu canto superior direito. Ao clicar, a condição é excluída, resolvendo de forma elegante a gestão visual de requisitos de vitória.

UC06 & UC07: Testar Nível e Retornar (Preview / Play / Stop)

- **Ambiguidade:** Quando o teste rola, bombas explodem e cordas partem. Como o jogo retorna ao estado perfeitamente inicial após parar o teste?
- **Resolução (User Journey):**
 - O usuário clica em "Play" (Espaço). Imediatamente, a engine *serializa* a cena inteira em um buffer temporário em memória RAM (`temp_state.yaml`).
 - O `PhysicsSolver` assume o controle. O jogo simula a 60Hz. Objetos são consumidos e posições são alteradas agressivamente.
 - Ao detectar falha no plano ou concluir o teste, o usuário clica em "Stop" (ou aperta Espaço).
 - O editor purga e deleta 100% da simulação atual da memória RAM.
 - Ele executa um `Load` silencioso e instantâneo a partir do `temp_state.yaml`. Resultado: a tela pisca e tudo volta exatamente ao estado pixel-perfect de antes do "Play".

UC08: Manipulação em Massa (Box Select & Mover Cluster)

- **Ambiguidade:** O *Box Select* serve apenas para deletar, ou posso mover um grupo inteiro de engrenagens montadas sem quebrar as conexões delas?
 - **Resolução (User Journey):**
 - O usuário arrasta o clique esquerdo em uma área vazia criando um retângulo de seleção.
 - Todas as peças capturadas ficam com um contorno em realce (Seleção Ativa). As linhas de correia/corda interconectando este grupo ficam visíveis.
 - **Mover:** Ao clicar e segurar qualquer peça dentro dessa área, o usuário arrasta *todo o cluster* em uníssono. A engine verifica se *nenhuma* das peças vai colidir na nova posição de *snap-to-grid*. Se o local couber todas, o cluster é movido e as coordenadas de tudo se atualizam. Se alguma peça colidir com uma externa ao cluster, a silhueta inteira do cluster fica Vermelha e impede o drop.
 - **Deletar:** Apertar `Delete` apaga todas as peças do cluster de uma vez e gera **apenas 1 passo** no sistema de *UndoRedo*.
-

Questões em Aberto e Definições Futuras

Algumas dúvidas de arquitetura não cobertas pelas User Journeys básicas foram levantadas e respondidas diretamente cruzando os dados da engenharia reversa do formato original:

Q1. Diferenciação de Inventário (Parts Bin) vs Cenário Fixo

- **A Dúvida:** Quando estou criando um nível novo, como eu digo para a engine que a "Corda" que acabei de colocar deve ir para o inventário do jogador (Parts Bin) ao invés de aparecer travada no cenário?
- **Resposta Baseada no Clean Room (YAML):** A extração do *timgres* e os YAMLS de fase (`tim2_specs/user-journey-specification.md`) revelam que o comportamento da peça depende da sua máscara de bits. O `flag_1` contendo `0x0040` dita que a peça está *Locked* (travada).
- **Diretriz para a Godot:** O Editor deve possuir um botão de "Cadeado" no painel de propriedades ou um Handle de *Lock/Unlock* na peça. Se destrancada (sem o `0x0040`), a peça não aparece na tela na hora de jogar, mas sim contabilizada dentro do Parts Bin do jogador.

Q2. Ordem de Renderização (Z-Indexing Overlap)

- **A Dúvida:** Se o Level Designer colocar uma parede de cenário (ID 150) exatamente na mesma coordenada de uma Bola de Boliche (ID 4), quem renderiza na frente? A engine original dependia da ordem de *draw* (array order).
- **Diretriz para a Godot:** Para resolver definitivamente essa ambiguidade e evitar o *z-fighting*, o `PartContainer` deve atribuir ordens rigorosas via `CanvasItem.z_index`:
- `z_index = -1` para Cenários/Scenery (IDs ≥ 150).
- `z_index = 0` para peças estruturais fixas (Paredes, Canos).
- `z_index = 1` para peças ativas, molas e correias.
- `z_index = 2` para partículas, explosões e *overlays* de laser.

Q3. Limite de Vínculos Geométricos

- **A Dúvida:** O usuário pode puxar 3 ou 4 cordas a partir de uma mesma roldana usando o UC03?
- **Resposta Baseada no Clean Room: Não.** O struct binário original possui apenas 2 ponteiros para conexões mecânicas (`connected_1` e `connected_2`). A interface da Godot deve bloquear a tentativa de criar um 3º vínculo mecânico numa mesma peça, exibindo o cursor vermelho de erro.

Q4. Loop Infinito na Simulação

- **A Dúvida:** Se o Level Designer configurar uma máquina de moto-contínuo perfeita e apertar *Play* (UC06), a simulação rodará para sempre sem atingir o objetivo. O editor corre o risco de travar ou estourar memória se deixado ali?
- **Resposta Baseada no Clean Room:** Conforme catalogado no `user-journey-specification.md` (J08), o motor tem um limiar.
- **Diretriz para a Godot:** O `PhysicsSolver` do editor deve manter um contador global de *ticks*. Ao atingir `10.000 ticks` lógicos (~2.7 minutos de simulação) sem acionar o Win Condition, a engine deve disparar um Auto-Stop informando o *Timeout* da fase.

Roadmap Ágil de Implementação (14 Dias)

Para orientar a equipe de Engenharia (Track B) a desenvolver este editor partindo da "Baixa Interdependência" até a complexidade máxima, a implementação foi dividida em um Sprint de 6 Fases (14 dias):

Phase 1: Fundação & Câmera (Dias 1-2)

Objetivo: Ter um ambiente 2D funcional para testes de navegação. - **Entregáveis:** Setup do `LevelEditor.tscn`, `CanvasLayer` (UI), `SubViewport` (GameSpace), Controle de Câmera (Zoom/Pan) e Grid de 16x16. - **Leitura Obrigatória:** *Seção Estrutura da Árvore de Nós*.

Phase 2: Interface e Dicionários (Dias 3-4)

Objetivo: UI conectada a dados dinâmicos. - **Entregáveis:** Painel Lateral (Catálogo) fazendo parse do `part-name-catalog.json`. Geração de 149 `TextureButtons` iterativos. - **Leitura Obrigatória:** `build/phase-9/part-name-catalog.json` e `08_detailed_parts_catalog.md`.

Phase 3: Ferramentas Básicas (Dias 5-7)

Objetivo: O usuário pode colocar fantasmas das peças no grid, selecioná-las e apagá-las. - **Entregáveis:** Ferramenta `MODE_PLACE` (UC02) com validação visual de AABB overlap. Ferramenta `MODE_SELECT` (UC08) com Box Select e Delete. - **Leitura Obrigatória:** *Resoluções de Ambiguidades de Overlap para UC02 e UC08*.

Phase 4: Memória & Serialização (Dias 8-10)

Objetivo: Exportar/Importar as edições voláteis da tela para um `.YAML`. - **Entregáveis:** Módulo `LevelSerializer.gd`. Salvar (Export) e Carregar (Import) nativo usando `FileAccess`. - **Leitura Obrigatória:** `build/phase-5/yaml/ELEV1.yaml`.

Phase 5: Conexões Físicas e Win States (Dias 11-12)

Objetivo: Interações geométricas e Lógica de Vitória. - **Entregáveis:** Ferramenta `MODE_CONNECT` (UC03) com nós `Line2D` elásticos bloqueando ligações inválidas. Painel de Goal (UC05) com Target Rects verdes arrastáveis. - **Leitura Obrigatória:** *Questões Abertas Q3: Limites geométricos* e Jornada J06.

Phase 6: Integração Backend & Play (Dias 13-14)

Objetivo: Handover hermético para o motor de jogo. - **Entregáveis:** Botão **Play** (UC06) que salva um `temp_state.yaml` e despacha o `PhysicsSolver` de 60Hz. Botão **Stop** (UC07) que restaura silenciosamente o estado. Temporizador de *Timeout* de 10.000 ticks. - **Leitura Obrigatória:** `04_physics_engine.md` e *Q4. Loop infinito e regras de reset*.

12. Guia de Produção de Assets (Arte e Áudio)

Este documento é o manual definitivo para os Artistas (Pixel Art/2D) e Sound Designers responsáveis por criar a identidade visual e sonora da versão "Remake" do motor.

Como a arquitetura "Clean Room" dita que o código de simulação (`backend`) é **matematicamente idêntico ao jogo de 1994**, os novos assets de arte e áudio operam como uma "skin" sobre a física. Eles não podem alterar o comportamento do jogo.

1. A Regra de Ouro: A Arte Serve à Física

[!WARNING] Nunca modifique o *hitbox* (área de colisão matemática) de uma peça para acomodar um novo desenho. O seu desenho deve se acomodar ao hitbox!

Se a "Bola de Boliche" original possuía um raio de colisão rígido de 16px, a sua nova e ultra-detalhada bola de boliche pode ser renderizada com um sprite de 32x32px, mas a física continuará tratando-a como tendo raio de 16px. Se você desenhar uma bola gigantesca de 64x64px num arquivo `.PNG` com centro 16, ela vai "atravessar" as paredes visualmente.

Onde consultar as medidas exatas: - A base de dados principal é `docs/parts/`. - Toda peça possui uma tabela informando a dimensão em *pixels* (`sprite_w`, `sprite_h`). Use essas dimensões como o limite máximo do seu canvas.

2. Guia de Estilo (Pixel Art Moderna)

Para afastar o visual da paleta de 256-cores de 1994 sem perder o carisma, o guia estético dita:

2.1. Direção Estética

- **Slapstick Comedy & Steampunk:** As enghocas devem parecer saídas de um desenho animado do *Pica-Pau* ou *Tom & Jerry*, cruzado com máquinas a vapor extravagantes.
- **Formas e Silhuetas:** A legibilidade imediata é mais importante do que a textura. O jogador não pode confundir um dínamo com uma torradeira.

2.2. Restrições Técnicas de Imagem

- **Formato:** `.PNG` (fundo estritamente transparente via Alpha Channel).
- **Cor:** 32-bits (RGBA). Não há restrição de paleta, mas recomenda-se limitar as cores e focar no contraste (cel-shading ou iluminação direcional unificada) para evitar poluição visual nas fases mais caóticas.
- **Sombras Padrão:** Não crie sombras diretas (drop shadows) projetadas nos sprites de peças móveis! A engine da Godot aplicará shaders de oclusão e drop-shadow dinâmicos dependendo da iluminação da fase.

2.3. Alinhamento de Grade (Grid Snap)

A engine opera numa **grade base de 16x16 pixels**. Elementos que se encaixam na estrutura (como paredes, canos e esteiras) precisam ter *tiles* com arestas limpas (seamless tiles) que se alinhem perfeitamente nesses múltiplos (ex: 16x16, 32x32, 64x16).

3. Guia de Animação (Frame Sync)

[!IMPORTANT] A quantidade de *frames* (quadros) por animação **não é flexível**.

O antigo motor controlava as animações via máquinas de estado finitas (`.ANM`). O `state_counter` do backend roda a 60Hz.

- Se a tabela `docs/parts/part-004-balloon.md` diz que a transição de estado de explosão ("Burst") da bexiga dura **8 frames**, a sua nova animação precisa ser entregue em um *Sprite Sheet* contendo **exatamente 8 quadros**.
- Se você criar 12 quadros para deixar mais fluído, os 4 últimos quadros nunca serão tocados, e a peça já estará morta na engine física.

Formato de Entrega de Sprites: Recomendado usar *Sprite Sheets* de linha única (eixo horizontal) ordenados por estado. O *pivot/anchor* das imagens deve ser especificado no centro lógico.

4. Guia de Engenharia de Áudio (SFX e Música)

4.1. Música de Fundo

- **Formato e Qualidade:** `.OGG` ou `.WAV` (44.1kHz / 48kHz, 16-bit, Stereo). Substituindo o antigo sistema MIDI/OPL2 FM (11kHz).
- **Tonalidade:** As músicas precisam ter loops ininterruptos (*seamless looping*). Estilos recomendados vão do "Jazz Sintético" ao "Reggae Eletrônico". Devem ficar no fundo e não irritar, visto que os jogadores passarão horas na mesma tela pensando num quebra-cabeça.

4.2. Efeitos Sonoros (SFX) Reativos

- **Sem bloqueios (Non-blocking):** A física rege o jogo. Um som não pode ditar o atraso de uma colisão.
- O *AudioDispatcher* despacha eventos sonoros baseados em colisões.

- **Sobreposição (Concurrency):** Você precisa criar áudios curtos, secos e de forte "ataque". Se 15 bolas de boliche quicarem simultaneamente, os ecos longos se acumularão e estourarão as caixas de som do jogador. Se um som exigir decaimento (ex: sino grande), certifique-se de que a Godot utilize pools de áudio rotativos ou reduza os decaimentos do arquivo `.WAV`.
-

5. Diretrizes do Design de Níveis (UGC e YAML)

Se novos Level Designers forem contratados para adicionar desafios além dos 206 originais: 1. Trabalhem primeiramente no **Godot Sandbox Editor**. 2. As peças devem ser restritas àquelas já catalogadas em `build/phase-9/part-name-catalog.json`. 3. Evitem cenários que dependem de saltos milimétricos interconectados em redes de 20 peças; o hardware dos anos 90 permitia saltos por arredondamento impreciso que nós evitamos no remake (devido ao fix-point moderno). Confiem mais na lógica e em conectores de eletricidade/cordas do que em rebotes absurdos.

Capítulo 13: Especificações Técnicas e Formatos de Dados

Este capítulo condensa os detalhes de baixo nível extraídos durante o processo de engenharia reversa (Clean Room), servindo como guia definitivo para a reimplementação do motor físico, da máquina de estados e do decodificador de arquivos na Godot.

1. Máquinas de Estado (State Machines)

O The Incredible Machine 2 utiliza um sistema de estados composto por quatro conceitos interligados. Estas regras regem as 52 peças interativas e gerenciam animações e transições lógicas.

Conceitos Fundamentais

1. `state_counter`: Variável interna de física (INT16). Incrementada pelo motor a cada tick. Pode ser sobrescrita por colisões, gatilhos de proximidade ou eventos de som.
2. `current_state`: Cópia em runtime do `state_counter`, utilizada para checagem de resolução do puzzle e para buscar a animação correta no arquivo ANM.
3. `ANM state_id`: Identificador presente na Seção C dos arquivos de animação, que mapeia para o primeiro frame visual de um determinado estado. Existem 89 IDs de estado únicos no jogo.
4. **Gatilhos `SOLVE.RES (self / other)`**: Pares lógicos que determinam reações em cadeia. Quando a peça atual está no estado `self` e a peça colidida está no estado `other`, ocorre uma transição de estado.

Motor de Execução (Tick = 60Hz)

O motor avança os estados das peças móveis a cada tick físico. Para a maioria das peças: - O estado avança de um limite inferior (geralmente 0) até um limite superior (8). - Quando atinge o topo, sofre wrap-around (reseta para 0). - Estados iguais a zero (estado de repouso) muitas vezes possuem um delay randômico antes de começar a avançar, garantindo assincronicidade visual em peças repetidas (ex: velas piscando).

Mapeamento de Colisões

Quando duas peças (A e B) colidem, a engine lê ambos os `state_counter` e verifica o catálogo `SOLVE.RES`. Se a combinação satisfizer a regra de transição de uma das peças (ou de ambas), o `state_counter` é atualizado para o estado de destino (ex: Vela acesa encosta no pavio da Dinamite -> Dinamite passa para o estado "Acesa").

2. Formatos de Arquivo (Arquitetura de Dados)

Os arquivos originais extraídos (Levels e ANMs) possuem formatos binários Little-Endian específicos.

Formato de Fases (.LEV / .TIM)

Os arquivos de puzzle na versão TIM2/TIM3 geralmente usam o **Magic Number** `EF AC 13 01` (ou `14 01`). A estrutura de um arquivo de fase segue a ordem: 1. **Cabeçalho**: Magic Number + cor de fundo da paleta. 2. **Strings**: Título (null-terminated) e Descrição do Objetivo (null-terminated). 3. **Dicas (Hints)**: Array opcional contendo coordenadas e o texto da dica associada à Lupa. 4. **Header Físico**: Valores de Gravidade, Pressão de ar e ID da música. 5. **Blocos de Peças**: - Lista de peças fixas (não afetadas pela gravidade/mouse). - Lista de peças móveis (podem interagir livremente). 6. **Bloco de Solução**: As condições de vitória (qual peça deve estar em qual posição ou em qual estado).

Estrutura de Peça (Part Entry)

Cada instância de peça gravada na fase ocupa, via de regra, **48 bytes**. (Peças compostas como Correias e Cordas possuem tamanhos estendidos: 52 e 54 bytes, respectivamente). - **ID da Peça (INT16)**: Define o tipo. - **Flags_1**: Bitfield que define atributos. Ex: `0x40` (Locked/Inventário), `0x200` (Pode virar verticalmente), `0x400` (Horizontalmente), `0x1000` (Móvel). - **Flags_2**: Controle de conexões. Ex: `0x1` (Aceita correia), `0x4` (Aceita corda), `0x10/0x20` (Sprite invertido). - **Flags_3**: Atributos TIM2 específicos. Ex: `0x1` (Elétrico), `0x4` (Inflamável), `0x40` (Trancado no nível). - **Posição X/Y**: Coordenadas da peça em pixels, relativas ao grid de colisão. - **Ligações e Conexões**: Índices apontando para as IDs de outras peças com as quais estão conectadas por cabos ou correias (`belt_connected_part`, `rope_connected_part`).

Formato de Animação e Sprites (.ANM)

O jogo não guarda imagens soltas. Tudo fica dentro dos `.ANM`, que usam o padrão "Dynamix Chunk". A estrutura é dividida em blocos lógicos, onde os mais vitais para a renderização são: - **Sec B (Frames)**: Contém o tamanho (Width/Height) do sprite e o offset de textura. - **Sec C (State Map)**: Mapeia o `state_id` da lógica do jogo (ex: 4 = "Estourado") para a ID do frame visual na Seção B. - **Sec D (Draw Commands)**: Comandos de baixo nível (0 a 5) instruindo a paleta de cores, cópia em bloco ou transparência para compor a imagem em tela. O script de extração (Track A) transformou isso em `.png`. Na Godot, usaremos os PNGs extraídos mapeados pelos arquivos JSON gerados pelo extrator.

3. Detalhes Técnicos e Limitações

Determinismo Físico

O motor de The Incredible Machine opera estritamente de maneira **determinística a 60 FPS**. Qualquer variação de ponto flutuante ou frame-rate destrói a solução dos puzzles. - **Track B Requirement**: Na engine moderna (Godot), a física **deve** usar Inteiros (Ponto Fixo) ou `_physics_process` cravado a 60 Hz utilizando os mesmos multiplicadores do MS-DOS originais para Massa, Inércia, Gravidade (default: 272) e Pressão do ar (default: 67).

Z-Indexing Categórico

A ordem de renderização (Z-Index) das peças originais obedece implicitamente a categoria de flag delas: 1. Fundo (Background / Paredes) 2. Correias e Cordas 3. Peças do Cenário (Scenery) 4. Peças Móveis Interativas 5. Efeitos Especiais de Sobreposição (Explosões / Faíscas) 6. Cursor e UI.

Compatibilidade e Clean Room

Todo o mapeamento das reações (SOLVE.RES), propriedades de massa e centro de cintos (`belt_connect_pos_x/y`) já foram descompilados e serializados na pasta `build/phase-16/godot/`. A porta da Godot não precisa ler os arquivos binários diretamente em runtime, mas consumirá os JSONs limpos que agem como as tabelas de verdade.

14. Comportamento das Peças: Edição, Simulação e Gizmos

A arquitetura da Godot deve tratar as peças de forma bidimensional quanto ao seu ciclo de vida. Uma mesma peça possui **dois modos distintos de existência**: o Modo de Edição (`MODE_PLACE` / `MODE_SELECT`) e o Modo de Simulação (`MODE_PREVIEW`).

Este documento mapeia o comportamento das peças em cada um desses estados e categoriza todos os manipuladores visuais (*Gizmos*) exigidos pela ferramenta.

1. Modos de Vida da Peça

Modo de Edição (Edit Mode)

Enquanto a simulação está pausada e o usuário está arquitetando a engenhoca, o motor de física não atua. - **state_counter**: Fixo no estado inicial (geralmente 0, a não ser que alterado pelo usuário via Gizmo de Programação). - **Colisões Físicas**: Desativadas. A *hitbox* serve puramente para o evento de clique do Mouse (`input_event`) e validação estática de *overlap* na hora de instanciar a peça. - **Renderização de Gizmos**: A peça desenha *Overlays* ou instancia Nódulos Filhos (ex: botões de arrastar, âncoras de conexão elétricas piscando). - **Hover/Highlight**: O Mouse-over pinta a peça com um *outline* amarelo/branco e exibe o nome da peça (Tooltip).

Modo de Simulação (Simulation Mode)

Ativado assim que o usuário aperta *Play*. - **Obliteração de Gizmos**: Todas as alças, nós visuais de conexão, setas e highlights são destruídos ou ocultos (`hide()`). O jogador deve ver a tela limpa como em 1993. - **Loop de Física (60Hz)**: O `state_counter` é liberado para avançar. Temporizadores internos disparam. - **Colisões Resolvidas**: As máscaras de colisão passam a agir diretamente via `SOLVE.RES`. A sobreposição não é mais bloqueada pelo *mouse*, mas passa a gerar trancos, empurrões, destruição ou transições de estado.

2. Comportamento de Posicionamento (Placing Behavior)

O Godot lidará com a instância de peças através de uma matriz de restrições geométricas.

- **Snap-to-Grid**: Todo clique arredonda a posição global para o múltiplo mais próximo de 16 pixels.
 - **Validação de Sobreposição (Overlap)**: Antes da peça ser de fato fixada (`add_child`), uma consulta de shape (`PhysicsDirectSpaceState2D.intersect_shape`) testa a área. Se detectar uma peça da camada Ativa, o fantasma da peça sob o mouse fica **Vermelho** e a colocação é rejeitada.
 - **Exceções de Sobreposição (Scenery & Backgrounds)**: Peças do tipo Cenário (IDs ≥ 150) e Paredes de Tijolo têm máscaras ignoradas por esse teste e recebem `z_index = -1`, permitindo que o jogador crie fundos murados onde motores e ratos operam fisicamente por cima sem acusar erro de intercessão no editor.
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3. Arquétipos de Gizmos (Manipuladores Visuais)

Ao clicar numa peça no Modo de Edição, ela revela *Handles* (alças) ao seu redor. Abaixo, catalogamos os arquétipos e as peças que as utilizam.

3.1 Gizmos de Extensão (Stretch)

Peças que não têm tamanho fixo. Possuem handles circulares nas extremidades. Clicar e arrastar esses handles expande ou contrai a peça em blocos (geralmente repetindo o segmento do meio do sprite). * **Uso (Godot)**: Redimensiona o `.region_rect` do `Sprite2D` e recalcula o polígono de colisão para abranger o novo tamanho. * **Peças Aplicáveis**: * Paredes (Brick Wall, Caution Wall) * Rampas de Madeira (Wood Incline) * Correias de Transmissão (Belts) e Cordas (Ropes) - *O Stretch aqui não repete blocos, mas estica uma Line2D entre as polias.*

3.2 Gizmos de Orientação (Flip)

Peças direcionais. Ao ser selecionada, a peça revela ícones interativos flutuantes (`BUTTONS_f000.png` originais) que rodam ou espelham a mecânica da peça. * **Uso (Godot):** Altera a rotação visual, inverte as normais de colisão (eixo X ou Y) e salva a orientação desejada nos `flags` da peça. * **Peças Aplicáveis:** * Canhões (Cannons) - *Apontam para 6 direções diferentes.* * Lasers e Detectores - *Apontam para 4 direções ordinais.* * Luvas de Boxe (Boxing Gloves) e Aceleradores (Accelerators) - *Golpeiam na direção girada.* * Lupa (Magnifying Glass), Moinho de Vento (Pinwheel) e Espelhos. * Rampas inclinadas e Esteiras rolantes (Conveyor Belts) - *Muda a direção da força da esteira.*

3.3 Gizmos de Conexão (Anchors / Plugs)

Não são handles para serem arrastados isoladamente, mas "Tomadas" visuais. Durante o uso da ferramenta Corda (`MODE_CONNECT`), passar o mouse por cima ativa um brilho de atração (Snap Magnético). * **Uso (Godot):** São meros marcadores (`Marker2D` ou `Area2D` restritos). Quando o usuário arrasta uma linha até ele, o ID da peça é gravado em `connected_1` ou `connected_2`. * **Peças Aplicáveis:** * **Mecânicas:** Engrenagens, Motores, Polias. * **Elétricas:** Geradores, Tomadas (Outlets), Painéis Solares, Ventiladores. * **Tubulações:** Canos retos e curvos.

3.4 Gizmos de Programação (Program / Dialog)

Peças que operam logicamente ou têm atrasos customizados. Exibem um ícone de Computador. Ao clicar no gizmo, abre-se uma tela flutuante Modal. * **Uso (Godot):** Abre um painel GUI em `CanvasLayer` com botões e caixas de texto. * **Peças Aplicáveis:** * **Balões de Fala/Computador de Mensagem:** Campo de texto livre para *Strings*. * **Laser:** Rádio *Buttons* para selecionar a cor/comprimento de onda da luz. * **Timers (Egg Timer):** *Slider* ou número definindo o atraso em *ticks* da física. * **Interruptores de Parede:** Opção para já iniciar a simulação no estado Ligado (estado pré-setado ao invés do default).

3.5 Condições de Vitória (Goal Regions)

Não pertencem a uma peça específica, mas à arquitetura geométrica do nível. * **Uso (Godot):** Usam a ferramenta *Add Target Rect*. Elas funcionam exatamente como Gizmos de Extensão, gerando zonas (`Area2D`) flutuantes na tela que testam a presença de peças específicas quando a simulação entra em curso.

15. Especificação Comportamental (BDD)

Este capítulo documenta o comportamento esperado do motor de simulação e das ferramentas de edição usando cenários Gherkin (Given/When/Then). Os cenários servem como especificação executável — qualquer implementação (Python editor ou Godot) deve passar em todos os cenários listados.

Metodologia: Os cenários foram derivados do processo de engenharia reversa do formato binário (.LEV, .PAR, .ANM, SOLVE.RES). Cada cenário representa um comportamento observado ou deduzido a partir dos dados e do código de referência em `tim2_editor/`. Consulte `docs/clean_room_gdd/tim2_specs/state-machine-specification.md` para a referência completa da máquina de estados.

Feature: Physics Simulation Loop

O motor de simulação opera em exactly 60 ticks por segundo (60Hz), executando 7 passes em ordem estrita a cada tick.

Scenario: State counters advance each tick at 60Hz

- **Given** a level is running in Play mode
- **When** the simulation tick fires at 60Hz
- **Then** the engine executes all 7 passes in strict order: State Counters → Gravity → Behaviors → Viewport Clamp → Collisions → Proximity/Electrical → Constraints

- **And** no pass is skipped or reordered

Scenario: State counters update animation timers

- **Given** a part with `state_counter = N` and `LIMIT_UPPER = 8`
- **When** the State Counter pass executes
- **Then** if `state_counter != 0`, the engine increments `state_counter` by 1
- **And** if `state_counter >= LIMIT_UPPER`, it wraps to `LIMIT_RESET` (default 0)

Scenario: Zero-state parts apply random delay

- **Given** a part with `state_counter = 0` (idle state)
- **When** the State Counter pass executes
- **Then** the engine applies a random delay in the range [40, 239] ticks before allowing `state_counter` to advance
- **And** the delay is generated by the LCG RNG seeded per-part

Scenario: Sub-state countdown before state advance

- **Given** a part with `sub_counter > 0`
- **When** the State Counter pass executes
- **Then** the engine decrements `sub_counter` by 1
- **And** when `sub_counter` reaches 0, `state_counter` is set to `LIMIT_RESET`

Scenario: Gravity integration applies to unconstrained objects

- **Given** a moving part that is not connected to a rope or belt anchor
- **When** the Gravity Integration pass executes
- **And** the part is not marked as fixed (`flags_1` does not have 0x6000)
- **Then** the engine applies gravitational acceleration to `vel_y` based on the level's `gravity` value
- **And** the acceleration is looked up from the gravity tier table for the part's category

Scenario: Gravity direction respects level settings

- **Given** a level with `gravity` set to a specific value (default 272)
- **When** the Gravity Integration pass executes
- **Then** all non-fixed parts receive the same `gravity_y` direction (downward)
- **And** if `flags_3` bit 3 is set (wind), `gravity_x` may be non-zero

Scenario: Special behaviors execute per part type

- **Given** a part with a defined behavior handler (e.g., motor, cannon, animal AI)
- **When** the Behaviors pass executes
- **Then** the engine calls the appropriate behavior handler for that part type
- **And** the handler may modify `pos_x`, `pos_y`, `vel_x`, `vel_y`, or `state_counter`

Scenario: Viewport clamping prevents parts leaving screen bounds

- **Given** a moving part with position `pos_x`, `pos_y`
- **When** the Viewport Clamp pass executes
- **And** `pos_x < 0` or `pos_x > viewport_width` or `pos_y < 0` or `pos_y > viewport_height`
- **Then** the engine clamps the position to the viewport boundary
- **And** if the part hit the floor (`pos_y > max`), it bounces with COR applied to `vel_y`
- **And** if the part hit the ceiling (`pos_y < 0`), `vel_y` is negated

Scenario: Gravity field clamp at maximum values

- **Given** a level with gravity value
- **When** the gravity vector magnitude exceeds `±0x2600` or `±0x3000` depending on the phase
- **Then** the engine clamps the gravity to the maximum allowed value
- **And** the clamped value is used for all subsequent gravity integration passes

Scenario: AABB collision detection finds overlapping pairs

- **Given** two parts A and B with bounding boxes
- **When** the Collision pass executes
- **And** the AABB of A overlaps the AABB of B (axis-aligned bounding box intersection)
- **Then** the engine registers a collision event between A and B
- **And** it repositions both parts to eliminate the overlap along the minimum separation axis
- **And** it applies an impulse to both parts' velocities based on the Coefficient of Restitution (COR)

Scenario: Interconnected parts skip collision with each other

- **Given** two parts A and B where A is connected to B via rope or belt
- **When** the Collision pass executes
- **Then** the engine skips collision detection between A and B
- **And** the `connected_1` / `connected_2` fields are checked to determine skip eligibility

Scenario: Collision response applies COR-based velocity impulse

- **Given** two parts A and B in collision with overlap vector and normal
- **When** the Collision pass calculates the response
- **Then** the engine computes the relative velocity along the collision normal
- **And** applies an impulse magnitude of $J = -(1 + \text{COR}) * \text{rel_vel_n} / (1/mA + 1/mB)$
- **And** adds `J * normal` to each part's velocity

Scenario: Proximity detection activates electrical conduction

- **Given** a part with electrical capability (generator, outlet, switch)
- **When** the Proximity & Electrical pass executes
- **Then** the engine checks all connected parts in the electrical network
- **And** propagates the electrical state through the connection chain
- **And** activates any part whose `connected_N` index references an electrified source

Scenario: Laser beam traces line-of-sight to first obstacle

- **Given** a laser emitter (part type 91, 92, or 93) in active state
- **When** the Proximity & Electrical pass executes
- **Then** the engine casts a ray from the emitter in the laser's direction
- **And** checks for intersections with obstacle parts (walls, mirrors, detectors)
- **And** the beam travels until it hits a target or reaches maximum range

Scenario: Laser activates detector when beam intersects

- **Given** a laser beam path intersecting a laser-activated plug (part type 96)
 - **When** the Proximity & Electrical pass validates line-of-sight
 - **Then** the detector part transitions to active state
 - **And** any part connected to the detector receives electrical current
-

Feature: Constraint Systems

Os sistemas de constraint (cordas e correias) são resolvidos no passo 7 do loop de simulação.

Scenario: Rope maintains maximum distance between anchors

- **Given** a rope part (part type 10) connecting anchor A and anchor B
- **And** the rope has a defined maximum length (stored in the `behavior` field of the rope part)
- **When** the Constraint Solver pass executes
- **Then** the engine computes the current distance between A and B
- **And** if the distance exceeds the maximum rope length, it corrects the positions of A and/or B to satisfy the constraint
- **And** the correction moves the anchor that is not fixed (non-0x6000 flags)

Scenario: Single-pass rope solver corrects only when stretched beyond limit

- **Given** a rope connecting two parts A and B
- **When** the current distance between A and B is less than or equal to the rope's maximum length
- **Then** the engine makes no corrections (passive constraint)
- **And** when the distance exceeds the maximum, the engine applies a single correction step

Scenario: Pulley redirects rope force vector

- **Given** a pulley part (part type 7) with a rope passing over it
- **When** the Constraint Solver pass executes
- **Then** the engine treats the pulley as a redirection point for the rope
- **And** the effective distance becomes the sum of rope segments on each side of the pulley
- **And** force is transmitted through the pulley to the other rope segment

Scenario: Belt synchronizes angular velocity across all connected parts

- **Given** a belt part (part type 8) connecting multiple gears and motors
- **When** the Constraint Solver pass executes
- **Then** the engine sums the torques from all driving parts (motors with positive torque)
- **And** distributes the same angular velocity `±60px/s` standard to all driven parts
- **And** the sign of velocity depends on the relative orientation of the belt connection

Scenario: Belt angular velocity locked at ±60px/s standard

- **Given** a belt with at least one driving part (motor, pinwheel, etc.)
- **When** the Constraint Solver calculates the belt angular velocity
- **Then** the resulting angular velocity is locked to the standard `±60px/s`
- **And** all parts connected to the belt receive the same absolute angular velocity
- **And** the direction (sign) is preserved from the driving part's orientation

Scenario: Electrical current conducts through connected chain

- **Given** an electrical network with a power source (generator, solar panel, outlet)
 - **When** the Proximity & Electrical pass executes
 - **Then** the engine traverses the connection graph starting from the power source
 - **And** for each connected part, it sets the electrical state based on the source
 - **And** parts with `outlet_plugged_1` or `outlet_plugged_2` referencing the source become active
-

Feature: State Machine Execution

A máquina de estados governa transições visuais e lógicas das peças. Consulte `docs/clean_room_gdd/tim2_specs/state-machine-specification.md` para referência completa.

Scenario: `state_counter` increments each tick when not zero

- **Given** a part with `state_counter = N` where $N > 0$
- **When** the State Counter pass executes
- **And** the part's `sub_counter == 0`
- **Then** the engine increments `state_counter` by 1
- **And** if the new value reaches `LIMIT_UPPER`, it wraps to `LIMIT_RESET`

Scenario: `state_counter` wraps at `LIMIT_UPPER` to `LIMIT_RESET`

- **Given** a part with `state_counter = LIMIT_UPPER - 1` (typically 7 when `LIMIT_UPPER = 8`)
- **When** the State Counter pass increments the counter
- **Then** `state_counter` becomes equal to `LIMIT_RESET` (typically 0)
- **And** the animation state resets to the initial visual state

Scenario: `SOLVE.RES` trigger fires when self AND other states match

- **Given** a collision between part A and part B
- **And** part A is in state `self` (from `SOLVE.RES` entry)
- **And** part B is in state `other` (from `SOLVE.RES` entry)
- **When** the Collision pass resolves the interaction
- **Then** the engine looks up the `SOLVE.RES` table entry for the pair (A.state, B.state)
- **And** if a matching entry exists, it overrides `state_counter` of the target part to `anm_state`

Scenario: Collision with null `anm_state` uses trigger self value directly

- **Given** a `SOLVE.RES` entry where `anm_state` is null
- **When** the trigger condition is satisfied (self AND other states match)
- **Then** the target part's `state_counter` is set to the trigger's `self` value directly
- **And** no override to a different state occurs

Scenario: `SOLVE.RES` explicit `anm_state` overrides trigger chain

- **Given** a `SOLVE.RES` entry where `anm_state` is explicitly set
- **When** the trigger condition fires
- **Then** the target part jumps to the specific `anm_state` regardless of the current position in the trigger chain
- **And** this overrides the default sequential progression

Scenario: Orphan ANM states exist but are never reached by `SOLVE.RES`

- **Given** an ANM file with state IDs that are not referenced in any `SOLVE.RES` entry
- **When** the simulation runs
- **Then** those orphan states are never reached through collision triggers
- **And** they serve only as internal transition frames in continuous animations
- **And** the engine logs a warning about orphan states during level load (SM4 validation)

Scenario: Broken trigger chains do not crash the simulation

- **Given** a `SOLVE.RES` entry referencing a non-existent part type or state
- **When** the trigger condition would fire

- **Then** the engine gracefully skips the invalid entry
- **And** the simulation continues without crashing (SM5 validation)

Scenario: State limits vary by part type

- **Given** a part with custom state limits (e.g., `LIMIT_UPPER != 8`)
 - **When** the State Counter pass executes for that part
 - **Then** the engine uses the part-specific limit values from the lookup table at segment 0x5B41
 - **And** 17 parts are known to require custom limits beyond the default of 8
-

Feature: Solution Conditions

O sistema de avaliação de vitória verifica condições quando a simulação está em execução.

Scenario: `state_match` condition triggers when part reaches target state

- **Given** a solution condition of type `state_match` targeting part P with target state T
- **When** the simulation runs and part P's `state_counter` reaches T
- **Then** the engine marks the condition as satisfied
- **And** all solution conditions are checked at each tick

Scenario: `AABB_overlap` condition triggers when part center enters target region

- **Given** a solution condition of type `AABB_overlap` defining a target rectangle and a part P
- **When** the simulation runs and the center of part P falls within the target rectangle
- **Then** the engine marks the condition as satisfied
- **And** the target rectangle is defined in level coordinates (pixels)

Scenario: Multi-condition requires all conditions simultaneously satisfied

- **Given** a solution condition of type `multi_condition` containing N sub-conditions
- **When** the simulation evaluates the solution
- **Then** all N sub-conditions must be satisfied simultaneously
- **And** if any sub-condition fails, the overall condition is not met

Scenario: Level completes when all solution conditions are met

- **Given** a level with a set of solution conditions
 - **When** all conditions transition to satisfied state
 - **Then** the engine displays the victory screen
 - **And** the simulation halts (physics freezes, win dialog appears)
-

Feature: Editor Tool Modes

O editor de níveis em Godot (ou o protótipo Python) implementa modos de ferramenta que controlam a interação do usuário com o canvas.

Scenario: `MODE_PLACE` instances part at grid-snapped position

- **Given** the user is in `MODE_PLACE` (shortcut P)
- **When** the user clicks on the viewport canvas
- **Then** the editor snaps the click position to the nearest 16x16 pixel grid
- **And** instances a new part node at that position with default appearance
- **And** the part is added to the scene tree under `PartContainer`

Scenario: **MODE_PLACE** shows ghost preview following cursor

- **Given** the user is in **MODE_PLACE** with a part type selected
- **When** the mouse cursor moves over the viewport
- **Then** the editor displays a semi-transparent ghost sprite of the selected part
- **And** the ghost follows the cursor position in real-time

Scenario: **MODE_PLACE** rejects placement on overlap with active parts

- **Given** the user is in **MODE_PLACE** and hovers over an area occupied by an existing active part
- **When** the user attempts to click and place a new part
- **Then** the editor highlights the ghost in red (modulate = red)
- **And** rejects the placement, keeping the part in the catalog
- **And** parts with type ≥ 150 (scenery) are exempt from this check

Scenario: **MODE_SELECT** reveals contextual handles on selected part

- **Given** the user is in **MODE_SELECT** (shortcut S)
- **When** the user clicks on an existing part in the viewport
- **Then** the editor selects that part and displays contextual handles
- **And** the handles include Flip, Delete, and Program buttons appropriate to the part type

Scenario: Flip handle rotates part or mirrors axis per part type

- **Given** a part selected in **MODE_SELECT** with a Flip handle available
- **When** the user clicks the Flip handle
- **Then** the editor rotates the part by 90° or mirrors its axis (X or Y)
- **And** updates the `flags` field of the part to reflect the new orientation
- **And** the behavior changes accordingly (e.g., cannon fires in new direction)

Scenario: Delete handle removes part from scene

- **Given** a part selected in **MODE_SELECT** with a Delete handle available
- **When** the user clicks the Delete handle
- **Then** the editor removes the part node from the scene tree
- **And** frees the node memory with `queue_free()`

Scenario: Program handle opens modal dialog for logical settings

- **Given** a programmable part selected in **MODE_SELECT** (laser, balloon, timer, switch)
- **When** the user clicks the Program handle
- **Then** the editor opens a modal dialog appropriate to the part type
- **And** user settings (color, text, delay ticks, initial state) are saved to the part's properties

Scenario: **MODE_CONNECT** draws elastic line from anchor to cursor

- **Given** the user is in **MODE_CONNECT** (shortcut C)
- **When** the user clicks on a valid anchor point of part A (rope, belt, or electrical)
- **Then** the editor begins drawing an elastic Line2D from anchor A to the cursor
- **And** the line follows the cursor in real-time

Scenario: Valid connection anchor highlights green during drag

- **Given** the user is in **MODE_CONNECT** and dragging a connection line
- **When** the cursor passes over a valid connection anchor (compatible type)
- **Then** the anchor highlights green (snap magnetic assistance)

- **And** the line snaps to the anchor position

Scenario: Invalid connection target highlights red

- **Given** the user is in `MODE_CONNECT` and dragging a connection line
- **When** the cursor passes over an incompatible connection anchor
- **Then** the anchor highlights red
- **And** releasing the mouse does not create a connection

Scenario: `MODE_CONNECT` registers `belt_connected_part` or `rope_connected_part` IDs

- **Given** the user completes a connection in `MODE_CONNECT` by clicking on a valid target anchor
- **When** the connection is validated
- **Then** the editor writes the target part's ID to `connected_1` or `connected_2` of the source part
- **And** for ropes: `rope_connected_part` indices are updated
- **And** for belts: `belt_connected_part` indices are updated
- **And** for electrical: `outlet_plugged_1` / `outlet_plugged_2` indices are updated

Scenario: Box selection selects multiple parts in drag rectangle

- **Given** the user is in `MODE_SELECT` and drags a rectangular selection box
 - **When** the drag is released
 - **Then** all parts whose AABB intersects the selection box become selected
 - **And** multi-select handles appear for group operations (move, delete)
-

Feature: Simulation Controls

O controle de simulação gerencia a transição entre o modo de edição e o modo de jogo.

Scenario: Play serializes level state and starts 60Hz physics loop

- **Given** the editor is in edit mode with a level loaded
- **When** the user clicks the Play button (or presses Space in preview mode)
- **Then** the editor serializes the current parts array to `temp_state.yaml`
- **And** switches to simulation mode, starting the 60Hz physics loop
- **And** all gizmos and handles are hidden

Scenario: Stop halts physics and restores parts to exact pre-play positions

- **Given** the simulation is running (physics loop active)
- **When** the user clicks the Stop button
- **Then** the engine halts the physics loop
- **And** reloads the parts from `temp_state.yaml`
- **And** restores all positions, velocities, and states to the exact pre-play values
- **And** returns to edit mode with gizmos visible

Scenario: Fast Forward accelerates physics at 2x/4x speed

- **Given** the simulation is running
- **When** the user clicks the Fast Forward button
- **Then** the engine accelerates the tick rate to 2x (120Hz) or 4x (240Hz)
- **And** all physics calculations scale accordingly
- **And** the visual rendering updates at the accelerated rate

Scenario: Reset returns all parts to initial YAML state

- **Given** the simulation has run for N ticks
 - **When** the user clicks Reset (or Stop then reloads the level)
 - **Then** all parts are restored to their initial state from the loaded YAML
 - **And** `state_counter` values are reset to initial values
 - **And** `pos_x`, `pos_y`, `vel_x`, `vel_y` are restored to YAML values
-

Feature: Level Persistence

A persistência de níveis permite salvar e carregar níveis em formato YAML.

Scenario: Save exports current parts array to YAML with all fields

- **Given** the editor has a level in memory (loaded or newly created)
- **When** the user clicks Save (or Ctrl+S)
- **Then** the editor serializes the entire level to a `.yaml` file
- **And** includes: `level_info` (magic, background, title, goal), `environment` (gravity, pressure, music_track), `hints` array, `fixed_parts` array, `moving_parts` array
- **And** each part entry contains: `part_type`, `x`, `y`, `appearance`, `flags` (`flag_1`, `flag_2`, `flag_3`), `dimensions`, `physics` (behavior, belt_anchor, rope anchors, connections)

Scenario: Load reconstructs part instances from YAML with full fidelity

- **Given** a valid `.yaml` level file exists
- **When** the user clicks Load and selects the file
- **Then** the editor parses the YAML and reconstructs all part instances
- **And** validates the magic number (`EF AC 13 01` or `14 01`)
- **And** populates all fields: `part_type`, position, `appearance`, `flags`, physics properties
- **And** throws an error for invalid or corrupted YAML

Scenario: New warns when unsaved changes exist

- **Given** the editor has unsaved changes to the current level
- **When** the user clicks New (or Ctrl+N)
- **Then** the editor displays a confirmation dialog warning of data loss
- **And** if the user confirms, the current level is cleared
- **And** if the user cancels, the current level is preserved

Scenario: Level info editing preserves all metadata fields

- **Given** the user presses the I key to open the Level Info dialog
- **When** the dialog displays the current level metadata (title, goal, gravity, pressure, music_track, hints)
- **Then** the user can edit all 6 fields (title, goal_description, gravity, pressure, music_track, active_hints_count)
- **And** Tab navigates between fields, Enter confirms, Escape cancels
- **And** the changes are saved to the in-memory level state

Scenario: Level save uses YAML blocking structure matching decoder format

- **Given** the editor serializes a level for save
- **When** writing the YAML output
- **Then** the output matches the format consumed by `timgres/decoder/d_puzzle.py`
- **And** part entries use the blocking structure: `part_type`, `x`, `y`, `appearance`, `flags`, `flags_meanings`, `dimensions`, `physics`
- **And** connection fields are written as `connected_1`, `connected_2`, `outlet_plugged_1`, `outlet_plugged_2`

Scenario Count Summary

Feature	Scenarios
Physics Simulation Loop	14
Constraint Systems	7
State Machine Execution	8
Solution Conditions	4
Editor Tool Modes	12
Simulation Controls	4
Level Persistence	5
Total	54

Cross-References

This Chapter	Reference
Physics Loop (7 passes)	02_core_mechanics.md — "O Loop de Simulação (7 Passes)"
Constraint (rope/belt)	02_core_mechanics.md — "Sistemas de Restrição"
SOLVE.RES	03_part_catalog.md — "Pares de Gatilho (SOLVE.RES)"
State Machine	tim2_specs/state-machine-specification.md
Solution Conditions	05_level_design.md — "O Avaliador de Vitória"
Editor Tool Modes	11_godot_editor_specification.md — "Máquina de Estados da Ferramenta"
Part Behaviors	14_part_editor_behaviors.md
Validation	tim2_specs/validation-directives.md